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(54) **FN3 DOMAIN-SIRNA CONJUGATES AND
USES THEREOF**

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(57)

ABSTRACT

The present disclosure relates to compositions, such as
siRNA molecules and FN3 domains conjugated to the same,
as well as methods of making and using the molecules.

Specification includes a Sequence Listing.

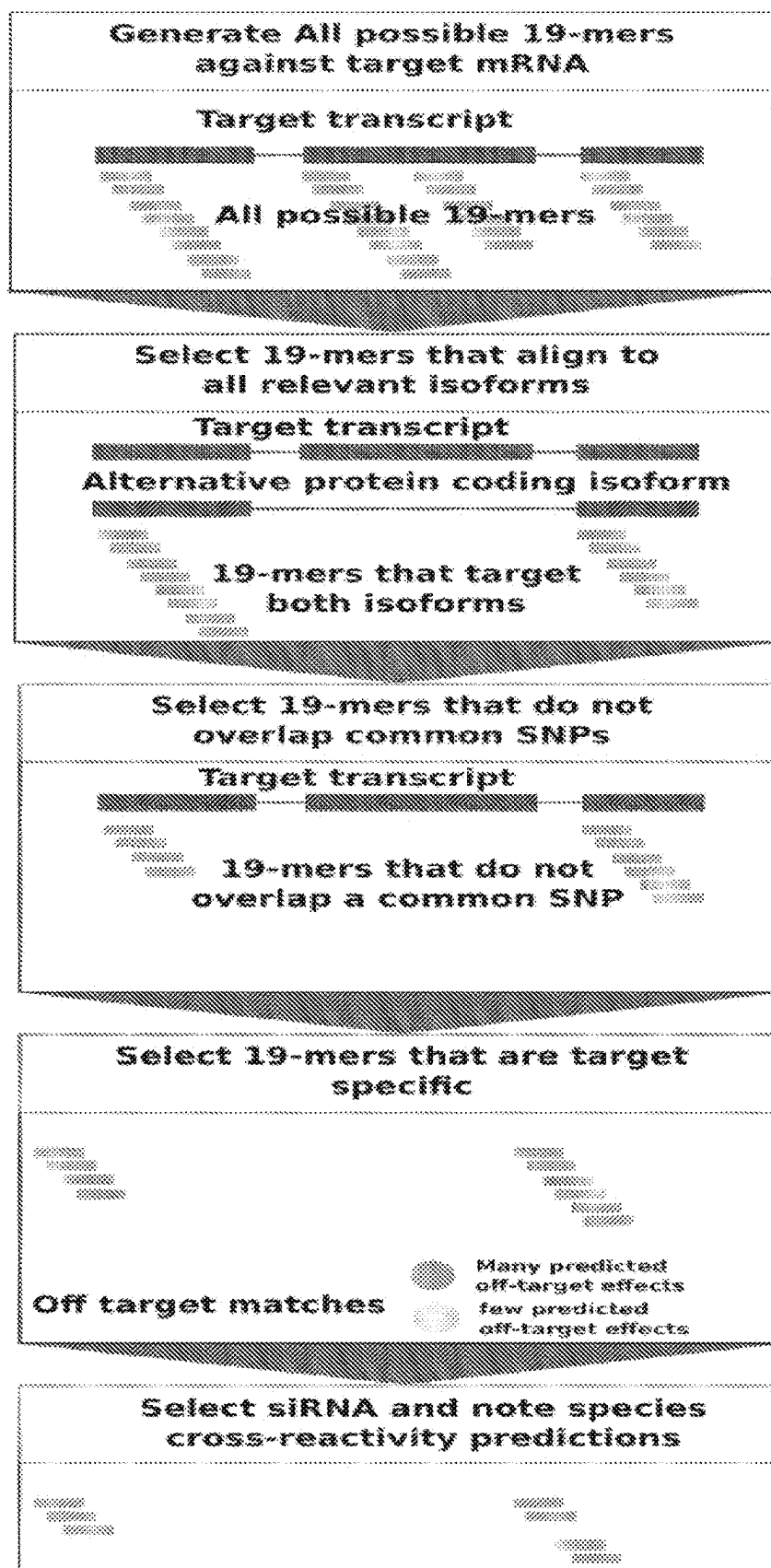


FIG. 1

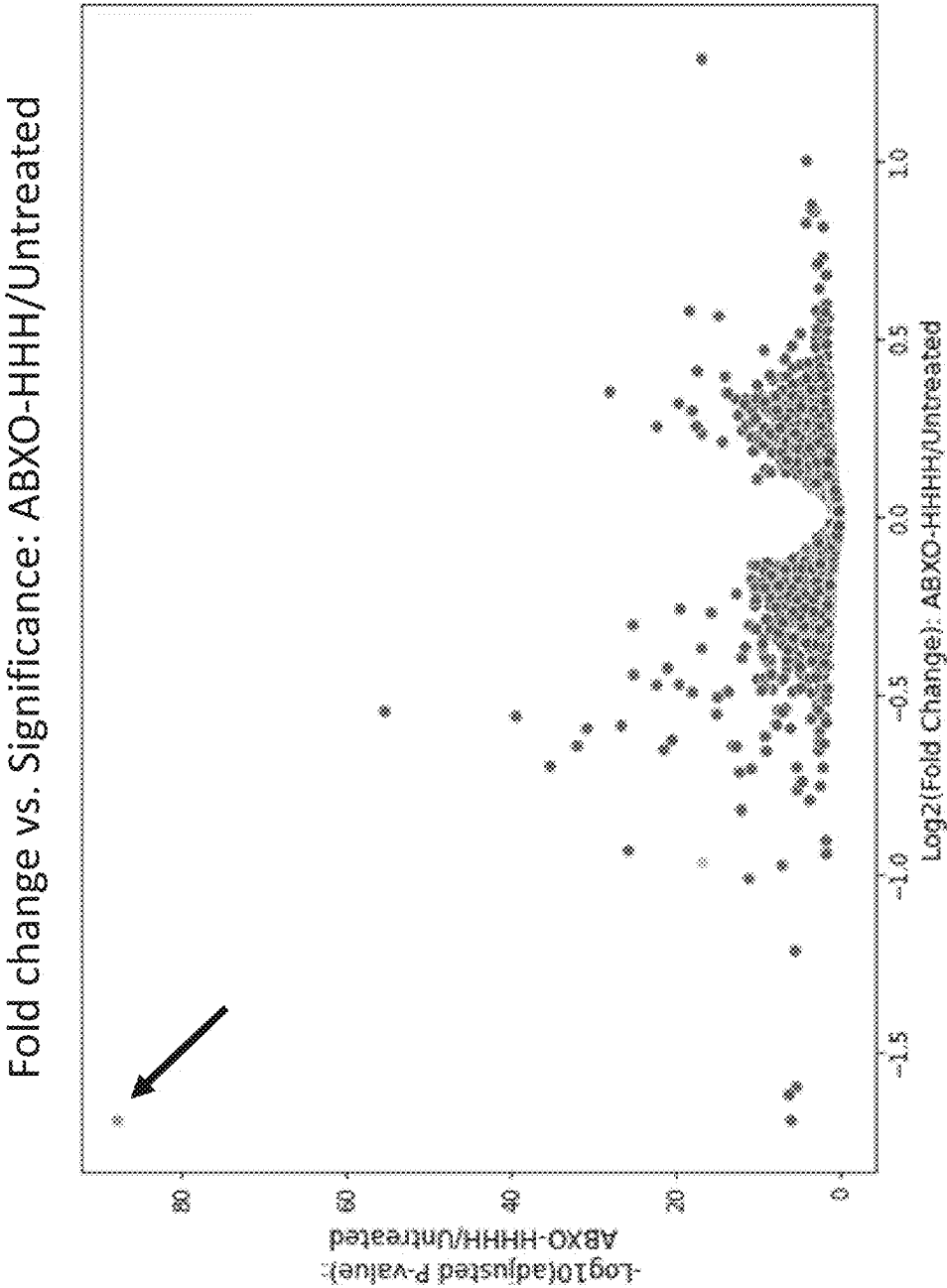


FIG.2

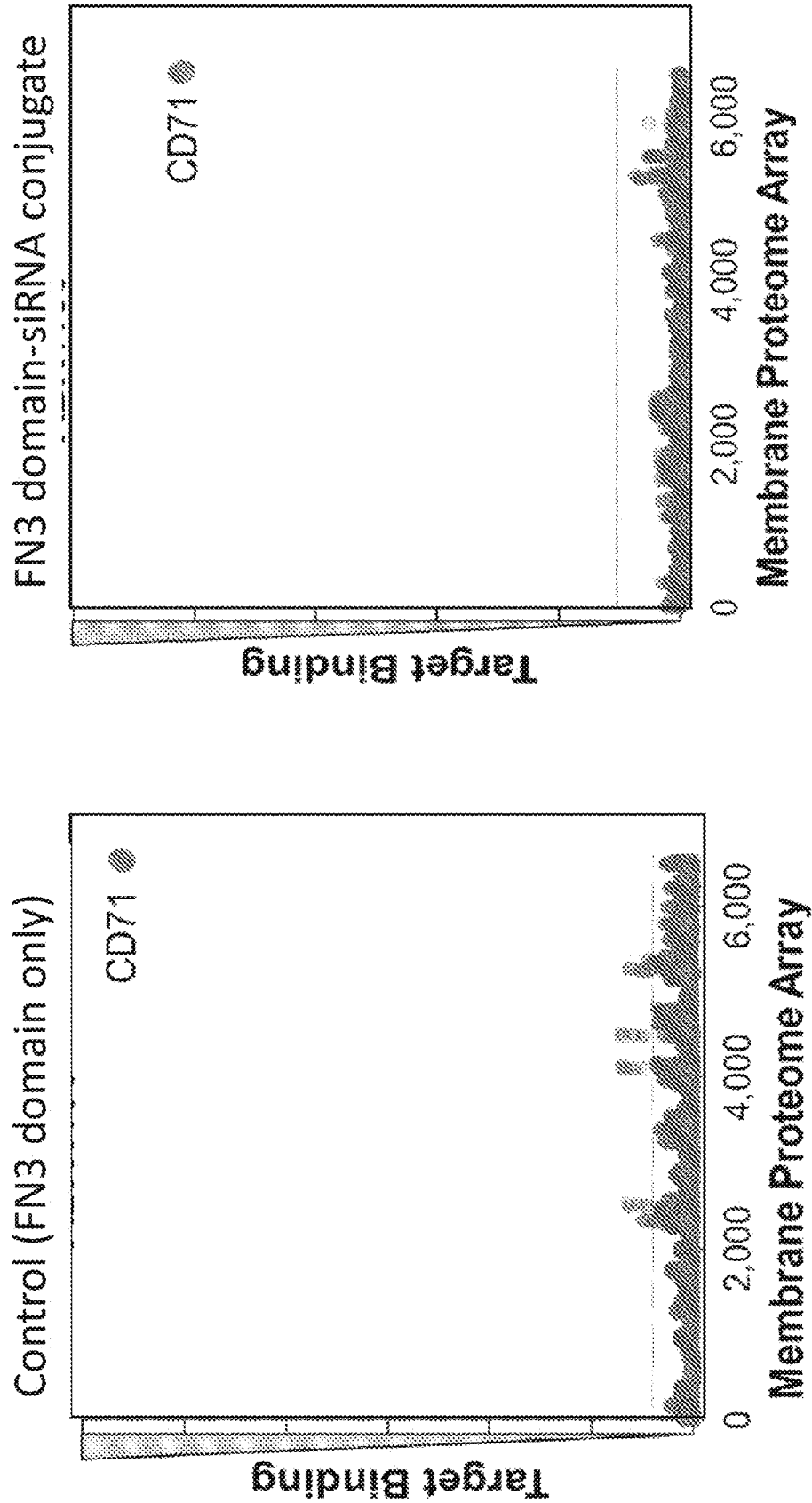


FIG.3

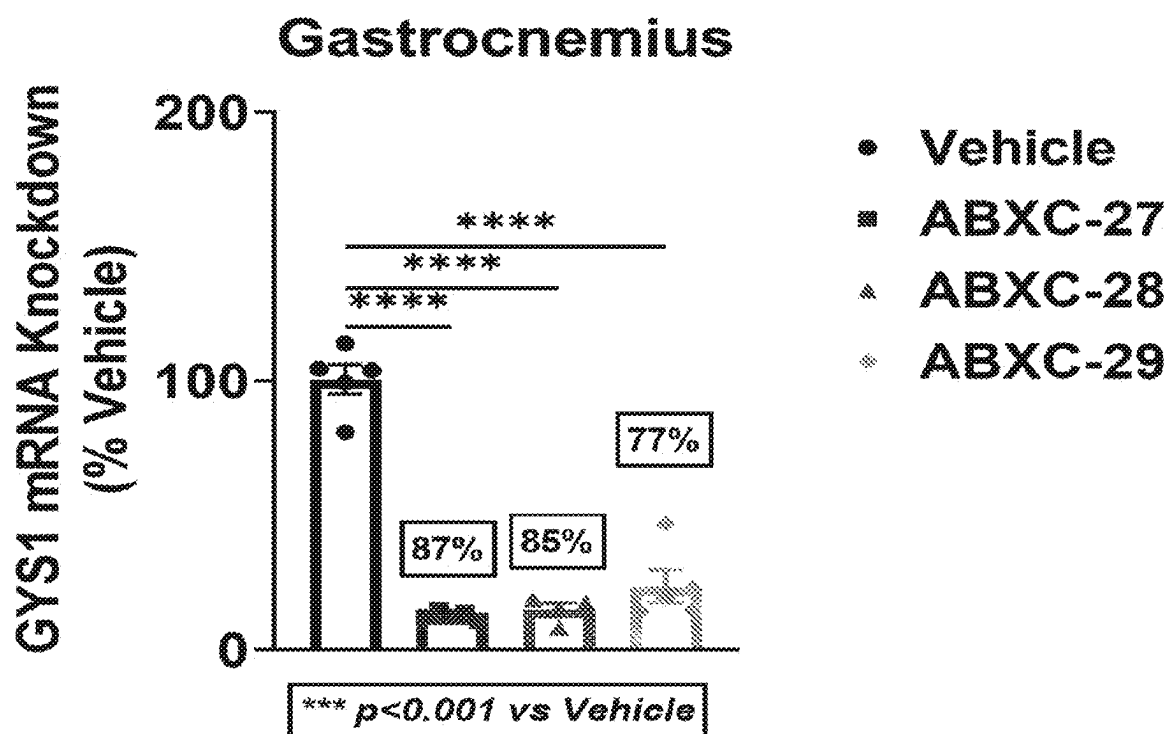


FIG. 4A

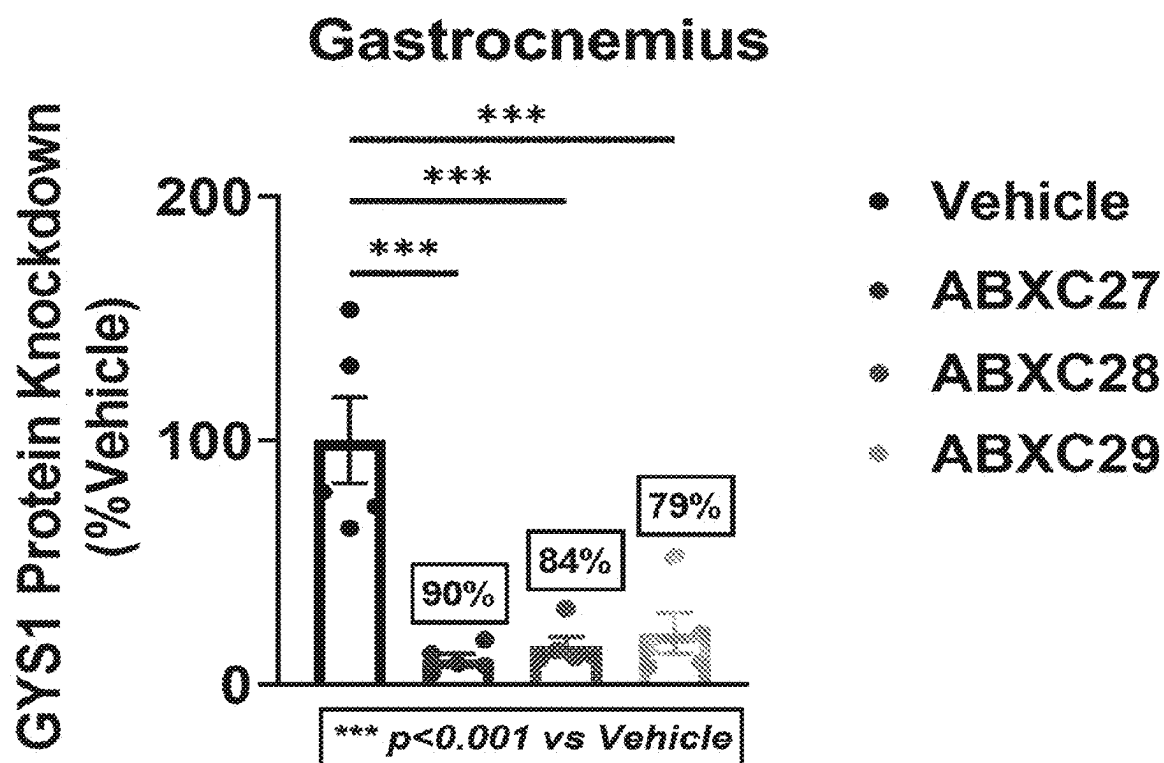


FIG. 4B

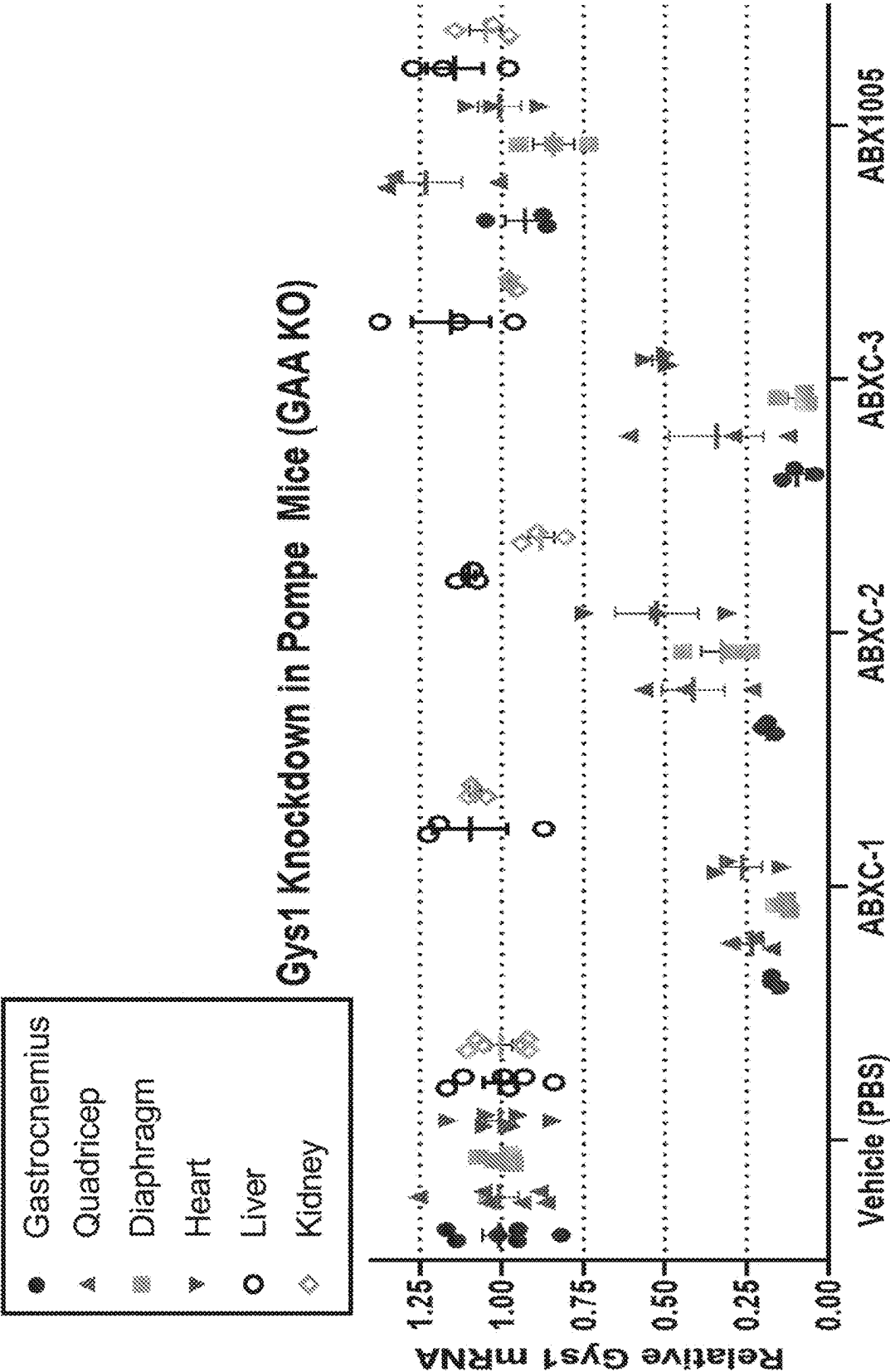


FIG. 5

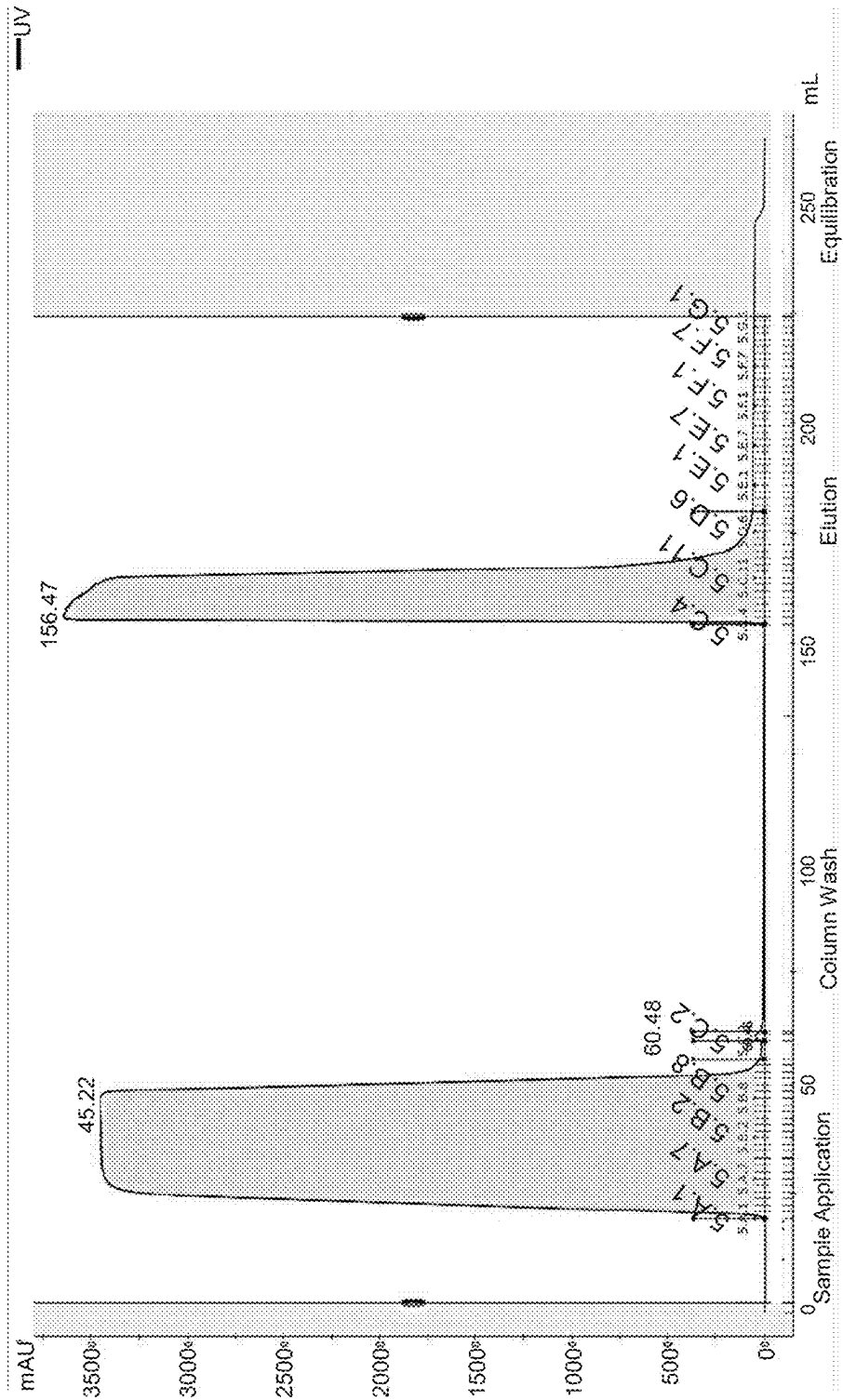


FIG. 6

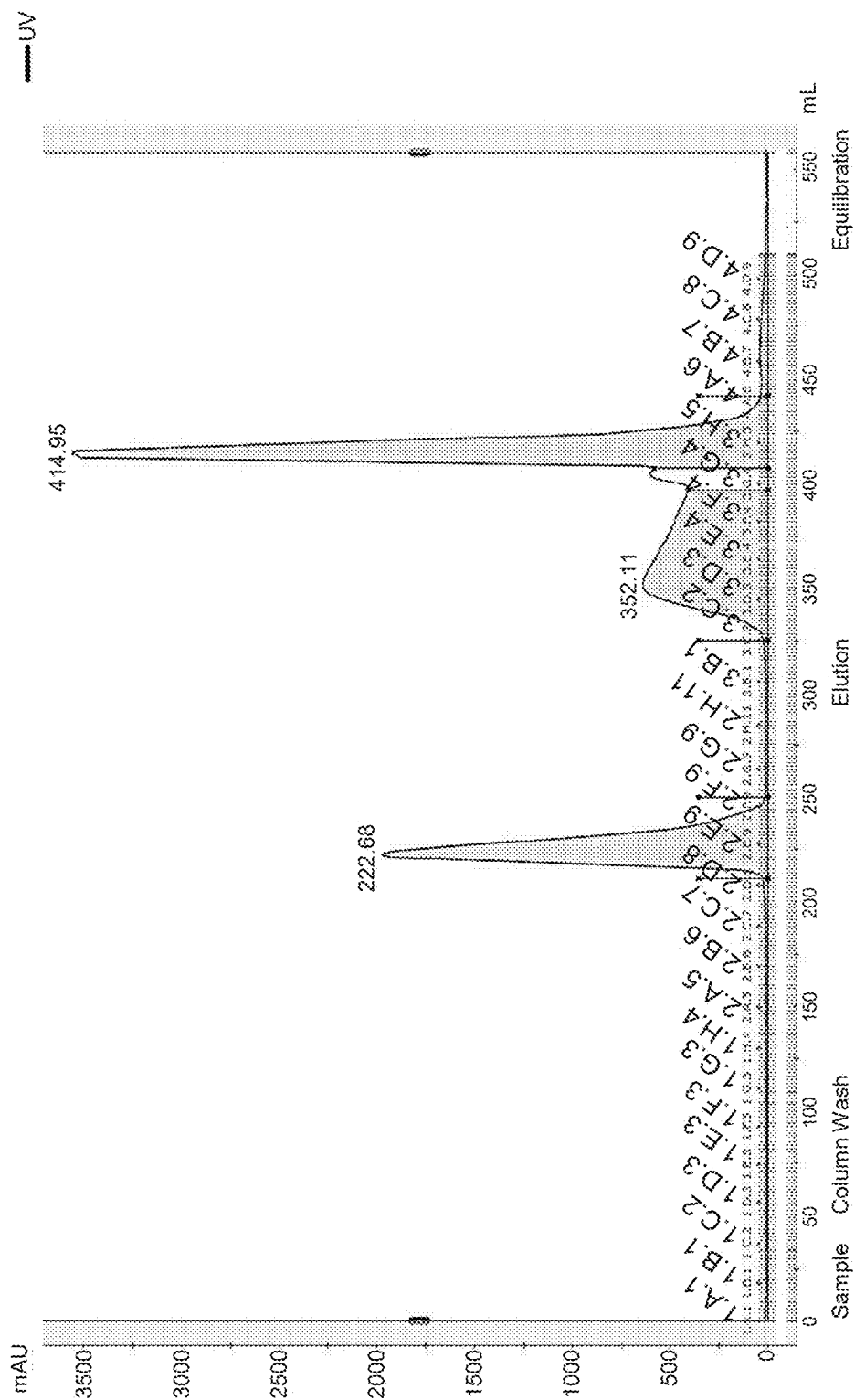


FIG. 7

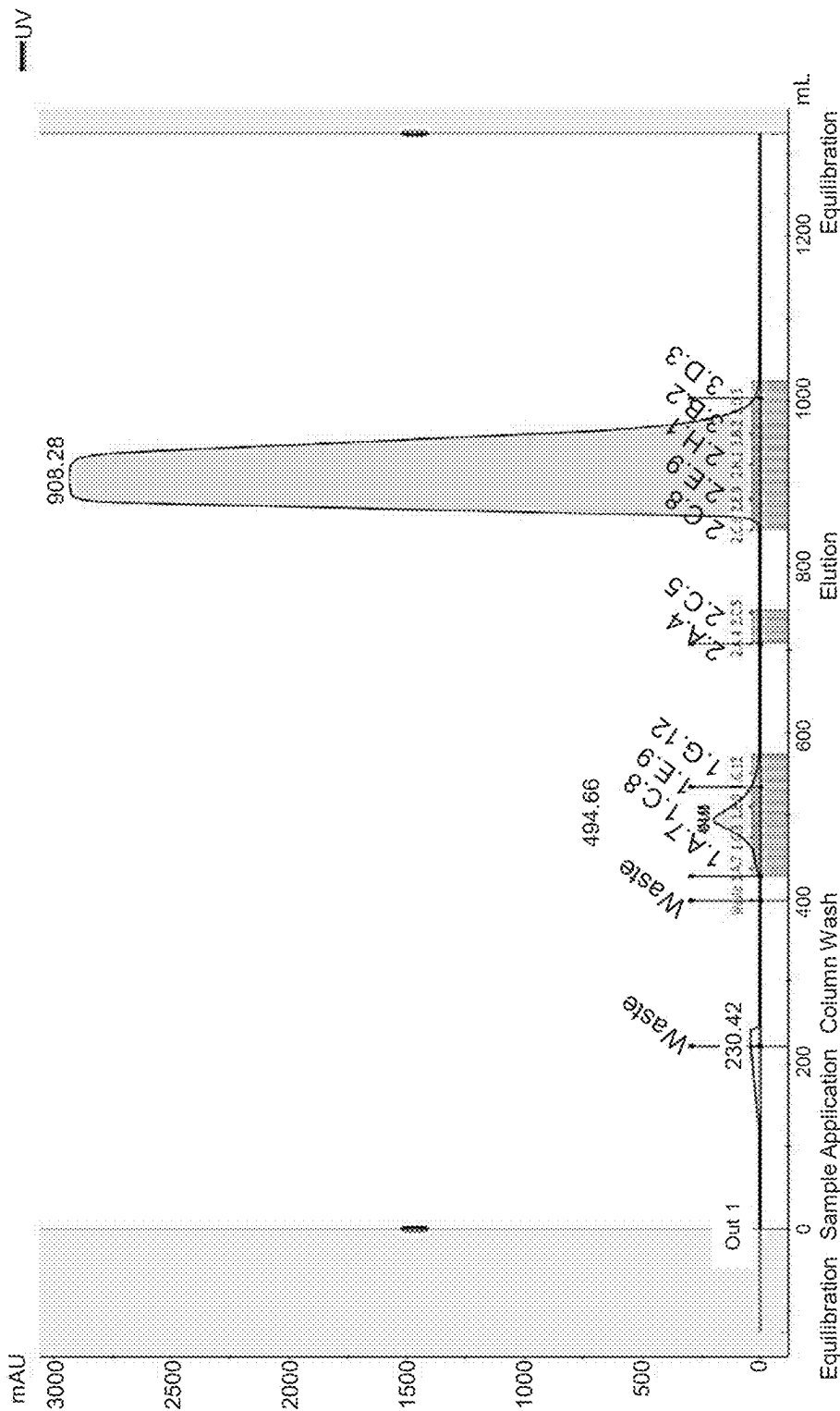


FIG. 8

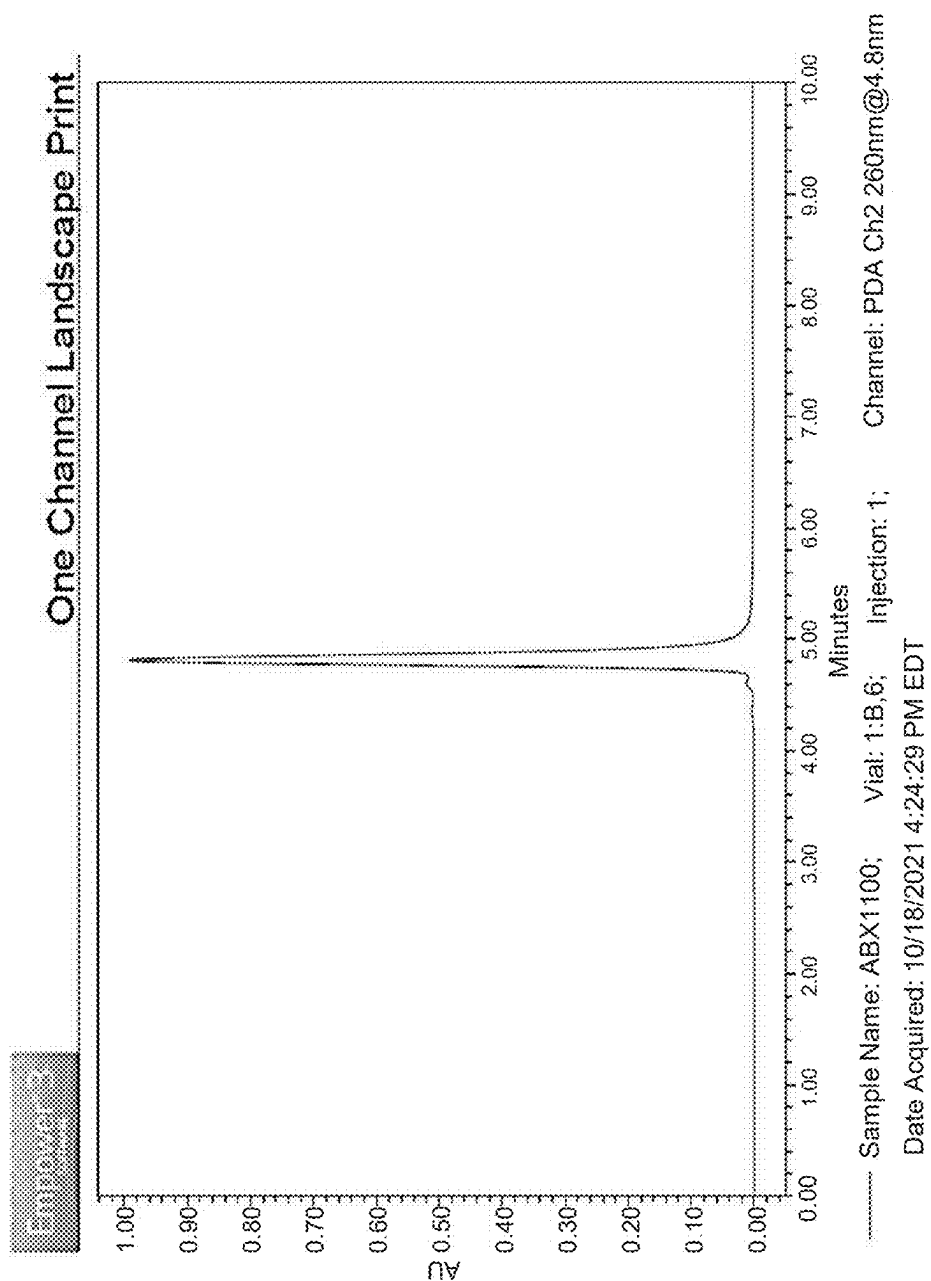


FIG. 9

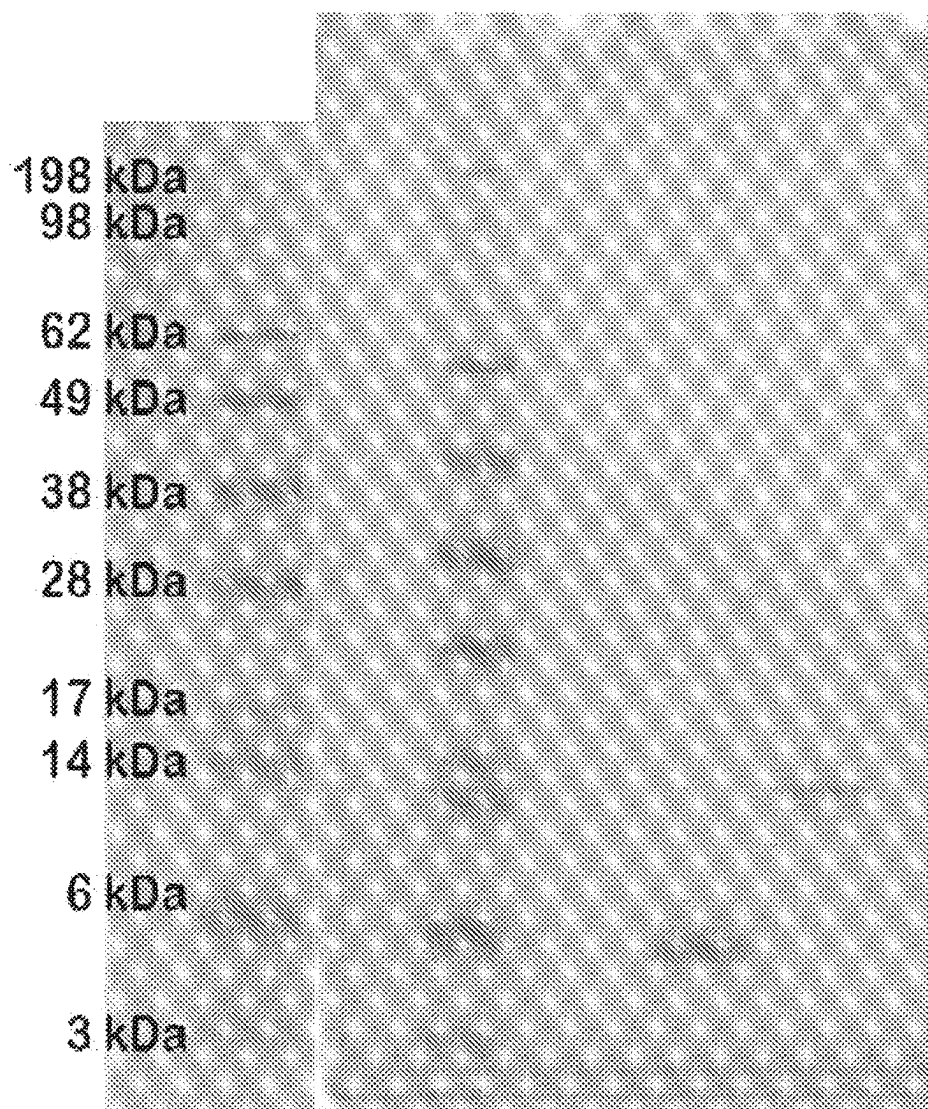


FIG. 10

FN3 DOMAIN-SIRNA CONJUGATES AND USES THEREOF

RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. Non-Provisional application Ser. No. 17/720,996, filed Apr. 14, 2022, now U.S. Pat. No. 12,239,710, which claims priority to U.S. Provisional Patent Application No. 63/174,776, filed Apr. 14, 2021, U.S. Provisional Patent Application No. 63/203,776, filed Jul. 30, 2021, and U.S. Provisional Application No. 63/324,437, filed Mar. 28, 2022, which are hereby incorporated by reference in their entirety.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on May 27, 2025, is named ROO-026USC1_SL.xml and is 2,082,415 bytes in size.

FIELD

[0003] The present embodiments relate to siRNA molecules that can be conjugated fibronectin type III domains (FN3) and methods of making and using the molecules.

BACKGROUND

[0004] Therapeutic nucleic acids include, e.g., small interfering RNA (siRNA), micro RNA (miRNA), antisense oligonucleotides, ribozymes, plasmids, immune stimulating nucleic acids, antisense, antagomir, antimir, microRNA mimic, supermir, U1 adaptor, and aptamer. In the case of siRNA or miRNA, these nucleic acids can down-regulate intracellular levels of specific proteins through a process termed RNA interference (RNAi). The therapeutic applications of RNAi are extremely broad, since siRNA and miRNA constructs can be synthesized with any nucleotide sequence directed against a target protein. To date, siRNA constructs have shown the ability to specifically down-regulate target proteins in both in vitro and in vivo models. In addition, siRNA constructs are currently being evaluated in clinical studies and have been approved for a variety of diseases.

[0005] However, two problems currently faced by siRNA constructs are, first, their susceptibility to nuclease digestion in plasma and, second, their limited ability to gain access to the intracellular compartment where they can bind the RISC (RNA-induced Silencing Complex) when administered systemically as the free siRNA or miRNA. Certain delivery systems, such as lipid nanoparticles formed from cationic lipids with other lipid components, such as cholesterol and PEG lipids, carbohydrates (such as GalNAc trimers) have been used to facilitate the cellular uptake of the oligonucleotides. However, these have not been shown to be successful in efficiently and effectively delivering siRNA to its intended target in tissues other than the liver.

[0006] There remains a need for compositions and methods for delivering siRNA to its intended cellular target. Further, what is needed is a FN3 domain with optimized properties for clinical use that can specifically bind to CD71, and methods of using such molecules for novel therapeutics that enable intracellular access via receptor mediated internalization of CD71. The present embodiments fulfill these needs as well as others.

SUMMARY

[0007] In some embodiments, siRNA conjugated to FN3 domains that bind CD71 protein are provided.

[0008] In some embodiments, FN3 domains are provided that comprise the amino acid sequence of any FN3 domain provided herein. In some embodiments, the FN3 domains bind to CD71. In some embodiments, the FN3 domains specifically bind to CD71.

[0009] In some embodiments, the composition comprises two FN3 domains connected by a linker, such as a flexible linker. In some embodiments, the two FN3 domains bind to different targets. In some embodiments, a first FN3 domain binds to CD71. In some embodiments, a second FN3 domain binds to a different target that is not CD71.

[0010] In some embodiments, oligonucleotides, such as dsRNA or siRNA molecules are provided herein. In some embodiments, the oligonucleotides have the sequences as provided herein, with or without the modifications provided herein. In some embodiments, the oligonucleotides are provided in a composition, such as a pharmaceutical composition. In some embodiments, the oligonucleotides are conjugated to a polypeptide.

[0011] In some embodiments, composition comprising one or more FN3 domains conjugated to a siRNA molecule are provided.

[0012] In some embodiments, a composition having a formula of $(X1)_n-(X2)_q-(X3)_y-L-X4$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; and X4 is an oligonucleotide molecule, wherein n, q, and y are each independently 0 or 1, are provided.

[0013] In some embodiments, a composition having a formula of $C-(X1)_n-(X2)_q-(X3)_y-L-X4$, wherein C is a polymer or albumin binding domain (ABD); X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; and X4 is an oligonucleotide molecule, wherein n, q, and y are each independently 0 or 1, are provided.

[0014] In some embodiments, a composition having a formula of $(X1)_n-(X2)_q-(X3)_y-L-X4-C$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer or albumin binding domain (ABD), wherein n, q, and y are each independently 0 or 1, are provided.

[0015] In some embodiments, a composition having a formula of $X4-L-(X1)_n-(X2)_q-(X3)_y$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; and X4 is an oligonucleotide molecule, wherein n, q, and y are each independently 0 or 1, are provided.

[0016] In some embodiments, a composition having a formula of $C-X4-L-(X1)_n-(X2)_q-(X3)_y$, wherein C is a polymer or albumin binding domain (ABD); X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; and X4 is an oligonucleotide molecule, wherein n, q, and y are each independently 0 or 1, are provided.

[0017] In some embodiments, a composition having a formula of $X4-L-(X1)_n-(X2)_q-(X3)_y-C$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer or albumin binding domain (ABD), wherein n, q, and y are each independently 0 or 1, are provided.

[0018] In some embodiments, a composition having a formula of $C-(X1)_n-(X2)_q-[L-X4]-(X3)_y$, wherein X1 is a

first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer or albumin binding domain (ABD), wherein n, q, and y are each independently 0 or 1, are provided.

[0019] In some embodiments, a composition having a formula of $(X1)_n-(X2)_q[L-X4]_y-(X3)_r-C$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer or albumin binding domain (ABD), wherein n, q, and y are each independently 0 or 1, are provided.

[0020] In some embodiments, pharmaceutical compositions comprising one or more of the compositions provided herein are provided.

[0021] In some embodiments, methods of treating Pompe Disease (GSD2, acid alpha-glucosidase (GAA) deficiency) in a subject in need thereof, the method comprising administering a composition provided herein are provided.

[0022] In some embodiments, methods of treating glycogen storage disease in a subject in need thereof, the method comprising administering a composition provided herein are provided. In some embodiments, the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease (GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type 11 Diabetes/diabetic nephropathy, Aldolase A Deficiency GSD12, Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), and adult polyglucosan body disease. In some embodiments, the glycogen storage disease is selected from the group consisting of Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11), Glucose transporter (GLUT2) deficiency, Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

[0023] In some embodiments, methods of treating cancer in a subject in need thereof, the method comprising administering to the subject a composition provided herein are provided.

[0024] In some embodiments, methods of treating a neurological condition and/or a brain tumor in a subject in need thereof, the method comprising administering to the subject a composition provided herein are provided. In some embodiments, the neurological condition is selected from the group consisting of Alzheimer's Disease, Amyotrophic Lateral Sclerosis, Parkinson's Disease, Lafora Disease, Pompe Disease, adult polyglucosan body disease, stroke, spinal cord injury, ataxia, Bell's Palsy, cerebral aneurysm, epilepsy, seizures, Guillain-Barre Syndrome, multiple sclerosis, muscular dystrophy, neurocutaneous syndromes, migraine, encephalitis, septicemia, and myasthenia gravis.

[0025] In some embodiments, methods of treating an autoimmune disease in a subject in need thereof, the method comprising administering to the subject a composition provided herein are provided. In some embodiments, the autoimmune disease is selected from the group consisting of rheumatoid arthritis, Hashimoto's autoimmune thyroiditis,

celiac disease, diabetes mellitus type 1, vitiligo, rheumatic fever, pernicious anemia/atrophic gastritis, alopecia areata, and immune thrombocytopenic purpura.

[0026] In some embodiments, a use of a composition as provided herein or of any of in the preparation of a pharmaceutical composition or medicament for treating cancer are provided. In some embodiments, the cancer is selected from the group consisting of acute myeloid leukemia, myelodysplastic syndromes, gastric cancer, clear cell renal cell carcinoma, clear cell carcinomas of the breast, clear cell carcinomas of the endometrium, clear cell carcinomas of the ovary, clear cell carcinomas of the uterus, hepatocellular carcinoma, pancreatic cancer, prostate cancer, soft tissue cancer, Ewings sarcoma, and non-small cell lung cancer.

[0027] In some embodiments, methods of reducing the expression of a target gene in a cell, the method comprising contacting the cell with a composition as provided herein are provided. In some embodiments, the cell is a tumor cell, a liver cell, a muscle cell, an immune cell, a dendritic cell, a heart cell, or a cell of the CNS.

[0028] In some embodiments, methods of selectively reducing GYS1 mRNA and protein in skeletal muscle. In certain embodiments, GYS1 mRNA and protein is not reduced in the liver and/or the kidney.

[0029] In some embodiments, isolated polynucleotides encoding the FN3 domains described herein are provided.

[0030] In some embodiments, a vector comprising the polynucleotides described herein are provided.

[0031] In some embodiments, a host cell comprising the vectors described herein are provided.

[0032] In some embodiments, methods of producing the FN3 domains are provided. In some embodiments, the method comprises culturing a host cell comprising a vector encoding or expressing the FN3 domain. In some embodiments, the method further comprises purifying the FN3 domain. In some embodiments, the FN3 domain binds CD71.

[0033] In some embodiments, pharmaceutical compositions comprising a FN3 domain that binds to CD71 linked to an oligonucleotide molecule and a pharmaceutically acceptable carrier are provided. In some embodiments, kits comprising one or more of the FN3 domains with or without the oligonucleotide molecules are provided.

DESCRIPTION OF THE DRAWINGS

[0034] FIG. 1 is a flow chart representing the properties assessed and considered for siRNA screening.

[0035] FIG. 2 is a graph of an RNA sequencing experiment which identifies the changes in the transcriptome after cells are transfected with siRNA pair HHH, wherein the arrow identifies the significant decrease in GYS1 transcript.

[0036] FIG. 3 provides the results of a target binding assay using over 6,000 receptors in the proteome array, wherein the data demonstrates that CD71 is the exclusive binding target of the FN3 domain.

[0037] FIG. 4A demonstrates the knockdown of GYS1 mRNA in mouse gastrocnemius muscle using 3 difference FN3 domain-siRNA conjugates compared with vehicle alone. FIG. 4B demonstrates the knockdown of GYS1 protein in mouse gastrocnemius muscle using 3 difference FN3 domain-siRNA conjugates compared with vehicle alone.

[0038] FIG. 5 demonstrates the GYS1 knockdown is highly specific for skeletal muscle using 3 different FN3 domain-siRNA conjugates compared with a siRNA to a different target (AHA-1).

[0039] FIG. 6 is an example of Histrap chromatogram for purification of conjugates (Tagged proteins).

[0040] FIG. 7 is an example of HIC chromatogram for purification of conjugate (tagless proteins).

[0041] FIG. 8 is an example of Ion-exchange chromatogram for purification of conjugate (Tagged/tagless protein).

[0042] FIG. 9 is an example of an analytical SEC of the centyrin-oligonucleotide conjugate.

[0043] FIG. 10 is an example of SDS PAGE Gel of the conjugate.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0044] As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to “a cell” includes a combination of two or more cells, and the like.

[0045] “Fibronectin type III (FN3) domain” (FN3 domain) refers to a domain occurring frequently in proteins including fibronectins, tenascin, intracellular cytoskeletal proteins, cytokine receptors and prokaryotic enzymes (Bork and Doolittle, Proc Nat Acad Sci USA 89:8990-8994, 1992; Meinke et al., J Bacteriol 175:1910-1918, 1993; Watanabe et al., J Biol Chem 265:15659-15665, 1990). Exemplary FN3 domains are the 15 different FN3 domains present in human tenascin C, the 15 different FN3 domains present in human fibronectin (FN), and non-natural synthetic FN3 domains as described for example in U.S. Pat. No. 8,278,419. Individual FN3 domains are referred to by domain number and protein name, e.g., the 3rd FN3 domain of tenascin (TN3), or the 10th FN3 domain of fibronectin (FN10). As used throughout, “centyrin” also refers to a FN3 domain. Further, FN3 domains as described herein are not antibodies as they do not have the structure of a variable heavy (V_H) and/or light (V_L) chain.

[0046] As used herein, “autoimmune disease” refers to disease conditions and states wherein the immune response of an individual is directed against the individual’s own constituents, resulting in an undesirable and often debilitating condition. As used herein, “autoimmune disease” is intended to further include autoimmune conditions, syndromes, and the like. Autoimmune diseases include, but are not limited to, Addison’s disease, allergy, allergic rhinitis, ankylosing spondylitis, asthma, atherosclerosis, autoimmune diseases of the ear, autoimmune diseases of the eye, autoimmune atrophic gastritis, autoimmune hepatitis, autoimmune hemolytic anemia, autoimmune parotitis, autoimmune uveitis, celiac disease, primary biliary cirrhosis, benign lymphocytic anitis, COPD, colitis, coronary heart disease, Crohn’s disease, diabetes (Type I), depression, diabetes, including Type 1 and/or Type 2 diabetes, epididymitis, glomerulonephritis, Goodpasture’s syndrome, Graves’ disease, Guillain-Barre syndrome, Hashimoto’s disease, hemolytic anemia, idiopathic thrombocytopenic purpura, inflammatory bowel disease (IBD), immune response to recombinant drug products, e.g., factor VII in hemophilia, juvenile idiopathic arthritis, systemic lupus erythematosus, lupus nephritis, male infertility, mixed connective tissue disease, multiple sclerosis, myasthenia gravis, oncology, osteoarthritis, pain, primary myxedema, pemphigus, pernicious anemia, polymyositis, psoriasis, psoriatic arthritis, reactive arthritis, rheumatic fever, rheumatoid arthritis, sarcoidosis, scleroderma, Sjogren’s syndrome, spondyloarthropathies, sympathetic ophthalmia, T-cell lymphoma, T-cell acute lymphoblastic leukemia, testicular antiocentric T-cell lymphoma, thyroiditis, transplant rejection, ulcerative

colitis, autoimmune uveitis, and vasculitis. Autoimmune diseases include, but are not limited to, conditions in which the tissue affected is the primary target, and in some cases, the secondary target. Such conditions include, but are not limited to, AIDS, atopic allergy, bronchial asthma, eczema, leprosy, schizophrenia, inherited depression, transplantation of tissues and organs, chronic fatigue syndrome, Alzheimer’s disease, Parkinson’s disease, myocardial infarction, stroke, autism, epilepsy. Arthus’s phenomenon, anaphylaxis, and alcohol and drug addiction.

[0047] The term “capture agent” refers to substances that bind to a particular type of cells and enable the isolation of that cell from other cells. Exemplary capture agents are magnetic beads, ferrofluids, encapsulating reagents, molecules that bind the particular cell type and the like.

[0048] “Sample” refers to a collection of similar fluids, cells, or tissues isolated from a subject, as well as fluids, cells, or tissues present within a subject. Exemplary samples are tissue biopsies, fine needle aspirations, surgically resected tissue, organ cultures, cell cultures and biological fluids such as blood, serum and serosal fluids, plasma, lymph, urine, saliva, cystic fluid, tear drops, feces, sputum, mucosal secretions of the secretory tissues and organs, vaginal secretions, ascites fluids, fluids of the pleural, pericardial, peritoneal, abdominal and other body cavities, fluids collected by bronchial lavage, synovial fluid, liquid solutions contacted with a subject or biological source, for example, cell and organ culture medium including cell or organ conditioned medium and lavage fluids and the like.

[0049] “Substituting” or “substituted” or “mutating” or “mutated” refers to altering, deleting or inserting one or more amino acids or nucleotides in a polypeptide or polynucleotide sequence to generate a variant of that sequence.

[0050] “Variant” refers to a polypeptide or a polynucleotide that differs from a reference polypeptide or a reference polynucleotide by one or more modifications for example, substitutions, insertions or deletions.

[0051] “Specifically binds” or “specific binding” refers to the ability of a FN3 domain to bind to its target, such as CD71, with a dissociation constant (K_D) of about 1×10^{-6} M or less, for example about 1×10^{-7} M or less, about 1×10^{-8} M or less, about 1×10^{-9} M or less, about 1×10^{-10} M or less, about 1×10^{-11} M or less, about 1×10^{-12} M or less, or about 1×10^{-13} M or less. Alternatively, “specific binding” refers to the ability of a FN3 domain to bind to its target (e.g CD71) at least 5-fold above a negative control in standard solution ELISA assay. Specific binding can also be demonstrated using the proteome array as described herein and shown in FIG. 3. In some embodiments, a negative control is an FN3 domain that does not bind CD71. In some embodiment, an FN3 domain that specifically binds CD71 may have cross-reactivity to other related antigens, for example to the same predetermined antigen from other species (homologs), such as *Macaca Fascicularis* (cynomolgous monkey, cyno) or Pan troglodytes (chimpanzee).

[0052] “Library” refers to a collection of variants. The library may be composed of polypeptide or polynucleotide variants.

[0053] “Stability” refers to the ability of a molecule to maintain a folded state under physiological conditions such that it retains at least one of its normal functional activities, for example, binding to a predetermined antigen such as CD71.

[0054] “CD71” refers to human CD71 protein having the amino acid sequence of SEQ ID NOs: 2 or 5. In some embodiments, SEQ ID NO: 2 is full length human CD71

protein. In some embodiments, SEQ ID NO: 5 is the extracellular domain of human CD71.

[0055] “Tencon” refers to the synthetic fibronectin type III (FN3) domain having the consensus sequence shown in SEQ ID NO:1

LPAPKNLVSEVTEDSLRLSWTAPDAAFDSFLIQYQSEKVGGEAINLTVP

GSERSYDLTGLKPGTEYTVSIYGVKGGRHSNPLSABFTT

and described in U.S. Pat. Publ. No. 2010/0216708.

[0056] A “cancer cell” or a “tumor cell” refers to a cancerous, pre-cancerous or transformed cell, either in vivo, ex vivo, and in tissue culture, that has spontaneous or induced phenotypic changes that do not necessarily involve the uptake of new genetic material. Although transformation can arise from infection with a transforming virus and incorporation of new genomic nucleic acid, or uptake of exogenous nucleic acid, it can also arise spontaneously or following exposure to a carcinogen, thereby mutating an endogenous gene. Transformation/cancer is exemplified by, e.g., morphological changes, immortalization of cells, aberrant growth control, foci formation, proliferation, malignancy, tumor specific markers levels, invasiveness, tumor growth or suppression in suitable animal hosts such as nude mice, and the like, in vitro, in vivo, and ex vivo (Freshney, Culture of Animal Cells: A Manual of Basic Technique (3rd ed. 1994)).

[0057] A “dendritic cell” refers to a type of antigen-presenting cell (APC) that form an important role in the adaptive immune system. The main function of dendritic cells is to present antigens to T lymphocytes, and to secrete cytokines that may further modulate the immune response directly or indirectly. Dendritic cells have the capacity to induce a primary immune response in the inactive or resting naïve T lymphocytes.

[0058] An “immune cell” refers to the cells of the immune system categorized as lymphocytes (T-cells, B-cells and NK cells), neutrophils, or monocytes/macrophages. These are all types of white blood cells.

[0059] “Vector” refers to a polynucleotide capable of being duplicated within a biological system or that can be moved between such systems. Vector polynucleotides typically contain elements, such as origins of replication, polyadenylation signal or selection markers that function to facilitate the duplication or maintenance of these polynucleotides in a biological system. Examples of such biological systems may include a cell, virus, animal, plant, and reconstituted biological systems utilizing biological components capable of duplicating a vector. The polynucleotide comprising a vector may be DNA or RNA molecules or a hybrid of these.

[0060] “Expression vector” refers to a vector that can be utilized in a biological system or in a reconstituted biological system to direct the translation of a polypeptide encoded by a polynucleotide sequence present in the expression vector.

[0061] “Polynucleotide” refers to a synthetic molecule comprising a chain of nucleotides covalently linked by a sugar-phosphate backbone or other equivalent covalent chemistry. cDNA is a typical example of a polynucleotide.

[0062] “Polypeptide” or “protein” refers to a molecule that comprises at least two amino acid residues linked by a peptide bond to form a polypeptide. Small polypeptides of less than about 50 amino acids may be referred to as “peptides”.

[0063] “Valent” refers to the presence of a specified number of binding sites specific for an antigen in a molecule. As such, the terms “monovalent”, “bivalent”, “tetravalent”, and “hexavalent” refer to the presence of one, two, four and six binding sites, respectively, specific for an antigen in a molecule.

[0064] “Subject” includes any human or nonhuman animal. “Nonhuman animal” includes all vertebrates, e.g., mammals and non-mammals, such as nonhuman primates, sheep, dogs, cats, horses, cows chickens, amphibians, reptiles, etc. Except when noted, the terms “patient” or “subject” are used interchangeably.

[0065] “Isolated” refers to a homogenous population of molecules (such as synthetic polynucleotides or a polypeptide such as FN3 domains) which have been substantially separated and/or purified away from other components of the system the molecules are produced in, such as a recombinant cell, as well as a protein that has been subjected to at least one purification or isolation step. “Isolated FN3 domain” refers to an FN3 domain that is substantially free of other cellular material and/or chemicals and encompasses FN3 domains that are isolated to a higher purity, such as to 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% purity.

[0066] In some embodiments, a composition comprising a polypeptide, such as a polypeptide comprising a FN3 domain, linked to an oligonucleotide molecule are provided. The oligonucleotide molecule can be, for example, a siRNA molecule.

[0067] Accordingly, in some embodiments, the siRNA is a double-stranded RNAi (dsRNA) agent capable of inhibiting the expression of a target gene. The dsRNA agent comprises a sense strand (passenger strand) and an antisense strand (guide strand). In some embodiments, each strand of the dsRNA agent can range from 12-40 nucleotides in length. For example, each strand can be from 14-40 nucleotides in length, 17-37 nucleotides in length, 25-37 nucleotides in length, 27-30 nucleotides in length, 17-23 nucleotides in length, 17-21 nucleotides in length, 17-19 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length.

[0068] In some embodiments, the sense strand and antisense strand typically form a duplex dsRNA. The duplex region of a dsRNA agent may be from 12-40 nucleotide pairs in length. For example, the duplex region can be from 14-40 nucleotide pairs in length, 17-30 nucleotide pairs in length, 25-35 nucleotides in length, 27-35 nucleotide pairs in length, 17-23 nucleotide pairs in length, 17-21 nucleotide pairs in length, 17-19 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotide pairs in length.

[0069] In some embodiments, the dsRNA comprises one or more overhang regions and/or capping groups of dsRNA agent at the 3'-end, or 5'-end or both ends of a strand. The overhang can be 1-10 nucleotides in length, 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with

the target mRNA or it can be complementary to the gene sequences being targeted or can be other sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

[0070] In some embodiments, the nucleotides in the overhang region of the dsRNA agent can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-F, 2'-O-methyl, 2'-O-(2-methoxyethyl), 2'-O-(2-methoxyethyl), 2'-O-(2-methoxyethyl), and any combinations thereof. For example, TT (UU) can be an overhang sequence for either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be other sequence.

[0071] The 5'- or 3'-overhangs at the sense strand, antisense strand or both strands of the dsRNA agent may be phosphorylated. In some embodiments, the overhang region contains two nucleotides having a phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate between the two nucleotides, where the two nucleotides can be the same or different. In one embodiment, the overhang is present at the 3'-end of the sense strand, antisense strand or both strands. In one embodiment, this 3'-overhang is present in the antisense strand. In one embodiment, this 3-overhang is present in the sense strand.

[0072] The dsRNA agent may comprise only a single overhang, which can strengthen the interference activity of the dsRNA, without affecting its overall stability. For example, the single-stranded overhang is located at the 3'-terminal end of the sense strand or, alternatively, at the 3'-terminal end of the antisense strand. The dsRNA may also have a blunt end, located at the 5'-end of the antisense strand (or the 3'-end of the sense strand) or vice versa. Generally, the antisense strand of the dsRNA has a nucleotide overhang at the 3-end, and the 5'-end is blunt. While not bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC. For example the single overhang comprises at least two, three, four, five, six, seven, eight, nine, or ten nucleotides in length.

[0073] In some embodiments, the dsRNA agent may also have two blunt ends, at both ends of the dsRNA duplex.

[0074] In some embodiments, every nucleotide in the sense strand and antisense strand of the dsRNA agent may be modified. Each nucleotide may be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens and/or of one or more of the linking phosphate oxygens; alteration of a constituent of the ribose sugar, e.g., of the 2 hydroxyl on the ribose sugar; wholesale replacement of the phosphate moiety with "dephospho" linkers; modification or replacement of a naturally occurring base, and replacement or modification of the ribose-phosphate backbone.

[0075] In some embodiments all or some of the bases in a 3' or 5' overhang may be modified. e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate modifications. Overhangs need not be homologous with the target sequence.

[0076] In some embodiments, each residue of the sense strand and antisense strand is independently modified with

LNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-deoxy, or 2'-fluoro. The strands can contain more than one modification. In one embodiment, each residue of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro.

[0077] In some embodiments, at least two different modifications are typically present on the sense strand and antisense strand. Those two modifications may be the 2'-deoxy, 2'-O-methyl or 2'-fluoro modifications, acyclic nucleotides or others.

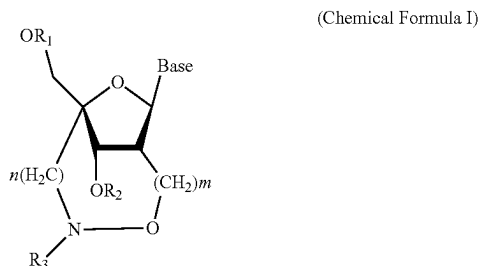
[0078] In one embodiment, the sense strand and antisense strand each comprises two differently modified nucleotides selected from 2'-fluoro, 2'-O-methyl or 2'-deoxy.

[0079] The dsRNA agent may further comprise at least one phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkage. The phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand or antisense strand or both in any position of the strand. For instance, the internucleotide linkage modification may occur on every nucleotide on the sense strand and/or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand comprises both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand.

[0080] In some embodiments, the dsRNA agent comprises the phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region comprises two nucleotides having a phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, mesyl phosphoramidate, or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. In some embodiments, these terminal three nucleotides may be at the 3'-end of the antisense strand.

[0081] In some embodiments, the dsRNA composition is linked by a modified base or nucleoside analogue as described in U.S. Pat. No. 7,427,672, which is incorporated herein by reference. In some embodiments, the modified base or nucleoside analogue is referred to as the linker or L in formulas described herein.

[0082] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and a salt thereof:



where Base represents an aromatic heterocyclic group or aromatic hydrocarbon ring group optionally having a substituent, R_1 and R_2 are identical or different, and each represent a hydrogen atom, a protective group for a hydroxyl group for nucleic acid synthesis, an alkyl group, an alkenyl group, a cycloalkyl group, an aryl group, an aralkyl group, an acyl group, a sulfonyl group, a silyl group, a phosphate group, a phosphate group protected with a protective group for nucleic acid synthesis, or $-P(R_4)R_5$ where R_4 and R_5 are identical or different, and each represent a hydroxyl group, a hydroxyl group protected with a protective group for nucleic acid synthesis, a mercapto group, a mercapto group protected with a protective group for nucleic acid synthesis, an amino group, an alkoxy group having 1 to 5 carbon atoms, an alkylthio group having 1 to 5 carbon atoms, a cyanoalkoxy group having 1 to 6 carbon atoms, or an amino group substituted by an alkyl group having 1 to 5 carbon atoms, R_3 represents a hydrogen atom, an alkyl group, an alkenyl group, a cycloalkyl group, an aryl group, an aralkyl group, an acyl group, a sulfonyl group, or a functional molecule unit substituent, and m denotes an integer of 0 to 2, and n denotes an integer of 0 to 3. In some embodiments, m and n are 0.

[0083] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein R_1 is a hydrogen atom, an aliphatic acyl group, an aromatic acyl group, an aliphatic or aromatic sulfonyl group, a methyl group substituted by one to three aryl groups, a methyl group substituted by one to three aryl groups having an aryl ring substituted by a lower alkyl, lower alkoxy, halogen, or cyano group, or a silyl group.

[0084] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein R_1 is a hydrogen atom, an acetyl group, a benzoyl group, a methanesulfonyl group, a p-toluenesulfonyl group, a benzyl group, a p-methoxybenzyl group, a trityl group, a dimethoxytrityl group, a monomethoxytrityl group, or a tert-butyldiphenylsilyl group.

[0085] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein R_2 is a hydrogen atom, an aliphatic acyl group, an aromatic acyl group, an aliphatic or aromatic sulfonyl group, a methyl group substituted by one to three aryl groups, a methyl group substituted by one to three aryl groups having an aryl ring substituted by a lower alkyl, lower alkoxy, halogen, or cyano group, a silyl group, a phosphoramidite group, a phosphonyl group, a

phosphate group, or a phosphate group protected with a protective group for nucleic acid synthesis.

[0086] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein R_2 is a hydrogen atom, an acetyl group, a benzoyl group, a methanesulfonyl group, a p-toluenesulfonyl group, a benzyl group, a p-methoxybenzyl group, a tert-butyldiphenylsilyl group, $-P(OC_2H_4CN)(N(i-Pr)_2)$, $-P(OCH_3)(N(i-Pr)_2)$, a phosphonyl group, or a 2-chlorophenyl- or 4-chlorophenylphosphate group.

[0087] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein R_3 is a hydrogen atom, a phenoxyacetyl group, an alkyl group having 1 to 5 carbon atoms, an alkenyl group having 1 to 5 carbon atoms, an aryl group having 6 to 14 carbon atoms, a methyl group substituted by one to three aryl groups, a lower aliphatic or aromatic sulfonyl group such as a methanesulfonyl group or a p-toluenesulfonyl group, an aliphatic acyl group having 1 to 5 carbon atoms such as an acetyl group, or an aromatic acyl group such as a benzoyl group.

[0088] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein the functional molecule unit substituent as R_3 is a fluorescent or chemiluminescent labeling molecule, a nucleic acid incision activity functional group, or an intracellular or nuclear transfer signal peptide.

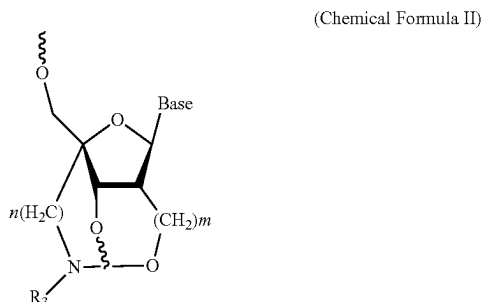
[0089] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein Base is a purin-9-yl group, a 2-oxypyrimidin-1-yl group, or a purin-9-yl group or a 2-oxypyrimidin-1-yl group having a substituent selected from the following α group; α group; A hydroxyl group, a hydroxyl group protected with a protective group for nucleic acid synthesis, an alkoxy group having 1 to 5 carbon atoms, a mercapto group, a mercapto group protected with a protective group for nucleic acid synthesis, an alkylthio group having 1 to 5 carbon atoms, an amino group, an amino group protected with a protective group for nucleic acid synthesis, an amino group substituted by an alkyl group having 1 to 5 carbon atoms, an alkyl group having 1 to 5 carbon atoms, and a halogen atom.

[0090] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein Base is 6-aminopurin-9-yl (i.e., adeninyl), 6-aminopurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2,6-diaminopurin-9-yl, 2-amino-6-chloropurin-9-yl, 2-amino-6-chloropurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-fluoropurin-9-yl, 2-amino-6-fluoropurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-bromopurin-9-yl, 2-amino-6-bromopurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-hydroxypurin-9-yl (i.e., guaninyl), 2-amino-6-hydroxypurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 6-amino-2-methoxypurin-9-yl, 6-amino-2-chloropurin-9-yl, 6-amino-2-fluoropurin-9-yl, 2,6-dimethoxypurin-9-yl, 2,6-dichloropurin-9-yl, 6-mercaptopurin-9-yl, 2-oxo-4-amino-1,2-dihydropyrimidin-1-yl (i.e., cytosinyl), 2-oxo-4-amino-1,2-dihydropyrimidin-1-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-oxo-4-amino-5-fluoro-1,2-dihydropyrimidin-1-yl, 2-oxo-4-amino-5-fluoro-1,2-dihydropyrimidin-1-yl having the amino group protected with a protective group for nucleic acid synthesis,

4-amino-2-oxo-5-chloro-1,2-dihydropyrimidin-1-yl, 2-oxo-4-methoxy-1,2-dihydropyrimidin-1-yl, 2-oxo-4-mercapto-1,2-dihydropyrimidin-1-yl, 2-oxo-4-hydroxy-1,2-dihydropyrimidin-1-yl (i.e., uraciny), 2-oxo-4-hydroxy-5-methyl-1,2-dihydropyrimidin-1-yl (i.e., thyminy), 4-amino-5-methyl-2-oxo-1,2-dihydropyrimidin-1-yl (i.e., 5-methylcytosiny), or 4-amino-5-methyl-2-oxo-1,2-dihydropyrimidin-1-yl having the amino group protected with a protective group for nucleic acid synthesis.

[0091] In some embodiments, the modified base or nucleoside analogue has the structure as shown in Chemical Formula I and salts thereof, wherein m is 0, and n is 1.

[0092] In some embodiments, the modified base or nucleoside analogue is a DNA oligonucleotide or RNA oligonucleotide analogue, containing one or two or more of one or more types of unit structures of nucleoside analogues having the structure as shown in Chemical Formula II, or a pharmacologically acceptable salt thereof, provided that a form of linking between respective nucleosides in the oligonucleotide analogue may contain one or two or more phosphorothioate bonds $[-OP(O)(S^-)O-]$, phosphorodithioate bonds $[-O_2PS_2-]$, phosphonate bonds $[-PO(OH)_2-]$, phosphoramidate bonds $[-O=P(OH)_2-]$, or mesyl phosphoramidate bonds $[-OP(O)(N)(SO_2)(CH_3)O-]$ aside from a phosphodiester bond $[-OP(O_2^-)O-]$ identical with that in a natural nucleic acid, and if two or more of one or more types of these structures are contained, Base may be identical or different between these structures:



where Base represents an aromatic heterocyclic group or aromatic hydrocarbon ring group optionally having a substituent. R_3 represents a hydrogen atom, an alkyl group, an alkenyl group, a cycloalkyl group, an aryl group, an aralkyl group, an acyl group, a sulfonyl group, a silyl group, or a functional molecule unit substituent, and m denotes an integer of 0 to 2, and n denotes an integer of 0 to 3. In some embodiments, m and n are 0.

[0093] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein R_1 is a hydrogen atom, an aliphatic acyl group, an aromatic acyl group, an aliphatic or aromatic sulfonyl group, a methyl group substituted by one to three aryl groups, a methyl group substituted by one to three aryl groups having an aryl ring substituted by a lower alkyl, lower alkoxy, halogen, or cyano group, or a silyl group.

[0094] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein R_1 is a hydrogen atom, an acetyl group, a benzoyl group, a methanesulfonyl group, a p-toluenesulfonyl group, a benzyl

group, a p-methoxybenzyl group, a trityl group, a dimethoxytrityl group, a monomethoxytrityl group, or a tert-butyldiphenylsilyl group.

[0095] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein R_2 is a hydrogen atom, an aliphatic acyl group, an aromatic acyl group, an aliphatic or aromatic sulfonyl group, a methyl group substituted by one to three aryl groups, a methyl group substituted by one to three aryl groups having an aryl ring substituted by a lower alkyl, lower alkoxy, halogen, or cyano group, a silyl group, a phosphoramidite group, a phosphonyl group, a phosphate group, or a phosphate group protected with a protective group for nucleic acid synthesis.

[0096] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein R_2 is a hydrogen atom, an acetyl group, a benzoyl group, a benzyl group, a p-methoxybenzyl group, a methanesulfonyl group, a p-toluenesulfonyl group, a tert-butyldiphenylsilyl group, $-P(OC_2H_4CN)(N(i-Pr)_2)$, $-P(OCH_3)(N(i-Pr)_2)$, a phosphonyl group, or a 2-chlorophenyl- or 4-chlorophenylphosphate group.

[0097] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein R_3 is a hydrogen atom, a phenoxyacetyl group, an alkyl group having 1 to 5 carbon atoms, an alkenyl group having 1 to 5 carbon atoms, an aryl group having 6 to 14 carbon atoms, a methyl group substituted by one to three aryl groups, a lower aliphatic or aromatic sulfonyl group such as a methanesulfonyl group or a p-toluenesulfonyl group, an aliphatic acyl group having 1 to 5 carbon atoms such as an acetyl group, or an aromatic acyl group such as a benzoyl group.

[0098] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein the functional molecule unit substituent as R_3 is a fluorescent or chemiluminescent labeling molecule, a nucleic acid incision activity functional group, or an intracellular or nuclear transfer signal peptide.

[0099] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein Base is a purin-9-yl group, a 2-oxopyrimidin-1-yl group, or a purin-9-yl group or a 2-oxopyrimidin-1-yl group having a substituent selected from the following α group; α group: A hydroxyl group, a hydroxyl group protected with a protective group for nucleic acid synthesis, an alkoxy group having 1 to 5 carbon atoms, a mercapto group, a mercapto group protected with a protective group for nucleic acid synthesis, an alkylthio group having 1 to 5 carbon atoms, an amino group, an amino group protected with a protective group for nucleic acid synthesis, an amino group substituted by an alkyl group having 1 to 5 carbon atoms, an alkyl group having 1 to 5 carbon atoms, and a halogen atom.

[0100] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein Base is 6-aminopurin-9-yl (i.e. adeniny), 6-aminopurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2,6-diaminopurin-9-yl, 2-amino-6-chloropurin-9-yl, 2-amino-6-chloropurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-fluoropurin-9-yl, 2-amino-6-fluoropurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-bro-

mopurin-9-yl, 2-amino-6-bromopurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-amino-6-hydroxypurin-9-yl (i.e., guaninyl), 2-amino-6-hydroxypurin-9-yl having the amino group protected with a protective group for nucleic acid synthesis, 6-amino-2-methoxypurin-9-yl, 6-amino-2-chloropurin-9-yl, 6-amino-2-fluoropurin-9-yl, 2,6-dimethoxypurin-9-yl, 2,6-dichloropurin-9-yl, 6-mercaptopurin-9-yl, 2-oxo-4-amino-1,2-dihydropyrimidin-1-yl (i.e., cytosinyl), 2-oxo-4-amino-1,2-dihydropyrimidin-1-yl having the amino group protected with a protective group for nucleic acid synthesis, 2-oxo-4-amino-5-fluoro-1,2-dihydropyrimidin-1-yl, 2-oxo-4-amino-5-fluoro-1,2-dihydropyrimidin-1-yl group having the amino group protected with a protective group for nucleic acid synthesis, 4-amino-2-oxo-5-chloro-1,2-dihydropyrimidin-1-yl, 2-oxo-4-methoxy-1,2-dihydropyrimidin-1-yl, 2-oxo-4-mercapto-1,2-dihydropyrimidin-1-yl, 2-oxo-4-hydroxy-1,2-dihydropyrimidin-1-yl (i.e., uraciny), 2-oxo-4-hydroxy-5-methyl-1,2-dihydropyrimidin-1-yl (i.e., thyminy), 4-amino-5-methyl-2-oxo-1,2-dihydropyrimidin-1-yl (i.e., 5-methylcytosinyl), or 4-amino-5-methyl-2-oxo-1,2-dihydropyrimidin-1-yl having the amino group protected with a protective group for nucleic acid synthesis.

[0101] In some embodiments, the oligonucleotide analogue or the pharmacologically acceptable salt thereof has the structure as shown in Chemical Formula II, wherein m is 0, and n is 1.

[0102] In some embodiments, compositions described herein further comprises a polymer (polymer moiety C). In some instances, the polymer is a natural or synthetic polymer, consisting of long chains of branched or unbranched monomers, and/or cross-linked network of monomers in two or three dimensions. In some instances, the polymer includes a polysaccharide, lignin, rubber, or polyalkylene oxide (e.g., polyethylene glycol). In some instances, the at least one polymer includes, but is not limited to, alpha-, omega-dihydroxypolyethyleneglycol, biodegradable lactone-based polymer, e.g. polyacrylic acid, polylactide acid (PLA), poly (glycolic acid) (PGA), polypropylene, polystyrene, polyolefin, polyamide, polycyanoacrylate, polyimide, polyethyl-enterephthalat (PET, PETG), polyethylene terephthalate (PETE), polytetramethylene glycol (PTG), or polyurethane as well as mixtures thereof. As used herein, a mixture refers to the use of different polymers within the same compound as well as in reference to block copolymers. In some cases, block copolymers are polymers wherein at least one section of a polymer is build up from monomers of another polymer. In some instances, the polymer comprises polyalkylene oxide. In some instances, the polymer comprises PEG. In some instances, the polymer comprises polyethylene imide (PEI) or hydroxy ethyl starch (HES).

[0103] In some instances, C is a PEG moiety. In some instances, the PEG moiety is conjugated at the 5' terminus of the oligonucleotide molecule while the binding moiety is conjugated at the 3' terminus of the oligonucleotide molecule. In some instances, the PEG moiety is conjugated at the 3' terminus of the oligonucleotide molecule while the binding moiety is conjugated at the 5' terminus of the oligonucleotide molecule. In some instances, the PEG moiety is conjugated to an internal site of the oligonucleotide molecule. In some instances, the PEG moiety, the binding moiety, or a combination thereof, are conjugated to an internal site of the oligonucleotide molecule. In some instances, the conjugation is a direct conjugation. In some instances, the conjugation is via native ligation.

[0104] In some embodiments, the polyalkylene oxide (e.g., PEG) is a polydispers or monodispers compound. In

some instances, polydispers material comprises disperse distribution of different molecular weight of the material, characterized by mean weight (weight average) size and dispersity. In some instances, the monodisperse PEG comprises one size of molecules. In some embodiments, C is poly- or monodispersed polyalkylene oxide (e.g., PEG) and the indicated molecular weight represents an average of the molecular weight of the polyalkylene oxide, e.g., PEG, molecules.

[0105] In some embodiments, the molecular weight of the polyalkylene oxide (e.g., PEG) is about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1450, 1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3250, 3350, 3500, 3750, 4000, 4250, 4500, 4600, 4750, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 10,000, 12,000, 20,000, 35,000, 40,000, 50,000, 60,000, or 100,000 Da.

[0106] In some embodiments, C is polyalkylene oxide (e.g., PEG) and has a molecular weight of about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1450, 1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3250, 3350, 3500, 3750, 4000, 4250, 4500, 4600, 4750, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 10,000, 12,000, 20,000, 35,000, 40,000, 50,000, 60,000, or 100,000 Da. In some embodiments, C is PEG and has a molecular weight of about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1450, 1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3250, 3350, 3500, 3750, 4000, 4250, 4500, 4600, 4750, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 10,000, 12,000, 20,000, 35,000, 40,000, 50,000, 60,000, or 100,000 Da. In some instances, the molecular weight of C is about 200 Da. In some instances, the molecular weight of C is about 300 Da. In some instances, the molecular weight of C is about 400 Da. In some instances, the molecular weight of C is about 500 Da. In some instances, the molecular weight of C is about 600 Da. In some instances, the molecular weight of C is about 700 Da. In some instances, the molecular weight of C is about 800 Da. In some instances, the molecular weight of C is about 900 Da. In some instances, the molecular weight of C is about 1000 Da. In some instances, the molecular weight of C is about 1100 Da. In some instances, the molecular weight of C is about 1200 Da. In some instances, the molecular weight of C is about 1300 Da. In some instances, the molecular weight of C is about 1400 Da. In some instances, the molecular weight of C is about 1450 Da. In some instances, the molecular weight of C is about 1500 Da. In some instances, the molecular weight of C is about 1600 Da. In some instances, the molecular weight of C is about 1700 Da. In some instances, the molecular weight of C is about 1800 Da. In some instances, the molecular weight of C is about 1900 Da. In some instances, the molecular weight of C is about 2000 Da. In some instances, the molecular weight of C is about 2100 Da. In some instances, the molecular weight of C is about 2200 Da. In some instances, the molecular weight of C is about 2300 Da. In some instances, the molecular weight of C is about 2400 Da. In some instances, the molecular weight of C is about 2500 Da. In some instances, the molecular weight of C is about 2600 Da. In some instances, the molecular weight of C is about 2700 Da. In some instances, the molecular weight of C is about 2800 Da. In some instances, the molecular weight of C is about 2900 Da. In some instances, the molecular weight of C is about 3000 Da. In some instances, the molecular weight of C is about 3250 Da. In some instances, the molecular weight of C is

about 3350 Da. In some instances, the molecular weight of C is about 3500 Da. In some instances, the molecular weight of C is about 3750 Da. In some instances, the molecular weight of C is about 4000 Da. In some instances, the molecular weight of C is about 4250 Da. In some instances, the molecular weight of C is about 4500 Da. In some instances, the molecular weight of C is about 4600 Da. In some instances, the molecular weight of C is about 4750 Da. In some instances, the molecular weight of C is about 5000 Da. In some instances, the molecular weight of C is about 5500 Da. In some instances, the molecular weight of C is about 6000 Da. In some instances, the molecular weight of C is about 6500 Da. In some instances, the molecular weight of C is about 7000 Da. In some instances, the molecular weight of C is about 7500 Da. In some instances, the molecular weight of C is about 8000 Da. In some instances, the molecular weight of C is about 10,000 Da. In some instances, the molecular weight of C is about 12,000 Da. In some instances, the molecular weight of C is about 20,000 Da. In some instances, the molecular weight of C is about 35,000 Da. In some instances, the molecular weight of C is about 40,000 Da. In some instances, the molecular weight of C is about 50,000 Da. In some instances, the molecular weight of C is about 60,000 Da. In some instances, the molecular weight of C is about 10,000 Da.

[0107] In some embodiments, the polyalkylene oxide (e.g., PEG) is a discrete PEG, in which the discrete PEG is a polymeric PEG comprising more than one repeating ethylene oxide units. In some instances, a discrete PEG (dPEG) comprises from 2 to 60, from 2 to 50, or from 2 to 48 repeating ethylene oxide units. In some instances, a dPEG comprises about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 24, 26, 28, 30, 35, 40, 42, 48, 50 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 2 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 3 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 4 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 5 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 6 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 7 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 8 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 9 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 10 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 11 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 12 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 13 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 14 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 15 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 16 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 17 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 18 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 19 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 20 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 22 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 24 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 26 or more repeating ethylene oxide units. In some

instances, a dPEG comprises about 28 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 30 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 35 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 40 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 42 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 48 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 50 or more repeating ethylene oxide units. In some cases, a dPEG is synthesized as a single molecular weight compound from pure (e.g., about 95%, 98%, 99%, or 99.5%) starting material in a step-wise fashion. In some cases, a dPEG has a specific molecular weight, rather than an average molecular weight. In some cases, a dPEG described herein is a dPEG from Quanta Biodesign, LMD.

[0108] In some embodiments, C is an albumin binding domain. In certain aspects, the albumin binding domain specifically binds to serum albumin, e.g., human serum albumin (HSA) to prolong the half-life of the domain or of another therapeutic to which the albumin-binding domain is associated or linked with. In some embodiments, the human serum albumin-binding domain comprises an initiator methionine (Met) linked to the N-terminus of the molecule. In some embodiments, the human serum albumin-binding domain comprise a cysteine (Cys) linked to a C-terminus or the N-terminus of the domain. The addition of the N-terminal Met and/or the C-terminal Cys may facilitate expression and/or conjugation to another molecule, which can be another half-life extending molecules, such as PEG, a Fc region, and the like.

[0109] In some embodiments, the albumin binding domain comprises the amino acid sequence of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119, provided in Table 8. In some embodiments, the albumin binding domain (protein) is isolated. In some embodiments, the albumin binding domain comprises an amino acid sequence that is at least, or is, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the albumin binding domain comprises an amino acid sequence that is at least, or is, 85%, 86%, 87%, 88%, 89%, 90%, 901%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119 provided that the protein has a substitution that corresponds to position 10 of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the substitution is A10V. In some embodiments, the substitution is A10G, A10L, A10I, A10T, or A10S. In some embodiments, the substitution at position 10 is any naturally occurring amino acid. In some embodiments, the isolated albumin binding domain comprises an amino acid sequence that has 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, or 14 substitutions when compared to the amino acid sequence of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the substitution is at a position that corresponds to position 10 of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, FN3 domains provided comprises a cysteine residue in at least one residue position corresponding to residue positions

6, 11, 22, 25, 26, 52, 53, 61, 88 or positions 6, 8, 10, 11, 14, 15, 16, 20, 30, 34, 38, 40, 41, 45, 47, 48, 53, 54, 59, 60, 62, 64, 70, 88, 89, 90, 91, or 93 of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119, or at a C-terminus. Although the positions are listed in a series, each position can also be chosen individually. In some embodiments, the cysteine is at a position that corresponds to position 6, 53, or 88. In some embodiments, additional examples of albumin binding domains can be found in U.S. Pat. No. 10,925,932, which hereby incorporated by reference.

TABLE 8

SEQ ID NO: SEQUENCE	
101	MLPAPKNLVASRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
102	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
103	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFAIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
104	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
105	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
106	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
107	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
108	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGAGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
109	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIAGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
110	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IALRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
111	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLAVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIARFTT
112	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYAVWIHGVKGGASSPPLIARFTT
113	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVAIHGVKGGASSPPLIARFTT
114	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIAGVKGGASSPPLIARFTT
115	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGSSPPLIARFTT
116	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGAASPPLIARFTT
117	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASAPLIARFTT
118	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLAARFTT

TABLE 8-continued

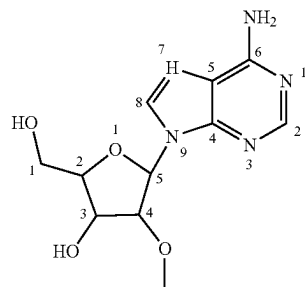
SEQ ID NO: SEQUENCE	
119	MLPAPKNLVSVRVTEDSARLSWTAPDAAFDSFNIAYWEPGIGGEE IWLVRVPGSERSYDLTGLKPGTEYKVWIHGVKGGASSPPLIAAFTT

[0110] In some embodiments, the dsRNA agent comprises mismatch(es) with the target, within the duplex, or combinations thereof. The mismatch can occur in the overhang region or the duplex region. The base pair can be ranked on the basis of their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C, G:U is preferred over G:C; and I:C is preferred over G:C (I=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings.

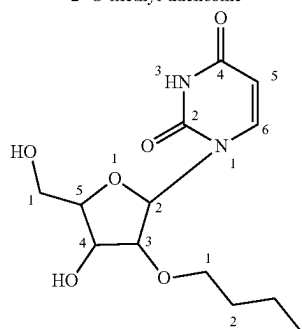
[0111] In some embodiments, the dsRNA agent can comprise a phosphorus-containing group at the 5'-end of the sense strand or antisense strand. The 5'-end phosphorus-containing group can be 5'-end phosphate (5'-P), 5'-end phosphorothioate (5'-PS), 5'-end phosphorodithioate (5'-PS₂), 5'-end vinylphosphonate (5'-VP), 5'-end methylphosphonate (MePhos), 5'-end mesyl phosphoramidate (5'-MsPA), or 5'-deoxy-5'-C-malonyl. When the 5'-end phosphorus-containing group is 5'-end vinylphosphonate (5'-VP), the 5'-VP can be either 5'-E-VP isomer, such as trans-vinylphosphate or cis-vinylphosphate, or mixtures thereof. Representative structures of these modifications can be found in, for example, U.S. Pat. No. 10,233,448, which is hereby incorporated by reference in its entirety.

[0112] In some embodiments, nucleotide analogues or synthetic nucleotide base comprise a nucleic acid with a modification at a 2' hydroxyl group of the ribose moiety. In some instances, the modification includes an H, OR, R, halo, SH, SR, NH₂, NHR, NR₂, or CN, wherein R is an alkyl moiety. Exemplary alkyl moiety includes, but is not limited to, methyl, ethyl, n-propyl, iso-propyl, n-butyl, iso-butyl, tert-butyl, C₁-C₁₀ chain lengths both linear and branched. In some instances, the alkyl moiety further comprises a modification. In some instances, the modification comprises an azo group, a keto group, an aldehyde group, a carboxyl group, a nitro group, a nitroso group, a nitrile group, a heterocycle (e.g., imidazole, hydrazine or hydroxylamino) group, an isocyanate or cyanate group, or a sulfur containing group (e.g., sulfoxide, sulfone, sulfide, and disulfide). In some instances, the alkyl moiety further comprises additional hetero atom such as O, S, N, Se and each of these hetero atoms can be further substituted with alkyl groups as described above. In some instances, the carbon of the heterocyclic group is substituted by a nitrogen, oxygen or sulfur. In some instances, the heterocyclic substitution includes but is not limited to, morpholino, imidazole, and pyrrolidino.

[0113] In some instances, the modification at the 2' hydroxyl group is a 2'-O-methyl modification or a 2'-O-methoxyethyl (2'-O-MOE) modification. Exemplary chemical structures of a 2'-O-methyl modification of an adenosine molecule and 2'-O-methoxyethyl modification of an uridine are illustrated below.

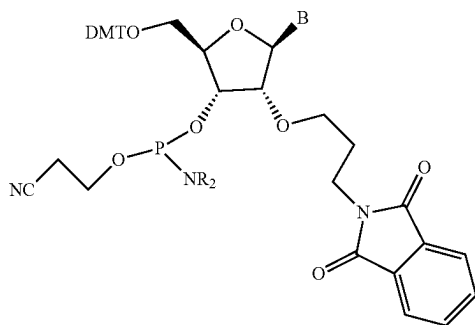


2'-O-methyl-adenosine



2'-O-methoxyethyl uridine

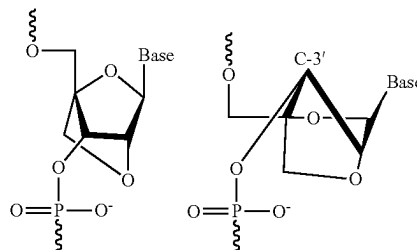
[0114] In some instances, the modification at the 2' hydroxyl group is a 2'-O-aminopropyl modification in which an extended amine group comprising a propyl linker binds the amine group to the 2' oxygen. In some instances, this modification neutralizes the phosphate derived overall negative charge of the oligonucleotide molecule by introducing one positive charge from the amine group per sugar and thereby improves cellular uptake properties due to its zwitterionic properties. An exemplary chemical structure of a 2'-O-aminopropyl nucleoside phosphoramidite is illustrated below.



2'-O-aminopropyl nucleoside phosphoramidite

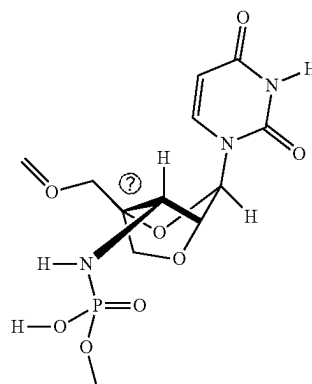
[0115] In some instances, the modification at the 2' hydroxyl group is a locked or bridged ribose modification (e.g., locked nucleic acid or LNA) in which the oxygen molecule bound at the 2' carbon is linked to the 4' carbon by a methylene group, thus forming a 2'-C,4'-C-oxy-methylene-linked bicyclic ribonucleotide monomer. Exemplary representations of the chemical structure of LNA are illustrated below. The representation shown to the left highlights the chemical connectivities of an LNA monomer. The represen-

tation shown to the right highlights the locked 3'-endo (3E) conformation of the furanose ring of an LNA monomer.

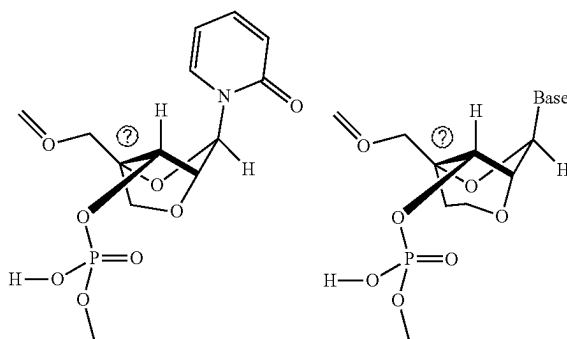


LNA (Locked Nucleic Acids)

[0116] In some instances, the modification at the 2' hydroxyl group comprises ethylene nucleic acids (ENA) such as for example 2'-4'-ethylene-bridged nucleic acid, which locks the sugar conformation into a C3'-endo sugar puckering conformation. ENA are part of the bridged nucleic acids class of modified nucleic acids that also comprises LNA. Exemplary chemical structures of the ENA and bridged nucleic acids are illustrated below.

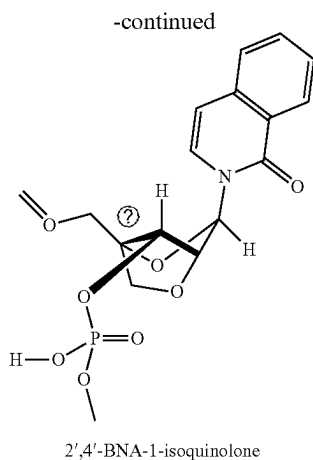


3'-amino-2',4'-BNA



2',4'-BNA-2-pyridine

2',4'-ENA



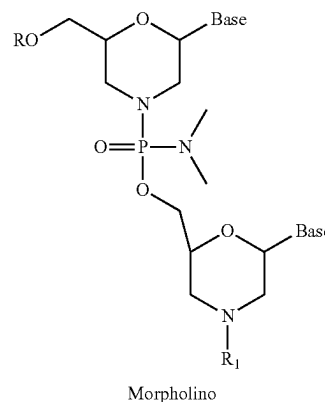
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[0117] In some embodiments, additional modifications at the 2' hydroxyl group include 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE), or 2'-O-N-methylacetamido (2'-O-NMA).

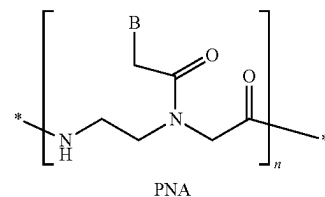
[0118] In some embodiments, nucleotide analogues comprise modified bases such as, but not limited to, 5-propynyluridine, 5-propynylcytidine, 6-methyladenine, 6-methylguanine, N, N-dimethyladenine, 2-propyladenine, 2-propylguanine, 2-aminoadenine, 1-methylinosine, 3-methyluridine, 5-methylcytidine, 5-methyluridine and other nucleotides having a modification at the 5 position, 5-(2-amino) propyl uridine, 5-halocytidine, 5-halouridine, 4-acetylcytidine, 1-methyladenosine, 2-methyladenosine, 3-methylcytidine, 6-methyluridine, 2-methylguanosine, 7-methylguanosine, 2, 2-dimethylguanosine, 5-methylaminoethyluridine, 5-methoxyuridine, deazanucleotides such as 7-deaza-adenosine, 6-azouridine, 6-azocytidine, 6-azothymidine, 5-methyl-2-thiouridine, other thio bases such as 2-thiouridine and 4-thiouridine and 2-thiocytidine, dihydrouridine, pseudouridine, queuosine, archaeosine, naphthyl and substituted naphthyl groups, any O- and N-alkylated purines and pyrimidines such as N6-methyladenosine, 5-methylcarbonylmethyluridine, uridine 5-oxyacetic acid, pyridine-4-one, pyridine-2-one, phenyl and modified phenyl groups such as aminophenol or 2,4, 6-trimethoxy benzene, modified cytosines that act as G-clamp nucleotides, 8-substituted adenines and guanines, 5-substituted uracils and thymines, azapyrimidines, carboxyhydroxyalkyl nucleotides, carboxyalkylaminoalkyl nucleotides, and alkylcarbonylalkylated nucleotides. Modified nucleotides also include those nucleotides that are modified with respect to the sugar moiety, as well as nucleotides having sugars or analogs thereof that are not ribosyl. For example, the sugar moieties, in some cases are or be based on, mannoses, arabinoses, glucopyranoses, galactopyranoses, 4'-thioribose, and other sugars, heterocycles, or carbocycles. The term nucleotide also includes what are known in the art as universal bases. By way of example, universal bases include but are not limited to 3-nitropyrrole, 5-nitroindole, or nebularine.

[0119] In some embodiments, nucleotide analogues further comprise morpholinos, peptide nucleic acids (PNAs), methylphosphonate nucleotides, thiophosphonate nucleotides, 2-fluoro N3-P5'-phosphoramidites, 1', 5'-anhydrohexitol nucleic acids (HNAs), or a combination thereof. Mor-

pholino or phosphorodiamidate morpholino oligo (PMO) comprises synthetic molecules whose structure mimics natural nucleic acid structure by deviates from the normal sugar and phosphate structures. In some instances, the five-member ribose ring is substituted with a six member morpholino ring containing four carbons, one nitrogen and one oxygen. In some cases, the ribose monomers are linked by a phosphordiamidate group instead of a phosphate group. In such cases, the backbone alterations remove all positive and negative charges making morpholinos neutral molecules capable of crossing cellular membranes without the aid of cellular delivery agents such as those used by charged oligonucleotides.



[0120] In some embodiments, peptide nucleic acid (PNA) does not contain sugar ring or phosphate linkage and the bases are attached and appropriately spaced by oligoglycine-like molecules, therefore, eliminating a backbone charge.



[0121] In some embodiments, one or more modifications optionally occur at the internucleotide linkage. In some instances, modified internucleotide linkage include, but is not limited to, phosphorothioates, mesyl phosphoramidate, phosphorodithioates, methylphosphonates, 5'-alkylenephosphonates, 5'-methylphosphonate, 3'-alkylene phosphonates, borontrifluoridates, borano phosphate esters and selenophosphates of 3'-5' linkage or 2'-5' linkage, phosphotriesters, thionoalkylphosphotriesters, hydrogen phosphonate linkages, alkyl phosphonates, alkylphosphonothioates, arylphosphonothioates, phosphoroselenoates, phosphorodiselenoates, phosphinates, phosphoramidates, 3'-alkylphosphoramidates, aminoalkylphosphoramidates, thionophosphoramidates, phosphoropiperazidates, phosphoroanilothioates, phosphoroanilidates, ketones, sulfones, sulfonamides, carbonates, carbamates, methylenehydrazos, methylenedimethylhydrazos, formacetals, thioformacetals, oximes, methyleneiminos, methylenemethyliminos, thioamides, linkages with riboacetyl groups, aminoethyl glycine, silyl or siloxane linkages, alkyl or

cycloalkyl linkages with or without heteroatoms of, for example, 1 to 10 carbons that are saturated or unsaturated and/or substituted and/or contain heteroatoms, linkages with morpholino structures, amides, polyamides wherein the bases are attached to the aza nitrogens of the backbone directly or indirectly, and combinations thereof. Phosphorothioate antisense oligonucleotides (PS ASO) are antisense oligonucleotides comprising a phosphorothioate linkage. Methyl phosphoramidate antisense oligonucleotides (MsPA ASO) are antisense oligonucleotides comprising a methyl phosphoramidate linkage.

[0122] In some instances, the modification is a methyl or thiol modification such as methylphosphonate, methyl phosphoramidate, or thiolphosphonate modification. In some instances, a modified nucleotide includes, but is not limited to, 2'-fluoro N3-P5'-phosphoramidites.

[0123] In some instances, a modified nucleotide includes, but is not limited to, hexitol nucleic acid (or 1', 5-anhydro-hexitol nucleic acids (HNA)).

[0124] In some embodiments, one or more modifications further optionally include modifications of the ribose moiety, phosphate backbone and the nucleoside, or modifications of the nucleotide analogues at the 3' or the 5' terminus. For example, the 3' terminus optionally include a 3' cationic group, or by inverting the nucleoside at the 3-terminus with a 3'-3' linkage. In another alternative, the 3'-terminus is optionally conjugated with an aminoalkyl group, e.g., a 3' C5-aminoalkyl dT. In an additional alternative, the 3'-terminus is optionally conjugated with an abasic site, e.g., with an apurinic or apyrimidinic site. In some instances, the 5'-terminus is conjugated with an aminoalkyl group, e.g., a 5'-O-alkylamino substituent. In some cases, the 5'-terminus is conjugated with an abasic site, e.g., with an apurinic or apyrimidinic site.

[0125] In some embodiments, the oligonucleotide molecule comprises one or more of the synthetic nucleotide analogues described herein. In some instances, the oligonucleotide molecule comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 25, or more of the synthetic nucleotide analogues described herein. In some embodiments, the synthetic nucleotide analogues include 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl, 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE), or 2'-O—N-methylacetamido (2'-O-NMA) modified, LNA, ENA, PNA, HNA, morpholino, methylphosphonate nucleotides, thiolphosphonate nucleotides, 2'-fluoro N3-P5'-phosphoramidites, or a combination thereof. In some instances, the oligonucleotide molecule comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 25, or more of the synthetic nucleotide analogues selected from 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl, 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE), or 2'-O—N-methylacetamido (2'-O-NMA) modified, LNA, ENA, PNA, HNA, morpholino, methylphosphonate nucleotides, thiolphosphonate nucleotides, 2'-fluoro N3-P5'-phosphoramidites, or a combination thereof. In some instances, the oligonucleotide molecule comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 25, or more of 2'-O-methyl modified nucleotides. In some instances, the oligonucleotide molecule comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 25, or more of 2'-O-methoxyethyl (2'-O-MOE) modified nucleotides. In

some instances, the oligonucleotide molecule comprises 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 20, 25, or more of thiolphosphonate nucleotides.

[0126] In some instances, the oligonucleotide molecule comprises at least one of: from about 5% to about 100% modification, from about 10% to about 100% modification, from about 20% to about 100% modification, from about 30% to about 100% modification, from about 40% to about 100% modification, from about 50% to about 100% modification, from about 60% to about 100% modification, from about 70% to about 100% modification, from about 80% to about 100% modification, and from about 90% to about 100% modification. In some instances, the oligonucleotide molecule comprises 100% modification.

[0127] In some cases, the oligonucleotide molecule comprises at least one of: from about 10% to about 90% modification, from about 20% to about 90% modification, from about 30% to about 90% modification, from about 40% to about 90% modification, from about 50% to about 90% modification, from about 60% to about 90% modification, from about 70% to about 90% modification, and from about 800% to about 100% modification.

[0128] In some cases, the oligonucleotide molecule comprises at least one of: from about 10% to about 80% modification, from about 20% to about 80% modification, from about 30% to about 80% modification, from about 40% to about 80% modification, from about 50% to about 80% modification, from about 60% to about 80% modification, and from about 70% to about 80% modification.

[0129] In some instances, the oligonucleotide molecule comprises at least one of: from about 10% to about 70% modification, from about 20% to about 70% modification, from about 30% to about 70% modification, from about 40% to about 70% modification, from about 50% to about 70% modification, and from about 60% to about 70% modification.

[0130] In some instances, the oligonucleotide molecule comprises at least one of: from about 10% to about 60% modification, from about 20% to about 60% modification, from about 30% to about 60% modification, from about 40% to about 60% modification, and from about 50% to about 60% modification.

[0131] In some cases, the oligonucleotide molecule comprises at least one of: from about 10% to about 50% modification, from about 20% to about 50% modification, from about 30% to about 50% modification, and from about 40% to about 50% modification.

[0132] In some cases, the oligonucleotide molecule comprises at least one of: from about 10% to about 40% modification, from about 20% to about 40% modification, and from about 30% to about 40% modification.

[0133] In some cases, the oligonucleotide molecule comprises at least one of: from about 10% to about 30% modification, and from about 20% to about 30% modification.

[0134] In some cases, the oligonucleotide molecule comprises from about 10% to about 20% modification.

[0135] In some cases, the oligonucleotide molecule comprises from about 15% to about 90%, from about 20% to about 80%, from about 30% to about 70%, or from about 40% to about 60% modifications.

[0136] In additional cases, the oligonucleotide molecule comprises at least about 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, or 99% modification.

[0137] In some embodiments, the oligonucleotide molecule comprises at least about 1, about 2, about 3, about 4, about 5, about 6, about 7, about 8, about 9, about 10, about

gonucleotide molecule comprises about 16 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 17 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 18 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 19 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 20 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 21 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 22 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 23 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 24 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 25 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 26 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 27 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 28 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 29 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 30 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 31 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 32 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 33 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 34 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 35 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 36 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 37 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 38 modifications in

which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 39 modifications in which the modifications comprise a synthetic nucleotide analogue described herein. In some embodiments, the oligonucleotide molecule comprises about 40 modifications in which the modifications comprise a synthetic nucleotide analogue described herein.

[0141] In some embodiments, an oligonucleotide molecule is assembled from two separate polynucleotides wherein one polynucleotide comprises the sense strand and the second polynucleotide comprises the antisense strand of the oligonucleotide molecule. In other embodiments, the sense strand is connected to the antisense strand via a linker molecule, which in some instances is a polynucleotide linker or a non-nucleotide linker.

[0142] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, wherein pyrimidine nucleotides in the sense strand comprises 2'-O-methylpyrimidine nucleotides and purine nucleotides in the sense strand comprise 2'-deoxy purine nucleotides. In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, wherein pyrimidine nucleotides present in the sense strand comprise 2'-deoxy-2'-fluoro pyrimidine nucleotides and wherein purine nucleotides present in the sense strand comprise 2'-deoxy purine nucleotides.

[0143] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, wherein the pyrimidine nucleotides when present in said antisense strand are 2'-deoxy-2'-fluoro pyrimidine nucleotides and the purine nucleotides when present in said antisense strand are 2'-O-methyl purine nucleotides.

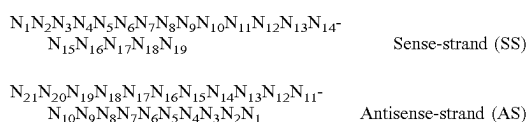
[0144] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, wherein the pyrimidine nucleotides when present in said antisense strand are 2'-deoxy-2'-fluoro pyrimidine nucleotides and wherein the purine nucleotides when present in said antisense strand comprise 2'-deoxy-purine nucleotides.

[0145] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, and at least one of sense strand and antisense strands has a plurality of (e.g., two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, etc) 2'-O-methyl or 2'-deoxy-2'-fluoro modified nucleotides. In some embodiments, at least two, three, four, five, six, or seven out of the a plurality of 2'-O-methyl or 2'-deoxy-2'-fluoro modified nucleotides are consecutive nucleotides. In some embodiments, consecutive 2'-O-methyl or 2'-deoxy-2'-fluoro modified nucleotides are located at the 5'-end of the sense strand and/or the antisense strand. In some embodiments, consecutive 2'-O-methyl or 2'-deoxy-2'-fluoro modified nucleotides are located at the 3'-end of the sense strand and/or the antisense strand. In some embodiments, the sense strand of oligonucleotide molecule includes at least four, at least five, at least six consecutive 2'-O-methyl modified nucleotides at its 5' end and/or 3' end, or both. Optionally, in such embodiments, the sense strand of oligonucleotide molecule includes at least one, at least two, at least three, at least four 2'-deoxy-2'-fluoro modified nucleotides at the 3' end of the at least four, at least five, at least six consecutive 2'-O-methyl modified nucleotides at the polynucleotides' 5' end, or at the 5' end of the at least four, at least five, at least six consecutive 2'-O-methyl modified nucleotides at the polynucleotides' 3' end. Also optionally, such at least two, at least three, at least four 2'-deoxy-2'-fluoro modified nucleotides are consecutive nucleotides.

[0146] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, and at least one of sense strand and antisense strand has 2'-O-methyl modified nucleotide located at the 5'-end of the sense strand and/or the antisense strand. In some embodiments, at least one of sense strand and antisense strands has 2'-O-methyl modified nucleotide located at the 3'-end of the sense strand and/or the antisense strand. In some embodiments, the 2'-O-methyl modified nucleotide located at the 5'-end of the sense strand and/or the antisense strand is a purine nucleotide. In some embodiments, the 2'-O-methyl modified nucleotide located at the 5'-end of the sense strand and/or the antisense strand is a pyrimidine nucleotide.

[0147] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, and one of sense strand and antisense strand has at least two consecutive 2'-deoxy-2'-fluoro modified nucleotides located at the 5'-end, while another strand has at least two consecutive 2'-O-methyl modified nucleotides located at the 5'-end. In some embodiments, where the strand has at least two consecutive 2'-deoxy-2'-fluoro modified nucleotides located at the 5'-end, the strand also includes at least two, at least three consecutive 2'-O-methyl modified nucleotides at the 3'-end of the at least two consecutive 2'-deoxy-2'-fluoro modified nucleotides. In some embodiments, one of sense strand and antisense strand has at least two, at least three, at least four, at least five, at least six, or at least seven consecutive 2'-O-methyl modified nucleotides that are linked to a 2'-deoxy-2'-fluoro modified nucleotide on its 5'-end and/or 3'-end. In some embodiments, one of sense strand and antisense strand has at least four, at least five nucleotides that have alternating 2'-O-methyl modified nucleotide and 2'-deoxy-2'-fluoro modified nucleotide.

[0148] In some embodiments, the oligonucleotide molecule, such as a siRNA, has the formula as illustrated in Formula I:



wherein each nucleotide represented by N, is independently, A, U, C, or G or a modified nucleotide base, such as those provided for herein. The N₁ nucleotides of the sense strand and the antisense strand represent the 5' end of the respective strands. For clarity, although Formula I utilizes N₁, N₂, N₃, etc. in both the sense and the antisense strand, the nucleotide bases do not need to be the same and are not intended to be the same. The siRNA that is illustrated in Formula I would be complementary to a target sequence.

[0149] For example, in some embodiments, the sense strand comprises a 2'-O-methyl modified nucleotide with a phosphorothioate (PS) modified backbone at N₁ and N₂, a 2'-fluoro modified nucleotide at N₃, N₇, N₈, N₉, N₁₂, and N₁₇, and a 2'-O-methyl modified nucleotide at N₄, N₅, N₆, N₁₀, N₁₁, N₁₃, N₁₄, N₁₅, N₁₆, N₁₈, and N₁₉.

[0150] In some embodiments, the antisense strand comprises a vinylphosphonate moiety attached to N₁, a 2'-fluoro modified nucleotide with a phosphorothioate (PS) modified backbone at N₂, a 2'-O-methyl modified nucleotide at N₃, N₄, N₅, N₆, N₇, N₈, N₉, N₁₀, N₁₁, N₁₂, N₁₃, N₁₅, N₁₆, N₁₇, N₁₈, and N₁₉, a 2'-fluoro modified nucleotide at N₁₄, and a 2'-O-methyl modified nucleotide with a phosphorothioate (PS) modified backbone at N₂₀ and N₂₁.

[0151] In some embodiments, an oligonucleotide molecule comprises a sense strand and antisense strand, wherein the sense strand includes a terminal cap moiety at the 5'-end,

the 3'-end, or both of the 5' and 3' ends of the sense strand. In other embodiments, the terminal cap moiety is an inverted deoxy abasic moiety.

[0152] In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, wherein the antisense strand comprises a glyceryl modification at the 3' end of the antisense strand.

[0153] In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, in which the sense strand comprises one or more, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or about one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the sense strand; and in which the antisense strand comprises about 1 to about 10 or more, specifically about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the antisense strand. In other embodiments, one or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more, pyrimidine nucleotides of the sense and/or antisense strand are chemically-modified with 2'-deoxy, 2'-O-methyl and/or 2'-deoxy-2'-fluoro nucleotides, with or without one or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more, phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages and/or a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends, being present in the same or different strand.

[0154] In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, in which the sense strand comprises about 1 to about 25, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the sense strand; and in which the antisense strand comprises about 1 to about 25 or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the antisense strand. In other embodiments, one or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more, pyrimidine nucleotides of the sense and/or antisense strand are chemically-modified with 2'-deoxy, 2'-O-methyl and/or 2'-deoxy-2'-fluoro nucleotides, with

or without about 1 to about 25 or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages and/or a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends, being present in the same or different strand.

[0155] In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, in which the antisense strand comprises one or more, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or about one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the sense strand and/or antisense strand, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the sense strand. In some embodiments, the antisense strand comprises about 1 to about 10 or more, specifically about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the antisense strand. In other embodiments, one or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more pyrimidine nucleotides of the sense and/or antisense strand are chemically-modified with 2'-deoxy, 2'-O-methyl and/or 2'-deoxy-2'-fluoro nucleotides, with or without one or more, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages and/or a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3' and 5'-ends, being present in the same or different strand.

[0156] In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, in which the antisense strand comprises about 1 to about 25 or more, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the sense strand; and the antisense strand comprises about 1 to about 25 or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) 2'-deoxy, 2'-O-methyl, 2'-deoxy-2'-fluoro, and/or one or more (e.g., about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more) universal base modified nucleotides, and optionally a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of the antisense strand. In other embodiments, one or more, for example about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more pyrimidine nucleotides of the sense

and/or antisense strand are chemically-modified with 2'-deoxy, 2'-O-methyl and/or 2'-deoxy-2'-fluoro nucleotides, with or without about 1 to about 5, for example about 1, 2, 3, 4, 5 or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages and/or a terminal cap molecule at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends, being present in the same or different strand.

[0157] In some embodiments, an oligonucleotide molecule described herein is a chemically-modified short interfering nucleic acid molecule having about 1 to about 25, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate internucleotide linkages in each strand of the oligonucleotide molecule. In some embodiments, an oligonucleotide molecule comprises a sense strand and an antisense strand, and the antisense strand comprises a phosphate backbone modification at the 3' end of the antisense strand. Alternatively and/or additionally, an oligonucleotide molecule comprises a sense strand and an antisense strand, and the sense strand comprises a phosphate backbone modification at the 5' end of the antisense strand. In some instances, the phosphate backbone modification is a phosphorothioate. In some instances, the phosphate backbone modification is a phosphorodithioate. In some instances, the phosphate backbone modification is a phosphonate. In some instances, the phosphate backbone modification is a phosphoramidate. In some instances, the phosphate backbone modification is a mesyl phosphoramidate. In some embodiments, the sense or antisense strand has three consecutive nucleosides that are coupled via two phosphorothioate backbone. In some embodiments, the sense or antisense strand has three consecutive nucleosides that are coupled via two phosphorodithioate backbone. In some embodiments, the sense or antisense strand has three consecutive nucleosides that are coupled via two phosphonate backbone. In some embodiments, the sense or antisense strand has three consecutive nucleosides that are coupled via two phosphoramidate backbone. In some embodiments, the sense or antisense strand has three consecutive nucleosides that are coupled via two mesyl phosphoramidate backbone.

[0158] In another embodiment, an oligonucleotide molecule described herein comprises 2'-5' internucleotide linkages. In some instances, the 2'-5' internucleotide linkage(s) is at the 3'-end, the 5'-end, or both of the 3'- and 5'-ends of one or both sequence strands. In addition instances, the 2'-5' internucleotide linkage(s) is present at various other positions within one or both sequence strands, for example, about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more including every internucleotide linkage of a pyrimidine nucleotide in one or both strands of the oligonucleotide molecule comprise a 2'-5' internucleotide linkage, or about 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more including every internucleotide linkage of a purine nucleotide in one or both strands of the oligonucleotide molecule comprise a 2'-5' internucleotide linkage.

[0159] In some embodiments, an oligonucleotide molecule is a single stranded molecule that mediates RNAi activity in a cell or reconstituted in vitro system, wherein the oligonucleotide molecule comprises a single stranded polynucleotide having complementarity to a target nucleic acid sequence, and wherein one or more pyrimidine nucleotides present in the oligonucleotide molecule are 2'-deoxy-2'-fluoro pyrimidine nucleotides (e.g., wherein all pyrimidine nucleotides are 2'-deoxy-2'-fluoro pyrimidine nucleotides or alternately a plurality of pyrimidine nucleotides are 2'-deoxy-2'-fluoro pyrimidine nucleotides), and wherein any

purine nucleotides present in the oligonucleotide molecule are 2'-deoxy purine nucleotides (e.g., wherein all purine nucleotides are 2'-deoxy purine nucleotides or alternately a plurality of purine nucleotides are 2'-deoxy purine nucleotides), and a terminal cap modification, that is optionally present at the 3'-end, the 5'-end, or both of the 3' and 5'-ends of the antisense sequence, the oligonucleotide molecule optionally further comprising about 1 to about 4 (e.g., about 1, 2, 3, or 4) terminal 2'-deoxynucleotides at the 3'-end of the oligonucleotide molecule, wherein the terminal nucleotides further comprise one or more (e.g., 1, 2, 3, or 4) phosphorothioate or mesyl phosphoramidate internucleotide linkages, and wherein the oligonucleotide molecule optionally further comprises a terminal phosphate group, such as a 5'-terminal phosphate group.

[0160] In some cases, one or more of the synthetic nucleotide analogues described herein are resistant toward nucleases such as for example ribonuclease such as RNase H, deoxyribonuclease such as DNase, or exonuclease such as 5'-3' exonuclease and 3'-5' exonuclease when compared to natural polynucleic acid molecules and endonucleases. In some instances, synthetic nucleotide analogues comprising 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl, 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE), or 2'-O-N-methylacetamido (2'-O-NMA) modified, LNA, ENA, PNA, HNA, morpholino, methylphosphonate nucleotides, thiolphosphonate nucleotides, 2'-fluoro N3-P5'-phosphoramidites, or combinations thereof are resistant toward nucleases such as for example ribonuclease such as RNase H, deoxyribonuclease such as DNase, or exonuclease such as 5'-3' exonuclease and 3'-5' exonuclease. In some instances, 2'-O-methyl modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-methoxyethyl (2'-O-MOE) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-aminopropyl modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-deoxy modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-deoxy-2'-fluoro modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-aminopropyl (2'-O-AP) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-dimethylaminoethyl (2'-O-DMAOE) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-dimethylaminopropyl (2'-O-DMAP) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, 2'-O-N-methylacetamido (2'-O-NMA) modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, LNA modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance).

exonuclease resistance). In some instances, ENA modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, HNA modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, morpholinos is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, PNA modified oligonucleotide molecule is resistant to nucleases (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, methylphosphonate nucleotides modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, thiolphosphonate nucleotides modified oligonucleotide molecule is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, oligonucleotide molecule comprising 2'-fluoro N3-P5'-phosphoramidites is nuclease resistance (e.g., RNase H, DNase, 5'-3' exonuclease or 3'-5' exonuclease resistance). In some instances, the 5' conjugates described herein inhibit 5'-3' exonucleolytic cleavage. In some instances, the 3' conjugates described herein inhibit 3'-5' exonucleolytic cleavage.

[0161] In some embodiments, one or more of the synthetic nucleotide analogues described herein have increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. The one or more of the synthetic nucleotide analogues comprising 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl, 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE), or 2'-O-N-methylacetamido (2'-O-NMA) modified, LNA, ENA, PNA, HNA, morpholino, methylphosphonate nucleotides, thiolphosphonate nucleotides, or 2'-fluoro N3-P5'-phosphoramidites have increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-methyl modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-methoxyethyl (2'-O-MOE) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-aminopropyl modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-deoxy modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-deoxy-2'-fluoro modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-aminopropyl (2'-O-AP) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-dimethylaminoethyl (2'-O-DMAOE) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-dimethylaminopropyl (2'-O-DMAP) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O-dimethylaminoethoxyethyl (2'-O-DMAEOE) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule.

DMAEOE) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, 2'-O—N-methylacetamido (2'-O-NMA) modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, LNA modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, ENA modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, PNA modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, HNA modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, morpholino modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, methylphosphonate nucleotides modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, thiolphosphonate nucleotides modified oligonucleotide molecule has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some instances, oligonucleotide molecule comprising 2'-fluoro N3-P5'-phosphoramidites has increased binding affinity toward their mRNA target relative to an equivalent natural polynucleic acid molecule. In some cases, the increased affinity is illustrated with a lower K_d, a higher melt temperature (T_m), or a combination thereof.

[0162] In some embodiments, an oligonucleotide molecule described herein is a chirally pure (or stereo pure) polynucleic acid molecule, or a polynucleic acid molecule comprising a single enantiomer. In some instances, the oligonucleotide molecule comprises L-nucleotide. In some instances, the oligonucleotide molecule comprises D-nucleotides. In some instance, an oligonucleotide molecule composition comprises less than 30%, 25%, 20%, 15%, 10%, 5%, 4%, 3%, 2%, 1%, or less of its mirror enantiomer. In some cases, an oligonucleotide molecule composition comprises less than 30%, 25%, 20%, 15%, 10%, 5%, 4%, 3%, 2%, 1%, or less of a racemic mixture.

[0163] In some embodiments, an oligonucleotide molecule described herein is further modified to include an aptamer conjugating moiety. In some instances, the aptamer conjugating moiety is a DNA aptamer conjugating moiety. In some instances, the aptamer conjugating moiety is Alphamer, which comprises an aptamer portion that recognizes a specific cell-surface target and a portion that presents a specific epitopes for attaching to circulating antibodies.

[0164] In additional embodiments, an oligonucleotide molecule described herein is modified to increase its stability. In some embodiment, the oligonucleotide molecule is RNA (e.g., siRNA). In some instances, the oligonucleotide molecule is modified by one or more of the modifications described above to increase its stability. In some cases, the oligonucleotide molecule is modified at the 2' hydroxyl position, such as by 2'-O-methyl, 2'-O-methoxyethyl (2'-O-MOE), 2'-O-aminopropyl, 2'-deoxy, 2'-deoxy-2'-fluoro, 2'-O-aminopropyl (2'-O-AP), 2'-O-dimethylaminoethyl (2'-O-DMAOE), 2'-O-dimethylaminopropyl (2'-O-DMAP), 2'-O-dimethylanunoethoxyethyl (2'-O-DMAEOE), or

2'-O—N-methylacetamido (2'-O-NMA) modification or by a locked or bridged ribose conformation (e.g., LNA or ENA). In some cases, the oligonucleotide molecule is modified by 2'-O-methyl and/or 2'-O-methoxyethyl ribose. In some cases, the oligonucleotide molecule also includes morpholinos, PNAs, HNA, methylphosphonate nucleotides, thiolphosphonate nucleotides, and/or 2'-fluoro N3-P5'-phosphoramidites to increase its stability. In some instances, the oligonucleotide molecule is a chirally pure (or stereo pure) oligonucleotide molecule. In some instances, the chirally pure (or stereo pure) oligonucleotide molecule is modified to increase its stability. Suitable modifications to the RNA to increase stability for delivery will be apparent to the skilled person.

[0165] In some embodiments, the oligonucleotide molecule comprises 2' modifications. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 3, 7, 8, 9, 12, and 17 from the 5' end of the sense strand are not modified with a 2'-methyl modification. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 3, 7, 8, 9, 12, and 17 from the 5' end of the sense strand are modified with a 2'-fluoro modification. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 2 and 14 from the 5' end of the anti-sense strand are not modified with a 2'-methyl modification. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 2 and 14 from the 5' end of the anti-sense strand are modified with a 2'-fluoro modification. In some embodiments, any of the nucleotides may further comprise a 5'-phosphorothioate group modification. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 1 and 2 from the 5' end of the sense strand are modified with a 5'-phosphorothioate group modification. In some embodiments, the nucleotides of the oligonucleotide molecule at positions 1, 2, 20, and 21 from the 5' end of the antisense strand are modified with a 5'-phosphorothioate group modification. In some embodiments, the 5' end of the sense or antisense strand of the oligonucleotide molecule may further comprise a vinylphosphonate modification. In some embodiments, the nucleotide of the oligonucleotide molecule at position 1 from the 5' end of the antisense strand is modified with a vinylphosphonate modification.

[0166] In some instances, the oligonucleotide molecule is a double-stranded polynucleotide molecule comprising self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof. In some instances, the oligonucleotide molecule is assembled from two separate polynucleotides, where one strand is the sense strand and the other is the antisense strand, wherein the antisense and sense strands are self-complementary (e.g., each strand comprises nucleotide sequence that is complementary to nucleotide sequence in the other strand; such as where the antisense strand and sense strand form a duplex or double stranded structure, for example wherein the double stranded region is about 19, 20, 21, 22, 23, or more base pairs); the antisense strand comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense strand comprises nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof. Alternatively, the oligonucleotide molecule is assembled from a single oligonucleotide, where the self-complementary sense

and antisense regions of the oligonucleotide molecule are linked by means of a nucleic acid based or non-nucleic acid-based linker(s).

[0167] In some cases, the oligonucleotide molecule is a polynucleotide with a duplex, asymmetric duplex, hairpin or asymmetric hairpin secondary structure, having self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a separate target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof. In other cases, the oligonucleotide molecule is a circular single-stranded polynucleotide having two or more loop structures and a stem comprising self-complementary sense and antisense regions, wherein the antisense region comprises nucleotide sequence that is complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof and the sense region having nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof, and wherein the circular polynucleotide is processed either *in vivo* or *in vitro* to generate an active oligonucleotide molecule capable of mediating RNAi. In additional cases, the oligonucleotide molecule also comprises a single-stranded polynucleotide having nucleotide sequence complementary to nucleotide sequence in a target nucleic acid molecule or a portion thereof (for example, where such oligonucleotide molecule does not require the presence within the oligonucleotide molecule of nucleotide sequence corresponding to the target nucleic acid sequence or a portion thereof), wherein the single stranded polynucleotide further comprises a terminal phosphate group, such as a 5'-phosphate, or 5', 3'-diphosphate.

[0168] In some instances, an asymmetric hairpin is a linear oligonucleotide molecule comprising an antisense region, a loop portion that comprises nucleotides or non-nucleotides, and a sense region that comprises fewer nucleotides than the antisense region to the extent that the sense region has enough complementary nucleotides to base pair with the antisense region and form a duplex with loop. For example, an asymmetric hairpin oligonucleotide molecule comprises an antisense region having length sufficient to mediate RNAi in a cell or *in vitro* system (e.g., about 19 to about 22 nucleotides) and a loop region comprising about 4 to about 8 nucleotides, and a sense region having about 3 to about 18 nucleotides that are complementary to the antisense region. In some cases, the asymmetric hairpin oligonucleotide molecule also comprises a 5'-terminal phosphate group that is chemically modified. In additional cases, the loop portion of the asymmetric hairpin oligonucleotide molecule comprises nucleotides, non-nucleotides, linker molecules, or conjugate molecules.

[0169] In some embodiments, an asymmetric duplex is an oligonucleotide molecule having two separate strands comprising a sense region and an antisense region, wherein the sense region comprises fewer nucleotides than the antisense region to the extent that the sense region has enough complementary nucleotides to base pair with the antisense region and form a duplex. For example, an asymmetric duplex oligonucleotide molecule comprises an antisense region having length sufficient to mediate RNAi in a cell or *in vitro* system (e.g., about 19 to about 22 nucleotides) and a sense region having about 3 to about 19 nucleotides that are complementary to the antisense region.

[0170] In some cases, a universal base refers to nucleotide base analogs that form base pairs with each of the natural DNA/RNA bases with little discrimination between them.

Non-limiting examples of universal bases include C-phenyl, C-naphthyl and other aromatic derivatives, inosine, azole carboxamides, and nitroazole derivatives such as 3-nitropyrrrole, 4-nitroindole, 5-nitroindole, and 6-nitroindole as known in the art.

[0171] In some embodiments, the dsRNA agents are 5' phosphorylated or include a phosphoryl analog at the 5' prime terminus. 5'-phosphate modifications include those which are compatible with RISC mediated gene silencing. Suitable modifications include: 5'-monophosphate ((HO)₂(O)P—O-5'); 5'-diphosphate ((HO)₂(O)P—O—P(HO)(O)—O-5'); 5'-triphosphate ((HO)₂(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-guanosine cap (7-methylated or non-methylated) (7m-G-O-5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-adenosine cap (Appp), and any modified or unmodified nucleotide cap structure (N—O-5'-(HO)(O)P—O—(HO)(O)P—O—P(HO)(O)—O-5'); 5'-monothiophosphate (phosphorothioate; (HO)₂(S)P—O-5'); 5'-monodithiophosphate (phosphorodithioate; (HO)(HS)(S)P—O-5'), 5'-phosphorothiolate ((HO)₂(O)P—S-5'); phosphorodithioate [$\text{—O}_2\text{PS}_2\text{—}$]; phosphonate [$\text{—PO(OH)}_2\text{—}$]; phosphoramidate [$\text{—O—P(OH)}_2\text{—}$]; mesyl phosphoramidate (CH₃)(SO₂)(N)P(O)₂—O-5'); any additional combination of oxygen/sulfur replaced monophosphate, diphosphate and triphosphates (e.g. 5'-alpha-thiotriphosphate, 5'-gamma-thiotriphosphate, etc.), 5'-phosphoramidates ((HO)₂(O)P—NH-5', (HO)(NH₂)(O)P—O-5'), 5'-alkylphosphonates (R=alkyl=methyl, ethyl, isopropyl, propyl, etc., e.g. RP(OH)(O)—O-5'-, 5'-alkenylphosphonates (i.e. vinyl, substituted vinyl), (OH)₂(O)P-5'-CH₂-), 5'-alkyletherphosphonates (R=alkylether=methoxymethyl (MeOCH₂-), ethoxymethyl, etc., e.g. RP(OH)(O)—O-5'-). In some embodiments, the modification can be placed in the antisense strand of a dsRNA agent.

[0172] In some embodiments, the sequence of the oligonucleotide molecule is at least 40%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 98%, 99%, or 99.5% complementary to a target sequence of GYS1. In some embodiments, the target sequence of GYS1 is a nucleic acid sequence of about 10-50 base pair length, about 15-50 base pair length, 15-40 base pair length, 15-30 base pair length, or 15-25 base pair length sequences in GYS1, in which the first nucleotide of the target sequence starts at any nucleotide in GYS1 mRNA transcript in the coding region, or in the 5' or 3'-untranslated region (UTR). For example, the first nucleotide of the target sequence can be selected so that it starts at the nucleic acid location (nal, number starting from the 5'-end of the full length of GYS1 mRNA, e.g., the 5'-end first nucleotide is nal. 1) 1, nal 2, nal 3, nal 4, nal 5, nal 6, nal 7, nal 8, nal 9, nal 10, nal 11, nal 12, nal 13, nal 14, nal 15, nal 16, nal 17, or any other nucleic acid location in the coding or noncoding regions (5' or 3'-untranslated region) of GYS1 mRNA. In some embodiments, the first nucleotide of the target sequence can be selected so that it starts at a location within, or between, nal 10-nal 15, nal 10-nal 20, nal 50-nal 60, nal 55-nal 65, nal 75-nal 85, nal 95-nal 105, nal 135-nal 145, nal 155-nal 165, nal 225-nal 235, nal 265-nal 275, nal 275-nal 245, nal 245-nal 255, nal 285-nal 335, nal 335-nal 345, nal 385-nal 395, nal 515-nal 525, nal 665-nal 675, nal 675-nal 685, nal 695-nal 705, nal 705-nal 715, nal 875-nal 885, nal 885-nal 895, nal 895-nal 905, nal 1035-nal 1045, nal 1045-nal 1055, nal 1125-nal 1135, nal 1135-nal 1145, nal 1145-nal 1155, nal 1155-nal 1165, nal 1125-nal 1135, nal 1155-nal 1165, nal 1225-nal 1235, nal 1235-nal 1245, nal 1275-nal 1245, nal 1245-nal 1255, nal 1265-nal 1275, nal 1125-nal 135, nal 1155-nal 1165, nal 1225-nal 1235, nal 1235-nal 1245, nal 1275-nal

1245, nal 1245-nal 1255, nal 1265-nal 1275, nal 1275-nal 1285, nal 1335-nal 1345, nal 1345-nal 1355, nal 1525-nal 1535, nal 1535-nal 1545, nal 1605-nal 1615, nal 1615-c. 1625, nal 1625-nal 1635, nal 1635-1735, nal 1735-1835, nal 1835-1935, nal. 1836-1856, nal 1935-2000, nal 2000-2100, nal 2100-2200, nal 2200-2260, nal 2260-2400, nal 2400-2500, nal 2500-2600, nal 2600-2700, nal 2700-2800, nal 2800-2500, nal 2500-2600, nal 2600-2700, nal 2700-2800, nal 2800-2860, etc. In some embodiments, the sequence of GYS1 mRNA is provided as NCBI Reference Sequence: NM_002103.

[0173] In some embodiments, the antisense strand of the dsRNA agent is 100% complementary to a target RNA to hybridize thereto and inhibits its expression through RNA interference. The target RNA can be any RNA expressed in a cell. In another embodiment, the cell is a tumor cell, a liver cell, a muscle cell, an immune cell, a dendritic cell, a heart cell, or a cell of the central nervous system. In another embodiment, the antisense strand of the dsRNA agent is at least 99%, at least 98%, at least 97%, at least 96%, 95%, at least 90%, at least 85%, at least 80%, at least 75%, at least 70%, at least 65%, at least 60%, at least 55%, or at least 50% complementary to a target RNA. In some embodiments, the target RNA is GYS1 RNA. In some embodiments, the siRNA molecule is a siRNA that reduces the expression of GYS1. In some embodiments, the siRNA molecule is a siRNA that reduces the expression of GYS and does not reduce the expression of other RNAs by more than 50% in an assay described herein at a concentration of no more than 200 nM as described herein.

[0174] The siRNA can be targeted against any gene or RNA (e.g. mRNA) transcript of interest.

[0175] Other modifications and patterns of modifications can be found in, for example, U.S. Pat. No. 10,233,448, which is hereby incorporated by reference.

[0176] Other modifications and patterns of modifications can be found in, for example, Anderson et al. Nucleic Acids Research 2021, 49 (16), 9026-9041, which is hereby incorporated by reference.

[0177] Other modifications and patterns of modifications can be found in, for example, PCT Publication No. WO2021/030778, which is hereby incorporated by reference.

[0178] Other modifications and patterns of modifications can be found in, for example, PCT Publication No. WO2021/030763, which is hereby incorporated by reference.

[0179] In some embodiments, the siRNA is linked to a protein, such as a FN3 domain. The siRNA can be linked to multiple FN3 domains that bind to the same target protein or different target proteins. In some embodiments, the linker is attached to the sense strand, which is used to facilitate the linkage of the sense strand to the FN3 domain.

[0180] In some embodiments, compositions are provided herein having a formula of $(X1)_n-(X2)_q-(X3)_y-L-X4$, wherein X1 is a first FN3 domain, X2 is second FN3 domain, X3 is a third FN3 domain or half-life extender molecule, L is a linker, and X4 is a nucleic acid molecule, such as, but not limited to a siRNA molecule, wherein n, q, and y are each independently 0 or 1. In some embodiments, X1, X2, and X3 bind to different target proteins. In some embodiments, y is 0. In some embodiments, n is 1, q is 0, and y is 0. In some embodiments, n is 1, q is 1, and y is 0. In some embodiments, n is 1, q is 1, and y is 1. In some embodiments, the third FN3 domain increases the half-life of the molecule as a whole as compared to a molecule without X3. In some embodiments, the half-life extending

moiety is a FN3 domain that binds to albumin. Examples of such FN3 domains include, but are not limited to, those described in U.S. Patent Application Publication No. 20170348397 and U.S. Pat. No. 9,156,887, which is hereby incorporated by reference in its entirety. The FN3 domains may incorporate other subunits for example via covalent interaction. In some embodiments, the FN3 domains further comprise a half-life extending moiety. Exemplary half-life extending moieties are albumin, albumin variants, albumin-binding proteins and/or domains, transferrin and fragments and analogues thereof, and Fc regions. Amino acid sequences of the human Fc regions are well known, and include IgG1, IgG2, IgG3, IgG4, IgM, IgA and IgE Fc regions. In some embodiments, the FN3 domains may incorporate a second FN3 domain that binds to a molecule that extends the half-life of the entire molecule, such as, but not limited to, any of the half-life extending moieties described herein. In some embodiments, the second FN3 domain binds to albumin, albumin variants, albumin-binding proteins and/or domains, and fragments and analogues thereof.

[0181] In some embodiments, compositions are provided herein having a formula of $(X1)-(X2)-L-(X4)$, wherein X1 is a first FN3 domain, X2 is second FN3 domain, L is a linker, and X4 is a nucleic acid molecule. In some embodiments, X4 is a siRNA molecule. In some embodiments, X1 is a FN3 domain that binds to one of CD71. In some embodiments, X2 is a FN3 domain that binds to one of CD71. In some embodiments X1 and X2 do not bind to the same target protein. In some embodiments, X1 and X2 bind to the same target protein, but at different binding sites on the protein. In some embodiments, X1 and X2 bind to the same target protein. In some embodiments, X1 and X2 are FN3 domains that bind to CD71. In some embodiments, the composition does not comprise (e.g. is free of) a compound or protein that binds to ASGPR.

[0182] In some embodiments, compositions are provided herein having a formula of $C-(X1)_n-(X2)_q-[L-X4]-(X3)_y$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1, are provided.

[0183] In some embodiments, compositions are provided herein having a formula of $(X1)_n-(X2)_q-[L-X4]-(X3)_y-C$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1, are provided.

[0184] In some embodiments, compositions are provided herein having a formula of $C-(X1)_n-(X2)_q-[L-X4]L-(X3)_y$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1, are provided.

[0185] In some embodiments, compositions are provided herein having a formula of $(X1)_n-(X2)_q-[L-X4]L-(X3)_y-C$, wherein X1 is a first FN3 domain; X2 is second FN3 domain; X3 is a third FN3 domain or half-life extender molecule; L is a linker; X4 is an oligonucleotide molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1, are provided.

[0186] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of $C-L_1-X_s$ and B_1 has a formula of $X_{AS}-L_2-F_1$, wherein.

[0187] C is a polymer, such as PEG;

[0188] L_1 and L_2 are each, independently, a linker;

[0189] X_s is a 5' to 3' oligonucleotide sense strand of a double stranded siRNA molecule;

[0190] X_{AS} is a 3' to 5' oligonucleotide antisense strand of a double stranded siRNA molecule;

[0191] F_1 is a polypeptide comprising at least one FN3 domain;

[0192] wherein X_s and X_{AS} form a double stranded oligonucleotide molecule to form the composition/complex.

[0193] In some embodiments, C can be a molecule that extends the half-life of the molecule. Examples of such moieties are described herein. In some embodiments, C can also be Endoportin, INF-7, TAT, polyarginine, polylysine, or an amphipathic peptide. These moieties can be used in place of or in addition to other half-life extending moieties provided for herein. In some embodiments, C can be a molecule that delivers the complex into the cell, the endosome, or the ER; said molecules are selected from those peptides listed in Table 9:

SEQ ID NO: Name	Sequence
750 TAT	RKKRRQRRR
751 Penetratin	RQIKIWFQNRRMKWKK
752 Transportan	GWTLSNAGYLLGKINKALAALAKKIL
753 MAP	KLALKLALKALKAAKLKA
754 Pep-1	KETWWETWWTEWSQPKKKRKV
755 KDEL	KDEL
756 GALA	WEAALAEALAEALAEHLAEALAEALAEALAA
757 HA2	GDIMGEWGNEIFGAIAAGFLGC
758 Aurine 1.2	GLFDIIEKKIAESF
759 MPG	GALFLGWLGAAGSTMGAPKSKRKV
760 TP-10	AGYLLGKINKALAALAKKIL
761 EB-1	LIRLWSHLIHIWFQNRRLKWKKK
762 HA2- Penetratin	GLFGAIAAGFIENGWEGMIDGRQIKIWFQNRR MKWKK
763 Endosomolytic	FPKKLAHALHLLALLALHLAHLAKKA
764 Endosomolytic	LFEAIEGFIENGWEGMIDGWYG
765 Endosomolytic	LFEAIEGFIENGWEGMIDGWYGRKKRRQRR
766 Endosomolytic	IGAVLKVLTTGLPALISWIKRKRQQ
767 ER Targeting	MKLAVTLTLVTLALSSSSASA
1061 ER Targeting	RLIEDICLPRWGCLWEDDKDEL
1062 ER Targeting	MIRTLTLLSTLVAGALSK

-continued

SEQ ID NO: Name	Sequence
1063 ER Targeting	ILSSLTVTQLLRRRLHQWIK
1064 ER Targeting	MIRTLTLLSTLVAGALSKDEL

[0194] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of X_s and B_1 has a formula of $X_{AS}-L_2-F_1$.

[0195] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of $C-L_1-X_s$ and B_1 has a formula of X_{AS} .

[0196] In some embodiments, the sense strand is a sense strand as provided for herein.

[0197] In some embodiments, the antisense strand is an antisense strand as provided for herein.

[0198] In some embodiments, the sense and antisense strand form a double stranded siRNA molecule that targets GYS1. In some embodiments, the double stranded oligonucleotide is about 21-23 nucleotides base pairs in length. In certain embodiments, C is optional.

[0199] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of $F_1-L_1-X_s$ and B_1 has a formula of $X_{AS}-L_2-C$, wherein:

[0200] F_1 is a polypeptide comprising at least one FN3 domain;

[0201] L_1 and L_2 are each, independently, a linker;

[0202] C is a polymer, such as PEG;

[0203] X_s is a 5' to 3' oligonucleotide sense strand of a double stranded siRNA molecule;

[0204] X_{AS} is a 3' to 5' oligonucleotide antisense strand of a double stranded siRNA molecule;

[0205] wherein X_s and X_{AS} form a double stranded oligonucleotide molecule to form the composition/complex. In certain embodiments, C is optional.

[0206] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of X , and B_1 has a formula of $X_{AS}-L_2-C$.

[0207] In some embodiments, compositions or complexes are provided having a formula of A_1-B_1 , wherein A_1 has a formula of $F_1-L_1-X_s$ and B_1 has a formula of X_{AS} .

[0208] In some embodiments, C is a natural or synthetic polymer, consisting of long chains of branched or unbranched monomers, and/or cross-linked network of monomers in two or three dimensions. In some instances, the polymer includes a polysaccharide, lignin, rubber, or polyalkylene oxide, which can be for example, polyethylene glycol. In some instances, the at least one polymer includes, but is not limited to, alpha-, omega-dihydroxypolyethylene glycol, biodegradable lactone-based polymer, e.g. polyacrylic acid, polylactide acid (PLA), poly(glycolic acid) (PGA), polypropylene, polystyrene, polyolefin, polyamide, polycyanoacrylate, polyimide, polyethyleneterephthalat (PET, PETG), polyethylene-B-terephthalate (PETE), polytetramethylene glycol (PTG), or polyurethane as well as mixtures thereof. As used herein, a mixture refers to the use of different polymers within the same compound as well as in reference to block copolymers. In some cases, block copolymers are polymers wherein at least one section of a polymer is build up from monomers of another polymer. In some instances, the polymer comprises polyalkylene oxide. In some instances, the polymer comprises PEG. In some

instances, the polymer comprises polyethylene imide (PEI) or hydroxy ethyl starch (HES).

[0209] In some embodiments, the polyalkylene oxide (e.g., PEG) is a polydispers or monodispers compound. In some instances, polydispers material comprises disperse distribution of different molecular weight of the material, characterized by mean weight (weight average) size and dispersity. In some instances, the monodisperse PEG comprises one size of molecules. In some embodiments, C is poly- or monodispersed polyalkylene oxide (e.g., PEG) and the indicated molecular weight represents an average of the molecular weight of the polyalkylene oxide, e.g., PEG, molecules.

[0210] In some embodiments, the molecular weight of the polyalkylene oxide (e.g., PEG) is about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1450, 1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3250, 3350, 3500, 3750, 4000, 4250, 4500, 4600, 4750, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 10,000, 12,000, 20,000, 35,000, 40,000, 50,000, 60,000, or 100,000 Da.

[0211] In some embodiments, C is polyalkylene oxide (e.g., PEG) and has a molecular weight of about 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1450, 1500, 1600, 1700, 1800, 1900, 2000, 2100, 2200, 2300, 2400, 2500, 2600, 2700, 2800, 2900, 3000, 3250, 3350, 3500, 3750, 4000, 4250, 4500, 4600, 4750, 5000, 5500, 6000, 6500, 7000, 7500, 8000, 10,000, 12,000, 20,000, 35,000, 40,000, 50,000, 60,000, or 100,000 Da. In some instances, the molecular weight of C is about 200 Da. In some instances, the molecular weight of C is about 300 Da. In some instances, the molecular weight of C is about 400 Da. In some instances, the molecular weight of C is about 500 Da. In some instances, the molecular weight of C is about 600 Da. In some instances, the molecular weight of C is about 700 Da. In some instances, the molecular weight of C is about 800 Da. In some instances, the molecular weight of C is about 900 Da. In some instances, the molecular weight of C is about 1000 Da. In some instances, the molecular weight of C is about 1100 Da. In some instances, the molecular weight of C is about 1200 Da. In some instances, the molecular weight of C is about 1300 Da. In some instances, the molecular weight of C is about 1400 Da. In some instances, the molecular weight of C is about 1450 Da. In some instances, the molecular weight of C is about 1500 Da. In some instances, the molecular weight of C is about 1600 Da. In some instances, the molecular weight of C is about 1700 Da. In some instances, the molecular weight of C is about 1800 Da. In some instances, the molecular weight of C is about 1900 Da. In some instances, the molecular weight of C is about 2000 Da. In some instances, the molecular weight of C is about 2100 Da. In some instances, the molecular weight of C is about 2200 Da. In some instances, the molecular weight of C is about 2300 Da. In some instances, the molecular weight of C is about 2400 Da. In some instances, the molecular weight of C is about 2500 Da. In some instances, the molecular weight of C is about 2600 Da. In some instances, the molecular weight of C is about 2700 Da. In some instances, the molecular weight of C is about 2800 Da. In

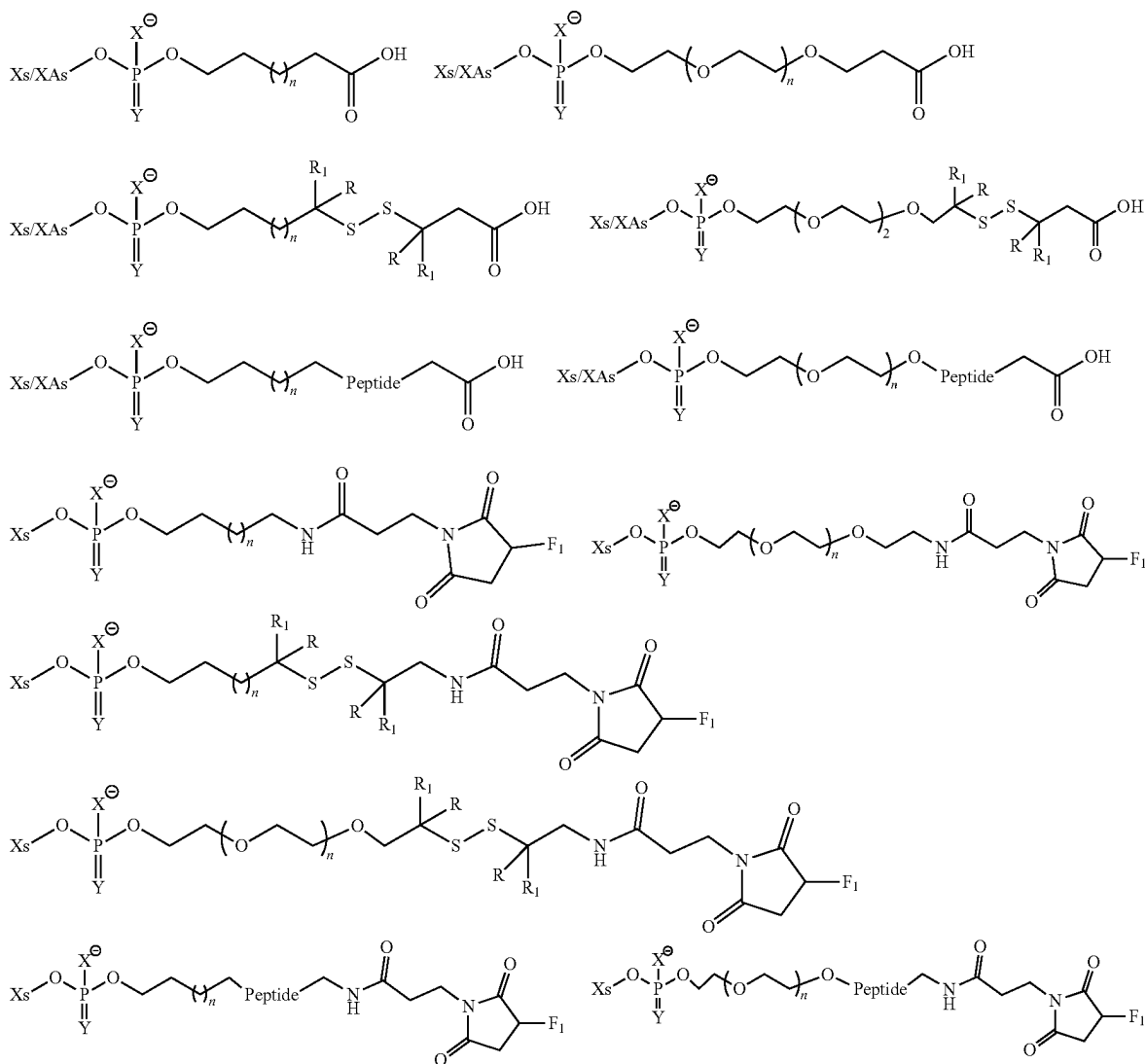
some instances, the molecular weight of C is about 2900 Da. In some instances, the molecular weight of C is about 3000 Da. In some instances, the molecular weight of C is about 3250 Da. In some instances, the molecular weight of C is about 3350 Da. In some instances, the molecular weight of C is about 3500 Da. In some instances, the molecular weight of C is about 3750 Da. In some instances, the molecular weight of C is about 4000 Da. In some instances, the molecular weight of C is about 4250 Da. In some instances, the molecular weight of C is about 4500 Da. In some instances, the molecular weight of C is about 4600 Da. In some instances, the molecular weight of C is about 4750 Da. In some instances, the molecular weight of C is about 5000 Da. In some instances, the molecular weight of C is about 5500 Da. In some instances, the molecular weight of C is about 6000 Da. In some instances, the molecular weight of C is about 6500 Da. In some instances, the molecular weight of C is about 7000 Da. In some instances, the molecular weight of C is about 7500 Da. In some instances, the molecular weight of C is about 8000 Da. In some instances, the molecular weight of C is about 10,000 Da. In some instances, the molecular weight of C is about 12,000 Da. In some instances, the molecular weight of C is about 20,000 Da. In some instances, the molecular weight of C is about 35,000 Da. In some instances, the molecular weight of C is about 40,000 Da. In some instances, the molecular weight of C is about 50,000 Da. In some instances, the molecular weight of C is about 60,000 Da. In some instances, the molecular weight of C is about 100,000 Da.

[0212] In some embodiments, the polyalkylene oxide (e.g., PEG) is a discrete PEG, in which the discrete PEG is a polymeric PEG comprising more than one repeating ethylene oxide units. In some instances, a discrete PEG (dPEG) comprises from 2 to 60, from 2 to 50, or from 2 to 48 repeating ethylene oxide units. In some instances, a dPEG comprises about 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 24, 26, 28, 30, 35, 40, 42, 48, 50 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 2 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 3 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 4 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 5 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 6 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 7 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 8 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 9 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 10 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 11 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 12 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 13 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 14 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 15 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 16 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 17 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 18 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 19 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 20 or more repeating

ethylene oxide units. In some instances, a dPEG comprises about 22 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 24 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 26 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 28 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 30 or more repeating ethylene oxide units. In some

molecular weight, rather than an average molecular weight. In some cases, a dPEG described herein is a dPEG from Quanta Biodesign, LMD.

[0213] In some embodiments, L_1 is any linker that can be used to link the polymer C to the sense strand X_S or to link the polypeptide of F_1 to the sense strand X_S . In some embodiments, L_1 has a formula of:



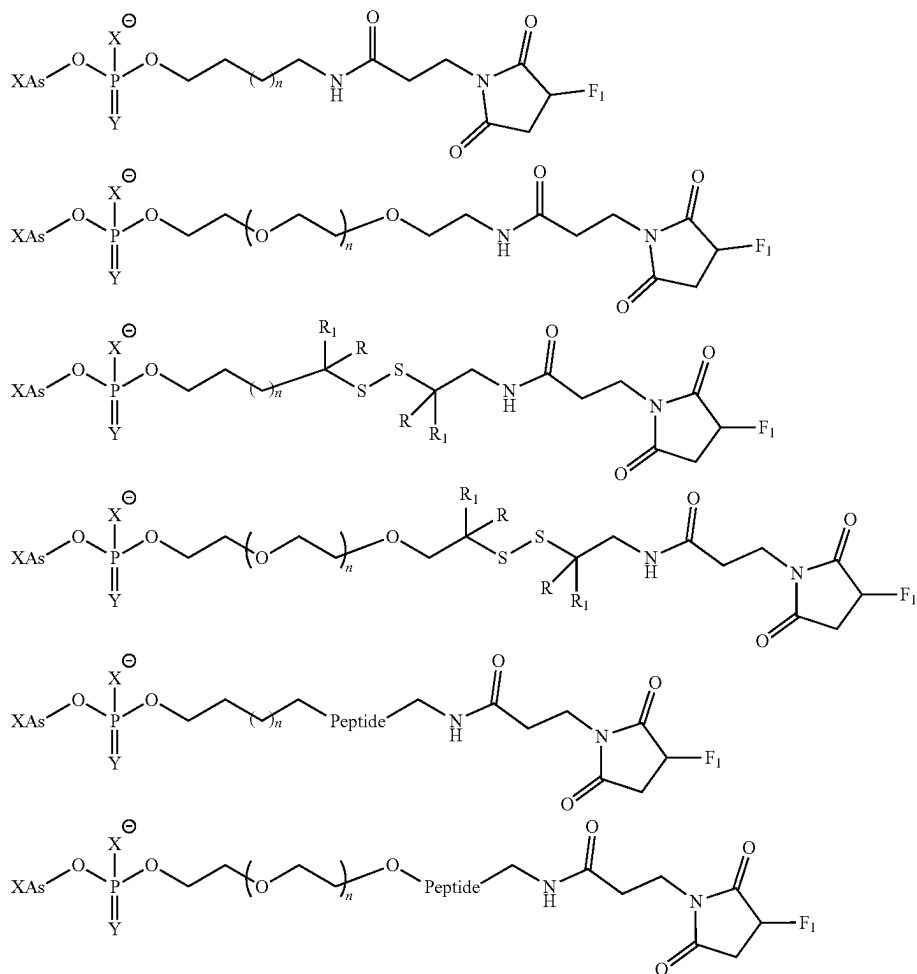
instances, a dPEG comprises about 35 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 40 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 42 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 48 or more repeating ethylene oxide units. In some instances, a dPEG comprises about 50 or more repeating ethylene oxide units. In some cases, a dPEG is synthesized as a single molecular weight compound from pure (e.g., about 95%, 98%, 99%, or 99.5%) starting material in a step-wise fashion. In some cases, a dPEG has a specific

wherein X_s , X_{As} , and F_1 are as defined above.

[0214] In some embodiments, $n=0-20$. In some embodiments, R and R_1 are independently methyl. In some embodiments, R and R_1 are independently present or both are absent. In some embodiments, X and Y are independently S . In some embodiments, X and Y are independently present or absent. In some embodiments, Peptide is an enzymatically cleavable peptide, such as, but not limited to, Val-Cit, Val-Ala etc.

[0215] In some embodiments, L_2 is any linker that can be used to link the polypeptide of F_1 to the antisense strand X_{As} or to link the polymer C to the antisense strand X_{As} .

[0216] In some embodiments, L_2 has a formula of in the complex of:

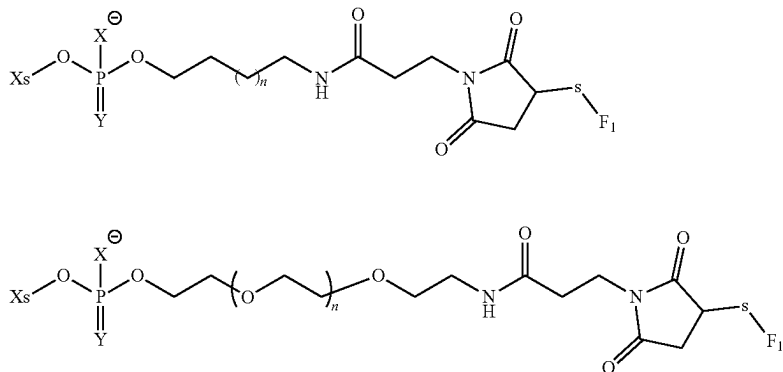


wherein X_{AS} and F_1 are as defined above.

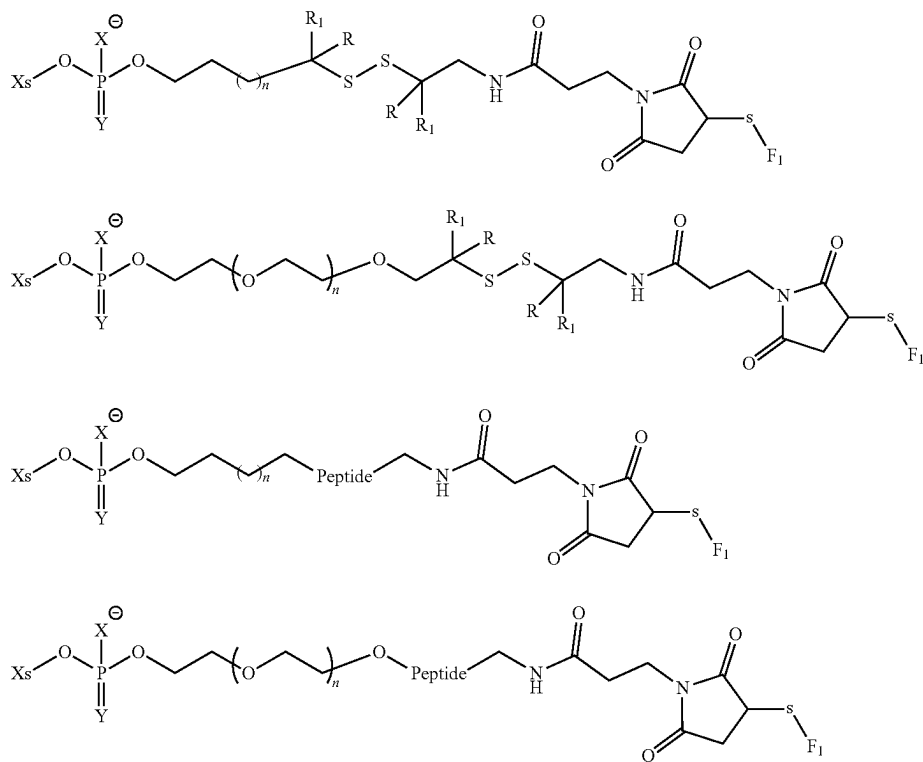
[0217] In some embodiments, $n=0-20$. In some embodiments, R and R_1 are independently methyl. In some embodiments, R and R_1 are independently present or both are absent. In some embodiments, X and Y are independently S . In some embodiments, X and Y are independently present or

absent. In some embodiments, Peptide is an enzymatically cleavable peptide, such as, but not limited to, Val-Cit, Val-Ala etc.

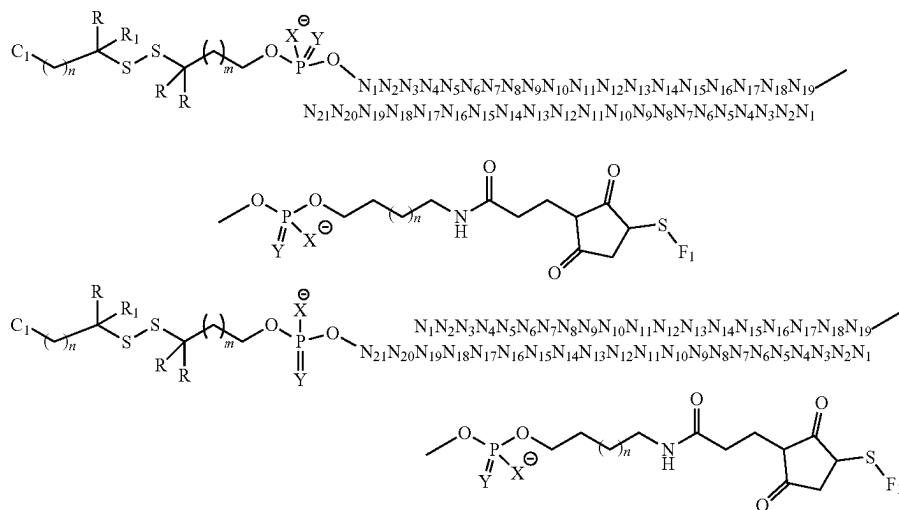
[0218] In some embodiments, the linker is covalently attached to F_1 through a cysteine residue present on F_1 , which can be illustrated as follows:



-continued



[0219] In some embodiments, A1-B1 has a formula of:



wherein C is the polymer, such as PEG, Endoport, INF-7, TAT, polyarginine, polylysine, an amphipathic peptide, or peptides listed in Table 9 as provided for herein, X_S is a 5' to 3' oligonucleotide sense strand of a double stranded siRNA molecule; X_{AS} is a 3' to 5' oligonucleotide antisense strand of a double stranded siRNA molecule; and F_1 is a polypeptide comprising at least one FN3 domain, wherein X_S and X_{AS} form a double stranded siRNA molecule. The

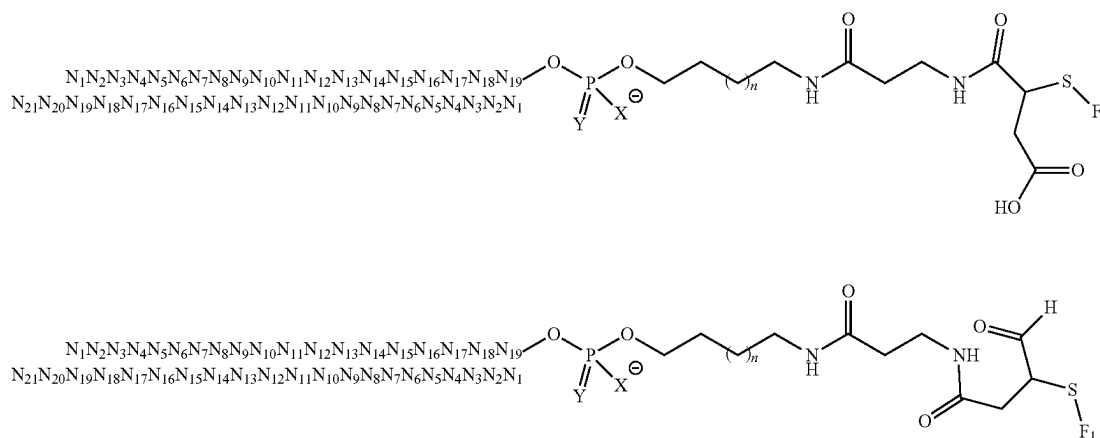
sense and antisense strands are represented by the "N" notations, wherein each nucleotide represented by N, is independently, A, U, C, or G or a modified nucleobase, such as those provided for herein. The N_1 nucleotides of the sense strand and the antisense strand represent the 5' end of the respective strands. For clarity, although Formula 1 utilizes N_1 , N_2 , N_3 , etc. in both the sense and the antisense strand, the nucleotide bases do not need to be the same and are not

intended to be the same. The siRNA that is illustrated in Formula I would be complementary to a target sequence.

[0220] For example, in some embodiments, the sense strand comprises a 2'-O-methyl modified nucleotide with a phosphorothioate (PS) modified backbone at N₁ and N₂, a 2'-fluoro modified nucleotide at N₃, N₇, N₈, N₉, N₁₂, and N₁₇, and a 2'-O-methyl modified nucleotide at N₄, N₅, N₆, N₁₀, N₁₁, N₁₃, N₁₄, N₁₅, N₁₆, N₁₈, and N₁₉.

[0221] In some embodiments, the antisense strand comprises a vinylphosphonate moiety attached to N₁, a 2'-fluoro-modified nucleotide with a phosphorothioate (PS) modified backbone at N₂, a 2'-O-methyl modified nucleotide at N₃, N₄, N₅, N₆, N₇, N₈, N₉, N₁₀, N₁₁, N₁₂, N₁₃, N₁₅, N₁₆, N₁₇, N₁₈, and N₁₉, a 2'-fluoro-modified nucleotide at N₁₄, and a 2'-O-methyl modified nucleotide with a phosphorothioate (PS) modified backbone at N₂₀ and N₂₁.

[0222] In some embodiments, a compound having a formula of.



wherein F₁ is a polypeptide comprising at least one FN3 domain and is conjugated to a linker, L₁, L₁ is linked to X_S, wherein X_S is a 5 to 3' oligonucleotide sense strand of a double stranded siRNA molecule and X_{AS} is a 3' to 5' oligonucleotide antisense strand of a double stranded siRNA molecule; and wherein X_S and X_{AS} form a double stranded siRNA molecule. The linker illustrated above, is a non-limiting example, and other types of linkers can be used.

[0223] In some embodiments, F₁ comprises polypeptide having a formula of (X₁)_n-(X₂)_q-(X₃)_y, wherein X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; wherein n, q, and y are each independently 0 or 1, provided that at least one of n, q, and y is 1. In some embodiments, n, q, and y are each 1. In some embodiments, n and q are 1 and y is 0. In some embodiments n and y are 1 and q is 0.

[0224] In some embodiment X₁ is a CD71 FN3 binding domain, such as one provided herein. In some embodiments, X₂ is a CD71 FN3 binding domain. In some embodiments, X₁ and X₂ are different CD71 FN3 binding domains. In some embodiments, the binding domains are the same. In some embodiments, X₃ is a FN3 domain that binds to human serum albumin. In some embodiments, X₃ is a Fc domain without effector function that extends the half-life of a protein. In some embodiments, X₁ is a first CD71 binding domain, X₂ is a second CD71 binding domain, and X₃ is a FN3 albumin binding domain. Examples of such polypep-

tides are provided herein and below. In some embodiments, compositions are provided herein having a formula of C-(X₁)_n-(X₂)_q-(X₃)_y-L-X₄, wherein C is a polymer, such as PEG, Endoport, INF-7, TAT, polyarginine, polylysine, an amphipathic peptide, or peptides provided in Table 9; X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; L is a linker; and X₄ is a nucleic acid molecule, wherein n, q, and y are each independently 0 or 1.

[0225] In some embodiments, compositions are provided herein having a formula of (X₁)_n-(X₂)_q-(X₃)_y-L-X₄-C, wherein X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; L is a linker; X₄ is a nucleic acid molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1.

[0226] In some embodiments, compositions are provided herein having a formula of X₄-L-(X₁)_n-(X₂)_q-(X₃)_y,

wherein X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; L is a linker; and X₄ is a nucleic acid molecule, wherein n, q, and y are each independently 0 or 1.

[0227] In some embodiments, compositions are provided herein having a formula of C-X₄-L-(X₁)_n-(X₂)_q-(X₃)_y, wherein C is a polymer; X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; L is a linker; and X₄ is a nucleic acid molecule, wherein n, q, and y are each independently 0 or 1.

[0228] In some embodiments, compositions are provided herein having a formula of X₄-L-(X₁)_n-(X₂)_q-(X₃)_y-C, wherein X₁ is a first FN3 domain; X₂ is second FN3 domain; X₃ is a third FN3 domain or half-life extender molecule; L is a linker; X₄ is a nucleic acid molecule; and C is a polymer, wherein n, q, and y are each independently 0 or 1.

[0229] In some embodiments, the GYS1 siRNA pair may follow the sequence: sense strand (5'-3') nsnsnnnn-NfNfnnnnnnnnnsnsa and antisense strand (5'-3') UfsNfsnnnnNfnnnnnnnnNfNfnnnsusu, wherein (n) is 2'-O-Me (methyl), (Nf) is 2'-F (fluoro), (s) is phosphorothioate backbone modification. Each nucleotide in both sense and antisense strands are modified independently or in combination at ribosugar and nucleobase positions.

[0230] In some embodiments, the siRNA molecule comprises a sequence pair from Tables 1A or 1B.

TABLE 1A

siRNA Sense and Anti-sense sequences					
siRNA Pair	SEQ ID NO:	Sense Strand 5'-3'	SEQ ID NO:	Antisense Strand 5'-3'	
A	10	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fU][mC][fG][mA][mA][T]	11	[mU][*fU][*mC][fG][mA][fA][fU][fU] [mC][fA][mU][mC][mC][fU][mC] [fC][mC][fA][mG][*fU][*mU]	
B	12	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fU][mA][mA][T]	13	[mU][*fU][*mA][fG][mA][fU][fG][fC] [mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]	
C	14	[fU][*mC][*fC][mG][fC][mA][fG] [fC][fC][mU][fG][mG][mA][mU] [fG][mA][fU][mU][mA][T]	15	[mU][*fA][*mA][fU][mC][fA][fU][fC] [mC][fA][mG][mG][mC][fU][mG] [fC][mG][fG][mA][*fU][*mU]	
D	16	[mG][*mC][*fG][mA][mC][mU][fG] [fC][fC][mU][mG][fU][mA][mG] [mC][mA][fA][mC][mA][idT]	17	[fU][*fG][*mU][mG][mC][mU] [mA][mC][mA][mG][mG][mC][fA] [mG][mU][mC][mG][mC][*mU] [*mU]	
E	18	[mU][*mU][*fU][mA][mU][mG] [fG][fG][fC][mA][mU][fC][mU] [mG][mG][mA][fC][mU][mA][idT]	19	[fU][*fA][*mG][mU][mC][mC][mA] [mG][mA][mU][mG][mC][mC][fC] [mA][mU][mA][mA][mA][*mU] [*mU]	
F	20	[mG][*mC][*fG][mC][mG][mG] [fA][fC][fC][mA][mA][fC][mA] [mA][mU][mU][fU][mC][mA][idT]	21	[fU][*fG][*mA][mA][mU][mU] [mG][mU][mU][mG][mG][mU][fC] [mC][mG][mC][mG][mC][*mU] [*mU]	
G	22	[mG][*mG][*fA][mC][mC][mA] [fA][fC][fA][mA][mU][fU][mU] [mC][mA][mA][fC][mG][mA][idT]	23	[fU][*fC][*mG][mU][mU][mG][mA] [mA][mA][mU][mU][mG][mU][fU] [mG][mG][mU][mC][mC][*mU] [*mU]	
H	24	[mA][*mC][*fC][mA][mA][mC] [fA][fA][fU][mU][mU][fC][mA] [mA][mC][mG][fU][mG][mA][idT]	25	[fU][*fC][*mA][mC][mG][mU][mU] [mG][mA][mA][mA][mU][mU][fG] [mU][mU][mG][mG][mU][*mU] [*mU]	
I	26	[mG][*mC][*fG][mC][mA][mA] [fA][fC][fA][mG][mC][fU][mU] [mU][mG][mG][fG][mA][mA][idT]	27	[fU][*fU][*mC][mC][mA][mA] [mA][mG][mC][mU][mG][mU][fU] [mU][mG][mC][mG][mC][*mU] [*mU]	
J	28	[mA][*mG][*fA][mG][mC][mC] [fA][fU][fC][mU][mU][fU][mG] [mC][mA][mA][fC][mG][mA][idT]	29	[fU][*fC][*mG][mU][mG][mC] [mA][mA][mA][mG][mA][mU][fG] [mG][mC][mU][mC][mU][*mU] [*mU]	
K	30	[mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mC][mA][idT]	31	[fU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
L	32	[mA][*mG][*fC][mC][mA][mU] [fC][fU][fU][mU][mG][fC][mA] [mA][mC][mG][fC][mA][mA][idT]	33	[fU][*fU][*mG][mC][mU][mU] [mG][mC][mA][mA][mA][mG][fA] [mU][mG][mG][mC][mU][*mU] [*mU]	
M	34	[mC][*mC][*fA][mU][mC][mU] [fU][fU][fG][mC][mA][fA][mC] [mG][mC][mA][fG][mC][mA][idT]	35	[fU][*fG][*mC][mU][mG][mC][mG] [mU][mU][mG][mC][mA][mA][fA] [mG][mA][mU][mG][mG][*mU] [*mU]	
N	36	[mA][*mC][*fG][mC][mA][mG] [fC][fG][fG][mC][mA][fG][mU] [mC][mU][mU][fU][mC][mA][idT]	37	[fU][*fG][*mA][mA][mG][mA] [mC][mU][mG][mC][mC][mG][fC] [mU][mG][mC][mG][mU][*mU] [*mU]	

TABLE 1A-continued

siRNA Sense and Anti-sense sequences				
siRNA Pair	SEQ ID NO:	Sense Strand 5'-3'	SEQ ID NO:	Antisense Strand 5'-3'
O	38	[mU][*mG][*fA][mC][mC][mA] [fC][fC][fA][mU][mC][fC][mG] [mC][mC][mG][fA][mA][mA][idT]	39	[fU][*fU][*mU][mC][mG][mC] [mG][mG][mA][mU][mG][mG][fU] [mG][mG][mU][mC][mA][*mU] [*mU]
P	40	[mC][*mG][*fA][mA][mU][mC] [fG][fG][fC][mC][mU][fC][mU] [mU][mC][mA][fA][mU][mA][idT]	41	[fU][*fA][*mU][mG][mA][mA] [mG][mA][mG][mG][mC][mC][fG] [mA][mU][mU][mC][mG][*mU] [*mU]
Q	42	[mC][*mU][*fU][mC][mA][mA] [fU][fA][fG][mC][mA][fG][mU] [mG][mC][mC][fG][mA][mA][idT]	43	[fU][*fU][*mC][mG][mG][mC][mA] [mC][mU][mG][mC][mU][mA][fU] [mU][mG][mA][mA][mG][*mU] [*mU]
R	44	[mC][*mA][*fG][mU][mA][mU] [fC][fU][fC][mC][mA][fC][mC] [mA][mA][mU][fC][mU][mA][idT]	45	[fU][*fA][*mG][mU][mU][mG] [mG][mU][mG][mG][mA][mG][fA] [mU][mA][mC][mU][mG][*mU] [*mU]
S	46	[mC][*mC][*fA][mC][mC][mA] [fA][fU][fC][mU][mC][fU][mC] [mC][mG][mG][fC][mU][mA][idT]	47	[fU][*fA][*mG][mC][mC][mG][mG] [mA][mG][mA][mG][mA][mU][fU] [mG][mG][mU][mG][mG][*mU] [*mU]
T	48	[mC][*mA][*fC][mC][mA][mA] [fU][fC][fU][mC][mU][fC][mC] [mG][mG][mC][fU][mU][mA][idT]	49	[fU][*fA][*mA][mG][mC][mC][mG] [mG][mA][mG][mA][mG][mA][fU] [mU][mG][mG][mU][mG][*mU] [*mU]
U	50	[mC][*mC][*fC][mC][mU][mC] [fA][fG][fC][mU][mU][fA][mC] [mG][mG][mU][fA][mU][mA][idT]	51	[fU][*fA][*mU][mA][mC][mC][mG] [mU][mA][mA][mG][mC][mU][fG] [mA][mG][mG][mG][mG][*mU] [*mU]
V	52	[mC][*mC][*fC][mU][mC][mA] [fG][fC][fU][mU][mA][fC][mG] [mG][mU][mA][fU][mC][mA][idT]	53	[fU][*fG][*mU][mA][mC][mC] [mG][mU][mA][mA][mG][mC][fU] [mG][mA][mG][mG][mG][*mU] [*mU]
W	54	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mA][idT]	55	[fU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]
X	56	[mG][*mG][*fA][mA][mC][mC] [fG][fC][fA][mC][mG][fG][mA] [mG][mC][mG][fC][mC][mA][idT]	57	[fU][*fG][*mG][mC][mC][mU] [mC][mC][mG][mU][mG][mC][fG] [mG][mU][mU][mC][mC][*mU] [*mU]
Y	58	[mA][*mC][*fG][mG][mA][mG] [fC][fG][fC][mC][mU][fC][mU] [mC][mC][mG][fA][mC][mA][idT]	59	[fU][*fG][*mU][mC][mG][mG][mA] [mG][mA][mG][mG][mC][mG][fC] [mU][mC][mC][mG][mU][*mU] [*mU]
Z	60	[mA][*mG][*fC][mA][mA][mG] [fC][fG][fC][mA][mA][fC][mU] [mC][mU][mG][fU][mG][mA][idT]	61	[fU][*fC][*mA][mC][mA][mA] [mG][mU][mU][mG][mC][mG][fC] [mU][mU][mG][mC][mU][*mU] [*mU]
AA	62	[mC][*mA][*fG][mA][mU][mG] [fG][fU][fC][mC][mU][fC][mC] [mA][mU][mU][fU][mC][mA][idT]	63	[fU][*fG][*mA][mA][mA][mU][mG] [mG][mA][mG][mG][mA][mC][fC] [mA][mU][mC][mU][mG][*mU] [*mU]
BB	64	[mU][*mC][*fU][mA][mG][mA] [fU][fC][fU][mG][mG][fA][mA] [mC][mC][mU][fU][mA][mA][idT]	65	[fU][*fU][*mA][mA][mG][mU] [mU][mC][mC][mA][mG][mA][fU] [mC][mU][mA][mG][mA][*mU] [*mU]

TABLE 1A-continued

siRNA Sense and Anti-sense sequences				
siRNA Pair	SEQ ID NO:	Sense Strand 5'-3'	SEQ ID NO:	Antisense Strand 5'-3'
CC	66	[mC][*mC][*fC][mA][mC][mA] [fU][fU][fU][mA][mC][fC][mU] [mC][mG][mA][fG][mG][mA][idT]	67	[fU][*fC][*mC][mU][mC][mG][mA] [mG][mG][mU][mA][mA][mA][fU] [mG][mU][mG][mG][mG][*mU] [*mU]
DD	68	[mA][*mC][*fC][mG][mC][mU] [fU][fG][fG][mU][mU][fU][mG] [mC][mA][mG][fU][mU][mA][idT]	69	[fU][*fA][*mA][mC][mU][mG][mC] [mA][mA][mA][mC][mC][mA][fA] [mG][mC][mG][mG][mU][*mU] [*mU]
EE	600	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fU][mC][fG][mG][mA][T]	601	[mU][*fU][*mC][fG][mA][fA][fU][fU] [mC][fA][mU][mC][mC][fU][mC] [fC][mC][fA][mG][*fU][*mU]
FF	602	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fU][mA][fG][mG][mA][T]	603	[mU][*fU][*mC][fG][mA][fA][fU][fU] [mC][fA][mU][mC][mC][fU][mC] [fC][mC][fA][mG][*fU][*mU]
GG	604	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fG][mA][fG][mG][mA][T]	605	[mU][*fU][*mC][fG][mA][fA][fU][fU] [mC][fA][mU][mC][mC][fU][mC] [fC][mC][fA][mG][*fU][*mU]
HH	606	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fU][mC][fG][mA][mA][T]	607	[fU][*fU][*fC][fG][fA][fA][fU][fU] [fC][fA][fU][fC][fC][fU][fC][fC][fC] [fA][fG][*fU][*fU]
II	608	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][mU] [fU][mC][fG][mA][mA][T]	609	[mU][*fU][*fC][fG][fA][fA][fU][fU] [mC][fA][mU][fC][fC][fU][fC][fC] [fC][fA][fG][*fU][*mU]
JJ	610	[fC][*mU][*fG][mG][fG][mA][fG] [fG][fA][mU][fG][mA][mA][*fU] [idT]	611	[mU][*fU][*mC][fG][mA][fA][fU] [fU][mC][fA][mU][mC][mC][fU][mC] [fC][mC][fA][*mG][*fU][*mU]
KK	612	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fU][mA][mA][idT]	613	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
LL	614	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fU][mA][mA][idT]	615	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
MM	616	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fU][mG][mA][idT]	617	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
NN	618	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fG][mA][mA][idT]	619	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
OO	620	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mA][fU][mA][mA][idT]	621	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
PP	622	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fG][mC][fU][mA][mA][idT]	623	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
QQ	624	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fG][mC][fU][mA][mA][idT]	625	[mU][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]
RR	626	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mA][mC][fG][mC][mA][idT]	627	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]
SS	628	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mA][mC][fG][mC][mA][idT]	629	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]

TABLE 1A-continued

siRNA Sense and Anti-sense sequences					
siRNA Pair	SEQ ID NO:	Sense Strand 5'-3'	SEQ ID NO:	Antisense Strand 5'-3'	
TT	630	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mA][mC][fG][mA][mA][idT]	631	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
UU	632	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mA][mC][fA][mC][mA][idT]	633	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
VV	634	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mA][mA][fG][mC][mA][idT]	635	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
WW	636	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mG][mC][fG][mC][mA][idT]	637	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
XX	638	[mG][*mA][*fG][mC][mC][mA][fU] [fC][fU][mU][mU][fG][mC][mA] [mG][mC][fG][mC][mG][idT]	639	[mU][*fG][*mC][mG][mU][mU][mG] [mC][mA][mA][mA][mG][mA][fU] [mG][mG][mC][mU][mC][*mU] [*mU]	
YY	640	[mU][*mU][*fA][mC][mG][mG][fU] [fA][fU][mC][mU][fA][mC][mA] [mU][mU][fC][mU][mA][idT]	641	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
ZZ	642	[mU][*mU][*fA][mC][mG][mG][fU] [fA][fU][mC][mU][fA][mC][mA] [mU][mU][fC][mU][mG][idT]	643	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
AAA	644	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mG][mA][idT]	645	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
BBB	646	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fG][mU][mA][idT]	647	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
CCC	648	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mG][fC][mU][mA][idT]	649	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
DDD	650	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mG][fC][mU][mA][idT]	651	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
EEE	652	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mG][mU][fC][mU][mA][idT]	653	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
FFF	654	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mG][mU][fC][mU][mG][idT]	655	[mU][*fA][*mG][mA][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	
GGG	656	[fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC][mA] [fU][mC][fU][mA][mG][idT]	657	[mU][*fU][*mA][fG][mA][fU][fG][fC] [mC][fA][mC][mC][mC][fA][mC] [fC][mU][fU][mG][*fU][*mU]	

TABLE 1A-continued

siRNA Sense and Anti-sense sequences					
siRNA Pair	SEQ ID NO:	Sense Strand 5'-3'	SEQ ID NO:	Antisense Strand 5'-3'	
HHH	704	[mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mA]	705	[mU][*fA][*mG][mA][mU][mG] [mU][mA][mG][mA][mU][mA][fC] [mC][mG][mU][mA][mA][*mU] [*mU]	

Abbreviations Key: (n/N = any nucleotide) mN = 2'-O-methyl residues, fN = 2'-F residues, * = phosphorothioate and (idT) = inverted Dt, (VP) 2'-O methyl vinyl phosphonate uridine. The brackets indicate the individual bases.

TABLE 1B

SEQ ID NO:	Sense strand (5'-3')	SEQ ID NO:	Antisense strand (5'-3')
801	caggggugcg gucuugcaa	861	uugcaagacc gcacccuguu
802	ggugcgggucu ugcaauagg	862	ccuauugcaa gaccgcaccu
803	gugcgggucu gcaauagga	863	uccuauugca agaccgcacu
804	agccaugccu uaaaaccgc	864	gcgguuuaaa ggcauggcuu
805	augccuuuaa accgcacuu	865	aagugcgguu uaaaggcauu
806	ugggaggaug aaucgcacc	866	ggucgaauc auccucccau
807	aacgcagugc ucuucgaag	867	cuucgaagag cacugcguuu
808	ggacgaaugg ggcgacaac	868	guugucgccc cauucguccu
809	ggcgacugcc uguagcaac	869	guugcuacag gcagucgccu
810	cacgcugcug gggcgcuac	870	guagcgcccc agcagcgugu
811	ugcgucacg ucuucacua	871	uagugaagac gugagcgcau
812	ccuauauacu cuuuauccg	872	gcgauaaaga aguauaaggu
813	cuuuauccgc ggccgcuaa	873	auagcggccg gcgauaaagu
814	ucgcgggccc cuaugagu	874	aacucauagc ggccggcgau
815	gccgcuauga guucuccaa	875	uuggagaacu cauagcggu
816	ccgcuaugag uucuccaac	876	guuggagaac ucauagcggu
817	ggcucggcuc aacuaucug	877	cagauaguug agccgagccu
818	gcucggcuca acuaucugc	878	gcagauagu gagccgagcu
819	ucggcucaac uaucugcuc	879	gagcagauag uugagccgau
820	ccagcgcgga ccaacaauu	880	aauguuggu ccgcgcuggu
821	agcgcggacc aacaauuuc	881	gaaauguug guccgcgcuu
822	ucgggaggaa gcuuuuga	882	ucauaaagcu uccucccgau
823	acgcagcggc agucuuucc	883	ggaaagacug ccgucgcuu
824	gaccaccauc cgccgaau	884	gauucggcg auggugguu
825	ccgaauccgc cucuucuu	885	auugaagagg ccgaauccgu
826	aaucggccuc uucaauagc	886	gcuaauaga aggcggaauu
827	caauagcagu gccgacagg	887	ccugucggca cugcuauugu
828	aggaguuuu ccguggcug	888	cagccacgga caaacuccuu
829	cggcugagug cacgguuau	889	auaacccguc acucagccgu

TABLE 1B-continued

SEQ ID NO: Sense strand (5'-3')		SEQ ID NO: Antisense strand (5'-3')	
830	ugggaauccc caguaucuc	890	gagauacugg ggauuccau u
831	cagaccccuc agcuuacgg	891	ccguaagcug aggggucugu u
832	ccccucagcu uacgguauc	892	gauaccguua gcugaggggu u
833	cucagcuuac gguaucuauc	893	guagauaccg uaagcugagu u
834	gcuuacggua ucuacauuc	894	gaauguagau accguaagcu u
835	ccggcggcag cguaucauc	895	gaugauacgc ugccgccggu u
836	agcguaucua ccagcggaa	896	uuccgcugga ugauacgcuu u
837	gcguaucauc cagcggaac	897	guuccgcugg augauacgcu u
838	uaccuaggcc gguaucuaa	898	uauaguaccg gccuagguau u
839	cuaggccggu acuaauaugu	899	acauauagua ccggccuagu u
840	agggccguac uauaugucu	900	agacauauag uaccggccuu u
841	ggccgguacu auaugucug	901	cagacauuaa guaccggccu u
842	gccgguacua uaugucugc	902	gcagacauau aguaccggcu u
843	cgguaucuaa ugucugcgc	903	gcgcagacau auaguaccgu u
844	gguaucuaau gucugcgcg	904	cgcgcagaca uauaguaccu u
845	cuauaugucu gcgcgccac	905	guggcgcgca gacauauagu u
846	uaugucugcg cgccacaug	906	cauguggcgc gcagacauu u
847	augucugcgc gccacaugg	907	ccauguggcg cgcagacauu u
848	ucgcccucgc ugucacgac	908	gucgugacag cgagggcgau u
849	cggcgagcgc uacgaugag	909	cucaucguag cgcucgccgu u
850	caaggaccgg cgcaacauc	910	gauguugcgc cgguccuugu u
851	acauccgugc accagagug	911	cacucuggug cacggauguu u
852	cugggcgagg agcguaacu	912	aguuacgcuc cucgcccagu u
853	gggcgaggag cguaacuaa	913	uuaguuacgc uccucgccc u
854	gcgaggagcg uaacuaagu	914	acuuaguuaac gcuccucgcu u
855	gaggagcgua acuaagucc	915	ggacuuaguu acgcuccucu u
856	aguccgcaa acacuccac	916	guggaguguu uggcggacuu u
857	ggcgaucaag uccagagcc	917	ggcucuggac uugaucgcu u
858	cccuuaccug gcuuauucc	918	ggaauaagcc agguuagggu u
859	ugugaaacca cuagguucu	919	agaaccuagu gguuucacau u
860	accacuaggu ucuaggucc	920	ggaccuagaa ccuagugguu u
921	caggggugcg gucuugcaa	981	uugcaagacc gcacccugcu u
922	ggugcggucu ugcaauaga	982	ucuauugcaa gaccgcaccu u
923	gugcggucuu gcaauagga	983	uccuauugca agaccgcacu u
924	agccaugccu uuaaaccca	984	ucgguuuuuu ggcuauggcuu u
925	augccuuuaa accgcacua	985	uagugcgguu uaaaggcauu u
926	ugggaggaug aaaucgaca	986	ugucgaauuc auccucccau u

TABLE 1B-continued

SEQ ID NO: Sense strand (5'-3')	SEQ ID NO: Antisense strand (5'-3')
927 aacgcagugc ucuucgaaa	987 uuucgaagag cacugcguuu u
928 ggacgaauug ggcgacaaa	988 uuugucgccc cauucguccu u
929 ggcgacugcc uguagcaaa	989 uuugcuacag gcagucgccu u
930 cacgcugcug gggcgcuaa	990 uuagcgcccc agcagcgugu u
931 ugcgucacg ucuucacua	991 uagugaagac gugagcgcau u
932 ccuuauacuu cuuuauucga	992 ucgauaaaaga aguaauaggu u
933 cuuuauccgc ggccgcuaa	993 uuagcggccg gcgauaaagu u
934 ucgcggcgcc cuaugagua	994 uacucauagc ggccggcgau u
935 gccgcuauga guucuccaa	995 uuggagaacu cauagcggu u
936 ccgcuaugag uucuccaaa	996 uuuggagAAC ucauagcggu u
937 ggucggcguc aacuaucua	997 uagauaguug agccgagccu u
938 gcucggcuca acuaucuga	998 ucagauaguu gagccgagcu u
939 ucggcucaac uaucugcua	999 uagcagauag uuagccggu u
940 ccagcgcgga ccaacaaua	1000 uauuguuggu ccgcgcuggu u
941 agcgcgggacc aacaauua	1001 uaaaauuguug guccgcgcuu u
942 ucgggaggaa gcuuuaua	1002 ucauaaagcu uccucccgau u
943 acgcagcggc agucuuuca	1003 ugaagagacug ccgcugcggu u
944 gaccaccauc cgccgaaua	1004 uauucggcgg auggugguu u
945 ccgaauccgc cucuucaaa	1005 uuugaagagg ccgauucggu u
946 aaucggccuc uucaauaga	1006 ucuauugaag aggcggaauu u
947 caauagcagu gccgacaga	1007 ucugucggca cugcuauugu u
948 aggaguuugu ccguggcua	1008 uagccacgga caaacuccuu u
949 cggcugagug cacgguuaa	1009 uuaaccgugc acucagccgu u
950 ugggaauccc caguaucua	1010 uagauacugg ggauucccau u
951 cagaccccuc agcuuacga	1011 ucguaagcug aggggucugu u
952 cccucagcu uacgguaa	1012 uauaccguaa gcugaggggu u
953 cucagcuuac gguaucuaa	1013 uuagauaccg uaagcugagu u
954 gcuuacggua ucuacauua	1014 uaauguagau accguaagcu u
955 ccggcggcag cguaucua	1015 uaugauacgc ugccgccggu u
956 agcguaucua ccagcgga	1016 uuccgcugga ugaucgcuu u
957 gcguaucauc cagcgga	1017 uuuccgcugg augauacgu u
958 uaccuaggcc gguaucua	1018 uauaguaccg gccuagguau u
959 cuaggccggu acuaauaga	1019 ucauauagua ccggccuagu u
960 aggcggguac uauauguca	1020 ugacauauag uaccggccuu u
961 ggccgguauc auaugucua	1021 uagacauua guaccggccu u
962 gccgguaucua uaugucuga	1022 ucagacauau aguaccggcu u
963 ccguacuaua ugucugcga	1023 ucgcagacau auaguaccgu u
964 gguacuauau gucugcgca	1024 ugcgcagaca uauaguaccu u

TABLE 1B-continued

SEQ ID NO: Sense strand (5'-3')		SEQ ID NO: Antisense strand (5'-3')	
965	cuauaugucu gcgcgccaa	1025	uuggcgcgca gacauauagu u
966	uaugucugcg cgccacaua	1026	uauugggcg gcagacauu u
967	augucugcg gccacaua	1027	ucaugggcg gcagacauu u
968	ucgcccugc ugucacgaa	1028	uucgugacag cgagggcgau u
969	cggcgagcg uacgaugaa	1029	uucaucguag cgcucggcu u
970	caaggaccgg cgcaacaua	1030	uauugcg cgguccuugu u
971	acaucgugc accagagua	1031	uacucggug cacggaugu u
972	cugggcgagg agcguaaca	1032	uguuacguc cugcccagu u
973	ggggcgagg cguaacuaa	1033	uuaguuacgc uccucggcu u
974	gcgaggagcg uaacuaaga	1034	ucuuaguac gcucccgcu u
975	gaggagcgua acuaaguca	1035	ugacuuagu acgcucccu u
976	aguccgcaa acacuccaa	1036	uuggagugu ugccggacu u
977	ggcgaucaag uccagagca	1037	ugcucggac ugaucgcu u
978	cccuaaccug gcuuuuca	1038	ugaauaagg agguuaggu u
979	ugugaaacca cuagguuca	1039	ugaaccuagu gguuucacu u
980	accacuaggu ucuagguca	1040	ugaccuagaa ccuagguu u
1041	uggacuuaaccuagacaa	1042	uugcuagguugaaguccauu
1043	cugggaggauaauucgaa	1044	uucgaaucaucccccaguu
1045	gggugacaacuauucua	1046	uagauaguauugucccuu
1047	cugggaggauaauucgaa	1048	uucgaaucaucccccaguu
1049	caaggugggugcaucuaa	1050	uuagaugccaccaccuuguu
1051	uccgcagccugaugauua	1052	uaaucauccaggcugcgauu

[0231] In some embodiments, the polynucleotides illustrated above include those that do not include a 2'-O methyl vinyl phosphonate uridine as the 5' nucleotide on the antisense strand of the siRNA.

[0232] In some embodiments, a polynucleotide is as provided for herein. In some embodiments, the polynucleotide comprises a first strand and a second strand to form a portion that comprises a duplex. In some embodiments, the polynucleotide comprises a sense strand and an antisense strand. In some embodiments, comprises the sequences as illustrated in Tables 1A or 1B. In some embodiments, comprises the sequences as illustrated in Tables 1A or 1B but without the base modifications. In some embodiments, a pharmaceutical composition comprises a siRNA pair as provided herein. In some embodiments, the siRNA pair is not conjugated to a FN3 domain.

[0233] In some embodiments, an oligonucleotide molecule described herein is constructed using chemical synthesis and/or enzymatic ligation reactions using procedures known in the art. For example, an oligonucleotide molecule is chemically synthesized using naturally occurring nucleotides or variously modified nucleotides designed to increase the biological stability of the molecules or to increase the physical stability of the duplex formed between the oligo-

nucleotide molecule and target nucleic acids. Alternatively, the oligonucleotide molecule is produced biologically using an expression vector into which an oligonucleotide molecule has been subcloned in an antisense orientation (i.e., RNA transcribed from the inserted oligonucleotide molecule will be of an antisense orientation to a target polynucleic acid molecule of interest).

[0234] In some embodiments, an oligonucleotide molecule is synthesized via a tandem synthesis methodology, wherein both strands are synthesized as a single contiguous oligonucleotide fragment or strand separated by a cleavable linker which is subsequently cleaved to provide separate fragments or strands that hybridize and permit purification of the duplex.

[0235] In some instances, an oligonucleotide molecule is also assembled from two distinct nucleic acid strands or fragments wherein one fragment includes the sense region and the second fragment includes the antisense region of the molecule.

[0236] In some instances, while chemical modification of the oligonucleotide molecule internucleotide linkages with phosphorothioate, phosphorodithioate, phosphonate, phosphoramidate, or mesyl phosphoramidate, linkages improves stability. Excessive modifications sometimes cause toxicity

or decreased activity. Therefore, when designing nucleic acid molecules, the amount of these internucleotide linkages in some cases is minimized. In such cases, the reduction in the concentration of these linkages lowers toxicity, increases efficacy and higher specificity of these molecules.

[0237] As described herein, in some embodiments, the nucleic acid molecules can be modified to include a linker at the 5' end of the of the sense strand of the dsRNA. In some embodiments, the nucleic acid molecules can be modified to include a vinyl phosphonate or modified vinyl phosphonate at the 5' end of the of the anti-sense strand of the dsRNA. In

some embodiments, the nucleic acid molecules can be modified to include a linker at the 3' end of the of the sense strand of the dsRNA. In some embodiments, the nucleic acid molecules can be modified to include a vinyl phosphonate at the 3' end of the of the anti-sense strand of the dsRNA. The linker can be used to link the dsRNA to the FN3 domain. The linker can covalently attach, for example, to a cysteine residue on the FN3 domain that is there naturally or that has been substituted as described herein, and for example, in U.S. Pat. No. 10,196,446, which is hereby incorporated by reference in its entirety. Non-limiting examples of such modified strands of the dsRNA are illustrated in Table 2.

TABLE 2

Pairs with Linker and/or vinyl phosphonate			
siRNA	SEQ ID	SEQ ID	L = linker
Pair	NO: Sense Strand 5'-3'	NO: Antisense Strand 5'-3'	
III	70 [mU][*mC][*fU][mC][fG][mU] [fG][mG][fC][mC][fU][mU][fA] [mA][fU][mG][fA][mA]-L	71 [VP][*fU][*mU][fC][mA][fU][mU] [fA][mA][fG][mG][fC][mC][fA] [mC][fG][mA][fG][mA][*mU] [*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
JJJ	72 [fC][*mU][*fG][mG][fG][mA] [fG][fG][fC][mU][fG][mA][mA] [mU][fU][mC][fG][mA][mA]-L	73 [VP][*fU][*mC][mG][mA][mA] [mU][mU][mC][mA][mU][mC] [mC][fU][mC][mC][mC][mA][mG] [*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
KKK	74 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fC][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mA]-L	75 [VP][*fU][*mA][fG][mA][fU][fG] [fC][mC][fA][mC][mC][mC][fA] [mC][fC][mU][fU][mG][*fU] [*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
LLL	76 [mG][*mC][*fG][mC][mG][mG] [fA][fC][fC][mA][mA][fC][mA] [mA][mU][mU][fU][mC][mA]-L	77 [VP][*fG][*mA][mA][mA][mU] [mU][mG][mU][mU][mG][mG] [mU][fC][mC][mG][mC][mG] [mC][*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
MMM	78 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mC][mA]-L	79 [VP][*fG][*mC][mG][mU][mU] [mG][mC][mA][mA][mA][mG] [mA][fU][mG][mG][mC][mU][mC] [*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
NNN	80 [mA][*mG][*fC][mC][mA][mU] [fC][fU][fU][mU][mG][fC][mA] [mA][mC][mG][fC][mA][mA]-L	81 [VP][*fU][*mG][mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][fA][mU][mG][mG][mC][mU] [*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
OOO	82 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mA]-L	83 [VP][*fA][*mG][mA][mA][mU] [mG][mU][mA][mG][mA][mU] [mA][fC][mC][mG][mU][mA] [mA][*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
OPPP	84 [mC][*mC][*fC][mA][mC][mA] [fU][fU][fU][mA][mC][fC][mU] [mC][mG][mA][fG][mG][mA]-L	85 [VP][*fC][*mC][mU][mC][mG] [mA][mG][mG][mU][mA][mA] [mA][fU][mG][mU][mG][mG] [mG][*mU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
QQQ	86 [mU][*mC][*fU][mC][fG][mU] [fG][mG][fC][mC][fU][mU][fA] [mA][fU][mG][mG][mA]-L	87 [VP][*fU][*mU][fC][mA][fU] [mU][fA][mA][fG][mG][fC][mC] [fA][mC][fG][mA][fG][mA][*mU] [*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
RRR	658 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mA][idT]- L	659 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
SSS	660 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mG][idT]- L	661 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1 - C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -

TABLE 2-continued

Pairs with Linker and/or vinyl phosphonate			
siRNA	SEQ ID	SEQ ID	L =
Pair	NO: Sense Strand 5'-3'	NO: Antisense Strand 5'-3'	linker
TTT	662 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mG][mA][i dT]- L	663 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
UUU	664 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fG][mA][mA][i dT]- L	665 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
VVV	666 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mA][fU][mA][mA][i dT]- L	667 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
WWW	668 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mA][i dT]- L	669 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
XXX	670 [fC][*mA][*fA][mG][fG][mU] [fG][fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mG][i dT]- L	671 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
YYY	672 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mC][mA][i dT]- L	673 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
ZZZ	674 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mC][mG][i dT]- L	675 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
AAAA	676 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mA][mA][i dT]- L	677 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
BBBB	678 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fA][mC][mA][i dT]- L	679 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
CCCC	680 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mA][mC][fG][mC][mA][i dT]- L	681 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
DDDD	682 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mG][mC][fG][mC][mA][i dT]- L	683 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
EEEE	684 [mG][*mA][*fG][mC][mC][mA] [fU][fC][fU][mU][mU][fG][mC] [mA][mG][mC][fG][mC][mG][i dT]- L	685 [VP][mU][*fG][*mC][mG][mU] [mU][mG][mC][mA][mA][mA] [mG][mA][fU][mG][mG][mC][mU] [mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
FFFF	686 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mA][i dT]- L	687 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
GGGG	688 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mG][i dT]- L	689 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -

TABLE 2-continued

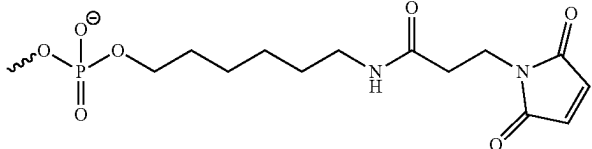
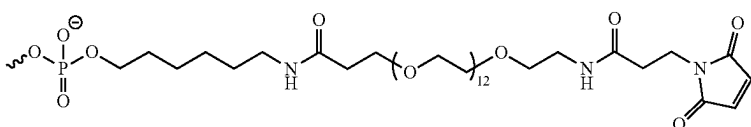
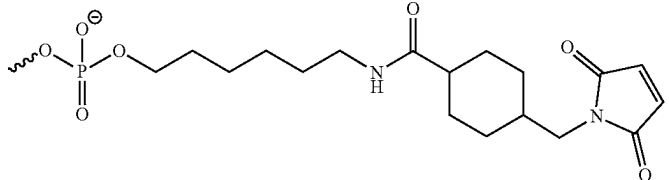
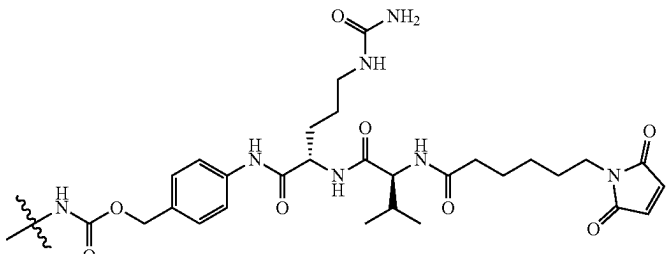
Pairs with Linker and/or vinyl phosphonate			
siRNA Pair	SEQ ID NO: Sense Strand 5'-3'	SEQ ID NO: Antisense Strand 5'-3'	L = linker
HHHH	690 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mG][mA][idT]- L	691 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
IIII	692 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fG][mU][mA][idT]- L	693 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
JJJJ	694 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mG][fC][mU][mA][idT]- L	695 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
KKKK	696 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mG][fC][mU][mA][idT]- L	697 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
LLLL	698 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mG][mU][fC][mU][mA][idT]- L	699 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
MMMM	700 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mG][mU][fC][mU][mG][idT]- L	701 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU] [mA][mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
NNNN	702 [fC][*mA][*fA][mG][fG][mU][fG] [fG][fG][mU][fG][mG][mC] [mA][fU][mC][fU][mA][mG][idT]- L	703 [VP][mU][*fU][*mA][fG][mA] [fU][fG][fC][mC][fA][mC][mC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
OOOO	706 [mU][*mU][*fA][mC][mG][mG] [fU][fA][fU][mC][mU][fA][mC] [mA][mU][mU][fC][mU][mA]-L	707 [VP][mU][*fA][*mG][mA][mA] [mU][mG][mU][mA][mG][mA] [mU][mA][fC][mC][mG][mU][mA] [mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
PPPP	711 [mU][*mG][*fG][mA][mC][mU] [fU][fC][fA][mA][mC][fC][mU] [mA][mG][mA][fC][mA][mA]-L	712 [VP][mU][*fU][*mG][mU][mC] [mU][mA][mG][mG][mU][mU] [mG][mA][fA][mG][mU][mC][mC] [mA][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
QQQQ	1053 [mG][*mG][*fG][mU][mG][mA] [fC][fA][fA][mC][mU][fA][mC] [mU][mA][mU][fC][mU][mA]	1054 [VP][mU][fA][*mG][*mA][mU] [mA][mG][mU][mA][mG][mU] [mU][mG][fU][mC][mA][mC] [mC][mC][*mU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
RRRR	1055 [fC][*mU][*fG][mG][fG][mA][fG] [mG][fA][mU][fG][mA][fA][mU] [fU][mC][fG][mA][fA]	1056 [VP][mU][*fU][*mC][fG][mA] [fA][mU][fU][mC][fA][mU][fC] [mC][fU][mC][fC][mC][fA][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
SSSS	1057 [fC][*mA][*fA][mG][fG][mU][fG] [mG][fG][mU][fG][mG][fC][mA] [fU][mC][fU][mA][fA]	1058 [VP][mU][*fU][*mA][fG][mA] [fU][mG][fC][mC][fA][mC][fC] [mC][fA][mC][fC][mU][fU][mG] [*fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -
TTTT	1059 [fU][*mC][*fC][mG][fC][mA][fG] [mC][fC][mU][fG][mG][fA][mU] [fG][mA][fU][mU][fA]	1060 [VP][mU][*fA][*mA][fU][mC] [fA][mU][fC][mC][fA][mG][fG] [mC][fU][mG][fC][mG][fG][mA] [fU][*mU]	Ma1- C ₂ H ₄ C (O) (NH) - (CH ₂) ₆ -

[0238] In some embodiments, the siRNA pairs of A to PPPP provided for above comprise a linker at the 3' end of the sense strand. In some embodiments, the siRNA pairs of A to PPPP provided for above comprise a vinyl phosphonate at the 5' end of the sense strand.

[0239] Abbreviations Key: (n/N=any nucleotide) mN=2'-O-methyl residues, fN=2'-F residues, *=phosphorothioate, (idT)=inverted Dt, (VP) 2'-O methyl vinyl phosphonate uridine, BMPS=propyl maleimide,

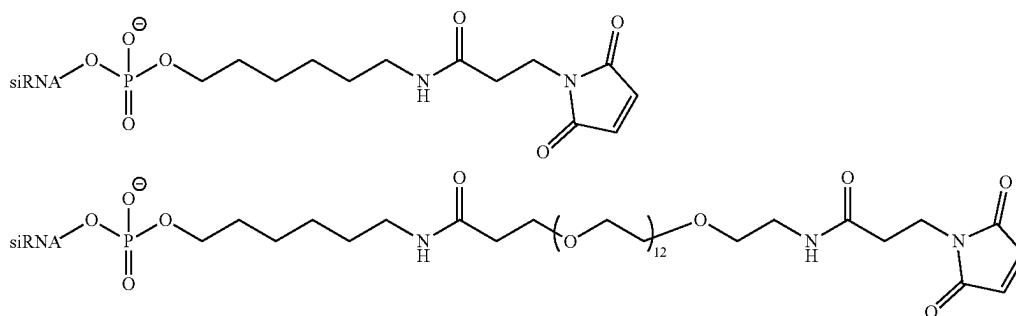
[0240] Structure of the linkers (L) are as follows in Table 3.

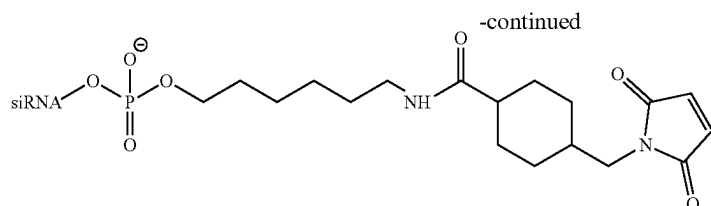
TABLE 3

Representative examples of Linkers (L)	
Linker Structure	Linker Name
	Mal- $C_2H_4C(O)(NH)-(CH_2)_6$
	Mal- $(PEG)_{12}(NH)(CH_2)_6$
	Mal-NH-(CH ₂) ₆ or Aminohexyl linker- (CH ₂) ₆ -
	Val-Cit-PABA

[0241] Other linkers can also be used, such as, linkers formed with click chemistry, amide coupling, reductive amination, oxime, enzymatic couplings such as transglutaminase and storage conjugations. The linkers provided here are exemplary in nature and other linkers made with other such methods can also be used.

[0242] When connected to the siRNA, the structures, L-(X4) can be represented by the following formulas:





[0243] Although certain siRNA sequences are illustrated herein with certain modified nucleobases, the sequences without such modifications are also provided herein. That is, the sequence can comprise the sequences illustrated in the tables provided herein without any modifications. The unmodified siRNA sequences can still comprise, in some embodiments, a linker at the 5' end of the of the sense strand of the dsRNA. In some embodiments, the nucleic acid molecules can be modified to include a vinyl phosphonate at the 5' end of the of the anti-sense strand of the dsRNA. In some embodiments, the nucleic acid molecules can be modified to include a linker at the 3' end of the of the sense strand of the dsRNA. In some embodiments, the nucleic acid molecules can be modified to include a vinyl phosphonate at the 3' end of the of the anti-sense strand of the dsRNA. The linker can be as provided herein.

[0244] In some embodiments, the FN3 proteins comprise a polypeptide comprising a polypeptide that binds CD71 are provided. In some embodiments, the polypeptide comprises a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence of SEQ ID NOs: 273, 288-291, 301-310, 312-572, 592-599, or 708-710 are provided. In some embodiments, the polypeptide that binds CD71 comprises a sequence of SEQ ID NOs: 301-301, 310, 312-572, 592-599, or 708-710. The sequence of CD71 protein that the polypeptides can bind to can be, for example, SEQ ID Nos: 2 or 3. In some embodiments, the FN3 domain that binds to CD71 specifically binds to CD71.

[0245] In some embodiments, the FN3 domain that binds CD71 is based on Tencon sequence of SEQ ID NO:1 or Tencon 27 sequence of SEQ ID NO:4 (LPAPKNLVSRVT-EDSARLSWTAPDAAF-DSFLIQYQESEKVGAEIVLTPGSERSYDLTGLKPGTEYTVSIYGVKGGHRSNPLSAIFTT), optionally having substitutions at residues positions 11, 14, 17, 37, 46, 73, or 86 (residue numbering corresponding to SEQ ID NO:4).

[0246] In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NOs: 273, 288-291, 301-310, 312-572, 592-599, or 708-710.

[0247] In some embodiments, proteins comprising a polypeptide comprising an amino acid sequence of SEQ ID NO: 273. SEQ ID NO: 273 is a consensus sequence based on the sequences of SEQ ID NO: 288, SEQ ID NO: 289, SEQ ID NO: 290, and SEQ ID NO: 291. The sequence of SEQ ID NO: 273 is MLAPKNLVSRVTEDSARLSWTAPDA-AFDSFX₂YX₃EX₄X₅X₆X₇GEAIX₈LX₉VPGSE RSYDLTGLKPGTEYX₁₀VX₁₁X₁₂X₁₃VKGGX₁₄X₁₅-SX₁₆PLX₁₇AX₁₈FTT wherein X₈, X₉, X₁₇, and X₁₈ are each, independently, any amino acid other than methionine or proline, and

[0248] X₁ is selected from D, F, Y, or H,

[0249] X₂ is selected from Y, G, A, or V,

[0250] X₃ is selected from I, T, L, A, or H,

[0251] X₄ is selected from S, Y or P.

[0252] X₅ is selected from Y, G, Q, or R,

[0253] X₆ is selected from G or P,

[0254] X₇ is selected from A, Y, P, D, or S,

[0255] X₁₀ is selected from W, N, S, or E,

[0256] X₁₁ is selected from L, Y, or G,

[0257] X₁₂ is selected from D, Q, H, or V,

[0258] X₁₃ is selected from G or S,

[0259] X₁₄ is selected from R, G, F, L, or D.

[0260] X₁₅ is selected from W, S, P, or L, and

[0261] X₁₆ is selected from T, V, M, or S.

[0262] In some embodiments:

[0263] X₁ is selected from D, F, Y, or H,

[0264] X₂ is selected from G, A, or V,

[0265] X₃ is selected from T, L, A, or H,

[0266] X₄ is selected from Y or P,

[0267] X₅ is selected from G, Q, or R,

[0268] X₆ is selected from G or P,

[0269] X₇ is selected from Y, P, D, or S,

[0270] X₁₀ is selected from W, N, S, or E,

[0271] X₁₁ is selected from L, Y, or G,

[0272] X₁₂ is selected from Q, H, or V,

[0273] X₁₃ is selected from G or S,

[0274] X₁₄ is selected from G, F, L, or D,

[0275] X₁₅ is selected from S, P, or L, and

[0276] X₁₆ is selected from V, M, or S.

[0277] In some embodiments, X₁, X₂, X₃, X₄, X₅, X₆, X₇, X₁₀, X₁₁, X₁₂, X₁₃, X₁₄, X₁₅, and X₁₆ are as shown in the sequence of SEQ ID NO: 288. In some embodiments, X₁, X₂, X₃, X₄, X₅, X₆, X₇, X₁₀, X₁₁, X₁₂, X₁₃, X₁₄, X₁₅, and X₁₆ are as shown in the sequence of SEQ ID NO: 289. In some embodiments, X₁, X₂, X₃, X₄, X₅, X₆, X₇, X₁₀, X₁₁, X₁₂, X₁₃, X₁₄, X₁₅, and X₁₆ are as shown in the sequence of SEQ ID NO: 290. In some embodiments, X₁, X₂, X₃, X₄, X₅, X₆, X₇, X₁₀, X₁₁, X₁₂, X₁₃, X₁₄, X₁₅, and X₁₆ are as shown in the sequence of SEQ ID NO: 291.

[0278] In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, phenylalanine, serine, threonine, tryptophan, tyrosine, or valine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, not alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, phenylalanine, serine, threonine, tryptophan, tyrosine, or valine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, alanine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, arginine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, asparagine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, aspartic acid. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, cysteine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, glutamine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, glutamic acid. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, glycine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, histidine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, isoleucine. In some embodiments, X₈, X₉, X₁₇, and X₁₈ is, independently, leucine. In some embodi-

ments, X_8 , X_9 , X_{17} , and X_{18} is, independently, lysine. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, phenylalanine. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, serine. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, threonine. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, tryptophan. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, tyrosine. In some embodiments, X_8 , X_9 , X_{17} , and X_{18} is, independently, valine.

[0279] In some embodiments, the sequence is set forth as shown in the sequence of SEQ ID NO: 288, except that the positions that correspond to the positions of X_8 , X_9 , X_{17} , and X_{18} can be any other amino acid residue as set forth above, except that in some embodiments, X_8 is not V, X_9 is not T, X_{17} is not S, and X_{18} is not I.

[0280] In some embodiments, the sequence is set forth as shown in the sequence of SEQ ID NO: 289, except that the positions that correspond to the positions of X_8 , X_9 , X_{17} , and X_{18} can be any other amino acid residue as set forth above, except that in some embodiments, X_8 is not V, X_9 is not T, X_{17} is not S, and X_{18} is not I.

[0281] In some embodiments, the sequence is set forth as shown in the sequence of SEQ ID NO: 29X, except that the positions that correspond to the positions of X_8 , X_9 , X_{17} , and X_{18} can be any other amino acid residue as set forth above, except that in some embodiments, X_8 is not V, X_9 is not T, X_{17} is not S, and X_{18} is not I.

[0282] In some embodiments, the sequence is set forth as shown in the sequence of SEQ ID NO: 291, except that the positions that correspond to the positions of X_8 , X_9 , X_{17} , and X_{18} can be any other amino acid residue as set forth above, except that in some embodiments, X_8 is not V, X_9 is not T, X_{17} is not S, and X_{18} is not I.

[0283] In some embodiments, proteins comprising a polypeptide comprising an amino acid sequence that is at least 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to a sequence of SEQ ID NO: 273. In some embodiments, the protein is at least 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to a sequence of SEQ ID NO: 273. In some embodiments, the protein is at least 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to a sequence of SEQ ID NO: 273. In some embodiments, the protein is at least 95%, 96%, 97%, 98% or 99% identical to a sequence of SEQ ID NO: 273.

[0284] Percent identity can be determined using the default parameters to align two sequences using BlastP available through the NCBI website.

[0285] In some embodiments, fibronectin type III (FN3) domains that bind or specifically bind human CD71 protein (SEQ ID Nos: 2 or 5) are provided. As provided herein, the FN3 domains can bind to the CD71 protein. Also provided, even if not explicitly stated is that the domains can also specifically bind to the CD71 protein. Thus, for example, a FN3 domain that binds to CD71 would also encompass a FN3 domain protein that specifically binds to CD71. These molecules can be used, for example, in therapeutic and diagnostic applications and in imaging. In some embodiments, polynucleotides encoding the FN3 domains disclosed herein or complementary nucleic acids thereof, vectors, host cells, and methods of making and using them are provided.

[0286] In some embodiments, an isolated FN3 domain that binds or specifically binds CD71 is provided.

[0287] In some embodiments, the FN3 domain comprises two FN3 domains connected by a linker. The linker can be a flexible linker. The linker can be a short peptide sequence, such as those described herein. For example, the linker can be a G/S linker and the like.

[0288] In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker, such as those provided for herein. Exemplary linker include, but are not limited to, (GS)₂, (SEQ ID NO: 720), (GGGS)₂ (SEQ ID NO: 721), (GGGS)₁₋₅ (SEQ ID NO: 1065), (AP)₁₋₂₀ (SEQ ID NO: 1066), (AP)₂ (SEQ ID NO: 723), (AP)₅ (SEQ ID NO: 724), (AP)₁₀ (SEQ ID NO: 725), (AP)₂₀ (SEQ ID NO: 726), A(EAAAK)₅AAA (SEQ ID NO: 727), or (EAAAK)₁₋₅ (SEQ ID NO: 728). In some embodiments, the linker comprises or is an amino acid sequence of; EAAAKEAAAKEAAAKEAAAK (SEQ ID NO: 729); GGGGSGGGGSGGGGSGGGGS (SEQ ID NO: 730); APAPAPAPAP (SEQ ID NO: 731); or EAAAK (SEQ ID NO: 732).

[0289] In some embodiments, the FN3 domain may bind CD71 with a dissociation constant (K_D) of less than about 1×10^{-7} M, for example less than about 1×10^{-8} M, less than about 1×10^{-9} M, less than about 1×10^{-10} M, less than about 1×10^{-11} M, less than about 1×10^{-12} M, or less than about 1×10^{-13} M as determined by surface plasmon resonance or the Kinexa method, as practiced by those of skill in the art. The measured affinity of a particular FN3 domain-antigen interaction can vary if measured under different conditions (e.g., osmolarity, pH). Thus, measurements of affinity and other antigen-binding parameters (e.g., K_D , K_{on} , K_{off}) are made with standardized solutions of protein scaffold and antigen, and a standardized buffer, such as the buffers described herein.

[0290] In some embodiments, the FN3 domain may bind CD71 at least 5-fold above the signal obtained for a negative control in a standard solution ELISA assay.

[0291] In some embodiments, the FN3 domain that binds or specifically binds CD71 comprises an initiator methionine (Met) linked to the N-terminus of the molecule. In some embodiments, the FN3 domain that binds or specifically binds CD71 comprises a cysteine (Cys) linked to a C-terminus of the FN3 domain. The addition of the N-terminal Met and/or the C-terminal Cys may facilitate expression and/or conjugation to extend half-life and to provide other functions of molecules.

[0292] The FN3 domain can also contain cysteine substitutions, such as those that are described in U.S. Pat. No. 10,196,446, which is hereby incorporated by reference in its entirety. Briefly, in some embodiments, the polypeptides provided herein can comprise at least one cysteine substitution at a position selected from the group consisting of residues 6, 8, 10, 11, 14, 15, 16, 20, 30, 34, 38, 40, 41, 45, 47, 48, 53, 54, 59, 60, 62, 64, 70, 88, 89, 90, 91, and 93 of the FN3 domain based on SEQ ID NO: 6 or SEQ ID NO: 1 of U.S. Pat. No. 10,196,446, and the equivalent positions in related FN3 domains. In some embodiments, the substitution is at residue 6. In some embodiments, the substitution is at residue 8. In some embodiments, the substitution is at residue 10. In some embodiments, the substitution is at residue 11. In some embodiments, the substitution is at residue 14. In some embodiments, the substitution is at residue 15. In some embodiments, the substitution is at residue 16. In some embodiments, the substitution is at residue 20. In some embodiments, the substitution is at residue 30. In some embodiments, the substitution is at residue 34. In some embodiments, the substitution is at residue 38. In some embodiments, the substitution is at

residue 40. In some embodiments, the substitution is at residue 41. In some embodiments, the substitution is at residue 45. In some embodiments, the substitution is at residue 47. In some embodiments, the substitution is at residue 48. In some embodiments, the substitution is at residue 53. In some embodiments, the substitution is at residue 54. In some embodiments, the substitution is at residue 59. In some embodiments, the substitution is at residue 60. In some embodiments, the substitution is at residue 62. In some embodiments, the substitution is at residue 64. In some embodiments, the substitution is at residue 70. In some embodiments, the substitution is at residue 88. In some embodiments, the substitution is at residue 89. In some embodiments, the substitution is at residue 90. In some embodiments, the substitution is at residue 91. In some embodiments, the substitution is at residue 93.

[0293] A cysteine substitution at a position in the domain or protein comprises a replacement of the existing amino acid residue with a cysteine residue. In some embodiments, instead of a substitution a cysteine is inserted into the sequence adjacent to the positions listed above. Other examples of cysteine modifications can be found in, for example, U.S. Patent Application Publication No. 20170362301, which is hereby incorporated by reference in its entirety. The alignment of the sequences can be performed using BlastP using the default parameters at, for example, the NCBI website.

[0294] In some embodiments, a cysteine residue is inserted at any position in the domain or protein.

[0295] In some embodiments, the FN3 domain that binds CD71 is internalized into a cell. In some embodiments, internalization of the FN3 domain may facilitate delivery of a detectable label or therapeutic into a cell. In some embodiments, internalization of the FN3 domain may facilitate delivery of a cytotoxic agent into a cell. The cytotoxic agent can act as a therapeutic agent. In some embodiments, internalization of the FN3 domain may facilitate the delivery of any detectable label, therapeutic, and/or cytotoxic agent disclosed herein into a cell. In some embodiments, internalization of the FN3 domain may facilitate delivery of an oligonucleotide into a cell. In some embodiments, the cell is a tumor cell. In some embodiments, the cell is a liver cell. In some embodiments, the cell is a muscle cell. In some embodiments, the cell is an immune cell. In some embodiments, the cell is a dendritic cell. In some embodiments, the cell is a cell of the central nervous system. In some embodiments, the cell is a heart cell.

[0296] In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NOs: 273, 288-291, 301-310, 312-572, 592-599, or 708-710.

[0297] In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:301. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:302. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:303. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:304. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:305. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:306. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:307. In some embodiments, an isolated FN3 domain that

binds CD71 comprises the amino acid sequence of SEQ ID NO:310. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:312. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:313. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:314. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:315. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:316. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:317. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:318. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:319. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:320. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:321. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:322. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:323. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:324. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:325. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:326. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:327. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:328. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:329. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:330. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:331. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:332. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:333. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:334. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:335. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:336. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:337. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:338. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:339. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:340. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:341. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:342. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO:343. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID

[illegible][illegible]

[illegible][illegible]

[illegible][illegible]

ID NO: 546. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 547. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 548. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 549. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 550. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 551. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 552. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 553. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 554. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 555. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 556. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 557. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 558. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 559. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 560. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 561. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 562. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 563. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 564. In some embodiments, an isolated FN3 domain

that binds CD71 comprises the amino acid sequence of SEQ ID NO: 565. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 566. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 567. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 568. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 569. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 570. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 571. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 572. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 708. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 709. In some embodiments, an isolated FN3 domain that binds CD71 comprises the amino acid sequence of SEQ ID NO: 710.

[0298] In some embodiments, the isolated FN3 domain that binds CD71 comprises an initiator methionine (Met) linked to the N-terminus of the molecule.

[0299] In some embodiments, the isolated FN3 domain that binds CD71 comprises an amino acid sequence that is 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% identical to one of the amino acid sequences of SEQ ID NOs: 273, 288-291, 301-310, 312-572, 592-599, or 708-710. Percent identity can be determined using the default parameters to align two sequences using BlastP available through the NCBI website. The sequences of the FN3 domains that bind to CD71 can be found, for example, in Table 4.

TABLE 4

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
301	MLPAPKKNLVSRVTEDSARLSWTAPDAAFDSFPITYIEAVVLGEAIVLTVPGSE RSYDLTGLKPGTEYVPVGISGVKGGHNSMPLSAIFTT
302	MLPAPKKNLVSEVTEDSARLSWGVARAFDSFMINYSELFWMGEAIVLTVPG SERSYDLTGLKPGTEYVVRIGVKGGKGSWPLHAHFTT
303	MLPAPKKNLVSRVTEDSARLSWTAPDAAFDSFNIEYAETRWYGEAIVLTVPGS ERSYDLTGLKPGTEYVVPIDGVKGGIASKPLSAIFTT
304	MLPAPKKNLVSEVTEDSARLSWGVARAFDSFLITYRDQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYVWVIQGVKGGSPSAPLSAESTT
305	MLPAPKKNLVSEVTEDSARLSWGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYVWYIINGVKGGKPSFPLRAGFTT
306	MLPAPKKNLVSRVTEDSARLSWTAPDAAFDSFPIIYMETFSRGEAIVLTVPGSE RSYDLTGLKPGTEYRVPVIGVKGGSSCPLSAIFTT
307	MLPAPKKNLVSDVTEDSARLSWGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYVWVIQGVKGGSPSAPLSAEFTT
310	MLPAPKKNLVSRVTEDSARLSWTAPDAAFDSFKIAYIETATRGEAIVLTVPGSE RSYDLTGLKPGTEYVVPVIGVKGGNTSSPLSAIFTT
312	MLPAPKKNLVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSPTGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGHLSPLSAISTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
313	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFLIAYPEDGFRGEAIVLTVPGSE RSYDLTGLKPGTEYPVPILGVKGGGSGPLSAIFTT
314	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIYYVENVVWGEAIVLTVPGS ERSYDLTGLKPGTEYWEV IIGVKGGQCSRPLSAIFTT
315	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTECPVWIQGVKGGSPSAPLSAEFTT
316	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFGIAYREFRPSGEAIVLTVPGSE RSYDLTVETGYRNEVVICGVKGGFWGSGPLSAIFTT
317	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPILYTECVYRGEAIVLTVPGSE RSYDLTGLKPGTEYHVPIITGVKGGGSGWPLSAIFTT
318	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFNIMYHEIIYVGEAIVLTVPGSE RSYDLTGLKPGTEYPVPIEGVKGGTSGPLSAIFTT
319	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAITYTEALCGEAIVLTVPGSE RSYDLTGLKPGTEYPVPINGVKGGTSGPLSAIFTT
320	MLPAPKNLVVARVTEDSARLSWTAPDAIDSFPIDYSEYWWGGEAIVLTVPGS ERSYDLTGLKPGTEYPVLITGVKGGYRSGPLSAIFTT
321	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFSIRYNEFIVAGEAIVLTVPGSE RSYDLTGLKPGTEYDVPIAGVKGGGSGWPLSAIVTT
322	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFWYLELQFAGEAIVLTVPGSER SYDLTGLKPGTEYNVPIITGVKGGIISFPLSAIFTT
323	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIWYHEWYGDGEAIVLTVPG SERSYDLTGPKPGTEYRVIRISGVKGGFESGPLSAIFTT
324	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFMIRYQEGTRWGEAIVLTVPGS ERSYDLTGLKPGTEYIVMIAGVKGGQISLPLSAIFTT
325	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIWYLEKSYQGEAIVLTVPGS ERSYDLTGLKPGTEYVPIIGVKGGRDSCPLSAIFTT
326	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYPVWIQGVKGGSPSAPLSAEFTT
327	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFRISYAETVRQGEAIVLTVPGSE RSYDLTVETGYRNWVMILGVKGGPGSLPLSAIFTT
328	MLPAPKNLVVSEVTEDSARLSWQGVVRAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYPVWIQGVKGGSPSAPLSAEFTT
329	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFWIEYWEAVGFGEAIVLTVPGS ERSYDLTGLKPGTEYFVGIIYGVKGGYLSAPLSAIFTT
330	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFI IHVYEQQLIGEIVLTVPGSE RSYDLTGLKPGTEYPVPITGVKGGACSWPLSAIFTT
331	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIEYSEHPIDGEAIPLFVPGSER SYDLTGLKPGTEYVRIHGVKGGWFSHPLWAFPTT
332	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYGVTIAGVKGGWRSKPLNAESTT
333	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFGIAYVESYWYGEAIVLTVPGS ERSYDLTGLKPGTEYNVPIYGVKGGDGSGLSAIFTT
334	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYITYVELNLAGEAIVLTVPGS ERSYDLTGLKPGTEYPVPIIGVKGGSLSQPLSAIFTT
335	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPISYIESIADGEAIVLTVPGSER SYDLTGLKPGTEYWVAIVGVKGGPFWSLSAIVTT
336	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVPTVPGSE RSYDLTGLKPGTEYPVPIAGVKGGGSPSAPLSAIFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
337	MLPAPKNLVVSRVTEDSARLSWTTTPDAAFDSFPIYYWEVITITGEAIYLSVPGSE RSYDLTGLKPGTEYVPVDIPGVKGGAAAPPLSAIFTT
338	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPILYLEHTVRSEAIVLTVPGSE RSYDLTGLKPGTEYCVPIDGVKGGRLSRPLSAIFTT
339	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPITYTEPPDPGEAIVLTVPGESE RSYDLTGLKPGTEYLVITILGVKGGMSVPLSAIFTT
340	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIDYWENRCPGEAIVLTVPGESE ERSYDLTGLKPGTEYCVWISGVKGGYSWPLSAIFTT
341	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGESE RSYDLTGLKPGTEYVPVWIQGVKGGHLSLPLSAIVTT
342	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFLIPYAETSPSGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGVKGGDYSEPLSAIFTT
343	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFMIVYYEYTRFGEAIVLTVPGESE ERSYDLTGLKPGTEYTPIDGVKGGGRSSPLSAIFTT
344	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSPTEGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGVKGGHLSLPLSAIVTT
345	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGESE RSYDLTGLKPGTEYVPVWIQGVKGGSPSAPLSAEFTT
346	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFIIPYAEVRPDGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGVKGGKLSLPLSAIFTT
347	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFGIVYLEMMVTGEAIVLTVPGESE ERSYDLTGLKPGTEYDVPILGVKGGTRSVPLSAIFTT
348	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFLIYYEEGYLEYYSGEAIVLT VPGSESYDLTGLKPGTEYVGVIVGVKGGGLSGPLSAISTT
349	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYAEPRPDGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGVKGGDWSLPLSAIFTT
350	MSLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIHYREFQLSGEAIVLTVPGESE ERSYDLTGLKPGTEYDVPIDGVKGGGSRPLSAIFTT
351	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGESE CSYDLTGLKPGTEYVPVWIQGVKGGSPSAPLSAEFTT
352	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFEIDYDELAIFYGEAIVLTVPGESE RSYDLTGLKPGTEYGVRIPGVKGGMPSLPLSAIVTT
353	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGESE RSYDLTGLKPGTEYVPVWIQGVKGGSPSAPLSAEFTT
354	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIAYGEHIVIGEAIVLTVPGSESE RSYDLTGLKPGTEYMPVPIAGVKGGPISLPLSAIFTT
355	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSPTEGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGVKGGHLSLPLSAIFTT
356	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFSIGYVELVLLGEAIVLTVPGESE RSYDLTGLKPGTEYDVLI PGVKGGSLSRPLSAIFTT
357	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIPYAELSRNGEAIVLTVPGESE RSYDLTGLKPGTEYTVLIHGVKGGCLSDPLSAIFTT
358	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHIEYLELSRHGEAIVLTVPGESE RSYDLTGLKPGTEYVWMI PGVKGGGPKPLSAIFTT
359	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFVYNEVHWIGEAIVLTVPGSER SYDLTGLKPGTEYFVGIVGVKGGHWSKPLSAIFTT
360	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFEIDYDELAIFYGEAIVLTVPGESE RSYDLTGLKPGTEYGVRI PGVKGGMPSLPLSAIVTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
361	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFQIVYSELWIKGEAIVLTVPGSE RSYDLTGLKPGTEYQVPIPGVKGGGRNSFPLSAIFTT
362	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIRYTETRSIGEIVLTVPGSE RSYDLTGLKPGTEYCVPIGGVKGGDSSWPLSAISTT
363	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFCISYYERMGRGEAIVLTVPGS ERSYDLTGLKPGTEYVMVYIFGVKGGGLNSLPLSAIFTT
364	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIVYAEPIPNGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGGRNSDPLSAIFTT
365	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYAEPRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGGLSSPLSAIFTT
366	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIDYDEPRSPGEAIVLTVPGSE RSYDLTGLKPGTEYRVFIWGIKGGDTSFPLSAIFTT
367	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTILYAEQAQFGEAIVLTVPGSE RSYDLTGLKPGTEYPI TGVKGGTRSGPLSAISTT
368	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFIIPYAEVRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGHLSLPLSAISTT
369	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHIAYEETATSGEAIYLRVPGSE RSYDLTGLKPGTEYGVIEIGVKGGARSRLYADFTT
370	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYPVWIQGVKGGDLNPLSAIFTT
371	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPISYLELSLYGEAIVLTVPGSE RSYDLTGLKPGTEYPVGIAGVKGGVSRPLSAIFTT
372	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIGYREWYWGGEAIVLTVPG SERSYDLTGLKPGTEYNVPI SGVKGGLDSPPLSAIFTT
373	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSE RSYDLTGLKPGTEYPVWIQGVKGGSPSAPLSAESTT
374	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFSITYLEWNLGEAIVLTVPGS ERSYDLTGLKPGTEYMTIPGVKGGMSYPLSAIFTT
375	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTISYGEEALIGEAIYLRVPGSE RSYDLTGLKPGTEYVYVHI EGVKGGSWQPLAAFTT
376	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFTIEYYENIGIGEIVLTVPGSE RSYDLTGLKPGTEYSVPIVGVKGGPYSHPLSAIFTT
377	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYAEPRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGGLSSPLSAIFTT
378	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFGIGYYEHKRFGEAIQLSVPGS ERSYDLTGLKPGTEYEVDI EGVKGGVLSWPLFAEFTT
379	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFVIEYTERFWSGEAIVLTVPGSE RSYDLTGLKPGTEYSVPIDGVKGGQCSPLSAIFTT
380	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFWIDYEEGVIGEAIYLHVP GS ERSYDLTGLKPGTEYVVKIHGVKGGHP SHPLVAVFTT
381	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYVELRHLGEAIVLTVPGS ERSYDLTGLKPGTEYPVWIQGVKGGSPSAPLSAEFTT
382	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFLIPYAETSPSGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGDYSSPLSAIFTT
383	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYAEPRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGGLSSPLSAISTT
384	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFSLYLELTPKGEAIVLTVPGSE RSYDLTGLKPGTEYPVWIQGVKGGSPSAPLSAEFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
385	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFIIIEYFEPIPIGEAIVLTVPGSERSYDLTGLKPGTEYAVNIYGVKGGYLSHPLSAIFTT
386	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSECSYDLTGLKPGTEYPVWIIQGVKGGSPSAPLSAEFTT
387	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIEYTEFLYSGEAIVLTVPGSERSYDLTGLKPGTEYGVPIINGVKGGFVSPPLSAIVTT
388	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIKYREVLRCGEAIVLTVPGSERSYDLTGLKPGTEYTVPIITGVKGGFGSSPLSAIFTT
389	MLPAPENLVVSRVTEDSARLSWTAPDAAFDSFWIEYYEGVIQGEAIVLTVPGSERSYDLTGLKPGTEYFVAIWGVKGGKWSVPLSAIFTT
390	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSERSYDLTGLKPGTEYPVWIIQGVKGGSPSAPLSAEFTT
391	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFQIHYWETQGFGEAIVLTVPGSERSYDLTGLKPGTEYVPLIPGVKGGPSSPLSAIFTT
392	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSPTEGEAIVLTVPGSERSYDLTGLKPGTEYSVLIHGVKGGHLSDDLPSAIFTT
393	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIEYYEVPVAGEAIYLDVPGSERSYDLTGLKPGTEYDVTIYGVKGGYSHPLFASFTT
394	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSERSYDLTGLKPGTEYPVWIIQGVKGGSPSAPLSAIFTT
395	MLPAPKNLVVSEVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSERSYDLTGLKPGTEYPVWIIQGVKGGSPSAPLSAEFTT
396	MSLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYLEVFEYGEAIVLTVPGSERSYDLTGLKPGTEYQVPIEGVKGGAMSLPLSAIFTT
397	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHIWYEEETTIGEAIYLVHPGSERSYDLTGLKPGTEYEVHITGVKGGPYSRPLFANFTT
398	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFGIAYDEWPEFGEAIVLTVPGSERSYDLTGLKPGTEYIIVEIYGVKGGWFSWPLSAIFTT
399	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSPTEGEAIVLTVPGSERSYDLTGLKPGTEYSVLIHGVKGGHLSDDLPSVIFTT
400	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIWEYEVMYLGEAIVLTVPGSERSYDLTGLKPGTEYNVPIPGVKGGHSSPPLSAIFTT
401	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHILYEELFLVGEAIVLTVPGSERSYDLTGLKPGTEYKVPIISGVKGGPVSRPLSAIFTT
402	MLPAPKNLVVSRVTEDSARLSWQGVARAFDSFLITYREQIFAGEVIVLTVPGSERSYDLTGLKPGTEYPVWIIQGVKGGSPSAPLSAEFTT
403	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHIVYHEPRPSGEAIWLHPGSEERSYDLTGLKPGTEYEVGIVSVKGGDLSPVPLVAFFTT
404	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLLVPGSERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLYAVFTT
405	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIFLVVPGSERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHANFTT
406	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLDVPGSERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLYASFTT
407	MSLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLVVPGSERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAISTT
408	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLRVPGSERSYDLTGLKPGTEYNVTIIGVKGGFPMPLSAIFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
409	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHISYEEDYTFGEAIYLRVPGSE RSYDLTGLKPGTEYRVVIGVKGGWFSEPLLAFTT
410	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIYLTVPGESE RSYDLTGLKPGTEYNVTIQGVKGGFSPPLDASFTT
411	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIDLGVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPMLDPLEAYFTT
412	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLLVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAFTT
413	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAINLQVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAFFTT
414	MLPAPKNLVVSRVTEDSARLSWTTPDAAFDSFFIGYLEPQPPGEAISLQVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPPLFAVFTT
415	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIELHVPGESE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLFTT
416	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLVVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAFTT
417	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAITLDVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAFTT
418	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLVPVGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVASFTT
419	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAINLDVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAEFTT
420	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLSVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAFTT
421	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIALWVPGESE RSYDLTGLKPGTEYNVTIQGVKGGFSPMLSAFTT
422	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILVVPGESE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAHFTT
423	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLWVPGESE RSYDLTGLKPGTEYNVTIQGVKGGFSPHPLGAVFTT
424	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLHVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLLASFTT
425	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALHVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAFFTT
426	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIHLHVPGESE RSYDLTGLKPGTEYNVTIQGVKGGFSPILHANFTT
427	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIFLGVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAFTT
428	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLRVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLIASFTT
429	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAINLWVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLDASFTT
430	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFVIEYFEWTLNGEAIVLTVPGS ERSYDLTGLKPGTEYSVQIYGVKGGCLSRPLSAFTT
431	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLWVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLIAHFTT
432	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIPYAEPSTGEAIVLTVPGESE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAHFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
433	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIYLYVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAFFTT
434	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLQVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
435	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLAVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAFHTT
436	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIWLHVPGE RSYDLTGLKPGTEYNVTIQGVKGGFPSIPLIAIFTT
437	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLDPVGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAEFTT
438	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLRVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAFHTT
439	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLGVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPSHPLNANFTT
440	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLEVPGE RSYDLTGLKPGTEYNVTIQGVKGGFPSMPLSAIFTT
441	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIFLGVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLIAFTT
442	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIGLQVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPSHPLKAQFTT
443	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLFVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAHFTT
444	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLYVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLGAFHTT
445	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLTAIFTT
446	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAITLHVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
447	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLEVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAHFTT
448	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALHVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLRAVFTT
449	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLWVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
450	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLIVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
451	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLWVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPSRPLQAHFTT
452	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAITLDVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLYAFTT
453	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALHVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
454	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLWVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLIAHFTT
455	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLVPVGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAFHTT
456	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIFLQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAAVFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
457	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLHVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAISTT
458	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAVFTT
459	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIYLVKPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAHFTT
460	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLLAYFTT
461	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILHVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLEAKFTT
462	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIKLEVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAIFTT
463	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIYLEVPGE RSYDLTGLKPGTEYNVTIQGVKGGFSPFLKAAFTT
464	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILRVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAIFTT
465	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLQVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAAWFTT
466	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIFLQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLNAFTT
467	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAILGVPGSER SYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAYTT
468	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLDVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
469	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLLVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAVFTT
470	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLLVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLLAHFTT
471	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLWVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAYFTT
472	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIVLTVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPMLASFTT
473	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFHIVYHEPRPSGEAIHLQVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPYPLSAFTT
474	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIQLWVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPMLSAIFTT
475	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFRISYCETFYHGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLIAKFTT
476	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLKVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLQANFTT
477	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLKVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLQANFTT
478	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLQVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIVTT
479	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFPIAYAEPRPDGEAIVLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAFTT
480	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALLVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAQFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
481	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILHVPVGS RSYDLTGLKPGTEYSVLIHGKGGGLSSPLEAKFTT
482	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIDLHVPVGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLHALFTT
483	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLDVPGSE RSYDLTGLKPGTEYSVLIHGKGGFPMPLSAIFTT
484	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIDLAVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLSFTT
485	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIYLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLRAKFTT
486	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLSAIFTT
487	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAISLLVPDSE RSYDLTGLKPGTEYNVTIQGVKGGFPMPLKFTT
488	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLDASFTT
489	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLTVPGSE RSYDLTGPKPGTEYWVLIQGVKGGSSVPLVAYFTT
490	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLDVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLEASFTT
491	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAILAVPGSER SYDLTGLKPGTEYNVTIQGVKGGFPLPLVASFTT
492	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFDIGYTEYGGYGEAIYLSVPGS ERSYDLTGLKPGTEYWVLIQGVKGGSSVPLSAIFTT
493	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLSVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLIANFTT
494	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALLVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLSAIVTT
495	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILDVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLSSIFTT
496	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLRASFTT
497	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIKLDVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLVAFFTT
498	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILEVPGSE RSYDLTGLKPGTEYSVLIHGKGGGLSSPLVAYFTT
499	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLHADFTT
500	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLVADFTT
501	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLVPGS ERSYDLTGLKPGTEYSVLIHGKGGGLSSPLLAHFTT
502	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFDIGYTEYGGYGEAILHVPVGS RSYDLTGLKPGTEYWVLIQGVKGGSSVPLSAIFTT
503	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIVLLVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPMPLAFTT
504	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAILLVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPMPLQFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
505	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAILLGVPGE RSYDLTGLKPGTEYNVTIQGVKGGFSPMLHPLVALFTT
506	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLDVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
507	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLSVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAAYFTT
508	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIVLAVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPYPLVAFTT
509	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLQVPGSCRSYDLTGLKP GTEYSVLIHGVKGGLLSSPLTAIFTT
510	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAINLQVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPPLSAVFTT
511	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLHVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAIFTT
512	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLAVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAQFTT
513	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLGVPGE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLFTT
514	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLQVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLCAEFTT
515	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLWVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLIAEFTT
516	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPPKFTT
517	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIILEVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLRAVFTT
518	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
519	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAISLKVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLEAIFTT
520	MLPAPKNPVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLLVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLKRLSPVVTITITMAVCRKPAEN LSQTL
521	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIFLDVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPPLTAFTT
522	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIVLDVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPSHPLAAFTT
523	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIGLAVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPVPLQANFTT
524	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLRVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAEFTT
525	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSASFTT
526	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLHVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLTASFTT
527	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLRVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAIFTT
528	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLRVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAASFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
529	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIQLLVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAHFTT
530	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIWLLVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVAFFTT
531	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIHLVYVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSDPLDAVFTT
532	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIYLDVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPSTFTT
533	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIQLFVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLKAYFTT
534	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIVLVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLSAIFTT
535	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIQLTVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSLPLSADFTT
536	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAILLQVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLDAEFTT
537	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIVLAVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLYASFTT
538	MLPAPKNLVVSRVTEDSARLSWTTPDAAFDSEFYIAYAEPRPDGEAIRLQVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLGFTT
539	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIQLVVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLYAIFTT
540	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAISLVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLHAKFTT
541	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIHLGVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPIPLFASFTT
542	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIQLLVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLYAFTT
543	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIHLAVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSPVLAFTT
544	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAISLVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLGAHFTT
545	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIALWVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLVASFTT
546	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIQLHVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLYAFTT
547	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAILLHVPGE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLRASFTT
548	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIWLGVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLHATFTT
549	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIVLEVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLHANFTT
550	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAILLRVPGSE RSYDLTGLKPGTEYSVLIHGKGGLLSSPLYAKFTT
551	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFFIGYLEPQPPGEAIGLVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFSDPLQAVFTT
552	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSEFYIAYAEPRPDGEAIVLHVPGS ERSYDLTGLKPGTEYSVLIHGKGGLLSSPLDAFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
553	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILHVPVGS RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAYFTT
554	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLAVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLSAKFTT
555	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAILLFVPGSE RSYDLTGLKPGTEYNVTIQGVKGGFPSTPLSAFTT
556	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLTVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLHAYFTT
557	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLGVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLRAYFTT
558	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLEVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAFFTT
559	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIQLGVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLLAFTT
560	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIHLRVPGS ERSYDLTGLKPGTEYNVTIQGVKGGFPMPLSAFTT
561	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAILLQVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLIAFTT
562	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLHVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLQAFTT
563	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIALVVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLAANFTT
564	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAINLSVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAYFTT
565	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIVLEVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLTASFTT
566	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIRLQVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLGASFTT
567	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIGLVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLVAYFTT
568	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIYLEVPGSE RSYDLTGLKPGTEYSVLIHGVKGGLLSSPLFTT
569	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFYIAYAEPRPDGEAIWLDVPGS ERSYDLTGLKPGTEYSVLIHGVKGGLLSSPLDAYFTT
570	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFAIEYCETKMCGEAIVLTVPGS ERSYDLTGLKPGTEYRVPVPGVKGGTASLPLSAFTT
571	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFFIYYIESYPAGEAIVLTVPGSE RSYDLTGLKPGTEYVWVIGVKGGRWSTPLSAFTT
572	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFFIYYIESYPAGEAIVLTVPGSC RSYDLTGLKPGTEYVWVIGVKGGRWSTPLSAFTT
592	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGS CRSYDLTGLKPGTEYEVVILGVKGGVHSPLSAIFTTAPAPAPAPAPLPAPKNL VVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGSERSYDLTGL KPGTEYEVVILGVKGGVHSPLSAIFTT
593	MLPAPKKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGS CRSYDLTGLKPGTEYEVVILGVKGGVHSPLSAIFTTGGGGSGGGSGGGSG GGSLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTV GSERSYDLTGLKPGTEYEVVILGVKGGVHSPLSAIFTT

TABLE 4-continued

CD71-binding FN3 domain sequences	
SEQ ID	Amino Acid sequence of FN3 domains that bind to CD71
594	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGS CRSYDLTGLKPGTEYEVVILGVKGGVHSYPLSAIFTTEAAAKEAAAKEAAAKE AAAKLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTV PGSERSYDLTGLKPGTEYEVVILGVKGGVHSYPLSAIFTT
595	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGS CRSYDLTGLKPGTEYEVVILGVKGGVHSYPLSAIFTTEAAAKLPAPKNLVVSR VTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGSERSYDLTGLKPGT EYEVVILGVKGGVHSYPLSAIFTT
596	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGS CRSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTTAPAPAPAPAPLPAPKNL VVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGSERSYDLTGL KPGTEYVVAIYGKGGWYSRPLSAIFTT
597	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGS CRSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTTGGGGSGGGSGGGGS GGGSLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLT VPGSERSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTT
598	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGS CRSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTTEAAAKEAAAKEAAAK EAAAKLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLT VPGSERSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTT
599	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGS CRSYDLTGLKPGTEYVVAIYGKGGWYSRPLSAIFTTEAAAKLPAPKNLVVSR VTEDSARLSWTAPDAAFDSFKIEYFEYVGYGEAIVLTVPGSERSYDLTGLKPGT EYVVAIYGKGGWYSRPLSAIFTT
708	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGSCRSYDLTGLKP GTEYEVVILGVKGGVHSYPLSAIFTT
709	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFAIVYHEPRPDGEAIVLTVPGSERSYDLTGLKP GTEYEVVILGVKGGVHSYPLSAIFTT
710	MLPAPKNLVVSRVTEDSARLSWTAPDAAFDSFFIGYLEPQPPGEAIHLGVPGSCRSYDLTGLKPG TEYNVTIQGVKGGFSPILFASFTT

[0300] As provided herein, in some embodiments, the FN3 domain that binds to CD71 binds to SEQ ID NO: 2 (human mature CD71) or SEQ ID NO: 5 (human mature CD71 extracellular domain), sequence of each provided below:

2
MTKEYQDLQHLNDESDHHQLRKGPFPQPLQLRCSGPRLLLLSLGLSL
LLLVVVCVIGSQNSQLQEELRGLRETFNSFTASTEAQVKGLSTQGGNVGR
KMKLESQLEKQKDLSEDHSSLLLVKQFVSDLRSLSCQMAALQNGSGE
RTCCPVNWVEHERSCYWFSRSGKAWADADNYCRLEDAHLVVVTSWEEQKF
VQHHIGPVNTWMGLHDQNGPWKWDGTDYETGFKNWRPEQDDWYGHGLG
GGEDCAHFTDDGRWNDDVCQRPYRWVCETELDKASQEPPLL

5
QNSQLQEELRGLRETFNSFTASTEAQVKGLSTQGGNVGRKMKLESQLEK
QKDLSEDHSSLLLVKQFVSDLRSLSCQMAALQNGSERTCCPVNWVEH
ERSCYWFSRSGKAWADADNYCRLEDAHLVVVTSWEEQKFVQHHIGPVNTW
MGLHDQNGPWKWDGTDYETGFKNWRPEQDDWYGHGLGGGEDCAHFTDD
GRWNDDVCQRPYRWVCETELDKASQEPPLL

[0301] In some embodiments, the FN3 domain comprises two FN3 domains connected by a linker. The linker can be a flexible linker. The linker can be a short peptide sequence, such as those described herein. For example, the linker can be a G/S or G/A linker and the like. As provided herein, the linker can be, for example, (GS)₂, (SEQ ID NO: 720), (GGGS)₂ (SEQ ID NO: 721), (GGGGS)₅ (SEQ ID NO: 722), (AP)₂₋₂₀ (SEQ ID NO: 1067), (AP)₂(SEQ ID NO: 723), (AP)₅ (SEQ ID NO: 724), (AP)₁₀ (SEQ ID NO: 725), (AP)₂₀ (SEQ ID NO: 726) and A(EAAAK)₅AAA (SEQ ID NO: 727) or (EAAAK)₁₋₅ (SEQ ID NO: 728). These are non-limiting examples and other linkers can also be used. The number of GGGGS (SEQ ID NO: 1068) or GGGGA (SEQ ID NO: 1069) repeats can also be 1, 2, 3, 4, or 5. In some embodiments, the linker comprises one or more GGGGS (SEQ ID NO: 1068) repeats and one or more GGGGA (SEQ ID NO: 1069) repeats. In some embodiments, the linker comprises EAAAKEAAAKEAAAKEAAAK (SEQ ID NO: 729); GGGGSGGGGSGGGGSGGGGS (SEQ ID NO: 730); APAPAPAPAP (SEQ ID NO: 731); or EAAAK (SEQ ID NO: 732).

[0302] In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 592. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 593. In some embodiments, the FN3 domain comprising two FN3

domains connected by a linker have the amino acid sequence of SEQ ID NO: 594. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 595. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 596. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 597. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 598. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of SEQ ID NO: 599. In some embodiments, the FN3 domain comprising two FN3 domains connected by a linker have the amino acid sequence of one of SEQ ID Nos: 592-599.

[0303] In some embodiments, the FN3 domains may bind CD71, as applicable, with a dissociation constant (K_D) of less than about 1×10^{-7} M, for example less than about 1×10^{-8} M, less than about 1×10^{-9} M, less than about 1×10^{-10} M, less than about 1×10^{-11} M, less than about 1×10^{-12} M, or less than about 1×10^{-13} M as determined by surface plasmon resonance or the Kinexa method, as practiced by those of skill in the art. The measured affinity of a particular FN3 domain-antigen interaction can vary if measured under different conditions (e.g., osmolarity, pH). Thus, measurements of affinity and other antigen-binding parameters (e.g., K_D , K_{on} , K_{off}) are made with standardized solutions of protein scaffold and antigen, and a standardized buffer, such as the buffers described herein.

[0304] In some embodiments, the FN3 domain may bind to its target protein at least 5-fold above the signal obtained for a negative control in a standard solution ELISA assay.

[0305] In some embodiments, the FN3 domain that binds or specifically binds its target protein comprises an initiator methionine (Met) linked to the N-terminus of the molecule. In some embodiments, the FN3 domain that binds or specifically binds to its target protein comprises a cysteine (Cys) linked to a C-terminus of the FN3 domain. The addition of the N-terminal Met and/or the C-terminal Cys may facilitate expression and/or conjugation of half-life extending molecules.

[0306] The FN3 domain can also contain cysteine substitutions, such as those that are described in U.S. Pat. No. 10,196,446, which is hereby incorporated by reference in its entirety. Briefly, in some embodiments, the polypeptide comprising an FN3 domain can have an FN3 domain that has a residue substituted with a cysteine, which can be referred to as a cysteine engineered fibronectin type III (FN3) domain. In some embodiments, the FN3 domain comprises at least one cysteine substitution at a position selected from the group consisting of residues 6, 8, 10, 11, 14, 15, 16, 20, 30, 34, 38, 40, 41, 45, 47, 48, 53, 54, 59, 60, 62, 64, 70, 88, 89, 90, 91, and 93 of the FN3 domain based on SEQ ID NO: 1 (LPAPKNLVVSEVTEDSLRLSWTAP-DAAFDSFLIQYQESEKVGGEAINLTPGSEERSYDLTG LKPGTEYTVSIYGVKGGHRSNPLSAEFTT) of U.S. Pat. No. 10,196,446, which is hereby incorporated by reference in its entirety, and the equivalent positions in related FN3 domains. A cysteine substitution at a position in the domain or protein comprises a replacement of the existing amino acid residue with a cysteine residue. Other examples of cysteine modifications can be found in, for example, U.S. Patent Application Publication No. 20170362301, which is hereby incorporated by reference in its entirety. The alignment of the sequences can be performed using BlastP using the default parameters at, for example, the NCBI website.

[0307] In some embodiments, the FN3 domain that binds to the target protein is internalized into a cell. In some embodiments, internalization of the FN3 domain may facilitate delivery of a detectable label or therapeutic into a cell. In some embodiments, internalization of the FN3 domain may facilitate delivery of a cytotoxic agent into a cell. The cytotoxic agent can act as a therapeutic agent. In some embodiments, internalization of the FN3 domain may facilitate the delivery of any detectable label, therapeutic, and/or cytotoxic agent disclosed herein into a cell. In some embodiments, the cell is a tumor cell. In some embodiments, the cell is a liver cell, a lung cell, muscle cell, an immune cell, a dendritic cell, a cell of the CNS, or a heart cell. In some embodiments, the therapeutic is a siRNA molecule as provided for herein. The FN3 domains that bind CD71 conjugated to a detectable label can be used to evaluate expression of CD71 on samples such as tumor tissue in vivo or in vitro. The FN3 domains that bind CD71 conjugated to a detectable label can be used to evaluate expression of CD71 on samples blood, immune cells, muscle cells, or dendritic cells in vivo or in vitro.

[0308] As provided herein, the different FN3 domains that are linked to the siRNA molecule can also be conjugated or linked to another FN3 domain that binds to a different target. This would enable the molecule to be multi-specific (e.g. bi-specific, tri-specific, etc.), such that it binds to a first target and another, for example, target. In some embodiments, the first FN3 binding domain is linked to another FN3 domain that binds to an antigen expressed by a tumor cell (tumor antigen).

[0309] In some embodiments, FN3 domains can be linked together by a linker to form a bivalent FN3 domain. The linker can be a flexible linker. In some embodiments, the linker is a G/S linker. In some embodiments the linker has 1, 2, 3, or 4 G/S repeats. A G/S repeat unit is four glycines followed by a serine, e.g. GGGGS (SEQ ID NO: 1068). Other examples of linkers are provided herein and can also be used.

[0310] In some embodiments, the linker is a polypeptide of (GS)₂, (SEQ ID NO: 720), (GGGS)₂ (SEQ ID NO: 721), (GGGS)₅ (SEQ ID NO: 722), (AP)₂₋₂₀ (SEQ ID NO: 1067), (AP)₂ (SEQ ID NO: 723), (AP)₅ (SEQ ID NO: 724), (AP)₁₀ (SEQ ID NO: 725), (AP)₂₀ (SEQ ID NO: 726) and A(EAAAK)₅AAA (SEQ ID NO: 727) or (EAAAK)₁₋₅ (SEQ ID NO: 728). These are non-limiting examples and other linkers can also be used. The number of GGGGS (SEQ ID NO: 1068) or GGGGA (SEQ ID NO: 1069) repeats can also be 1, 2, 3, 4, or 5. In some embodiments, the linker comprises one or more GGGGS (SEQ ID NO: 1068) repeats and one or more GGGGA (SEQ ID NO: 1069) repeats. In some embodiments, the linker comprises one or more GGGGS (SEQ ID NO: 1068) repeats and one or more EAAAK (SEQ ID NO: 732) repeats. In some embodiments, the linker comprises one or more GGGGS (SEQ ID NO: 1068) repeats and one or more "AP" repeats. In some embodiments, the linker comprises EAAAKEAAAKEAAAKEAAAK (SEQ ID NO: 729); GGGSGGGSGGGSGGGSGGGGS (SEQ ID NO: 730); APAPAPAPAP (SEQ ID NO: 731); or EAAAK (SEQ ID NO: 732).

[0311] Without being bound to any particular theory, in some embodiments, the FN3 domains that are linked to the nucleic acid molecule may be used in the targeted delivery of the therapeutic agent to cells that express the binding partner of the one or more FN3 domains (e.g. tumor cells), and lead intracellular accumulation of the nucleic acid molecule therein. This can allow the siRNA molecule to

properly interact with the cell machinery to inhibit the expression of the target gene, improve efficacy, and also avoid, in some embodiments, toxicity that may arise with untargeted administration of the same siRNA molecule.

[0312] The FN3 domain described herein that bind to their specific target protein may be generated as monomers, dimers, or multimers, for example, as a means to increase the valency and thus the avidity of target molecule binding, or to generate bi- or multispecific scaffolds simultaneously binding two or more different target molecules. The dimers and multimers may be generated by linking monospecific, bi- or multispecific protein scaffolds, for example, by the inclusion of an amino acid linker, for example a linker containing poly-glycine, glycine and serine, or alanine and proline. Exemplary linker include (GS)₂ (SEQ ID NO:720), (GGGS)₂ (SEQ ID NO:721), (GGGS)₃ (SEQ ID NO:722), (AP)₂₋₂₀ (SEQ ID NO: 1067), (AP)₂ (SEQ ID NO:723), (AP)₅ (SEQ ID NO:724), (AP)₁₀ (SEQ ID NO:725), (AP)₂₀ (SEQ ID NO:726) and A(EAAAK)₅AAA (SEQ ID NO:727) or (EAAAK)₁₋₅ (SEQ ID NO: 728). In some embodiments, the linker comprises or is an amino acid sequence of: EAAAKEAAAKEAAAKEAAAK (SEQ ID NO: 729); GGGGSGGGGSGGGGSGGGGSGGGG (SEQ ID NO: 730); APAPAPAPAP (SEQ ID NO: 731); or EAAAK (SEQ ID NO: 732).

[0313] The dimers and multimers may be linked to each other in a N- to C-direction. The use of naturally occurring as well as synthetic peptide linkers to connect polypeptides into novel linked fusion polypeptides is well known in the literature (Hallewell et al., *J Biol Chem* 264, 5260-5268, 1989; Alfthan et al., *Protein Eng.* 8, 725-731, 1995; Robinson & Sauer, *Biochemistry* 35, 109-116, 1996; U.S. Pat. No. 5,856,456). The linkers described in this paragraph may be also be used to link the domains provided in the formula provided herein and above.

Half-Life Extending Moieties

[0314] The FN3 domains may, also, in some embodiments, incorporate other subunits for example via covalent interaction. In some embodiments, the FN3 domains that further comprise a half-life extending moiety. Exemplary half-life extending moieties are albumin, albumin variants, albumin-binding proteins and/or domains, transferrin and fragments and analogues thereof, and Fc regions. Amino acid sequences of the human Fc regions are well known, and include IgG1, IgG2, IgG3, IgG4, IgM, IgA and IgE Fc regions. In some embodiments, the FN3 domain binds to albumin, albumin variants, albumin-binding proteins and/or domains, and fragments and analogues thereof, extending the half-life of the entire molecule.

[0315] In some embodiments, the albumin binding domain comprises the amino acid sequence of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the albumin binding domain (protein) is isolated. In some embodiments, the albumin binding domain comprises an amino acid sequence that is at least, or is, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the albumin binding domain comprises an amino acid sequence that is at least, or is, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119 provided that

the protein has a substitution that corresponds to position 10 of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the substitution is A10V. In some embodiments, the substitution is A10G, A10L, A10I, A10T, or A10S. In some embodiments, the substitution at position 10 is any naturally occurring amino acid. In some embodiments, the isolated albumin binding domain comprises an amino acid sequence that has 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, or 14 substitutions when compared to the amino acid sequence of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, the substitution is at a position that corresponds to position 10 of SEQ ID NOs: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119. In some embodiments, FN3 domains provided comprises a cysteine residue in at least one residue position corresponding to residue positions 6, 11, 22, 25, 26, 52, 53, 61, 88 or positions 6, 8, 10, 11, 14, 15, 16, 20, 30, 34, 38, 40, 41, 45, 47, 48, 53, 54, 59, 60, 62, 64, 70, 88, 89, 90, 91, or 93 of SEQ ID NO: 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, or 119, or at a C-terminus. Although the positions are listed in a series, each position can also be chosen individually. In some embodiments, the cysteine is at a position that corresponds to position 6, 53, or 88. In some embodiments, additional examples of albumin binding domains can be found in U.S. Pat. No. 10,925,932, which hereby incorporated by reference.

[0316] All or a portion of an antibody constant region may be attached to the FN3 domain to impart antibody-like properties, especially those properties associated with the Fc region, such as Fc effector functions such as C1q binding, complement dependent cytotoxicity (CDC), Fc receptor binding, antibody-dependent cell-mediated cytotoxicity (ADCC), phagocytosis, down regulation of cell surface receptors (e.g., B cell receptor; BCR), and may be further modified by modifying residues in the Fc responsible for these activities (for review; see Strohl, *Curr Opin Biotechnol.* 20, 685-691, 2009).

[0317] Additional moieties may be incorporated into the FN3 domains such as polyethylene glycol (PEG) molecules, such as PEG5000 or PEG20,000, fatty acids and fatty acid esters of different chain lengths, for example laurate, myristate, stearate, arachidate, behenate, oleate, arachidonate, octanedioic acid, tetradecanedioic acid, octadecanedioic acid, docosanedioic acid, and the like, polylysine, octane, carbohydrates (dextran, cellulose, oligo- or polysaccharides) for desired properties. These moieties may be direct fusions with the protein scaffold coding sequences and may be generated by standard cloning and expression techniques. Alternatively, well known chemical coupling methods may be used to attach the moieties to recombinantly produced molecules disclosed herein.

[0318] A PEG moiety may for example be added to the FN3 domain by incorporating a cysteine residue to the C-terminus of the molecule, or engineering cysteines into residue positions that face away from the binding face of the molecule, and attaching a PEG group to the cysteine using well known methods.

[0319] FN3 domains incorporating additional moieties may be compared for functionality by several well-known assays. For example, altered properties due to incorporation of Fc domains and/or Fc domain variants may be assayed in Fc receptor binding assays using soluble forms of the receptors, such as the FcγRI, FcγRII, FcγRIII or FcRn receptors, or using well known cell-based assays measuring

for example ADCC or CDC, or evaluating pharmacokinetic properties of the molecules disclosed herein in in vivo models.

[0320] The compositions provided herein can be prepared by preparing the FN3 proteins and the nucleic acid molecules and linking them together. The techniques for linking the proteins to a nucleic acid molecule are known and any method can be used. For example, in some embodiments, the nucleic acid molecule is modified with a linker, such as the linker provided herein, and then the protein is mixed with the nucleic acid molecule comprising the linker to form the composition. For example, in some embodiments, a FN3 domains is conjugated to a siRNA a cysteine using thiol-maleimide chemistry. In some embodiments, a cysteine-containing FN3 domain can be reduced in, for example, phosphate buffered saline (or any other appropriate buffer) with a reducing agent (e.g. tris(2-carboxyethyl) phosphine (TCEP)) to yield a free thiol. Then, in some embodiments, the free thiol containing FN3 domain was mixed with a maleimide linked-modified siRNA duplex and incubated under conditions to form the linked complex. In some embodiments, the mixture is incubated for 0-5 hr, or about 1, 2, 3, 4 or 5 hr at RT. The reaction can be, for example, quenched with N-ethyl maleimide. In some embodiments, the conjugates can be purified using affinity chromatography and ion exchange. Other methods can also be used and this is simply one non-limiting embodiment.

[0321] Methods of making FN3 proteins are known and any method can be used to produce the protein. Examples are provided in the references incorporated by reference herein.

[0322] In some embodiments, the FN3 domain specifically binding CD71 comprises the amino acid sequence of SEQ ID NOs: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, wherein a histidine tag has been appended to the N-terminal or C-terminal end of the polypeptide for ease of purification. In some embodiments, the histidine tag (His-tag) comprises six histidine residues (SEQ ID NO: 1070). In further embodiments, the His-tag to connected to the FN3 domain by at least one glycine residue or about 2 to about 4 glycine residues. Accordingly, after purification of the FN3 domain and cleavage of the His-tag from the polypeptide one or more glycine may be left on the N-terminus or C-terminus. In some embodiments, if the His-tag is removed from the N-terminus all of the glycines are removed. In some embodiments, if the His-tag is removed from the C-terminus one or more of the glycines are retained.

[0323] In some embodiments, the FN3 domain specifically binding CD71 comprises the amino acid sequence of SEQ ID NOs: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, wherein the N-terminal methionine is retained after purification of the FN3 domain.

Kits

[0324] In some embodiments, a kit comprising the compositions described herein are provided.

[0325] The kit may be used for therapeutic uses and as a diagnostic kit.

[0326] In some embodiments, the kit comprises the FN3 domain conjugated to the nucleic acid molecule.

Uses of the Conjugates FN3 Domains

[0327] The compositions provided for herein may be used to diagnose, monitor, modulate, treat, alleviate, help prevent

the incidence of, or reduce the symptoms of human disease or specific pathologies in cells, tissues, organs, fluid, or, generally, a host.

[0328] In some embodiments, methods of selectively reducing GYS1 mRNA and protein in skeletal muscle. In certain embodiments, GYS1 mRNA and protein is not reduced in the liver and/or the kidney.

[0329] In some embodiments, the reduction in the GYS1 mRNA and protein is sustained for about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 5 weeks, or greater than 5 weeks after administration of the conjugate described herein.

[0330] In some embodiments, the FN3 domain can facilitate delivery into CD71 positive tissues (eg, skeletal muscle, smooth muscle) for treatment of muscle diseases.

[0331] In some embodiments, the FN3 domain can facilitate delivery to activated lymphocytes, dendritic cells, or other immune cells for treatment of immunological diseases.

[0332] In some embodiments, the polypeptide that binds to CD71 is directed to immune cells. In some embodiments, the polypeptide that binds to CD71 is directed to dendritic cells. In some embodiments, methods of treating an autoimmune disease in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent. In some embodiments, the autoimmune disease is selected from the group consisting of rheumatoid arthritis, Hashimoto's autoimmune thyroiditis, celiac disease, diabetes mellitus type 1, vitiligo, rheumatic fever, pernicious anemia/atrophic gastritis, alopecia areata, and immune thrombocytopenic purpura.

[0333] In some embodiments, a method of treating a subject having Pompe Disease (GSD2, acid alpha-glucosidase (GAA) deficiency) is provided, the method comprising administering to the subject a composition provided for herein. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent.

[0334] In some embodiments, methods of treating glycogen storage disease in a subject in need thereof, the method comprising administering a composition provided herein are provided. In some embodiments, the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease (GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type II Diabetes/diabetic nephropathy, Aldolase A Deficiency GSD12, Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), and adult polyglucosan body disease. In some embodiments, the glycogen storage disease is selected from the group consisting of Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase

kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11), Glucose transporter (GLUT2) deficiency, Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

[0335] In some embodiments, a use of a composition as provided herein or of any of in the preparation of a pharmaceutical composition or medicament for treating cancer are provided. In some embodiments, the cancer is selected from the group consisting of acute myeloid leukemia, myelodysplastic syndromes, gastric cancer, clear cell renal cell carcinoma, clear cell carcinomas of the breast, clear cell carcinomas of the endometrium, clear cell carcinomas of the ovary, clear cell carcinomas of the uterus, hepatocellular carcinoma, pancreatic cancer, prostate cancer, soft tissue cancer, Ewings sarcoma, and non-small cell lung cancer.

[0336] In some embodiments, the CD71 cell is a cell involved in a CNS diseases, inflammatory/immune diseases, such as MS & infectious diseases of the brain. In some embodiments, the polypeptide that binds to CD71 is directed to the central nervous system. In some embodiments, methods of treating a neurological condition and/or a brain tumor in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent. In some embodiments, the brain tumor is selected from the group consisting of nonmalignant, benign, and malignant brain tumors. In some embodiments, the neurological condition is selected from the group consisting of Alzheimer's Disease, Amyotrophic Lateral Sclerosis, Parkinson's Disease, Lafora Disease, Pompe Disease, adult polyglucosan body disease, stroke, spinal cord injury, ataxia, Bell's Palsy, cerebral aneurysm, epilepsy, seizures, Guillain-Barre Syndrome, multiple sclerosis, muscular dystrophy, neurocutaneous syndromes, migraine, encephalitis, septicemia, and myasthenia gravis.

[0337] In some embodiments, a method of treating a subject having cancer is provided, the method comprising administering to the subject a composition provided for herein. In some embodiments, the method comprises administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent.

[0338] In some embodiments, the subject has a solid tumor.

[0339] In some embodiments, the solid tumor is a melanoma.

[0340] In some embodiments, the solid tumor is a lung cancer. In some embodiments, the solid tumor is a non-small cell lung cancer (NSCLC). In some embodiments, the solid tumor is a squamous non-small cell lung cancer (NSCLC). In some embodiments, the solid tumor is a non-squamous NSCLC. In some embodiments, the solid tumor is a lung adenocarcinoma.

[0341] In some embodiments, the solid tumor is a renal cell carcinoma (RCC).

[0342] In some embodiments, the solid tumor is a mesothelioma.

[0343] In some embodiments, the solid tumor is a nasopharyngeal carcinoma (NPC).

[0344] In some embodiments, the solid tumor is a colorectal cancer.

[0345] In some embodiments, the solid tumor is a prostate cancer. In some embodiments, the solid tumor is castration-resistant prostate cancer.

[0346] In some embodiments, the solid tumor is a stomach cancer.

[0347] In some embodiments, the solid tumor is an ovarian cancer.

[0348] In some embodiments, the solid tumor is a gastric cancer.

[0349] In some embodiments, the solid tumor is a liver cancer.

[0350] In some embodiments, the solid tumor is pancreatic cancer.

[0351] In some embodiments, the solid tumor is a thyroid cancer.

[0352] In some embodiments, the solid tumor is a squamous cell carcinoma of the head and neck.

[0353] In some embodiments, the solid tumor is a carcinomas of the esophagus or gastrointestinal tract.

[0354] In some embodiments, the solid tumor is a breast cancer.

[0355] In some embodiments, the solid tumor is a fallopian tube cancer.

[0356] In some embodiments, the solid tumor is a brain cancer.

[0357] In some embodiments, the solid tumor is an urethral cancer.

[0358] In some embodiments, the solid tumor is a genitourinary cancer.

[0359] In some embodiments, the solid tumor is an endometriosis.

[0360] In some embodiments, the solid tumor is a cervical cancer.

[0361] In some embodiments, the solid tumor is a metastatic lesion of the cancer.

[0362] In some embodiments, the subject has a hematological malignancy.

[0363] In some embodiments, the hematological malignancy is a lymphoma, a myeloma or a leukemia. In some embodiments, the hematological malignancy is a B cell lymphoma. In some embodiments, the hematological malignancy is Burkitt's lymphoma. In some embodiments, the hematological malignancy is Hodgkin's lymphoma. In some embodiments, the hematological malignancy is a non-Hodgkin's lymphoma.

[0364] In some embodiments, the hematological malignancy is a myelodysplastic syndrome.

[0365] In some embodiments, the hematological malignancy is an acute myeloid leukemia (AML). In some embodiments, the hematological malignancy is a chronic myeloid leukemia (CML). In some embodiments, the hematological malignancy is a chronic myelomonocytic leukemia (CMML).

[0366] In some embodiments, the hematological malignancy is a multiple myeloma (MM).

[0367] In some embodiments, the hematological malignancy is a plasmacytoma.

[0368] In some embodiments, the cancer is a soft tissue cancer. In some embodiments, the soft tissue cancer is Ewings sarcoma.

[0369] In some embodiments, methods of treating cancer in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject any composition provided herein. In some embodiments, a use of a composition as provided herein are provided in the preparation of a pharmaceutical composition or medicament for treating cancer. In some embodiments, the composition can be used for treating cancer.

[0370] In some embodiments, methods of treating Pompe Disease (GSD2, acid alpha-glucosidase (GAA) deficiency) in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject any composition provided herein. In some embodiments, a use of a composition as provided herein are provided in the preparation of a pharmaceutical composition or medicament for treating Pompe Disease (GSD2, acid alpha-glucosidase (GAA) deficiency). In some embodiments, the composition can be used for treating Pompe Disease (GSD2, acid alpha-glucosidase (GAA) deficiency).

[0371] In some embodiments, methods of treating glycogen storage disease in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject any composition provided herein. In some embodiments, a use of a composition as provided herein are provided in the preparation of a pharmaceutical composition or medicament for treating glycogen storage disease. In some embodiments, the composition can be used for treating glycogen storage disease.

[0372] In some embodiments, methods of treating glycogen storage disease in a subject in need thereof, the method comprising administering a composition provided herein are provided. In some embodiments, the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease (GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type II Diabetes/diabetic nephropathy. Aldolase A Deficiency GSD12. Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), and adult polyglucosan body disease. In some embodiments, the glycogen storage disease is selected from the group consisting of Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11, Glucose transporter (GLUT2) deficiency. Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

[0373] In some embodiments, the polypeptide that binds to CD71 is directed to the central nervous system. In some embodiments, methods of treating a neurological condition and/or a brain tumor in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent. In some embodiments, the brain tumor is selected from the

group consisting of nonmalignant, benign, and malignant brain tumors. In some embodiments, the neurological condition is selected from the group consisting of Alzheimer's Disease, Amyotrophic Lateral Sclerosis, Parkinson's Disease, Lafora Disease, Pompe Disease, adult polyglucosan body disease, stroke, spinal cord injury, ataxia, Bell's Palsy, cerebral aneurysm, epilepsy, seizures, Guillain-Barre Syndrome, multiple sclerosis, muscular dystrophy, neurocutaneous syndromes, migraine, encephalitis, septicemia, and myasthenia gravis. In some embodiments, a method of treating a neurological condition and/or a brain tumor in a subject, the method comprising administering to the subject a FN3 domain that binds CD71 and the FN3 domain is conjugated to a therapeutic agent (e.g. cytotoxic agent, an oligonucleotide, such as a siRNA, ASO, and the like, a FN3 domain that binds to another target, and the like).

[0374] In some embodiments, methods of treating Pompe disease in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent. In some embodiments, a method of treating a Pompe disease in a subject, the method comprising administering to the subject a FN3 domain that binds CD71 and the FN3 domain is conjugated to a therapeutic agent (e.g. cytotoxic agent, an oligonucleotide, such as a siRNA, ASO, and the like, a FN3 domain that binds to another target, and the like).

[0375] In some embodiments, the polypeptide that binds to CD71 is directed to immune cells. In some embodiments, the polypeptide that binds to CD71 is directed to dendritic cells. In some embodiments, methods of treating an autoimmune disease in a subject in need thereof are provided. In some embodiments, the methods comprise administering to the subject a polypeptide or the pharmaceutical composition that binds to CD71. In some embodiments, that the polypeptide is a FN3 domain that binds to CD71. In some embodiments, the polypeptide comprises a sequence such as SEQ ID Nos: 301-301, 310, 312-519, 521-572, 592-599, or 708-710, or a polypeptide as provided herein that is linked to or conjugated to a therapeutic agent. In some embodiments, the autoimmune disease is selected from the group consisting of rheumatoid arthritis, Hashimoto's autoimmune thyroiditis, celiac disease, diabetes mellitus type 1, vitiligo, rheumatic fever, pernicious anemia/atrophic gastritis, alopecia areata, and immune thrombocytopenic purpura. In some embodiments, a method of treating an autoimmune disease in a subject, the method comprising administering to the subject a FN3 domain that binds CD71 and the FN3 domain is conjugated to a therapeutic agent (e.g. cytotoxic agent, an oligonucleotide, such as a siRNA, ASO, and the like, a FN3 domain that binds to another target, and the like).

[0376] In some embodiments, methods of reducing the expression of a target gene in a cell are provided. In some embodiments, the methods comprise delivering to the cell with a composition or a pharmaceutical composition as provided herein. In some embodiments, the cell is ex-vivo. In some embodiments, the cell is in-vivo. In some embodiments, the target gene is GYS1. The target gene, however, can be any target gene as the evidence provided herein demonstrates that siRNA molecules can be delivered efficiently when conjugated to a FN3 domain. In some embodiments, the siRNA targeting GYS1 is linked to a FN3

domain. In some embodiments, the FN3 polypeptide (domain) is one that binds to CD71. In some embodiments, the FN3 polypeptide is as provided for herein or as provided for in PCT Application No. PCT/US20/55509, U.S. application Ser. No. 17/070,337, PCT Application No. PCT/US20/55470, or U.S. application Ser. No. 17/070,020, each of which is hereby incorporated by reference in its entirety. In some embodiments, the siRNA is not conjugated to a FN3 domain.

[0377] In some embodiments, methods of reducing the expression of a target gene in a cell are provided. In some embodiments, the methods comprise delivering to the cell with a composition or a pharmaceutical composition as provided herein. In some embodiments, the cell is ex-vivo. In some embodiments, the cell is in-vivo. In some embodiments, a method of reducing the expression of a target gene results in a reduction of about 99%, 90-99%, 50-90%, or 10-50% in the expression of the target gene.

[0378] In some embodiments, a method of reducing the expression of GYS1 is provided. In some embodiments, the reduced expression is the expression (amount) of GYS1 mRNA. In some embodiments, a method of reducing the expression of GYS1 results in a reduction of about 99%, 90-99%, 50-90%, or 10-50% in the expression of GYS1. In some embodiments, the reduced expression is the expression (amount) of GYS1 protein. In some embodiments, the reduced protein is glycogen. In some embodiments, reduction of glycogen occurs in muscle cells. In some embodiments, reduction of glycogen occurs in heart cells. In some embodiments, the method comprises delivering to a cell with a siRNA molecule as provided herein that targets GYS1. In some embodiments, the siRNA is conjugated to a FN3 domain. In some embodiments, the FN3 domain is a FN3 domain that binds to CD71. In some embodiments, the FN3 domain is as provided for herein. In some embodiments, the FN3 domain is a dimer of two FN3 domains that bind to CD71. In some embodiments, the FN3 domains are the same. In some embodiments, the two FN3 domains are different, i.e., bind to different regions or amino acid residues of CD71, i.e. a different epitope. In some embodiments, the method comprises administering to a subject (patient) a GYS1 siRNA molecule, such as those provided herein. In some embodiments, the GYS1 siRNA administered to the subject is conjugated or linked to a FN3 domain. In some embodiments, the FN3 domain is a FN3 domain that binds to CD71. In some embodiments, the FN3 domain is as provided for herein. In some embodiments, the FN3 domain is a dimer of two FN3 domains that bind to CD71. In some embodiments, the FN3 domains are the same. In some embodiments, the two FN3 domains are different, i.e., bind to different regions or amino acid residues of CD71, i.e. a different epitope. In some embodiments, the CD71 binding domain is a polypeptide as provided for herein.

[0379] In some embodiments, methods of delivering a siRNA molecule to a cell in a subject are provided. In some embodiments, the methods comprise administering to the subject a pharmaceutical composition comprising a composition as provided for herein. In some embodiments, the cell is a CD71 positive cell. The term "positive cell" in reference to a protein refers to a cell that expresses the protein. In some embodiments, the protein is expressed on the cell surface. In some embodiments, the cell is a tumor cell, a liver cell, an immune cell, a dendritic cell, a heart cell, a muscle cell, a cell of the CNS, or a cell inside the blood brain barrier. In some embodiments, the siRNA downregulates the expression of a target gene in the cell. In some embodiments, the target gene is GYS1.

[0380] In some embodiments, the compositions provided herein may be used to diagnose, monitor, modulate, treat, alleviate, help prevent the incidence of, or reduce the symptoms of human disease or specific pathologies in cells, tissues, organs, fluid, or, generally, a host, also exhibit the property of being able to cross the blood brain barrier. The blood-brain barrier (BBB) prevents most macromolecules (e.g., DNA, RNA, and polypeptides) and many small molecules from entering the brain. The BBB is principally composed of specialized endothelial cells with highly restrictive tight junctions, consequently, passage of substances, small and large, from the blood into the central nervous system is controlled by the BBB. This structure makes treatment and management of patients with neurological diseases and disorders (e.g., brain cancer) difficult as many therapeutic agents cannot be delivered across the BBB with desirable efficiency. Additional conditions that involve disruptions of the BBB include: stroke, diabetes, seizures, hypertensive encephalopathy, acquired immunodeficiency syndrome, traumatic brain injuries, multiple sclerosis, Lafora Disease, Pompe Disease, adult polyglucosan body disease, Parkinson's disease (PD) and Alzheimer disease. This ability is especially useful for treating brain cancers including for example: astrocytoma, medulloblastoma, glioma, ependymoma, germinoma (pinealoma), glioblastoma multiform, oligodendroglioma, schwannoma, retinoblastoma, and congenital tumors; or a cancer of the spinal cord, e.g., neurofibroma, meningioma, glioma, and sarcoma. This is further useful in treating Pompe Disease and/or glycogen storage disease. In certain embodiments, the compositions provided for herein can be used to deliver a therapeutic or cytotoxic agent, for example, across the blood brain barrier. In certain embodiments, the compositions provided for herein can be used to deliver a therapeutic or cytotoxic agent, for example, into the muscle.

[0381] In some embodiments, methods of treating glycogen storage disease in a subject in need thereof, the method comprising administering a composition provided herein are provided. In some embodiments, the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease (GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type II Diabetes/diabetic nephropathy, Aldolase A Deficiency (GSD12, Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), and adult polyglucosan body disease. In some embodiments, the glycogen storage disease is selected from the group consisting of Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11, Glucose transporter (GLUT2) deficiency, Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

[0382] In some embodiments, the compositions or pharmaceutical compositions provided herein can be used to treat muscle diseases, such as muscular dystrophy, DMD, and the like.

[0383] In some embodiments, the compositions or pharmaceutical compositions provided herein may be adminis-

tered alone or in combination with other therapeutics, that is, simultaneously or sequentially. In some embodiments, the other or additional therapeutics are other anti-tumor agent or therapeutics. Different tumor types and stages of tumors can require the use of various auxiliary compounds useful for treatment of cancer. For example, the compositions provided herein can be used in combination with various chemotherapeutics such as taxol, tyrosine kinase inhibitors, leucovorin, fluorouracil, irinotecan, phosphatase inhibitors, MEK inhibitors, among others. The composition may also be used in combination with drugs which modulate the immune response to the tumor such as anti-PD-1 or anti-CTLA-4, among others. Additional treatments can be agents that modulate the immune system, such antibodies that target PD-1 or PD-L1.

[0384] In some embodiments, the compositions or pharmaceutical compositions provided herein may be administered in combination with GAA enzyme replacement therapy (ERT).

[0385] “Treat” or “treatment” refers to the therapeutic treatment and prophylactic measures, wherein the object is to prevent or slow down (lessen) an undesired physiological change or disorder, such as the development or spread of cancer. In some embodiments, beneficial or desired clinical results include, but are not limited to, alleviation of symptoms, diminishment of extent of disease, stabilized (i.e., not worsening) state of disease, delay or slowing of disease progression, amelioration or palliation of the disease state, and remission (whether partial or total), whether detectable or undetectable. “Treatment” can also mean prolonging survival as compared to expected survival if not receiving treatment. Those in need of treatment include those already with the condition or disorder as well as those prone to have the condition or disorder or those in which the condition or disorder is to be prevented.

[0386] A “therapeutically effective amount” refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result. A therapeutically effective amount of the compositions provided herein may vary according to factors such as the disease state, age, sex, and weight of the individual. Exemplary indicators of an effective amount is improved well-being of the patient, decrease or shrinkage of the size of a tumor, arrested or slowed growth of a tumor, and/or absence of metastasis of cancer cells to other locations in the body.

Administration/Pharmaceutical Compositions

[0387] In some embodiments, pharmaceutical compositions of the compositions provided herein and a pharmaceutically acceptable carrier, are provided. For therapeutic use, the compositions may be prepared as pharmaceutical compositions containing an effective amount of the domain or molecule as an active ingredient in a pharmaceutically acceptable carrier. “Carrier” refers to a diluent, adjuvant, excipient, or vehicle with which the active compound is administered. Such vehicles can be liquids, such as water and oils, including those of petroleum, animal, vegetable or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. For example, 0.4% saline and 0.3% glycine can be used. These solutions are sterile and generally free of particulate matter. They may be sterilized by conventional, well-known sterilization techniques (e.g., filtration). The compositions may contain pharmaceutically acceptable auxiliary substances as required to approximate physiological conditions such as pH adjusting and buffering agents, stabilizing, thickening, lubricating and coloring agents, etc. The concentration of the molecules disclosed

herein in such pharmaceutical formulation can vary widely. i.e., from less than about 0.5%, usually at least about 1% to as much as 15 or 20% by weight and will be selected primarily based on required dose, fluid volumes, viscosities, etc., according to the particular mode of administration selected. Suitable vehicles and formulations, inclusive of other human proteins, e.g., human serum albumin, are described, for example, in e.g. Remington: The Science and Practice of Pharmacy, 21st Edition, Troy, D. B. ed., Lipincott Williams and Wilkins, Philadelphia, PA 2006, Part 5, Pharmaceutical Manufacturing pp 691-1092, See especially pp. 958-989.

[0388] The mode of administration for therapeutic use of the compositions disclosed herein may be any suitable route that delivers the agent to the host, such as parenteral administration, e.g., intradermal, intramuscular, intraperitoneal, intravenous or subcutaneous, pulmonary; transmucosal (oral, intranasal, intravaginal, rectal), using a formulation in a tablet, capsule, solution, powder, gel, particle; and contained in a syringe, an implanted device, osmotic pump, cartridge, micropump; or other means appreciated by the skilled artisan, as well known in the art. Site specific administration may be achieved by for example intrarticular, intrabronchial, intraabdominal, intracapsular, intracartilaginous, intracavitary, intracelial, intracerebellar, intracerebroventricular, intracolic, intracervical, intragastric, intrahepatic, intracardial, intraosteal, intrapelvic, intrapericardiac, intraperitoneal, intrapleural, intraprostatic, intrapulmonary, intrarectal, intrarenal, intraretinal, intraspinal, intrasynovial, intrathoracic, intrauterine, intravascular, intravesical, intral-esional, vaginal, rectal, buccal, sublingual, intranasal, or transdermal delivery.

[0389] Pharmaceutical compositions can be supplied as a kit comprising a container that comprises the pharmaceutical composition as described herein. A pharmaceutical composition can be provided, for example, in the form of an injectable solution for single or multiple doses, or as a sterile powder that will be reconstituted before injection. Alternatively, such a kit can include a dry-powder disperser, liquid aerosol generator, or nebulizer for administration of a pharmaceutical composition. Such a kit can further comprise written information on indications and usage of the pharmaceutical composition.

EXAMPLES

[0390] The following examples are illustrative of the embodiments disclosed herein. These examples are provided for the purpose of illustration only and the embodiments should in no way be construed as being limited to these examples, but rather should be construed to encompass any and all variations which become evidence as a result of the teaching provided herein. Those of skill in the art will readily recognize a variety of non-critical parameters that could be changed or modified to yield essentially similar results.

Example 1: GYS1 siRNA Sequence ID and Characterization

[0391] siRNA In Silico Screening: In silico siRNA screening was performed to identify a human siRNA complementary to the human GYS1 mRNA, see FIG. 1. All possible 19-mer antisense sequences were generated from the human GYS1 siRNA sequence (NM_001161587) and each 19-mer was assessed for activity against other human GYS1 isoforms as well as potential cross-reactivity mouse, rat, and *Cynomolgus macaque*. Human siRNA target sites were

assessed for common human SNPs (MAF>1%) using dbSNP (b155 v2). Sequences that targeted a common allele were discarded. Next, siRNA off-target genes were assessed for sense and antisense strands in all relevant model organisms. This selection yielded 200 potential candidates, that were further defined by and in vitro knockdown screen. Lead siRNA candidates are in Table 1A and 1B. GYS1 siRNA linkers for conjugation to cysteine engineered Centyrins were prepared as described in Table 3.

[0392] HEK293T cells were lipofected with 10 nM ABXO-HHH (siRNA Pair HHH) for 24 hours alongside untreated control cells (6 replicates per treatment group). mRNA was polyA+ selected from cells and subjected to unstranded 2x150 bp paired end sequencing on the Illumina HiSeq platform to an average depth of >30 million reads.

[0393] FIG. 2 Volcano plot showing the DESeq2 results of ABXO-HHH versus untreated cells. The X axis represents the log₂ fold change in gene expression (ABXO-HHH divided by untreated) for all genes. The Y axis shows the negative log₁₀ transformed adjusted P value. Colored dots represent genes with significant complementarity to ABXO-HHH detected via BLAST.

[0394] RNA-seq library quality was inspected using FastQC. Libraries were pseudo aligned to the GRCh38 human transcriptome using Kallisto. We achieved exceptional read alignment with an average of >90% of reads aligning to the transcriptome. Differential expression was then assessed using DESeq2 comparing ABXO-HHH treated cells to lipofectamine only treated cells.

[0395] DESeq results showed substantial and significant knockdown of the GYS1 target. GYS1 expression in ABXO-HHH treated samples was downregulated to 31% of untreated samples (FIG. 2). GYS was the most downregulated protein coding gene for all genes that were differentially expressed (DE) under ABXO-HHH treatment. GYS1 was also by far the DE gene with the lowest p-value, with an adjusted p-value of 1e-87.7, over 30 orders of magnitude smaller than the next lowest adjusted p-value.

[0396] In silico prediction of potential off-target effects was performed using BLAST. The ABXO-HHH sense and antisense sequences were BLAST aligned against an internal BLAST database built from the GRCh38 transcriptome. Of all the potential off-target effects identified by BLAST, only RAP2C was significantly downregulated in the presence of ABXO-HHH (FIG. 2) and failed to surpass the DE threshold of log₂ fold change <-1.

[0397] Together these data demonstrate that ABXO-HHH is a highly specific siRNA even at relatively high concentrations in the human transcriptome.

[0398] Oligonucleotide synthesis, deprotection and annealing protocol:

[0399] Synthesis of oligonucleotides was performed on Mermade 12 synthesizer using standard phosphoramidite chemistry on 500 Å controlled pore glass (CPG) with phosphoramidites at a 0.1 M concentration in acetonitrile. 12 in THF/pyridine/water (0.02M) was the oxidizing agent with 0.6 M ETT (5-ethylthiotetrazole) as the activation agent. N,N-dimethyl-N'-(3-thioxo-3H-1,2,4-dithiazol-5-yl)methanimidamide (DDT), 0.09 M in pyridine, was used as the sulfurizing reagent for the introduction of phosphorothioate (PS) bonds. 3% (v/v) dichloroacetic acid in dichloromethane was used the deblocking solution. All single strands without maleimide were purified by ion exchange chromatography (IEX) with 20 mM phosphate at pH 8.5 as Buffer A and 20 mM phosphate pH 8.5 and 1 M sodium bromide as Buffer B. After purification, the oligonucleotide fractions were

pooled, concentrated, and desalted. Desalted samples were then lyophilized to dryness and stored at -20° C.

Deprotection of Antisense Strands

[0400] After synthesis, the support was washed with acetonitrile (ACN) and dried in the column under vacuum and transferred into a 1 mL screw cap that could be tightly sealed and shaken with a solution of 5% diethylamine in aqueous ammonia at 65° C. for 5 h. Cleavage and deprotection of crude oligo was checked by liquid chromatography-mass spectrometry (LC-MS) and was subsequently purified by IEX-HPLC.

Synthesis and Deprotection of Maleimide-Containing Oligonucleotides

[0401] Maleimide containing oligos were made using either a 3' amino-modified CPG solid support or a 5' amino modifier phosphoramidite. The support was transferred into a 1 mL vial that could be tightly sealed and incubated with 50/50 v/v 40% aqueous methyl amine and aqueous ammonia (AMA) at room temperature for 2 h or 65° C. for 10 minutes to cleave and deprotect. The single strand was purified by IEX chromatography and desalted the same conditions as the antisense before maleimide addition.

[0402] Approximately 20 mg/mL of the amine-modified sense strand in 0.05 M phosphate buffer at pH 7.1 was made to which 10 equivalents of the maleimide N-hydroxysuccinimide (NHS) ester, dissolved in ACN was added. The NHS ester solution was added to the aqueous oligonucleotide solution and shaken for 3 h at room temperature. The now maleimide conjugated oligonucleotides were purified by reverse-phase chromatography (20 mM triethyl ammonium acetate with 80% acetonitrile in Buffer B) to prevent the maleimide hydrolysis under ion exchange buffer conditions.

[0403] After purification, the oligonucleotide fractions were pooled, concentrated, and desalted.

[0404] To avoid hydrolysis of maleimide, duplexing of the sense and antisense strands was performed via freeze-drying using equimolar amounts of each desalted single strand.

[0405] Centyrin conjugation to siRNA, conjugate purification and analysis: Centyrins were conjugated to siRNA through cysteine-specific chemistry via maleimide. Cysteine-containing Centyrin in PBS at 50-200 µM were reduced with 10 mM tris(2-carboxyethyl) phosphine (TCEP) at room temperature (30 mins) to yield a free thiol. The free thiol containing Centyrin is then mixed with maleimide containing siRNA duplex in water immediately prior at a molar ratio of ~1.5:1 Centyrin:siRNA. After 2 hr incubation at RT or 37° C., reaction was quenched with N-ethyl maleimide (1 mM final NEM concentration in the reaction mixture) The conjugate was purified in two steps. Step I either immobilized metal affinity chromatography (for tagged proteins) or Hydrophobic interaction chromatography (for tagless proteins); to remove un-reacted siRNA linker. Step II-Capto-DEAE; to remove un-reacted centyrin. Fractions containing conjugate were pooled, exchanged into HBS by desalting using dialysis, and concentrated if necessary.

[0406] Analytical Characterization of Centyrin-siRNA conjugates: Centyrin-siRNA conjugates were characterized by a combination of analytical techniques. SDS-PAGE was used to compare amounts of conjugate to free protein. For SDS-PAGE, 4-20% Mini-PROTEAN® TGX Stain-Free™ Protein Gels (BioRad) were run in SDS buffer for one hour at 100 V. Gels were visualized under UV light. Analytical SEC (Superdex-75 5/150 GL column-GE) was used to

analyze purity and aggregation state of Centyrin-siRNA conjugates. Liquid chromatography/mass spectrometry (LCMS) was used to confirm identity and purity of the conjugates. Samples were analyzed using a Waters Acuity UPLC/Xevo G2-XS TOF mass spectrometer system. The instrument was operated in negative electro-spray ionization mode and scanned from m/z 200 to 3000. Conjugate was seen as two fragments; Antisense and Sense-Centyrin.

[0407] Centyrin conjugation to siRNA, conjugate purification and analysis: Centyrins were conjugated to siRNA through cysteine-specific chemistry via maleimide. Cysteine-containing Centyrin in PBS at 50-200 μ M were reduced with 10 mM tris(2-carboxyethyl) phosphine (TCEP) at room temperature (30 mins) to yield a free thiol. The free thiol containing Centyrin is then mixed with maleimide containing siRNA duplex in water immediately prior at a molar ratio of ~1.5:1 Centyrin:siRNA. After 2 hr incubation at RT or 37° C., reaction was quenched with N-ethyl maleimide (1 mM final NEM concentration in the reaction mixture) The conjugate was purified in two steps. Step I either immobilized metal affinity chromatography (for tagged proteins) or Hydrophobic interaction chromatography (for tagless proteins); to remove un-reacted SiRNA linker. Step II-Capto-DEAE; to remove un-reacted centyrin. Fractions containing conjugate were pooled, exchanged into HBS by desalting using dialysis, and concentrated if necessary.

[0408] FIG. 6: IMAC chromatography for Tagged protein. For IMAC, Histrap HP Columns (1 ml, 5 ml) from Cytiva were used. Histrap Buffer A (Binding Buffer) was 50 mM Tris pH7.4, 500 mM NaCl and 10 mM imidazole in Type 1 H2O, Histrap Buffer B (Elution Buffer): 50 mM Tris pH7.4, 500 mM NaCl and 250 mM imidazole in Type 1 H2O. Reaction sample was directly injected onto the column through sample loop or sample pump. Followed by sample application, column is washed with 5-10 CV of Buffer A. Elution is typically started with step gradient (0-100%). Fractions were collected when UV reading is 50 mAU and above using fraction collector and peak fragmentation.

[0409] FIG. 7 HIC—for tagless proteins (removal of excess of siRNA). For HIC, HiTrap Butyl HP Columns (1 ml, 5 ml) from Cytiva were used. HIC Buffer A (Binding Buffer) was 2M Ammonium Sulfate, 25 mM Sodium Phosphate pH 7.0 in Type 1 H2O, while, HIC Buffer B (Elution Buffer): 25 mM Sodium Phosphate pH 7.0 in Type 1 H2O. Conjugation reaction sample is diluted with Buffer A in ratio of 1:1. Sample prepared above was injected onto the column through sample loop or sample pump. Followed by sample application, column is washed with 5 CV of Buffer A. Elution is typically started with 0.0% B and then a gradient as Table 10 below. Fractions collected when UV reading is 50 mAU and above using fraction collector and peak fragmentation.

TABLE 10

	Type	% B	Length (CV)
1	Linear	30 to 90	3.00
2	Linear	35.0	10.00
3	Linear	100.0	10.00
4	Linear	100.0	5.00

[0410] Ring-opening-To avoid loss of cargo via retro-Michael reaction, maleimide ring hydrolysis is performed. Pooled fractions from either histrap (tagged protein) or from HIC (tagless protein) is dialyzed into 25 mM TRIS pH 8.9

buffer. In this buffer the reaction is placed in an incubator shaker at 37 degrees C. for 72 hrs. Reaction monitored for completion by LC-MS.

[0411] FIG. 8: Ion-exchange chromatography (IEX)—for tagged and tagless proteins (removal of excess unreacted centyrin). For IEX chromatography, HiTrap Capto DEAE Columns (1 ml, 5 ml) from Cytiva were used. Capto DEAE Buffer A (Binding Buffer) was 25 mM Tris pH 8.8 in Type 1 H2O, while, Capto DEAE Buffer B (Elution Buffer) used was 25 mM Tris pH 8.8, 1M NaCl in Type 1 H2O.

[0412] Sample after ring opening was directly injected onto the column through sample loop or sample pump. Followed by sample application, column is washed with 5-10 CV of Buffer A. Elution is typically started with 0.0% B and then a gradient as Table 11 below.

[0413] Unreacted protein is eluted in the flowthrough which is typically the first peak and second peak is typically pure conjugate. All the fractions are collected and pooled. Concentration of the pool is determined by measuring A260 using Nanodrop for yield calculation.

TABLE 11

1	Linear	20	5
2	Linear	100	15
3	Linear	100	5

[0414] FIG. 9: Example of an analytical SEC of the centyrin-oligonucleotide conjugate. Centyrin-siRNA conjugates were characterized by a combination of analytical techniques. Analytical SEC was used to analyze purity and aggregation state of Centyrin-siRNA conjugates. Waters H-class UPLC along with Waters ACQUITY UPLC Protein BEH SEC Column, 125A, 1.7 μ m, 4.6x150 mm was used for routine SEC analysis. Typically, 2-5 μ l sample was injected using the autosampler, flow rate was 0.25 ml/min and mobile phase used was either 1xPBS or 100 mM, pH 7.2 phosphate buffer.

[0415] FIG. 10: Example of SDS PAGE Gel of the conjugate. SDS-PAGE was used to compare amounts of conjugate to free protein. For performing the SDS-PAGE Gel; Invitrogen Mini Gel tank. PowerEase™ Touch 600 W Power Supply and 115 VAC was used. Invitrogen-NuPAGE™ 4-12% Bis-Tris Protein Gels, 1.0 mm were used along with 20xNuPAGE® MES SDS Running Buffer and SeeBlue™ Pre-stained Protein Standard. Samples normalized to around 0.5 mg/mL by ultrapure water. For non-reducing gel, 10 μ L of normalized samples are mixed with 10 μ L of 2x Laemmli Sample Buffer in 1 to 1 ratio. For reducing gel, 2x Laemmli Sample Buffer is mixed with β -Mercaptoethanol in 95 to 1 ratio and then mixed with 10 μ L of samples in 1 to 1 ratio. The resulting sample mixtures were boiled at 95° C. for 5 minutes and then cool down in the Thermocycler. 15 μ L of samples and Protein Ladder into appropriate wells were loaded, the voltage was set at 200V and run time was set for around 30 minutes. Once the gel run was finished, it was visualized by Coomassie blue or SYBR Green or methylene blue. Left most lane is Protein Ladder, next is Centyrin by itself and last is the centyrin-siRNA conjugate.

[0416] Candidate siRNA sequences were transfected into human cells (H358) at a range of concentrations. RNA was harvested 24 hours post-transfection and analyzed for GYS1 levels via quantitative reverse transcription polymerase chain reaction (RT-PCR). 18S ribosomal RNA was used as the RT-PCR endogenous control gene. Levels of knockdown were compared to untreated cells. EC50 values were calcu-

lated using Graphpad Prism software and Emax values represent the maximum percent knockdown of GYS1 mRNA observed (Table 5).

TABLE 5			
siRNA	Ref.	EC50 (pM)	Emax (%)
A	ABXO8	3.841	70
B	ABXO9	10.02	70.2
C	ABXO12	12.62	71.5
D	ABXO110	21.73	87.9
E	ABXO113	23.74	82.1
F	ABXO114	6.716	75.5
G	ABXO115	4.5	75.2
H	ABXO116	0.7542	69.4
I	ABXO117	20.87	93.8
J	ABXO122	8.93	65.6
K	ABXO123	4.049	75.5
L	ABXO124	2.642	70.5
M	ABXO125	0.8761	53.3
N	ABXO128	2.549	76.3
O	ABXO129	0.4961	64.3
P	ABXO139	0.6143	70.6
Q	ABXO142	23.19	80.3
R	ABXO150	58.88	96.1
S	ABXO151	1.502	75.6
T	ABXO152	24.76	91.1
U	ABXO155	5.893	60.2
V	ABXO156	0.6893	57.6
W	ABXO160	0.787	80.1
X	ABXO168	8.662	75.7
Y	ABXO171	38.82	69.2
Z	ABXO191	54.2	75.4
AA	ABXO197	19.8	77.1
BB	ABXO201	7.1	75.6
CC	ABXO203	5.7	74.9
DD	ABXO205	9.8	80.4

[0417] Other siRNAs have also been tested as described above and their EC50 are provided in Table 6.

TABLE 6	
siRNA Pair	EC50 (pM)
EE	1.8
FF	6
GG	5.9
HH	15.1
II	8.2
JJ	25.2
KK	
LL	1.35
MM	3.74
NN	5.66
OO	5.56
PP	6.34
QQ	11.76
RR	2.85
SS	3.35

TABLE 6-continued	
siRNA Pair	EC50 (pM)
UU	0.4
VV	3.72
WW	3.15
XX	2.57
YY	1.64
ZZ	1.56
AAA	ND
BBB	4.87
CCC	2.65
DDD	2.66
EEE	1.21
FFF	1.31
GGG	NA

[0418] Selectivity of candidate siRNA sequences was evaluated by transfection of siRNAs into human cells (HEK-293) at a range of concentrations. Cell viability was assessed 72 hours post-transfection using CellTiterGlo. Emax values are reported as the maximum percent reduction in cell viability at the highest siRNA concentration tested (10 nM) (Table 7).

TABLE 7		
siRNA Pair	Ref.	Emax, % (at 10 nM)
A	ABXO8	5.52
B	ABXO9	-0.3
C	ABXO12	31.67
D	ABXO110	25.68
E	ABXO113	22.83
F	ABXO114	-1.1
G	ABXO115	30.85
H	ABXO116	29.27
I	ABXO117	35.19
J	ABXO122	28.25
K	ABXO123	17.21
L	ABXO124	24.09
M	ABXO125	71.45
N	ABXO128	42.41
O	ABXO129	58.02
P	ABXO139	34.44
Q	ABXO142	8.94
R	ABXO150	16.06
S	ABXO151	60.95
T	ABXO152	42.84
U	ABXO155	54.4
V	ABXO156	34.1
W	ABXO160	-4.59
X	ABXO168	51.66
Y	ABXO171	4.79
Z	ABXO191	46.41
AA	ABXO197	25.01
BB	ABXO201	46.92
CC	ABXO203	15.18
DD	ABXO205	60.13

Example 2: Selection of Fibronectin Type III (FN3) Domains that Bind CD71

[0419] Panning and Biochemical Screening Methods for Identifying FN3 domains that bind to CD71 that do not inhibit transferrin binding to CD71. To screen for FN3 domains that specifically bind CD71 and do not inhibit transferrin binding to CD71, streptavidin-coated Maxisorp plates (Nunc catalog 436110) are blocked for 1 hour in Starting Block T20 (Pierce) and then are coated with biotinylated CD71 (using same antigen as in panning) or negative controls (an unrelated Fc-fused recombinant protein and human serum albumin) for 1 hour in the presence of transferrin or with FN3 protein that binds to the CD71 transferrin binding site. The concentration of transferrin is up to 35 μ M. Without being bound to any particular theory, the inclusion of the transferrin or the FN3 protein that binds to the CD71 transferrin binding site pushes the selection of the FN3 domains to those that do not compete or inhibit with transferrin binding to CD71. Plates are rinsed with TBST and diluted lysate is applied to plates for 1 hour. Following additional rinses, wells are treated with HRP-conjugated anti-V5 tag antibody (Abcam, ab1325), for 1 hour and then are assayed with POD (Roche, 11582950001). The DNA from FN3 domain lysates with signals at least 10-fold ELISA signal above that of streptavidin controls are sequenced resulting in FN3 domain sequences isolated from the screening.

Example 3: Selection of Fibronectin Type III (FN3) Domains that Bind CD71 and are not Competitive with Transferrin

[0420] To identify CD71 binding FN3 domains that were either not competitive or minimally competitive with transferrin a biased CIS-display strategy was designed. In short, using the output recovered after 5 rounds of panning on the ECD of human CD71 (Example 3). Additional rounds of off-rate selection were performed as described in Example 3 with the addition of either 1) a wash step with human holo transferrin to elute Centyrins that bound at the same site as transferrin before the final elution step or 2) elution of FN3 domain binders with monoclonal antibody OKT9. FN3 domains recovered from the transferrin wash strategy and the OKT9 elution strategy were PCR amplified and cloned into pET vector as previously described. 228 FN3 domains that specifically bound huCD71 were confirmed by ELISA for binding to huCD71 ECD. A subset of the unique binders was analyzed by SEC, conjugated to MMAF and assessed for internalization via cell viability assay in SKBR-3 cells +/- holo human transferrin. The polypeptides were found to be internalized by the receptor.

[0421] Integral Molecular performed Membrane Proteome Array (MPA) assay to profile the specificity of ABX1198 (SEQ ID NO: 509) and ABX1100 (SEQ ID NO: 509 plus siRNA pair with linker number 0000) against the library of human membrane proteins. The MPA library contains over 6000 human membrane proteins, including 94% of all single-pass, multi-pass and GPI anchored proteins including GPCRs, ion channels and transporters with each membrane protein uniquely expressed in an avian QT6 cell background. Flow cytometry is used to directly detect ligand (FN3 domain) binding to membrane proteins individually expressed in unfixed cells.

[0422] ABX1198 (SEQ ID NO: 509) and ABX1100 (SEQ ID NO: 509 plus siRNA pair with linker number 0000) were screened at the concentration with optimal signal/background noise ratio, 1.25 μ g/ml, 1.25 μ g/ml and 0.31 μ g/ml

respectively, against the MPA. Membrane protein targets identified in screening were followed up in validation procedure using ligand serial dilution and cells individually transfected with identified targets.

Example 4. Knockdown of mRNA in Muscle Cells Using CD71 FN3 Domain-Oligonucleotide Conjugates

[0423] muCD71 binding FN3 domains are conjugated to siRNA oligonucleotides or antisense oligonucleotides (ASOs) using maleimide chemistry via a cysteine that is uniquely engineered into the FN3 domain. The cysteine substitutions can be one such as those provided for herein and also as provided for in U.S. Patent Application Publication No. 20150104808, which is hereby incorporated by reference in its entirety. siRNAs or ASOs are modified with standard chemical modifications and confirmed to enable knockdown of the targeted mRNA in vitro. FN3 domain-oligonucleotide conjugates are dosed intravenously in mice at doses up to 10 mg/kg oligonucleotide payload. At various time points following dosing, mice are sacrificed; skeletal muscle, heart muscle and various other tissues will be recovered and stored in RNeasy Lysis Buffer (Qiagen) until needed. Target gene knockdown is assessed using standard qPCR $\Delta\Delta C_T$ methods and primers specific for the target gene and a control gene. The target gene is found to be knock downed in the muscles and such knockdown is enhanced by conjugating the siRNA or ASO to the CD71 FN3 binding domain.

[0424] FN3-siRNA conjugates tested are as described in table 8

TABLE 8

Conjugate	CD71 FN3 domain (Centyrin)	GYS1 siRNA Pair
ABXC-1	SEQ ID NO: 572	RRRR
ABXC-2	SEQ ID NO: 572	SSSS
ABXC-3	SEQ ID NO: 572	TTTT
ABX1005	SEQ ID NO: 572	Non-GYS1 target/control
ABXC-27	SEQ ID NO: 572	A
ABXC-28	SEQ ID NO: 572	QQQQ
ABXC-29	SEQ ID NO: 572	PPPP

[0425] FIG. 4A demonstrates the knockdown of GYS1 mRNA in mouse gastrocnemius muscle using 3 different FN3 domain-siRNA conjugates compared with vehicle alone. For efficacy studies, male GAA-/- mice (at the ages of 4-5 weeks) were obtained from Jackson Laboratories. All animals were treated in accordance with IACUC protocols. Five animals received a single tail vein intravenous bolus injection of either 5.4 mg/kg of three different FN3 domain-siRNA conjugates (3 mpk Gys1siRNA) or vehicle. Four weeks after the single dose, the mice were euthanized, gastrocnemius muscle were collected in RNeasy Lysis Buffer, stored at 4 C overnight and were frozen at -80 C. Total RNA was isolated from the gastrocnemius using Qiagen's RNeasy Fibrous Tissue kit. Expression levels of the target Gys1 and the endogenous controls (Pgk1, Ubc, Hprt1 and Aha1) were analyzed using real-time, quantitative PCR Data were analyzed using the $\Delta\Delta C_T$ method normalizing to control animals

dosed with vehicle alone. The percentage knockdown of Gys1 mRNA in the FN3 domain-siRNA conjugate treatment groups were measured by subtracting the percentage remaining Gys1 mRNA levels by 100. Statistical significance was calculated using One-way ANOVA with Dunnett's multiple comparison tests in the GraphPad Prism software. Statistical significance is displayed on the figure with asterisk *** $p < 0.001$.

[0426] FIG. 4B demonstrates the knockdown of GYS1 protein in mouse gastrocnemius muscle using 3 different FN3 domain-siRNA conjugates compared with vehicle alone. Gys1 protein quantification in gastrocnemius was performed by homogenizing gastrocnemius in RIPA buffer. Protein concentrations in the gastrocnemius were measured using the Bradford assay. Gys1 levels were quantified using the manufacturer's standard method for 12-230 kDa Jess separation modules (SM-W004). The proteins were separated by immobilizing on capillaries using protein Simple's proprietary photoactivated capture chemistry. Anti-Gys1 primary antibody were used at 1:100 dilution. The chemiluminescent revelations were established using peroxide/Auminol-S. A digital image of the capillaries' chemiluminescence was captured using Compass' Simple Western software, which automatically measures height (chemiluminescence intensity), area, and signal/noise ratio. An internal system was included in each run. The peak area values of FN3 domain-siRNA conjugate treatment groups were normalized to the vehicle treated tissues and the percentage knockdown of Gys1 protein in the treatment groups were measured by subtracting the percentage remaining Gys1 protein levels by 100. Statistical significance was calculated using One-way ANOVA with Dunnett's multiple comparison tests in the GraphPad Prism software. Statistical significance is displayed on the figure with asterisk *** $p < 0.001$.

[0427] FIG. 5 demonstrates the GYS1 knockdown is highly specific for skeletal muscle using 3 different FN3 domain-siRNA conjugates compared with a siRNA to a different target (Aha1). Male GAA^{-/-} mice (at the ages of 8-9 weeks) were obtained from Jackson Laboratories. All animals were treated in accordance with IACUC protocols. Three animals received a single tail vein intravenous bolus injection of either 17.9 mg/kg of three different FN3 domain-siRNA conjugates (10 mpk Gys1siRNA), 17.9 mg/kg of one FN3 domain-siRNA conjugate (10 mpk Aha1 siRNA) or vehicle. Two weeks after the single dose, the mice were euthanized. Gastrocnemius, quadriceps, diaphragm, heart, liver, and kidney tissues were collected in RNAlater, stored at 4 C overnight and were frozen at -80 C. Total RNA was isolated from the tissues using Qiagen's RNeasy Fibrous Tissue kit. Expression levels of the target Gys1/Aha1 and the endogenous control (Pgk1) were analyzed using real-time, quantitative PCR. Data were analyzed using the $\Delta\Delta C_t$ method normalizing to control animals dosed with vehicle alone.

Example 5: RNA-Seq Experiments Comparing ABX-HHH Treated and Untreated Cells

[0428] mRNA sequencing was performed on polyA⁺ selected mRNA after HEK293T cells were lipofected with 10 nM ABXO-HHH for 24 h. 6 replicates per treatment were sequenced to an average depth >30 million reads. Libraries were pseudoaligned to the GRCh38 transcriptome using Kallisto with a mapping rate >90%. The values in this volcano plot were generated using DESeq2 to compare ABXO-HHH treated RNA-seq libraries versus untreated cells. Each dot in the plot in FIG. 2 represents the measured expression change in a gene. The X axis represents the log

2 fold change in gene expression (ABXO-HHH divided by untreated) for all genes. The Y axis shows the negative $\log_{10}$ transformed adjusted P value. The black arrow points to GYS1 on the plot.

Example 6: CD71 FN3 Domain siRNA Conjugate Binding Specificity

[0429] Integral Molecular (www.integralmolecular.com) performed their proprietary Membrane Proteome Array (MPA) assay to profile the specificity of CD71 FN3 domain and CD71 FN3 domain siRNA conjugate against the library of human membrane proteins (FIG. 3). The MPA contains over 6000 human membrane proteins, covering 94% of all single-pass, multi-pass and GPI anchored proteins including GPCRs, ion channels and transporters with each membrane protein uniquely expressed in an avian QT6 cell background. Flow cytometry was used to directly detect FN3 domain binding to membrane proteins individually expressed in unfixed cells.

[0430] FN3 domain and FN3 domain-siRNA conjugate were screened at the concentration with optimal signal/background noise ratio, 1.25 ug/ml or 0.31 ug/ml respectively, against the MPA. Membrane protein targets identified in screening were validated using ligand serial dilution on cells uniquely expressing the identified targets.

GENERAL METHODS

[0431] Standard methods in molecular biology are described Sambrook, Fritsch and Maniatis (1982 & 1989 2nd Edition, 2001 3rd Edition) *Molecular Cloning, A Laboratory Manual*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY; Sambrook and Russell (2001) *Molecular Cloning*, 3rd ed. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY; Wu (1993) *Recombinant DNA*, Vol. 217, Academic Press, San Diego, CA). Standard methods also appear in Ausbel, et al. (2001) *Current Protocols in Molecular Biology*, Vols. 1-4. John Wiley and Sons, Inc. New York, NY, which describes cloning in bacterial cells and DNA mutagenesis (Vol. 1), cloning in mammalian cells and yeast (Vol. 2), glycoconjugates and protein expression (Vol. 3), and bioinformatics (Vol. 4).

[0432] Methods for protein purification including immunoprecipitation, chromatography, electrophoresis, centrifugation, and crystallization are described (Coligan, et al. (2000) *Current Protocols in Protein Science*, Vol. 1, John Wiley and Sons, Inc., New York). Chemical analysis, chemical modification, post-translational modification, production of fusion proteins, glycosylation of proteins are described (see, e.g., Coligan, et al. (2000) *Current Protocols in Protein Science*, Vol. 2, John Wiley and Sons, Inc., New York; Ausubel, et al. (2001) *Current Protocols in Molecular Biology*, Vol. 3, John Wiley and Sons, Inc., NY, NY, pp. 16.0.5-16.22.17; Sigma-Aldrich, Co. (2001) *Products for Life Science Research*, St. Louis, MO; pp. 45-89; Amersham Pharmacia Biotech (2001) *BioDirectory*, Piscataway, N.J., pp. 384-391). Production, purification, and fragmentation of polyclonal and monoclonal antibodies are described (Coligan, et al. (2001) *Current Protocols in Immunology*, Vol. 1, John Wiley and Sons, Inc., New York; Harlow and Lane (1999) *Using Antibodies*, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY; Harlow and Lane, supra). Standard techniques for characterizing ligand/receptor interactions are available (see, e.g., Coligan, et al. (2001) *Current Protocols in Immunology*, Vol. 4, John Wiley, Inc., New York).

incorporation by reference. Citation of the references herein is not intended as an admission that the reference is pertinent prior art, nor does it constitute any admission as to the contents or date of these publications or documents.

[0434] The present embodiments are not to be limited in scope by the specific embodiments described herein. Indeed, various modifications of the embodiments in addition to those described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are intended to fall within the scope of the appended claims.

[0435] The foregoing written specification is considered to be sufficient to enable one skilled in the art to practice the embodiments. Various modifications of the embodiments in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description and fall within the scope of the appended claims.

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                        organism = Homo sapiens
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KMSLBSQL	QKQKDLSE	120	120
SSLLHLVKQF	VSDLRLSLSC	MAALQGNSE	RTCCPVNWVE
HERSCYWFSR	SGKAWADADN	180	180
YCRLEDAHLV	VVTSENEQKF	VQHII GPVNT	WMGLHDQNGP
WKNVGDGTDYE	TGFKNWRPEQ	240	240
PDDYVGHGLG	GGEDCAHFTD	DGRWNDDVCQ	RPYRIVCETE
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                     organism = Homo sapiens
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CRLEDAHLVV	VTSWEEQKVF	QHHIGPVNTW	MGLHDQNGPW	KVDGTDYET	GFKNWRPEQP	180
DDWYGHGLGG	GEDCAHFTGD	TRWDNDVDCOR	PYRWVCETEL	DKASOEPLDL		230

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SEQUENCE: 7
000
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SEQ ID NO: 8          moltype =      length =
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-continued

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SEQUENCE: 8
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SEQ ID NO: 9          moltype =   length =
SEQUENCE: 9
000

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                    mol_type = other DNA
                    organism = synthetic construct
modified_base        1
                    mod_base = OTHER
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modified_base        2
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base        3
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        4
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modified_base        5
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        6
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base        7..9
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        10
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base        11
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        12..14
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base        15
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        16
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base        17
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base        18..19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
misc_feature         1..19
                    note = RNA
misc_feature         20
                    note = DNA
SEQUENCE: 10
ctgggaggat gaattcgaat                                20

SEQ ID NO: 11         moltype = RNA  length = 21
FEATURE              Location/Qualifiers
source               1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base        1
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base        2
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base        3
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base        4

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-continued

	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	6..8	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	11..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 11		
ttcgaattca tcctcccagt t		21
SEQ ID NO: 12	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	

-continued

modified_base	12..14	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	1..19	
	note = RNA	
misc_feature	20	
	note = DNA	
SEQUENCE: 12		
caaggtgggt ggcattcaat		20
SEQ ID NO: 13	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	6..8	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	11..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21	

-continued

	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 13		
ttagatgccca cccaccttgt t		21
SEQ ID NO: 14	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
misc_feature	1..19	
	note = RNA	
misc_feature	20	
	note = DNA	
SEQUENCE: 14		
tccgcagcct ggatgattat		20
SEQ ID NO: 15	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	

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modified_base      5
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      6..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     11..13
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     21
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide

SEQUENCE: 15
taatcatcca ggctgcggat t                                     21

SEQ ID NO: 16      moltype = RNA length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     4..6
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     7..9
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     10..11
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     12
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     13..16
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     17
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     18..19
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
misc_feature      19
                  note = Connected to an inverted thymidine via a 3'-3' bond

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-continued

SEQUENCE: 16
gcgactgcct gtagcaaca 19

SEQ ID NO: 17 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

modified_base 1..2
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide phosphorothioate

modified_base 3..13
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

modified_base 14
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide

modified_base 15..18
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

modified_base 19..20
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 21
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

SEQUENCE: 17
tgttgctaca ggcagtcgct t 21

SEQ ID NO: 18 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base 1..2
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 3
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide

modified_base 4..6
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

modified_base 7..9
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide

modified_base 10..11
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

modified_base 12
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide

modified_base 13..16
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

modified_base 17
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide

modified_base 18..19
 mod_base = OTHER
 note = 2'O-Methyl modified nucleotide

misc_feature 19
 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 18
tttatgggca tctggacta 19

SEQ ID NO: 19 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

modified_base 1..2
 mod_base = OTHER
 note = 2'Fluoro modified nucleotide phosphorothioate

modified_base 3..13
 mod_base = OTHER

-continued

modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER	
SEQUENCE: 19	note = 2'-Methyl modified nucleotide	
tagtccagat gcccataaat t		21
SEQ ID NO: 20	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	note = 2'-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 20		
gcgcggacca acaatttca		19
SEQ ID NO: 21	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER	
	note = 2'-Methyl modified nucleotide	

-continued

SEQUENCE: 21
tgaaattggt ggtcgcgcgt t 21

SEQ ID NO: 22 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 3
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 4..6
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 7..9
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 10..11
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 12
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 13..16
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 17
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 18..19
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

misc_feature 19
note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 22
ggaccaacaa tttcaacga 19

SEQ ID NO: 23 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'Fluoro modified nucleotide phosphorothioate

modified_base 3..13
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 14
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 15..18
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 19..20
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 21
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

SEQUENCE: 23
tcgttgaaat tggtggtcct t 21

SEQ ID NO: 24 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 3
mod_base = OTHER

-continued

modified_base	note = 2'Fluoro modified nucleotide 4..6 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 24
accaacaatt tcaacgtga 19

SEQ ID NO: 25
FEATURE
source
mol_type = RNA length = 21
Location/Qualifiers
1..21
mol_type = other RNA
organism = synthetic construct

modified_base	1..2 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 15..18 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 19..20 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide

SEQUENCE: 25
tcacgttgaa attgttggtt t 21

SEQ ID NO: 26
FEATURE
source
mol_type = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct

modified_base	1..2 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 4..6 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide

-continued

modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 26		
gcgcaaacag ctttgggaa		19
SEQ ID NO: 27	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 27		
ttcccaaagc tgtttgcgct t		21
SEQ ID NO: 28	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 28		
agagccatct ttgcaacga		19

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SEQ ID NO: 29      moltype = RNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     3..13
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     14
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     15..18
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     19..20
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     21
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
SEQUENCE: 29
tcgttgcaaa gatggctctt t                               21

SEQ ID NO: 30      moltype = RNA  length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     4..6
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     7..9
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     10..11
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     12
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     13..16
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     17
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     18..19
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
misc_feature      19
                  note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 30
gagccatctt tgcaacgca                               19

SEQ ID NO: 31      moltype = RNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     3..13
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     14

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modified_base      mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
15..18
modified_base      mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
19..20
modified_base      mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate
21
modified_base      mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
SEQUENCE: 31
tgcgttgcaa agatggctct t                               21

SEQ ID NO: 32      moltype = RNA length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     4..6
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     7..9
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     10..11
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     12
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     13..16
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     17
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     18..19
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
misc_feature      19
                  note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 32
agccatcttt gcaacgcaa                               19

SEQ ID NO: 33      moltype = RNA length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     3..13
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     14
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     15..18
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     19..20
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     21
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
SEQUENCE: 33
ttgcgttgca aagatggctt t                               21

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-continued

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SEQ ID NO: 34      moltype = RNA  length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     4..6
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     7..9
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     10..11
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     12
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     13..16
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     17
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     18..19
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
misc_feature      19
                  note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 34
ccatctttgc aacgcagca                               19

SEQ ID NO: 35      moltype = RNA  length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     3..13
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     14
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     15..18
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
modified_base     19..20
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     21
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide
SEQUENCE: 35
tgctgcgttg caaagatggt t                             21

SEQ ID NO: 36      moltype = RNA  length = 19
FEATURE           Location/Qualifiers
source            1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1..2
                  mod_base = OTHER
                  note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                  mod_base = OTHER
                  note = 2'Fluoro modified nucleotide
modified_base     4..6

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-continued

	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 36		
acgcagcggc agtctttca		19
SEQ ID NO: 37	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 37		
tgaaagactg ccgctgcggtt t		21
SEQ ID NO: 38	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	

-continued

modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 38	
tgaccacat cgcgcaaa	19
SEQ ID NO: 39	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide
SEQUENCE: 39	
tttcggcgga tgggtgcat t	21
SEQ ID NO: 40	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 4..6 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 40	
cgaatcggcc tcttcaata	19
SEQ ID NO: 41	moltype = RNA length = 21

-continued

FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 41		
tattgaagag gccgattcgt t		21
SEQ ID NO: 42	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 42		
cttcaatagc agtgccgaa		19
SEQ ID NO: 43	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	

-continued

modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 43		
ttcggcactg ctattgaagt t		21
SEQ ID NO: 44	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 44		
cagtatctcc accaatcta		19
SEQ ID NO: 45	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 45		
tagattgggtg gagatactgt t		21
SEQ ID NO: 46	moltype = RNA length = 19	

-continued

FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	4..6 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	7..9 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	10..11 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	13..16 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	17 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	18..19 mod_base = OTHER note = 2'O-Methyl modified nucleotide
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 46	
ccaccaatct ctcgggcta	19
SEQ ID NO: 47	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3..13 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	15..18 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	19..20 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	21 mod_base = OTHER note = 2'O-Methyl modified nucleotide
SEQUENCE: 47	
tagccggaga gattggtggt t	21
SEQ ID NO: 48	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	4..6 mod_base = OTHER note = 2'O-Methyl modified nucleotide

-continued

modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 48		
caccaatctc tccggctta		19
SEQ ID NO: 49	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 49		
taagccggag agattggtgt t		21
SEQ ID NO: 50	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	

-continued

	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 50		
cccctcagct tacggtata		19
SEQ ID NO: 51	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 51		
tataaccgtaa gctgaggggt t		21
SEQ ID NO: 52	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 52		
ccctcagctt acggtatca		19
SEQ ID NO: 53	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	

-continued

	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 53		
tgataaccgta agctgagggt t		21
SEQ ID NO: 54	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 54		
ttacgggtatc tacattcta		19
SEQ ID NO: 55	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	

-continued

modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide	
SEQUENCE: 55		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 56	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide	
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 56		
ggaaccgcac ggagcgcca		19
SEQ ID NO: 57	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER	
modified_base	note = 2'-Methyl modified nucleotide	
SEQUENCE: 57		
tggcgctccg tgcggttct t		21
SEQ ID NO: 58	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	

-continued

	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 58		
acggagcgcc tctccgaca		19
SEQ ID NO: 59	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 59		
tgtcggagag gcgtccggt t		21
SEQ ID NO: 60	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	

-continued

modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	12 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	13..16 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	17 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	18..19 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 60		
agcaagcgca actctgtga		19
SEQ ID NO: 61	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	14 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	15..18 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	19..20 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
SEQUENCE: 61		
tcacagagtt gcgcttgctt t		21
SEQ ID NO: 62	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	4..6 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	7..9 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	10..11 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	12 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	13..16 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	17 mod_base = OTHER note = 2'Fluoro modified nucleotide	

-continued

modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 62		
cagatggtcc tccatttca		19
SEQ ID NO: 63	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 63		
tgaaatggag gaccatctgt t		21
SEQ ID NO: 64	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 64		
tctagatctg gaaccttaa		19
SEQ ID NO: 65	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	

-continued

modified_base	1..2	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	14	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	15..18	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	19..20	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
SEQUENCE: 65			
ttaagggtcc agatctagat t			21
SEQ ID NO: 66	moltype = RNA length = 19		
FEATURE	Location/Qualifiers		
source	1..19	mol_type = other RNA	
		organism = synthetic construct	
modified_base	1..2	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	4..6	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	7..9	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	10..11	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	12	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	13..16	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	17	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	18..19	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
misc_feature	19	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 66			
cccacattta cctcgagga			19
SEQ ID NO: 67	moltype = RNA length = 21		
FEATURE	Location/Qualifiers		
source	1..21	mol_type = other RNA	
		organism = synthetic construct	
modified_base	1..2	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	14	mod_base = OTHER	
		note = 2'Fluoro modified nucleotide	
modified_base	15..18	mod_base = OTHER	
		note = 2'O-Methyl modified nucleotide	
modified_base	19..20		

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	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 67		
tcctcgaggt aaatgtgggt t		21
SEQ ID NO: 68	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 68		
accgcttggt ttgcagtta		19
SEQ ID NO: 69	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 69		
taactgcaaa ccaagcggtt t		21
SEQ ID NO: 70	moltype = RNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other RNA	
	organism = synthetic construct	

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modified_base      1..2
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base      7
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base      8
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base     11
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base     12
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base     13
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'-Fluoro modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'-O-Methyl modified nucleotide

SEQUENCE: 70
tctcgtggcc ttaatgaa

SEQ ID NO: 71      moltype = RNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other RNA
                  organism = synthetic construct
modified_base     1
                  mod_base = OTHER
                  note = Vinyl phosphonate 2'-fluoro modified nucleotide
                  phosphorothioate
modified_base     2
                  mod_base = OTHER
                  note = 2'-O-methyl modified nucleotide
modified_base     3
                  mod_base = OTHER
                  note = 2'-fluoro modified nucleotide
modified_base     4
                  mod_base = OTHER
                  note = 2'-O-methyl modified nucleotide
modified_base     5
                  mod_base = OTHER
                  note = 2'-fluoro modified nucleotide
modified_base     6

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-continued

	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 71		
ttcattaagg ccacgagatt		20
SEQ ID NO: 72	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	

-continued

modified_base	11	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 72		
ctgggaggat gaattcgaa		19
SEQ ID NO: 73	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide	
	phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	17	

-continued

	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 73		
tcgaattcat cctcccagtt		20
SEQ ID NO: 74	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 74		
caaggtgggt ggcattctaa		19
SEQ ID NO: 75	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

-continued

modified_base	3	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	4	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	5	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	6	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	7	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	8	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	9	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	10	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	11	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	12	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	14	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	15	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	16	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	18	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide phosphorothioate
modified_base	19	mod_base = OTHER
		note = 2'-fluoro modified nucleotide phosphorothioate
modified_base	20	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

SEQUENCE: 75
tagatgccac ccaccttggt

SEQ ID NO: 76
FEATURE
source

moltype = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base 3
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 4..6
mod_base = OTHER
note = 2'O-Methyl modified nucleotide

modified_base 7..9
mod_base = OTHER
note = 2'Fluoro modified nucleotide

modified_base 10..11
mod_base = OTHER

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-continued

modified_base	note = 2'-O-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER	
modified_base	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 76		
gcgcggacca acaatttca		19
SEQ ID NO: 77	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1 mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	8 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	10 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	12 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13 mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	14 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	15 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	17 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18 mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	

-continued

modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 77		
gaaattgttg gtcgcgctt		20
SEQ ID NO: 78	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 78		
gagccatctt tgcaacgca		19
SEQ ID NO: 79	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide	
	Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	

-continued

modified_base	note = 2'-O-methyl modified nucleotide 10 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 11 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 13 mod_base = OTHER	
modified_base	note = 2'-fluoro modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 15 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 16 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 18 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide Phosphorothioate 19 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide Phosphorothioate 20 mod_base = OTHER	
SEQUENCE: 79		
gcgttgcaaa gatggctctt		20
SEQ ID NO: 80	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 80		
agccatcttt gcaacgcaa		19
SEQ ID NO: 81	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other RNA	

-continued

modified_base	organism = synthetic construct
	1
	mod_base = OTHER
	note = Vinyl phosphonate 2'-fluoro modified nucleotide
	Phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	3
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	10
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	11
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	13
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	15
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	16
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	18
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	19
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
SEQUENCE: 81	
tcggttgcaa agatggcttt	20
SEQ ID NO: 82	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = other RNA
	organism = synthetic construct
modified_base	1..2
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide

-continued

modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 82		
ttacggtatc tacattcta		19
SEQ ID NO: 83	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide	
	Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16	

-continued

	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 83		
agaatgtaga taccgtaatt		20
SEQ ID NO: 84	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 84		
cccacattta cctcgagga		19
SEQ ID NO: 85	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

-continued

modified_base	7	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	8	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	9	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	10	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	11	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	12	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	14	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	15	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	19	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

SEQUENCE: 85
cctcgaggta aatgtgggtt

20

SEQ ID NO: 86	moltype = RNA length = 18
FEATURE	Location/Qualifiers
source	1..18
	mol_type = other RNA
	organism = synthetic construct
modified_base	1..2
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	10
	mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide	
	11	
	mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide	
	12	
	mod_base = OTHER	
modified_base	note = 2'-O-Methyl modified nucleotide	
	13	
	mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide	
	14	
	mod_base = OTHER	
modified_base	note = 2'-O-Methyl modified nucleotide	
	15	
	mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide	
	16..18	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 86		
tctcgtggcc ttaatgga		18
SEQ ID NO: 87	moltype = RNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-fluoro modified nucleotide	
	Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	3	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

-continued

modified_base	17	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 87		
ttcattaagg ccacgagatt		20
SEQ ID NO: 88	moltype =	length =
SEQUENCE: 88		
000		
SEQ ID NO: 89	moltype =	length =
SEQUENCE: 89		
000		
SEQ ID NO: 90	moltype =	length =
SEQUENCE: 90		
000		
SEQ ID NO: 91	moltype =	length =
SEQUENCE: 91		
000		
SEQ ID NO: 92	moltype =	length =
SEQUENCE: 92		
000		
SEQ ID NO: 93	moltype =	length =
SEQUENCE: 93		
000		
SEQ ID NO: 94	moltype =	length =
SEQUENCE: 94		
000		
SEQ ID NO: 95	moltype =	length =
SEQUENCE: 95		
000		
SEQ ID NO: 96	moltype =	length =
SEQUENCE: 96		
000		
SEQ ID NO: 97	moltype =	length =
SEQUENCE: 97		
000		
SEQ ID NO: 98	moltype =	length =
SEQUENCE: 98		
000		
SEQ ID NO: 99	moltype =	length =
SEQUENCE: 99		
000		
SEQ ID NO: 100	moltype =	length =
SEQUENCE: 100		
000		
SEQ ID NO: 101	moltype = AA	length = 90
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 101		
MLPAPKQLVA SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT		60
GLKPGTEYKV WIHGVKGGAS SPPLIARFTT		90

-continued

SEQ ID NO: 102	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 102			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 103	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 103			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFAIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 104	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 104			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNISYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 105	moltype = AA	length = 89	
FEATURE	Location/Qualifiers		
source	1..89		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 105			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNIAYAEPPG IGGEAIWLRV PGSERSYDLTG 60
LKPGTEYKVV	IHGVKGGASS	PPLIARFTT	89
SEQ ID NO: 106	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 106			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNIAYWEAG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 107	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 107			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNIAYWEPA IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 108	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 108			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPNIAYWEPG AGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 109	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		
	organism = synthetic construct		
SEQUENCE: 109			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFNIAYWEPG IAGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV	WIHGVKGGAS	SPPLIARFTT	90
SEQ ID NO: 110	moltype = AA	length = 90	
FEATURE	Location/Qualifiers		
source	1..90		
	mol_type = protein		

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                                organism = synthetic construct
SEQUENCE: 110
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIALRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAS SPPLIARFTT 90

SEQ ID NO: 111                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 111
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLAV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAS SPPLIARFTT 90

SEQ ID NO: 112                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 112
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYAV WIHGVKGGAS SPPLIARFTT 90

SEQ ID NO: 113                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 113
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV AIHGVKGGAS SPPLIARFTT 90

SEQ ID NO: 114                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 114
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIAGVKGGAS SPPLIARFTT 90

SEQ ID NO: 115                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 115
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGSS SPPLIARFTT 90

SEQ ID NO: 116                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 116
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAA SPPLIARFTT 90

SEQ ID NO: 117                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 117
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAS SAPLIARFTT 90

SEQ ID NO: 118                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                             mol_type = protein
                             organism = synthetic construct
SEQUENCE: 118
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAS SPPLAARFTT 90

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SEQ ID NO: 119 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 119
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFNIAYWEPG IGGEAIWLRV PGSERSYDLT 60
GLKPGTEYKV WIHGVKGGAS SPPLIAAFTT 90

SEQ ID NO: 120 moltype = length =
SEQUENCE: 120
000

SEQ ID NO: 121 moltype = length =
SEQUENCE: 121
000

SEQ ID NO: 122 moltype = length =
SEQUENCE: 122
000

SEQ ID NO: 123 moltype = length =
SEQUENCE: 123
000

SEQ ID NO: 124 moltype = length =
SEQUENCE: 124
000

SEQ ID NO: 125 moltype = length =
SEQUENCE: 125
000

SEQ ID NO: 126 moltype = length =
SEQUENCE: 126
000

SEQ ID NO: 127 moltype = length =
SEQUENCE: 127
000

SEQ ID NO: 128 moltype = length =
SEQUENCE: 128
000

SEQ ID NO: 129 moltype = length =
SEQUENCE: 129
000

SEQ ID NO: 130 moltype = length =
SEQUENCE: 130
000

SEQ ID NO: 131 moltype = length =
SEQUENCE: 131
000

SEQ ID NO: 132 moltype = length =
SEQUENCE: 132
000

SEQ ID NO: 133 moltype = length =
SEQUENCE: 133
000

SEQ ID NO: 134 moltype = length =
SEQUENCE: 134
000

SEQ ID NO: 135 moltype = length =
SEQUENCE: 135
000

SEQ ID NO: 136 moltype = length =
SEQUENCE: 136

-continued

000

SEQ ID NO: 137 moltype = length =
SEQUENCE: 137
000

SEQ ID NO: 138 moltype = length =
SEQUENCE: 138
000

SEQ ID NO: 139 moltype = length =
SEQUENCE: 139
000

SEQ ID NO: 140 moltype = length =
SEQUENCE: 140
000

SEQ ID NO: 141 moltype = length =
SEQUENCE: 141
000

SEQ ID NO: 142 moltype = length =
SEQUENCE: 142
000

SEQ ID NO: 143 moltype = length =
SEQUENCE: 143
000

SEQ ID NO: 144 moltype = length =
SEQUENCE: 144
000

SEQ ID NO: 145 moltype = length =
SEQUENCE: 145
000

SEQ ID NO: 146 moltype = length =
SEQUENCE: 146
000

SEQ ID NO: 147 moltype = length =
SEQUENCE: 147
000

SEQ ID NO: 148 moltype = length =
SEQUENCE: 148
000

SEQ ID NO: 149 moltype = length =
SEQUENCE: 149
000

SEQ ID NO: 150 moltype = length =
SEQUENCE: 150
000

SEQ ID NO: 151 moltype = length =
SEQUENCE: 151
000

SEQ ID NO: 152 moltype = length =
SEQUENCE: 152
000

SEQ ID NO: 153 moltype = length =
SEQUENCE: 153
000

SEQ ID NO: 154 moltype = length =
SEQUENCE: 154
000

SEQ ID NO: 155 moltype = length =
SEQUENCE: 155

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SEQ ID NO: 156 moltype = length =
SEQUENCE: 156
000

SEQ ID NO: 157 moltype = length =
SEQUENCE: 157
000

SEQ ID NO: 158 moltype = length =
SEQUENCE: 158
000

SEQ ID NO: 159 moltype = length =
SEQUENCE: 159
000

SEQ ID NO: 160 moltype = length =
SEQUENCE: 160
000

SEQ ID NO: 161 moltype = length =
SEQUENCE: 161
000

SEQ ID NO: 162 moltype = length =
SEQUENCE: 162
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SEQ ID NO: 163 moltype = length =
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SEQ ID NO: 271 moltype = length =
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SEQ ID NO: 272 moltype = length =
SEQUENCE: 272
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SEQ ID NO: 273 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct
VARIANT 33
 note = D, F, Y or H
VARIANT 35
 note = Y, G, A, or V
VARIANT 37
 note = I, T, L, A or H
VARIANT 39
 note = S, Y or P
VARIANT 40
 note = Y, G, Q or R
VARIANT 41
 note = G or P
VARIANT 42
 note = A, Y, P, D or S
VARIANT 47
 note = A, C, D, E, F, G, H, I, K, L, N, Q, R, S, T, V, W,
 or Y
VARIANT 49
 note = A, C, D, E, F, G, H, I, K, L, N, Q, R, S, T, V, W,
 or Y
VARIANT 69
 note = W, N, S or E
VARIANT 71
 note = L, Y or G
VARIANT 73
 note = D, Q, H or V
VARIANT 74
 note = G or S
VARIANT 79
 note = R, G, F, L or D
VARIANT 80
 note = W, S, P or L
VARIANT 82
 note = T, V, M or S
VARIANT 85
 note = A, C, D, E, F, G, H, I, K, L, N, Q, R, S, T, V, W,
 or Y
VARIANT 87
 note = A, C, D, E, F, G, H, I, K, L, N, Q, R, S, T, V, W,
 or Y

SEQUENCE: 273
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFXIXYXEXX XXGEAIXLXV PGSERSYDLT 60
GLKPGTEYXV XIXXVKGXGX SXPLXAXFTT 90

SEQ ID NO: 274 moltype = length =
SEQUENCE: 274
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SEQ ID NO: 275 moltype = length =
SEQUENCE: 275
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SEQ ID NO: 276 moltype = length =
SEQUENCE: 276
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SEQ ID NO: 277 moltype = length =

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SEQUENCE: 277
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SEQ ID NO: 278 moltype = length =
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SEQ ID NO: 296 moltype = length =

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SEQ ID NO: 297 moltype = length =
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SEQ ID NO: 298 moltype = length =
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SEQ ID NO: 299 moltype = length =
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SEQ ID NO: 300 moltype = length =
SEQUENCE: 300
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SEQ ID NO: 301 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 301
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFPITYIEAV VLGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV GISGVKGGHN SMPLSAIFTT 90

SEQ ID NO: 302 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 302
MLPAPKQLVV SEVTEDSARL SWQGVARAFD SFMINYSELF WMGEAIVLTV PGSERSYDLT 60
GLKPGTEYVV RIKGVKGGKG SWPLHAHFTT 90

SEQ ID NO: 303 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 303
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFNIEYAETR WYGEAIVLTV PGSERSYDLT 60
GLKPGTEYVV PIDGVKGGIA SKPLSAIFTT 90

SEQ ID NO: 304 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 304
MLPAPKQLVV SEVTEDSARL SWQGVARAFD SFLITYRDQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYPV WIQGVKGGSP SAPLSAESTT 90

SEQ ID NO: 305 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 305
MLPAPKQLVV SEVTEDSARL SWQGVARAFD SFLITYREQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYVW YIIGVKGGKP SFPLRAGFTT 90

SEQ ID NO: 306 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 306
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFPIIMETF SRGEAIVLTV PGSERSYDLT 60
GLKPGTEYRV PIGGVKGGSS SCPLSAIFTT 90

SEQ ID NO: 307 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90

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mol_type = protein
organism = synthetic construct

SEQUENCE: 307
MLPAPKKNLVV SDVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAEFTT 90

SEQ ID NO: 308 moltype = length =
SEQUENCE: 308
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SEQ ID NO: 309 moltype = length =
SEQUENCE: 309
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SEQ ID NO: 310 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 310
MLPAPKKNLVV SRVTEDSARL SWTAPDAAFD SFKIAYIETA TRGEAIVLTV PGSERSYDLT 60
GLKPGTEYVV PIPGVKGGNT SSPLSAIFTT 90

SEQ ID NO: 311 moltype = length =
SEQUENCE: 311
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SEQ ID NO: 312 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 312
MLPAPKKNLVV SRVTEDSARL SWTAPDAAFD SFFIPYAEPS PTGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGHL SDPLSAISTT 90

SEQ ID NO: 313 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 313
MLPAPKKNLVV SRVTEDSARL SWTAPDAAFD SFLIAYPEDG FRGEAIVLTV PGSERSYDLT 60
GLKPGTEYFPV PILGVKGGGG SGPLSAIFTT 90

SEQ ID NO: 314 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 314
MLPAPKKNLVV SRVTEDSARL SWTAPDAAFD SFPIYYVENV VWGEAIVLTV PGSERSYDLT 60
GLKPGTEYWE VIIGVKGGQC SRPLSAIFTT 90

SEQ ID NO: 315 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 315
MLPAPKKNLVV SRVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTECPV WIQGVKGGSP SAPLSAEFTT 90

SEQ ID NO: 316 moltype = AA length = 89
FEATURE Location/Qualifiers
source 1..89
mol_type = protein
organism = synthetic construct

SEQUENCE: 316
MLPAPKKNLVV SRVTEDSARL SWTAPDAAFD SFGIAYREFR PSGEAIVLTV PGSERSYDLT 60
VETGYRNEVV ICGVKGGPWS GPLSAIFTT 89

SEQ ID NO: 317 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 317
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFPILYTECV YRGEAIVLTV PGSERSYDLT 60
GLKPGTEYHV PITGVKGGGG SWPLSAIFTT 90

SEQ ID NO: 318      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 318
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFNIMYHEII YVGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV PIEGVKGGGT SGPLSAIFTT 90

SEQ ID NO: 319      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 319
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAITYTEAA LCGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV PINGVKGGGT SGPLSAIFTT 90

SEQ ID NO: 320      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 320
MLPAPKNLVV ARVTEDSARL SWTAPDAAID SFPIDYSEYW WGGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV LITGVKGGYR SGPLSAIFTT 90

SEQ ID NO: 321      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 321
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFSIRYNEFI VAGEAIVLTV PGSERSYDLT 60
GLKPGTEYDV PIAGVKGGGA SWPLSAIVTT 90

SEQ ID NO: 322      moltype = AA length = 88
FEATURE            Location/Qualifiers
source             1..88
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 322
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFWYLELQFA GEAILTVPG SERSYDLTGL 60
KPGTEYNVPI TGVKGGIISF PLSAIFTT 88

SEQ ID NO: 323      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 323
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAIWYHEWY GDGEAIVLTV PGSERSYDLT 60
GPKPGTEYRV RISGVKGGFE SGPLSAIFTT 90

SEQ ID NO: 324      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 324
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFMIRYQEGT RWGEAIVLTV PGSERSYDLT 60
GLKPGTEYIV MIAGVKGGQI SLPLSAIFTT 90

SEQ ID NO: 325      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 325
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFPWIWLEKS YQGEAIVLTV PGSERSYDLT 60
GLKPGTEYVV PIIGVKGGRD SCPLSAIFTT 90

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SEQ ID NO: 326      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 326
MLPAPKQLVV SEVTEDSARL SWQGVARAFD SFLITYREQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAEFTT                                90

SEQ ID NO: 327      moltype = AA  length = 89
FEATURE            Location/Qualifiers
source              1..89
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 327
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFRISYAETV RQGEAIVLTV PGSERSYDLT 60
VETGYRNWVM ILGVKGGPGS LPLSAIFTT                                89

SEQ ID NO: 328      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 328
MLPAPKQLVV SEVTEDSARL SWQGVVRAFD SFLITYREQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAEFTT                                90

SEQ ID NO: 329      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 329
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFWIEYWEAV GFGEAIVLTV PGSERSYDLT 60
GLKPGTEYFPV GIYGVKGGYL SAPLSAIFTT                                90

SEQ ID NO: 330      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 330
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFIIHYVEQQ LIGEAIPLTV PGSERSYDLT 60
GLKPGTEYFPV PITGVKGGAC SWPLSAIFTT                                90

SEQ ID NO: 331      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 331
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFTIEYSEHP IDGEAIPLFV PGSERSYDLT 60
GLKPGTEYFPV RIHGVKGGWF SHPLWAFFTT                                90

SEQ ID NO: 332      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 332
MLPAPKQLVV SRVTEDSARL SWQGVARAFD SFLITYREQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYGV TIAGVKGGWR SKPLNAESTT                                90

SEQ ID NO: 333      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 333
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFGIAYVESY WYGEAIVLTV PGSERSYDLT 60
GLKPGTEYNV PIYGVKGGDG SGPLSAIFTT                                90

SEQ ID NO: 334      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90

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mol_type = protein
organism = synthetic construct

SEQUENCE: 334
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPYITYVELN LAGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV PILGVKGGSL SQPLSAIFTT 90

SEQ ID NO: 335 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 335
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPPISYIESI ADGEAIVLTV PGSERSYDLT 60
GLKPGTEYVW AIVGVKGGPF SWSLSAIVTT 90

SEQ ID NO: 336 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 336
MLPAPKNLVV SEVTEDSARL SWQGVARAFD SPLITYREQI FAGEVIVPTV PGSERSYDLT 60
GLKPGTEYPV PIAGVKGGGP SAPLSAIFTT 90

SEQ ID NO: 337 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 337
MLPAPKNLVV SRVTEDSARL SWTTPDAAFD SPPIYYWEVT ITGEAIVLSV PGSERSYDLT 60
GLKPGTEYPV DIPGVKGGAA SPPLSAIFTT 90

SEQ ID NO: 338 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 338
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPPILYLEHT VRSEAIVLTV PGSERSYDLT 60
DLKPGTEYCV PIDGVKGGLR SRPLSAIFTT 90

SEQ ID NO: 339 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 339
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPPIPYTEPP DPGEAIVLTV PGSERSYDLT 60
GLKPGTEYLV TILGVKGGSM SVPLSAIFTT 90

SEQ ID NO: 340 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 340
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFTIDYWENR CPGEAIVLTV PGSERSYDLT 60
GLKPGTEYCV WISGVKGGYS SWPLSAIFTT 90

SEQ ID NO: 341 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 341
MLPAPKNLVV SRVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYPV WIQGVKGGHL SDPLSAIVTT 90

SEQ ID NO: 342 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 342
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPLIPYAETS PSGEAIVLTV PGSERSYDLT 60

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GLKPGTEYSV LIHGVKGGDY SEPLSAIFTT	90
SEQ ID NO: 343 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 343	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFMIVYYEYT RFGEAIVLTV PGSERSYDLT	60
GLKPGTEYTV PIDGVKGGGR SSPLSAIFTT	90
SEQ ID NO: 344 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 344	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIPYAEPS PTGEAIVLTV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGHL SDPLSAIVTT	90
SEQ ID NO: 345 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 345	
MLPAPKNLVV SEVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT	60
GLKPGTEYPV WIQGVKGGSP SAPLSAEFTT	90
SEQ ID NO: 346 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 346	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFIIPYAEVR PDGEAIVLTV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGKL SLPLSAIFTT	90
SEQ ID NO: 347 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 347	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFGIVYLEMM VTGEAIVLTV PGSERSYDLT	60
GLKPGTEYDV PILGVKGGTR SVPLSAIFTT	90
SEQ ID NO: 348 moltype = AA length = 94	
FEATURE Location/Qualifiers	
source 1..94	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 348	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFLIYYEEGY LEYYYSGEAI VLTVPGSERS	60
YDLTGLKPGT EYYVGIVGVK GGGLSGPLSA ISTT	94
SEQ ID NO: 349 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source 1..90	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 349	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFPIAYAEPR PDGEAIVLTV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGDW SLPLSAIFTT	90
SEQ ID NO: 350 moltype = AA length = 91	
FEATURE Location/Qualifiers	
source 1..91	
mol_type = protein	
organism = synthetic construct	
SEQUENCE: 350	
MSLPAPKNLV VSRVTEDSAR LSWTAPDAAF DSFTIHYREF QLSGEAIVLT VPGSERSYDL	60
TGLKPGTEYD VPIEGVKGGP GSRPLSAIFT T	91
SEQ ID NO: 351 moltype = AA length = 90	
FEATURE Location/Qualifiers	

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source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 351
MLPAPKQNLVV SEVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSECSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAEFTT                                     90

SEQ ID NO: 352        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 352
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFEIDYDELA IYGEAIVLTV PGSERSYDLT 60
GLKPGTEYGV RIPGVKGGMP SLPLSAIVTT                                     90

SEQ ID NO: 353        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 353
MLPAPENLVV SEVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAESTT                                     90

SEQ ID NO: 354        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 354
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFAIAYGEHI VIGEIVLTV PGSERSYDLT 60
GLKPGTEYMV PIAGVKGGPI SLPLSAIFTT                                     90

SEQ ID NO: 355        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 355
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIPYAEPS PTGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGHL SDPLSAIFTT                                     90

SEQ ID NO: 356        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 356
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFSIGYVELV LLGEAIVLTV PGSERSYDLT 60
GLKPGTEYDV LIPGVKGGSL SRPLSAIFTT                                     90

SEQ ID NO: 357        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 357
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFAIPYAELS RNGEAIVLTV PGSERSYDLT 60
GLKPGTEYTV LIHGVKGGCL SDPLSAIFTT                                     90

SEQ ID NO: 358        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 358
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFHIEYLELS RHGEAIVLTV PGSERSYDLT 60
GLKPGTEYWV MIFGVKGGGP SKPLSAIFTT                                     90

SEQ ID NO: 359        moltype = AA length = 88
FEATURE              Location/Qualifiers
source                1..88
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 359

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MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPVYNEVHWI GEIVLTVPG SERSYDLTGL 60
 KPGTEYFVGI YGVKGGHWSK PLSAIFTT 88

SEQ ID NO: 360 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 360
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPEIDYDELA IYGEIVLTV PGSERSYDLT 60
 GLKPGTEYGV RIPGVKGGMP SLPLSAIVTT 90

SEQ ID NO: 361 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 361
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPQIVYSELW IKGEIVLTV PGSERSYDLT 60
 GLKPGTEYQV PIPGVKGGRN SFPLSAIFTT 90

SEQ ID NO: 362 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 362
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPKIRYTETR SIGEIVLTV PGSERSYDLT 60
 GLKPGTEYCV PIGGVKGGDS SWPLSAISTT 90

SEQ ID NO: 363 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 363
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFCISYYERM GRGEIVLTV PGSERSYDLT 60
 GLKPGTEYMV YIFGVKGGLN SLPLSAIFTT 90

SEQ ID NO: 364 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 364
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIVYAEPI PNGEIVLTV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGRN SDPLSAIFTT 90

SEQ ID NO: 365 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 365
 MLPAPKNLVV SRVTKDSARL SWTAPDAAFD SFPIAYAEPR PDGEIVLTV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 366 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 366
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFTIDYDEPR SPGEIVLTV PGSERSYDLT 60
 GLKPGTEYRV FIWGIKGGDT SFPLSAIFTT 90

SEQ ID NO: 367 moltype = AA length = 88
 FEATURE Location/Qualifiers
 source 1..88
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 367
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFTILYAEQA QFGEIVLTV PGSERSYDLT 60
 GLKPGTEYPI TGVKGGTRSG PLSAISTT 88

SEQ ID NO: 368 moltype = AA length = 90

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FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 368		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFIIPYAEVR PDGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGHL	SDPLSAISTT 90
SEQ ID NO: 369		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 369		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFHIAYEETA TSGEAIYLRV PGSERSYDLT 60
GLKPGTEYGV	EIEGVKGGAR	SRPLYADFTT 90
SEQ ID NO: 370		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 370		
MLPAPKNLVV	SRVTEDSARL	SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYPV	WIQGVKGGDL	SNPLSAIFTT 90
SEQ ID NO: 371		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 371		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFPISYLELS LYGEAIVLTV PGSERSYDLT 60
GLKPGTEYPV	GIAGVKGGVV	SRPLSAIFTT 90
SEQ ID NO: 372		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 372		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFTIGYREWY WYGEAIVLTV PGSERSYDLT 60
GLKPGTEYNV	PISGVKGGDL	SFPLSAIFTT 90
SEQ ID NO: 373		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 373		
MLPAPKNLVV	SEVTEDSARL	SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYPV	WIQGVKGGSP	SAPLSAESTT 90
SEQ ID NO: 374		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 374		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFSITYLEWW NLGEAIVLTV PGSERSYDLT 60
GLKPGTEYMV	TIPGVKGGMS	SYPLSAIFTT 90
SEQ ID NO: 375		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 375		
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD SFTISYGEEA LIGEAIYLRV PGSERSYDLT 60
GLKPGTEYV	HIEGVKGGSW	SQPLAAAFIT 90
SEQ ID NO: 376		
	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 376
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFTIEYYENI GIGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV PIVGVKGGPY SHPLSAIFTT 90

SEQ ID NO: 377 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 377
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFPIAYAEPR PDGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 378 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 378
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFGIGYYEHK RFGEAIQLSV PGSERSYDLT 60
GLKPGTEYEV DIEGVKGGVL SWPLFAEFTT 90

SEQ ID NO: 379 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 379
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFVIEYTERF WSGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV PIDGVKGGQC STPLSAIFTT 90

SEQ ID NO: 380 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 380
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFWIDYEEEG VIGEAIYLVH PGSERSYDLT 60
GLKPGTEYVV KIHGVKGGHP SHPLVAVFTT 90

SEQ ID NO: 381 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 381
MLPAPKQLVV SRVTEDSARL SWQGVARAFD SFLITYVELR HLGEAIVLTV PGSERSYDLT 60
GLKPGTEYVP WIQGVKGGSP SAPLSAEFTT 90

SEQ ID NO: 382 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 382
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFLIPYAETS PSGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGDY SSPLSAIFTT 90

SEQ ID NO: 383 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 383
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFPIAYAEPR PDGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAISTT 90

SEQ ID NO: 384 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 384
MLPAPKQLVV SRVTEDSARL SWQGVARAFD SFSILYLELT PKGEAIVLTV PGSERSYDLT 60
GLKPGTEYVP WIQGVKGGSP SAPLSAEFTT 90

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SEQ ID NO: 385 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 385
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPIIEYFEPI PIGEAIVLTV PGSERSYDLT 60
 GLKPGTEYAV NIYGVKGGYL SHPLSAIFTT 90

SEQ ID NO: 386 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 386
 MLPAPKNLVV SEVTEDSARL SWQGVARAFD SPLITYREQI FAGEVIVLTV PGSECSYDLT 60
 GLKPGTEYPV WIQGVKGGSP SAPLSAEFTT 90

SEQ ID NO: 387 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 387
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPAIEYTEFL YSGEAIVLTV PGSERSYDLT 60
 GLKPGTEYGV PINGVKGGFV SPPLSAIVTT 90

SEQ ID NO: 388 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 388
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPPIKYREVL RCGEAIVLTV PGSERSYDLT 60
 GLKPGTEYTV PITGVKGGFG SSPLSAIFTT 90

SEQ ID NO: 389 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 389
 MLPAPENLVV SRVTEDSARL SWTAPDAAFD SFWIEYYEGV IQGEAIVLTV PGSERSYDLT 60
 GLKPGTEYFV AIWGVKGGKW SVPLSAIFTT 90

SEQ ID NO: 390 moltype = AA length = 86
 FEATURE Location/Qualifiers
 source 1..86
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 390
 MLPAPKNLVV SRVTEDSARL SWQGVARAFD SPLITYREQI FAGEVIVLTV PGSERSYDLT 60
 GLKPGTEYPV WIQGVKGGSP SAEFTT 86

SEQ ID NO: 391 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 391
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFQIHYWETQ GFGEAIVLTV PGSERSYDLT 60
 GLKPGTEYPV LIPGVKGGPS SLPLSAIFTT 90

SEQ ID NO: 392 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 392
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIPYAEPS PTGEAIVLTV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGHL SDPLSAIFTT 90

SEQ ID NO: 393 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 393
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFAIEYYEPV PAGEAIYLDV PGSERSYDLT 60
GLKPGTEYDV TIYGVKGGY SHPLFASFTT 90

SEQ ID NO: 394      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 394
MLPAPKNLVV SEVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYDV WIQGVKGGSP SAPLSAIFTT 90

SEQ ID NO: 395      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 395
MLPAPKNLVV SEVTEDSARL SWQGVARAFD SFLITYREQI FAGEVIVLTV PGSERSYDLT 60
GLKPGTEYDV WIQGVKGGSP SAPLSAEFTT 90

SEQ ID NO: 396      moltype = AA length = 91
FEATURE            Location/Qualifiers
source             1..91
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 396
MSLPAPKNLV VSRVTEDSAR LSWTAPDAAF DSFPIAYLEV FYEGEAIVLT VPGSERSYDL 60
TGLKPGTEYQ VPIEGVKGGA MSLPLSAIFT T 91

SEQ ID NO: 397      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 397
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFHIWYEEET TIGEAIYLVH PGSERSYDLT 60
GLKPGTEYEV HITGVKGGPY SRPLFANFTT 90

SEQ ID NO: 398      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 398
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFGIAYDEWP EFGEAIVLTV PGSERSYDLT 60
GLKPDTEYIV EIYGVKGGWF SWPLSAIFTT 90

SEQ ID NO: 399      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 399
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFFIPYAEPS PTGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGHL SDPLSVIFTT 90

SEQ ID NO: 400      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 400
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFYIWYEEVM LVGEAIVLTV PGSERSYDLT 60
GLKPGTEYNV PIPGVKGGHS SPPLSAIFTT 90

SEQ ID NO: 401      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 401
MLPAPKNLVV SRVTEDSARL SWTAPDAAF SFHILYEELE LVGEAIVLTV PGSERSYDLT 60
GLKPGTEYKV PISGVKGGPV SRPLSAIFTT 90

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SEQ ID NO: 402      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 402
MLPAPKQNLVV SRVTEDSARL SWQGVARAFD SFLITYREQI PAGEVIVLTV PGSERSYDLT 60
GLKPGTEYFPV WIQGVKGGSP SAPLSAEFTT                                90

SEQ ID NO: 403      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 403
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFHIVYHEPR PSGEAIWLHV PGSERSYDLT 60
GLKPGTEYEV  GIVSVKGGDL SVPLVAFFTT                                90

SEQ ID NO: 404      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 404
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLLV PGSERSYDLT 60
GLKPGTEYSV  LIHGVKGGLL SSPLYAVFTT                                90

SEQ ID NO: 405      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 405
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIFLVV PGSERSYDLT 60
GLKPGTEYSV  LIHGVKGGLL SSPLHANFTT                                90

SEQ ID NO: 406      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 406
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLDV PGSERSYDLT 60
GLKPGTEYSV  LIHGVKGGLL SSPLYASFSTT                                90

SEQ ID NO: 407      moltype = AA  length = 91
FEATURE            Location/Qualifiers
source             1..91
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 407
MSLPAPKQNLV VSRVTEDSAR LSWTAPDAAF DSFYIAYAEPR RPDGEAISLY VPGSERSYDL 60
TGLKPGTEYS VLIHGVKGG LSSPLSAIST T                                91

SEQ ID NO: 408      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 408
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIQLRV PGSERSYDLT 60
GLKPGTEYNV  TIQGVKGGFP SMPLSAIFTT                                90

SEQ ID NO: 409      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 409
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFHISYEEDY TFGEAIIYLRV PGSERSYDLT 60
GLKPGTEYRV  VIGGVKGGWF SEPLLAFTT                                90

SEQ ID NO: 410      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90

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mol_type = protein
organism = synthetic construct

SEQUENCE: 410
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIYLTV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SYPLDASFTT 90

SEQ ID NO: 411 moltype = AA length = 93
FEATURE Location/Qualifiers
source 1..93
mol_type = protein
organism = synthetic construct

SEQUENCE: 411
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIDLGV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SMPLDPLEAY FTT 93

SEQ ID NO: 412 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 412
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 413 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 413
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAINLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAFFTT 90

SEQ ID NO: 414 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 414
MLPAPKNLVV SRVTEDSARL SWTTPDAAFD SFFIGYLEPQ PPGEAISLQV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SSPLFAVFTT 90

SEQ ID NO: 415 moltype = AA length = 87
FEATURE Location/Qualifiers
source 1..87
mol_type = protein
organism = synthetic construct

SEQUENCE: 415
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIELHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLFTT 87

SEQ ID NO: 416 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 416
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 417 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 417
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAITLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 418 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
mol_type = protein
organism = synthetic construct

SEQUENCE: 418
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLTV PGSERSYDLT 60

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GLKPGTEYSV LIHGVKGGLL SSPLVASFTT	90
SEQ ID NO: 419 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 419	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAINLDV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGLL SSPLVAEFTT	90
SEQ ID NO: 420 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 420	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGLL SSPLVAIFTT	90
SEQ ID NO: 421 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 421	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIALWV PGSERSYDLT	60
GLKPGTEYNV TIQGVKG GPP SMPLSAIFTT	90
SEQ ID NO: 422 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 422	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIIILV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGLL SSPLSAHFTT	90
SEQ ID NO: 423 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 423	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIQLWV PGSERSYDLT	60
GLKPGTEYNV TIQGVKG GPP SHPLGAVFTT	90
SEQ ID NO: 424 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 424	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLHV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGLL SSPLLASFTT	90
SEQ ID NO: 425 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 425	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALHV PGSERSYDLT	60
GLKPGTEYSV LIHGVKGGLL SSPLVAFFTT	90
SEQ ID NO: 426 moltype = AA length = 90	
FEATURE Location/Qualifiers	
source	1..90
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 426	
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIIHLV PGSERSYDLT	60
GLKPGTEYNV TIQGVKG GPP SIPLHANFTT	90
SEQ ID NO: 427 moltype = AA length = 90	
FEATURE Location/Qualifiers	

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source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 427
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIFLGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAIFTT 90

SEQ ID NO: 428        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 428
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLRV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLIASFTT 90

SEQ ID NO: 429        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 429
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAINLWV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLDASFTT 90

SEQ ID NO: 430        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 430
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFVIEYFEWT LNGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV QIYGVKGGCL SRPLSAIFTT 90

SEQ ID NO: 431        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 431
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLWV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLIAHFTT 90

SEQ ID NO: 432        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 432
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIPYAEPS PTGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAHFTT 90

SEQ ID NO: 433        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 433
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIYLYV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLDAFFTT 90

SEQ ID NO: 434        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 434
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAIFTT 90

SEQ ID NO: 435        moltype = AA length = 90
FEATURE              Location/Qualifiers
source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 435

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MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPFYIAYAEPR	PDGEAIQLAV	PGSERSYDLT	60
GLKPGTEYSV	LIHGVKGGLL	SSPLHAFFTT				90
SEQ ID NO: 436 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 436						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIGYLEPQ	PPGEAIWLHV	PGSERSYDLT	60
GLKPGTEYNV	TIQGVKGGFP	SIPLIAIFTT				90
SEQ ID NO: 437 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 437						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPFYIAYAEPR	PDGEAIVLDV	PGSERSYDLT	60
GLKPGTEYSV	LIHGVKGGLL	SSPLVAEFTT				90
SEQ ID NO: 438 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 438						
MLPTPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPFYIAYAEPR	PDGEAIQLRV	PGSERSYDLT	60
GLKPGTEYSV	LIHGVKGGLL	SSPLHASFTT				90
SEQ ID NO: 439 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 439						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIGYLEPQ	PPGEAIQLGV	PGSERSYDLT	60
GLKPGTEYNV	TIQGVKGGFP	SHPLNANFTT				90
SEQ ID NO: 440 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 440						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIGYLEPQ	PPGEAIQLEV	PGSERSYDLT	60
GLKPGTEYNV	TIQGVKGGFP	SMPLSAIFTT				90
SEQ ID NO: 441 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 441						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPFYIAYAEPR	PDGEAIFLGV	PGSERSYDLT	60
GLKPGTEYSV	LIHGVKGGLL	SSPLIAFFTT				90
SEQ ID NO: 442 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 442						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIGYLEPQ	PPGEAIGLQV	PGSERSYDLT	60
GLKPGTEYNV	TIQGVKGGFP	SHPLKAQFTT				90
SEQ ID NO: 443 moltype = AA length = 90						
FEATURE Location/Qualifiers						
source 1..90						
mol_type = protein						
organism = synthetic construct						
SEQUENCE: 443						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SPFYIAYAEPR	PDGEAILLFV	PGSERSYDLT	60
GLKPGTEYSV	LIHGVKGGLL	SSPLVAHFTT				90
SEQ ID NO: 444 moltype = AA length = 90						

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FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 444		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLYV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLGAFFTT		90
SEQ ID NO: 445	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 445		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLQV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLTAIFTT		90
SEQ ID NO: 446	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 446		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAITLHV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT		90
SEQ ID NO: 447	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 447		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLEV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLDAHFTT		90
SEQ ID NO: 448	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 448		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALHV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLRAVFTT		90
SEQ ID NO: 449	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 449		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLWV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT		90
SEQ ID NO: 450	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 450		
MLPAPKNLVV SRVTEDSARL SRTAPDAAFD SFYIAYAEPR PDGEAIVLIV PGSERSYDLT		60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT		90
SEQ ID NO: 451	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 451		
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIQLWV PGSERSYDLT		60
GLKPGTEYNV TIQGVKGGLL SRPLQAHFTT		90
SEQ ID NO: 452	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 452
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAITLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYAFFTT 90

SEQ ID NO: 453      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 453
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 454      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 454
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLIAHFTT 90

SEQ ID NO: 455      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 455
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHARFTT 90

SEQ ID NO: 456      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 456
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIFLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAAVFTT 90

SEQ ID NO: 457      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 457
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAISTT 90

SEQ ID NO: 458      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 458
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAVFTT 90

SEQ ID NO: 459      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 459
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIYLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAHFTT 90

SEQ ID NO: 460      moltype = AA length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 460
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLLAYFTT 90

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SEQ ID NO: 461 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 461
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLKFTT 90

SEQ ID NO: 462 moltype = AA length = 89
FEATURE Location/Qualifiers
source 1..89
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 462
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIKLEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAIFTT 89

SEQ ID NO: 463 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 463
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIYLEV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGFP SFPLKAAFTT 90

SEQ ID NO: 464 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 464
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILRV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAIFTT 90

SEQ ID NO: 465 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 465
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAAWFTT 90

SEQ ID NO: 466 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 466
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIFLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLNAFTT 90

SEQ ID NO: 467 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 467
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAILGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHAYSTT 90

SEQ ID NO: 468 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 468
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 469 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 469
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAVFTT 90

SEQ ID NO: 470      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 470
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLLAHFTT 90

SEQ ID NO: 471      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 471
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLWV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAYFTT 90

SEQ ID NO: 472      moltype = AA length = 89
FEATURE            Location/Qualifiers
source             1..89
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 472
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIVLTV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SMPLASFTT 89

SEQ ID NO: 473      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 473
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFHIVYHEPR PSGEAIHLQV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SYPLSAFTT 90

SEQ ID NO: 474      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 474
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIQLWV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SMPLSAFTT 90

SEQ ID NO: 475      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 475
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFRISYCETF YHGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLIAKFTT 90

SEQ ID NO: 476      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 476
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLKV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLQANFTT 90

SEQ ID NO: 477      moltype = AA length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 477
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLKV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLQANFTT 90

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SEQ ID NO: 478      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 478
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIVTT 90

SEQ ID NO: 479      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 479
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFPIAYAEPR PDGEAIVLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAFFTT 90

SEQ ID NO: 480      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 480
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAQFTT 90

SEQ ID NO: 481      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 481
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIIHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLKFTT 90

SEQ ID NO: 482      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 482
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIDLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHALFTT 90

SEQ ID NO: 483      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 483
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SMPLSAIFTT 90

SEQ ID NO: 484      moltype = AA  length = 88
FEATURE            Location/Qualifiers
source             1..88
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 484
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIDLAV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSFTT 88

SEQ ID NO: 485      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 485
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIIYLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLRAKFTT 90

SEQ ID NO: 486      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90

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mol_type = protein
organism = synthetic construct

SEQUENCE: 486
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAIFTT 90

SEQ ID NO: 487 moltype = AA length = 88
FEATURE Location/Qualifiers
source 1..88
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 487
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAISLLV PDSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SMPLKFTT 88

SEQ ID NO: 488 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 488
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLDASFTT 90

SEQ ID NO: 489 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 489
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLTV PGSERSYDLT 60
GPKPGTEYVW LIQGVKGGGS SVPLVAYFTT 90

SEQ ID NO: 490 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 490
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLASFTT 90

SEQ ID NO: 491 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 491
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAILAV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SLPLVASFTT 90

SEQ ID NO: 492 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 492
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFDIGYTEYG GYGEAIYLSV PGSERSYDLT 60
GLKPGTEYVW LIQGVKGGGS SVPLSAIFTT 90

SEQ ID NO: 493 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 493
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLSV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLIANFTT 90

SEQ ID NO: 494 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 494
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALLV PGSERSYDLT 60

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GLKPGTEYSV LIHGVKGGLL SSPLSAIVTT 90

SEQ ID NO: 495 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 495
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIIILDV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLSSIFTT 90

SEQ ID NO: 496 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 496
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLWV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLRASFTT 90

SEQ ID NO: 497 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 497
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIKLDV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLVAFFTT 90

SEQ ID NO: 498 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 498
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIILEV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLVAYFTT 90

SEQ ID NO: 499 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 499
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLWV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLHADFTT 90

SEQ ID NO: 500 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 500
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLEV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLVADFTT 90

SEQ ID NO: 501 moltype = AA length = 90
 FEATURE Location/Qualifiers
 source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 501
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLWV PGSERSYDLT 60
 GLKPGTEYSV LIHGVKGGLL SSPLLAHFTT 90

SEQ ID NO: 502 moltype = AA length = 89
 FEATURE Location/Qualifiers
 source 1..89
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 502
 MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFDIGYTEYG GYGAILHVP GSERSYDLTG 60
 LKPGTEYWVL IQGVKGGSS VPLSAIFTT 89

SEQ ID NO: 503 moltype = AA length = 90
 FEATURE Location/Qualifiers

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source                1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 503
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIVLLV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SVPLAAFFTT 90

SEQ ID NO: 504        moltype = AA length = 85
FEATURE              Location/Qualifiers
source               1..85
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 504
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAILLV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SQFTT 85

SEQ ID NO: 505        moltype = AA length = 93
FEATURE              Location/Qualifiers
source               1..93
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 505
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAILLV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SMPLHPLVAL FTT 93

SEQ ID NO: 506        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 506
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAIFTT 90

SEQ ID NO: 507        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 507
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLSV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLAAYFTT 90

SEQ ID NO: 508        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 508
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIVLAV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SYPLVAAFTT 90

SEQ ID NO: 509        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 509
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLQV PGSCRSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLTAIFTT 90

SEQ ID NO: 510        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 510
MLPAPKQNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAINLQV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SFPLSAVFTT 90

SEQ ID NO: 511        moltype = AA length = 90
FEATURE              Location/Qualifiers
source               1..90
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 511

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MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAIFTT 90

SEQ ID NO: 512      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 512
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLAV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHAQFTT 90

SEQ ID NO: 513      moltype = AA  length = 87
FEATURE            Location/Qualifiers
source             1..87
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 513
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLFTT 87

SEQ ID NO: 514      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 514
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLCAEFTT 90

SEQ ID NO: 515      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 515
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLWV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLIAEFTT 90

SEQ ID NO: 516      moltype = AA  length = 88
FEATURE            Location/Qualifiers
source             1..88
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 516
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLSV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPPKFTT 88

SEQ ID NO: 517      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 517
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIILEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLRAVFTT 90

SEQ ID NO: 518      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 518
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 519      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 519
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLKV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAIFTT 90

SEQ ID NO: 520      moltype = AA  length = 114

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FEATURE	Location/Qualifiers	
source	1..114	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 520		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFYIAYAEPR PDGEAIHLLV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGLL	SSPLKRLSPP VVTITITMAV CRKPVAENLS QTLS 114
SEQ ID NO: 521	moltype = AA length = 89	
FEATURE	Location/Qualifiers	
source	1..89	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 521		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFFIGYLEPQ PPGEAIFLDV PGSERSYDLT 60
GLKPGTEYNV	TIQGVKGGFP	SPLTAFFTT 89
SEQ ID NO: 522	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 522		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFFIGYLEPQ PPGEAIVLDV PGSERSYDLT 60
GLKPGTEYNV	TIQGVKGGFP	SHPLAAFTT 90
SEQ ID NO: 523	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 523		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFFIGYLEPQ PPGEAIGLAV PGSERSYDLT 60
GLKPGTEYNV	TIQGVKGGFP	SVPLQANFTT 90
SEQ ID NO: 524	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 524		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFYIAYAEPR PDGEAILLRV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGLL	SSPLVAEFTT 90
SEQ ID NO: 525	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 525		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFYIAYAEPR PDGEAILLQV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGLL	SSPLSASFTT 90
SEQ ID NO: 526	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 526		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFYIAYAEPR PDGEAIGLHV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGLL	SSPLTASFTT 90
SEQ ID NO: 527	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 527		
MLPAPKPNLVV	SRVTEDSARL	SWTAPDAAFD SFYIAYAEPR PDGEAIGLRV PGSERSYDLT 60
GLKPGTEYSV	LIHGVKGGLL	SSPLSAIFTT 90
SEQ ID NO: 528	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 528
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLRV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAASFTT 90

SEQ ID NO: 529      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 529
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAHFTT 90

SEQ ID NO: 530      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 530
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAFFTT 90

SEQ ID NO: 531      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 531
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIHLVY PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SDPLDAVFTT 90

SEQ ID NO: 532      moltype = AA  length = 85
FEATURE            Location/Qualifiers
source              1..85
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 532
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIYLDV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL STFTT 85

SEQ ID NO: 533      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 533
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLFV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLKAYFTT 90

SEQ ID NO: 534      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 534
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLSAIFTT 90

SEQ ID NO: 535      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 535
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIQLTV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SLPLSADFTT 90

SEQ ID NO: 536      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source              1..90
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 536
MLPAPKQLV SVRTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAEFTT 90

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SEQ ID NO: 537 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 537
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLAV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYASFTT 90

SEQ ID NO: 538 moltype = AA length = 88
FEATURE Location/Qualifiers
source 1..88
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 538
MLPAPKNLVV SRVTEDSARL SWTTPDAAFD SFYIAYAEPR PDGEAIRLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLGFTT 88

SEQ ID NO: 539 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 539
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYAIFTT 90

SEQ ID NO: 540 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 540
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLSV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHAKFTT 90

SEQ ID NO: 541 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 541
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIHLGV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SIPLFASFTT 90

SEQ ID NO: 542 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 542
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLLV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYAAFTT 90

SEQ ID NO: 543 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 543
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIHLAV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGFP SVPLAAVFTT 90

SEQ ID NO: 544 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 544
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAISLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLGAHFTT 90

SEQ ID NO: 545 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 545
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALWV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVASFTT 90

SEQ ID NO: 546 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 546
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYAFFTT 90

SEQ ID NO: 547 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 547
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLRASFTT 90

SEQ ID NO: 548 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 548
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHATFTT 90

SEQ ID NO: 549 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 549
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLHANFTT 90

SEQ ID NO: 550 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 550
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLRV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLYAKFTT 90

SEQ ID NO: 551 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 551
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIGLWV PGSERSYDLT 60
GLKPGTEYNV TIQGVKGGLL SDPLQAVFTT 90

SEQ ID NO: 552 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 552
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAFFTT 90

SEQ ID NO: 553 moltype = AA length = 90
FEATURE Location/Qualifiers
source 1..90
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 553
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAYFTT 90

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SEQ ID NO: 554      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 554
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLAV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLSAKFTT 90

SEQ ID NO: 555      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 555
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAILLFV PGSERSYDLT 60
GLKPGTEYINV TIQGVKGGFP STPLSASFTT 90

SEQ ID NO: 556      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 556
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLTV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLHAYFTT 90

SEQ ID NO: 557      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 557
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLRAYFTT 90

SEQ ID NO: 558      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 558
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLVAFFTT 90

SEQ ID NO: 559      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 559
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIQLGV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLLAFTT 90

SEQ ID NO: 560      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 560
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIHLRV PGSERSYDLT 60
GLKPGTEYINV TIQGVKGGFP SMPLSAIFTT 90

SEQ ID NO: 561      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 561
MLPAPKQLVSV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAILLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGGL SSPLIAKFTT 90

SEQ ID NO: 562      moltype = AA  length = 90
FEATURE            Location/Qualifiers
source             1..90

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                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 562
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLHV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLQAIFTT 90

SEQ ID NO: 563                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 563
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIALVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLAANFTT 90

SEQ ID NO: 564                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 564
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAINLSV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAYFTT 90

SEQ ID NO: 565                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 565
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIVLEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLTASFTT 90

SEQ ID NO: 566                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 566
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIRLQV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLGASFTT 90

SEQ ID NO: 567                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 567
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIGLVV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLVAYFTT 90

SEQ ID NO: 568                moltype = AA length = 87
FEATURE                      Location/Qualifiers
source                      1..87
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 568
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIYLEV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLFTT 87

SEQ ID NO: 569                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 569
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFYIAYAEPR PDGEAIWLDV PGSERSYDLT 60
GLKPGTEYSV LIHGVKGGLL SSPLDAYFTT 90

SEQ ID NO: 570                moltype = AA length = 90
FEATURE                      Location/Qualifiers
source                      1..90
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 570
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SPAIEYCETK MCGEAIIVLTV PGSERSYDLT 60

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GLKPGTEYRV	PIPGVKGGTA	SLPLSAIFTT	90
SEQ ID NO: 571 moltype = AA length = 90			
FEATURE Location/Qualifiers			
source 1..90			
mol_type = protein			
organism = synthetic construct			
SEQUENCE: 571			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIYYIESY
		PAGEAIVLTV	PGSERSYDLT
GLKPGTEYVW	GIDGVKGGRW	STPLSAIFTT	60
			90
SEQ ID NO: 572 moltype = AA length = 90			
FEATURE Location/Qualifiers			
source 1..90			
mol_type = protein			
organism = synthetic construct			
SEQUENCE: 572			
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFFIYYIESY
		PAGEAIVLTV	PGSCRSYDLT
GLKPGTEYVW	GIDGVKGGRW	STPLSAIFTT	60
			90
SEQ ID NO: 573 moltype = length =			
SEQUENCE: 573			
000			
SEQ ID NO: 574 moltype = length =			
SEQUENCE: 574			
000			
SEQ ID NO: 575 moltype = length =			
SEQUENCE: 575			
000			
SEQ ID NO: 576 moltype = length =			
SEQUENCE: 576			
000			
SEQ ID NO: 577 moltype = length =			
SEQUENCE: 577			
000			
SEQ ID NO: 578 moltype = length =			
SEQUENCE: 578			
000			
SEQ ID NO: 579 moltype = length =			
SEQUENCE: 579			
000			
SEQ ID NO: 580 moltype = length =			
SEQUENCE: 580			
000			
SEQ ID NO: 581 moltype = length =			
SEQUENCE: 581			
000			
SEQ ID NO: 582 moltype = length =			
SEQUENCE: 582			
000			
SEQ ID NO: 583 moltype = length =			
SEQUENCE: 583			
000			
SEQ ID NO: 584 moltype = length =			
SEQUENCE: 584			
000			
SEQ ID NO: 585 moltype = length =			
SEQUENCE: 585			
000			
SEQ ID NO: 586 moltype = length =			
SEQUENCE: 586			
000			

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SEQ ID NO: 587 moltype = length =
SEQUENCE: 587
000

SEQ ID NO: 588 moltype = length =
SEQUENCE: 588
000

SEQ ID NO: 589 moltype = length =
SEQUENCE: 589
000

SEQ ID NO: 590 moltype = length =
SEQUENCE: 590
000

SEQ ID NO: 591 moltype = length =
SEQUENCE: 591
000

SEQ ID NO: 592 moltype = AA length = 189
FEATURE Location/Qualifiers
source 1..189
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 592
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT 60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT APAPAPAPAP LPAPKNLVVS RVTEDSARLS 120
WTAPDAAFD FAIVYHEPRP DGEAIVLTVP GSERSYDLTG LKPGTEYEVV ILGVKGGVHS 180
YPLSAIFTT 189

SEQ ID NO: 593 moltype = AA length = 199
FEATURE Location/Qualifiers
source 1..199
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 593
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT 60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT GGGGSGGGGS GGGGSGGGGS LPAPKNLVVS 120
RVTEDSARLS WTAPDAAFD FAIVYHEPRP DGEAIVLTVP GSERSYDLTG LKPGTEYEVV 180
ILGVKGGVHS YPLSAIFTT 199

SEQ ID NO: 594 moltype = AA length = 199
FEATURE Location/Qualifiers
source 1..199
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 594
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT 60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT EAAAKEAAK EAAAKEAAK LPAPKNLVVS 120
RVTEDSARLS WTAPDAAFD FAIVYHEPRP DGEAIVLTVP GSERSYDLTG LKPGTEYEVV 180
ILGVKGGVHS YPLSAIFTT 199

SEQ ID NO: 595 moltype = AA length = 184
FEATURE Location/Qualifiers
source 1..184
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 595
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT 60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT EAAAKLPAPK NLVVSRTED SARLSWTAPD 120
AARDFAIVY HEPRPDGEAI VLTVPGSERS YDLTGLKPGT EYEVVILGVK GGVHSYPLSA 180
IFTT 184

SEQ ID NO: 596 moltype = AA length = 189
FEATURE Location/Qualifiers
source 1..189
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 596
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFKIEYFEYV GYGEAIVLTV PGSCRSYDLT 60
GLKPGTEYEV AIYGVKGGWY SRPLSAIFTT APAPAPAPAP LPAPKNLVVS RVTEDSARLS 120
WTAPDAAFD FKIEYFEYV YGEAIVLTVP GSERSYDLTG LKPGTEYVVA IYGVKGGWYS 180
RPLSAIFTT 189

SEQ ID NO: 597 moltype = AA length = 199

FEATURE	Location/Qualifiers					
source	1..199					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 597						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFKIEYFEYV	GYGEAIVLTV	PGSCRSYDLT	60
GLKPGTEYYV	AIYGVKGGWY	SRPLSAIFTT	GGGGSGGGGS	GGGGSGGGGS	LPAPKNLVVS	120
RVTEDSARLS	WTAPDAAFD	FKIEYFEYVG	YGEAIVLTVP	GSERSYDLTG	LKPGTEYYVA	180
IYGVKGGWYS	RPLSAIFTT					199
SEQ ID NO: 598					moltype = AA length = 199	
FEATURE	Location/Qualifiers					
source	1..199					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 598						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFKIEYFEYV	GYGEAIVLTV	PGSCRSYDLT	60
GLKPGTEYYV	AIYGVKGGWY	SRPLSAIFTT	EAAAKEAAAK	EAAAKEAAAK	LPAPKNLVVS	120
RVTEDSARLS	WTAPDAAFD	FKIEYFEYVG	YGEAIVLTVP	GSERSYDLTG	LKPGTEYYVA	180
IYGVKGGWYS	RPLSAIFTT					199
SEQ ID NO: 599					moltype = AA length = 184	
FEATURE	Location/Qualifiers					
source	1..184					
	mol_type = protein					
	organism = synthetic construct					
SEQUENCE: 599						
MLPAPKNLVV	SRVTEDSARL	SWTAPDAAFD	SFKIEYFEYV	GYGEAIVLTV	PGSCRSYDLT	60
GLKPGTEYYV	AIYGVKGGWY	SRPLSAIFTT	EAAAKLPAPK	NLVVSRVTED	SARLSWTAPD	120
AAFDSFKIEY	FEYVGYGEAI	VLTVPGSERS	YDLTGLKPGT	EYYVAIYGVK	GGWYSRPLSA	180
IFFT						184
SEQ ID NO: 600					moltype = DNA length = 20	
FEATURE	Location/Qualifiers					
source	1..20					
	mol_type = other DNA					
	organism = synthetic construct					
modified_base	1					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide phosphorothioate					
modified_base	2					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide phosphorothioate					
modified_base	3					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					
modified_base	4					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide					
modified_base	5					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					
modified_base	6					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide					
modified_base	7..9					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					
modified_base	10					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide					
modified_base	11					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					
modified_base	12..14					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide					
modified_base	15					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					
modified_base	16					
	mod_base = OTHER					
	note = 2'O-Methyl modified nucleotide					
modified_base	17					
	mod_base = OTHER					
	note = 2'Fluoro modified nucleotide					

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modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	1..19	
	note = RNA	
misc_feature	20	
	note = DNA	
SEQUENCE: 600		
ctgggaggat gaattcggat		20
SEQ ID NO: 601	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	6..8	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	11..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 601		
ttcgaattca tcctccagct t		21
SEQ ID NO: 602	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	

-continued

modified_base	note = 2' Fluoro modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 4 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 5 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 6 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2' O-Methyl modified nucleotide 1..19 note = RNA
misc_feature	20 note = DNA

SEQUENCE: 602
ctgggaggat gaattaggat

SEQ ID NO: 603
FEATURE
source

moltype = RNA length = 21
Location/Qualifiers
1..21
mol_type = other RNA
organism = synthetic construct

modified_base	1 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 4 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 5 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 6..8 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 9 mod_base = OTHER
modified_base	note = 2' O-Methyl modified nucleotide 10 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide

20

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modified_base	11..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 603		
ttcgaattca tctctccagt t		21
SEQ ID NO: 604	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	

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misc_feature      note = 2'O-Methyl modified nucleotide
                  1..19
misc_feature      note = RNA
                  20
misc_feature      note = DNA
SEQUENCE: 604
ctgggaggat gaatgaggat                                20

SEQ ID NO: 605    moltype = RNA length = 21
FEATURE          Location/Qualifiers
source           1..21
                 mol_type = other RNA
                 organism = synthetic construct
modified_base    1
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base    2
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide phosphorothioate
modified_base    3
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    4
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    5
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    6..8
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    9
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    10
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    11..13
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    14
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    15
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    16
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    17
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base    18
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide
modified_base    19
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base    20
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide phosphorothioate
modified_base    21
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
SEQUENCE: 605
ttcgaattca tcctccagc t                                21

SEQ ID NO: 606    moltype = DNA length = 20
FEATURE          Location/Qualifiers
source           1..20
                 mol_type = other DNA
                 organism = synthetic construct
modified_base    1
                 mod_base = OTHER
                 note = 2' Fluoro modified nucleotide phosphorothioate
modified_base    2

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	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
misc_feature	1..19	
	note = RNA	
misc_feature	20	
	note = DNA	
SEQUENCE: 606		
ctgggaggat gaattcgaat		20
SEQ ID NO: 607	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	3..18	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
SEQUENCE: 607		
ttcgaattca tctctccagt t		21
SEQ ID NO: 608	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	

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modified_base      3
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      7..9
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      10
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      11
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      12..14
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      15
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      16
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      17
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      18..19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
misc_feature       1..19
                    note = RNA
misc_feature       20
                    note = DNA
SEQUENCE: 608
ctgggaggat gaattcgaat                                20

SEQ ID NO: 609      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      2..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      11
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      12..18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      19..20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      21
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
SEQUENCE: 609
ttcgaattca tcctcccagt t                                21

SEQ ID NO: 610      moltype = RNA  length = 14

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FEATURE	Location/Qualifiers
source	1..14 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	4 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	5 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	6 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	7..9 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	10 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	11 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	12..13 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'Fluoro modified nucleotide
misc_feature	14 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 610	
ctgggaggat gaat	14
SEQ ID NO: 611	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	2 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	5 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	6..8 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	9 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	10 mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	11..13 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER

-continued

modified_base	note = 2'Fluoro modified nucleotide 15 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 16 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 18 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 19 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 20 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 21 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 611		
ttcgaattca tcctccagtt t		21
SEQ ID NO: 612	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 2 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 4 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 5 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 6 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 10 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 11 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 12..14 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 15 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 16 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 612		
caagtggtgggt ggcattctaa		19

-continued

```

SEQ ID NO: 613      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                   mol_type = other RNA
                   organism = synthetic construct

modified_base      1
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base      2
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide phosphorothioate

modified_base      3
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      4
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      5
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      6..8
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      9
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      10
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      11..13
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      14
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      15
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      16
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      17
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

modified_base      18
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      19
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base      20
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide phosphorothioate

modified_base      21
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide

SEQUENCE: 613
ttagatgccca cccaccttgt t                                     21

```

```

SEQ ID NO: 614      moltype = RNA  length = 19
FEATURE            Location/Qualifiers
source              1..19
                   mol_type = other RNA
                   organism = synthetic construct

modified_base      1
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide phosphorothioate

modified_base      2
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate

modified_base      3
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide

modified_base      4
                   mod_base = OTHER

```

-continued

modified_base	note = 2'O-Methyl modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 6 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 614
caagggtgggt ggcatttag

19

SEQ ID NO: 615	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 6..8 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 9 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 11..13 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 16 mod_base = OTHER

-continued

modified_base	note = 2'Fluoro modified nucleotide 17 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	18 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	19 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
SEQUENCE: 615		
ttagatgccca cccaccttgt t		21
SEQ ID NO: 616	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	4 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	5 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	6 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	7..9 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	10 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	11 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	12..14 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	15 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	16 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	17 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	18..19 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 616		
caaggtgggt ggcactctga		19
SEQ ID NO: 617	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	

-continued

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modified_base      1
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      6..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     11..13
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     21
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide

SEQUENCE: 617
ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 618      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source              1..19
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      6
                    mod_base = OTHER

```

-continued

modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 618
caaggtgggt ggcacgaa

19

SEQ ID NO: 619	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 6..8 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 9 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 11..13 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 18 mod_base = OTHER

-continued

modified_base	note = 2'Fluoro modified nucleotide 19 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 20 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 21 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 619		
ttagatgccca cccaccttgt t		21
SEQ ID NO: 620	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 2 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 4 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 5 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 6 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 10 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 11 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 12..14 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 15 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 16 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 620		
caaggtgggt ggcataataa		19
SEQ ID NO: 621	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate	

-continued

```

modified_base      3
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      6..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     11..13
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     21
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide

SEQUENCE: 621
ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 622      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                  mol_type = other RNA
                  organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      7..9
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     10
                    mod_base = OTHER

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-continued

modified_base	note = 2'-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 622
caaggtgggt ggcagctaa

19

SEQ ID NO: 623	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 6..8 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 9 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 11..13 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 18 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 19 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 20 mod_base = OTHER

-continued

modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 21 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
SEQUENCE: 623		
ttagatgccca cccaccttgt t		21
SEQ ID NO: 624	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	4 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	5 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	6 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	7..9 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	10 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	11 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	12..14 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	15 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	16 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	17 mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	18..19 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 624		
caaggtgggt ggcagctag		19
SEQ ID NO: 625	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3 mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	4 mod_base = OTHER note = 2'Fluoro modified nucleotide	

-continued

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modified_base      5
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      6..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     11..13
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base     21
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide

SEQUENCE: 625
ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 626      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = other RNA
                   organism = synthetic construct
modified_base     1..2
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base     3
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base     4..6
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base     7..9
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base     10..11
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base     12
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base     13..16
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base     17
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base     18..19
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
misc_feature      19
                   note = Connected to an inverted thymidine via a 3'-3' bond

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-continued

SEQUENCE: 626
gagccatctt tgcaacgca 19

SEQ ID NO: 627 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

modified_base 1
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

modified_base 2
mod_base = OTHER
note = 2'-Fluoro modified nucleotide phosphorothioate

modified_base 3..13
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 14
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 15..18
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 19..20
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

modified_base 21
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

SEQUENCE: 627
tgcgttgcaa agatggctct t 21

SEQ ID NO: 628 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

modified_base 3
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 4..6
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 7..9
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 10..11
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 12
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 13..16
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 17
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 18..19
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

misc_feature 19
note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 628
gagccatctt tgcaacgcg 19

SEQ ID NO: 629 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
mol_type = other RNA
organism = synthetic construct

modified_base 1
mod_base = OTHER

-continued

modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 15..18 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 19..20 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 629		
tcgcttgcaa agatggctct t		21
SEQ ID NO: 630	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 630		
gagccatctt tgcaacgaa		19
SEQ ID NO: 631	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide	

-continued

modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 631		
tcggttgcaa agatggctct t		21
SEQ ID NO: 632	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 632		
gagccatctt tgcaacaca		19
SEQ ID NO: 633	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	

-continued

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mod_base = OTHER
note = 2'O-Methyl modified nucleotide
SEQUENCE: 633
tgcgttgcaa agatggctct t                                     21

SEQ ID NO: 634
FEATURE
source
moltype = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct
modified_base
1..2
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base
3
mod_base = OTHER
note = 2'Fluoro modified nucleotide
modified_base
4..6
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
modified_base
7..9
mod_base = OTHER
note = 2'Fluoro modified nucleotide
modified_base
10..11
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
modified_base
12
mod_base = OTHER
note = 2'Fluoro modified nucleotide
modified_base
13..16
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
modified_base
17
mod_base = OTHER
note = 2'Fluoro modified nucleotide
modified_base
18..19
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
misc_feature
19
note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 634
gagccatctt tgcaaagca                                     19

SEQ ID NO: 635
FEATURE
source
moltype = RNA length = 21
Location/Qualifiers
1..21
mol_type = other RNA
organism = synthetic construct
modified_base
1
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base
2
mod_base = OTHER
note = 2'Fluoro modified nucleotide phosphorothioate
modified_base
3..13
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
modified_base
14
mod_base = OTHER
note = 2'Fluoro modified nucleotide
modified_base
15..18
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
modified_base
19..20
mod_base = OTHER
note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base
21
mod_base = OTHER
note = 2'O-Methyl modified nucleotide
SEQUENCE: 635
tgcgttgcaa agatggctct t                                     21

SEQ ID NO: 636
FEATURE
source
moltype = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct

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-continued

modified_base	1..2	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	4..6	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	7..9	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	10..11	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	12	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	13..16	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	18..19	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
misc_feature	19	
		note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 636		
gagccatctt tgcagcgca		19
SEQ ID NO: 637		moltype = RNA length = 21
FEATURE		Location/Qualifiers
source	1..21	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	2	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3..13	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	14	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	15..18	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	19..20	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	21	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
SEQUENCE: 637		
tgcgttgcaa agatggctct t		21
SEQ ID NO: 638		moltype = RNA length = 19
FEATURE		Location/Qualifiers
source	1..19	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1..2	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	4..6	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	7..9	

-continued

	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 638		
gagccatctt tgcagcgcg		19
SEQ ID NO: 639	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 639		
tgcgttgcaa agatggctct t		21
SEQ ID NO: 640	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	

-continued

modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 640	
ttacggtatc tacattcta	19
SEQ ID NO: 641	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide
SEQUENCE: 641	
tagaatgtag ataccgtaat t	21
SEQ ID NO: 642	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 4..6 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide
misc_feature	19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 642	

-continued

ttacggtatc tacattctg	19
SEQ ID NO: 643 moltype = RNA length = 21	
FEATURE Location/Qualifiers	
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3..13
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	15..18
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	19..20
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	21
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
SEQUENCE: 643	
tagaatgtag ataccgtaat t	21
SEQ ID NO: 644 moltype = RNA length = 19	
FEATURE Location/Qualifiers	
source	1..19
	mol_type = other RNA
	organism = synthetic construct
modified_base	1..2
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	4..6
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	7..9
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	10..11
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	13..16
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	18..19
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
misc_feature	19
	note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 644	
ttacggtatc tacattcga	19
SEQ ID NO: 645 moltype = RNA length = 21	
FEATURE Location/Qualifiers	
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide phosphorothioate

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modified_base	2	mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	14	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	15..18	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	19..20	mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
SEQUENCE: 645			
tagaatgtag ataccgtaat t			21
SEQ ID NO: 646	moltype = RNA length = 19		
FEATURE	Location/Qualifiers		
source	1..19	mol_type = other RNA organism = synthetic construct	
modified_base	1..2	mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	4..6	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	7..9	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	10..11	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	12	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	13..16	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	17	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	18..19	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
misc_feature	19	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 646			
ttacggtatc tacattgta			19
SEQ ID NO: 647	moltype = RNA length = 21		
FEATURE	Location/Qualifiers		
source	1..21	mol_type = other RNA organism = synthetic construct	
modified_base	1	mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	mod_base = OTHER note = 2'O-Methyl modified nucleotide	
modified_base	14	mod_base = OTHER note = 2'Fluoro modified nucleotide	
modified_base	15..18		

-continued

	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 647		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 648	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 648		
ttacggtatc tacatgcta		19
SEQ ID NO: 649	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 649		
tagaatgtag ataccgtaat t		21

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SEQ ID NO: 650      moltype = RNA  length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1..2
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide phosphorothioate
                   3
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide
                   4..6
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide
                   7..9
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide
                   10..11
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide
                   12
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide
                   13..16
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide
                   17
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide
                   18..19
                   mod_base = OTHER
misc_feature       note = 2'O-Methyl modified nucleotide
                   19
                   note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 650
ttacggtatc tacatgcta                                     19

SEQ ID NO: 651      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide phosphorothioate
                   2
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide phosphorothioate
                   3..13
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide
                   14
                   mod_base = OTHER
modified_base      note = 2'Fluoro modified nucleotide
                   15..18
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide
                   19..20
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide phosphorothioate
                   21
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
SEQUENCE: 651
tagaatgtag ataccgtaat t                                   21

SEQ ID NO: 652      moltype = RNA  length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1..2
                   mod_base = OTHER
modified_base      note = 2'O-Methyl modified nucleotide phosphorothioate
                   3

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	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 652		
ttacggtatc tacagtcta		19
SEQ ID NO: 653	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide phosphorothioate	
modified_base	3..13	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	15..18	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	19..20	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
SEQUENCE: 653		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 654	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	

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modified_base	note = 2'-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 654	
ttacggtatc tacagtctg	19
SEQ ID NO: 655	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 3..13 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 15..18 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 19..20 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 21 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide
SEQUENCE: 655	
tagaatgtag ataccgtaat t	21
SEQ ID NO: 656	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 6 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'-Methyl modified nucleotide

-continued

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modified_base      11
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      12..14
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      15
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      16
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      17
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      18..19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
misc_feature       19
                    note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 656
caaggtgggt ggcacctag                                     19

SEQ ID NO: 657      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      6..8
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      10
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      11..13
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      14
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      15
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      16
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      17
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      18
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      20
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate

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-continued

modified_base	21	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 657		
ttagatgccacccaccttgt t		21
SEQ ID NO: 658	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 658		
caaggtgggt ggcatctaa		19
SEQ ID NO: 659	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	

-continued

modified_base	5	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	6	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	7	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	8	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	9	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	10	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	11	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	12	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	14	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	15	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	19	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	21	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

SEQUENCE: 659
ttagatgccca cccacctgtg t 21

SEQ ID NO: 660 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
mol_type = other RNA
organism = synthetic construct

modified_base	1	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	2	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	4	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	5	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	6	
		mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 660
caaggtgggt ggcatttag

19

SEQ ID NO: 661	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	6 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	7 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	8 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	10 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	11 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	13 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-O-methyl modified nucleotide

-continued

	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 661		
ttagatgccca cccaccttgt t		21
SEQ ID NO: 662	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	12..14	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 662		

-continued

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caaggtgggt ggcatttga                                     19

SEQ ID NO: 663      moltype = RNA  length = 21
FEATURE            Location/Qualifiers
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = Vinyl phosphonate 2'-O-methyl modified nucleotide
                    Phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      7
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      8
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     11
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     12
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     13
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base     20
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base     21
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide

SEQUENCE: 663
ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 664      moltype = RNA  length = 19
FEATURE            Location/Qualifiers

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-continued

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source          1..19
                 mol_type = other RNA
                 organism = synthetic construct
modified_base   1
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide phosphorothioate
modified_base   2
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base   3
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   4
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base   5
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   6
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base   7..9
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   10
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base   11
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   12..14
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base   15
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   16
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
modified_base   17
                 mod_base = OTHER
                 note = 2'Fluoro modified nucleotide
modified_base   18..19
                 mod_base = OTHER
                 note = 2'O-Methyl modified nucleotide
misc_feature    19
                 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 664
caaggtgggt ggcacgaa
19

SEQ ID NO: 665   moltype = RNA length = 21
FEATURE
source          1..21
                 mol_type = other RNA
                 organism = synthetic construct
modified_base   1
                 mod_base = OTHER
                 note = Vinyl phosphonate 2'-O-methyl modified nucleotide
                 Phosphorothioate
modified_base   2
                 mod_base = OTHER
                 note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base   3
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   4
                 mod_base = OTHER
                 note = 2'-fluoro modified nucleotide
modified_base   5
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   6
                 mod_base = OTHER
                 note = 2'-fluoro modified nucleotide
modified_base   7
                 mod_base = OTHER

```

-continued

modified_base	note = 2'-fluoro modified nucleotide 8 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 9 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 13 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 18 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 19 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide Phosphorothioate 20 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide Phosphorothioate 21 mod_base = OTHER

SEQUENCE: 665
ttagatgccca cccaccttgt t 21

SEQ ID NO: 666	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide phosphorothioate 2 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 6 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 11

-continued

	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	12..14
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	15
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	16
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide
modified_base	18..19
	mod_base = OTHER
	note = 2'O-Methyl modified nucleotide
misc_feature	19
	note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 666	
caaggtgggt ggcataaa	19

SEQ ID NO: 667	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide
	Phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	10
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	11
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	13
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	15
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	16
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide

-continued

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modified_base      17
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      18
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      19
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base      20
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base      21
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
SEQUENCE: 667
ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 668      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      2
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      4
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      7..9
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      10
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      11
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      12..14
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      15
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      16
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
modified_base      17
                    mod_base = OTHER
                    note = 2'Fluoro modified nucleotide
modified_base      18..19
                    mod_base = OTHER
                    note = 2'O-Methyl modified nucleotide
misc_feature       19
                    note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 668
caaggtgggt ggcagctaa                                     19

SEQ ID NO: 669      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base      1

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-continued

	mod_base = OTHER
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide
	Phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	10
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	11
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	13
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	15
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	16
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	18
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	19
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	21
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide

SEQUENCE: 669
ttagatgccca cccaccttgt t 21

SEQ ID NO: 670 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base	1
	mod_base = OTHER
	note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	2
	mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 4 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 5 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 6 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 12..14 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 670
caagggtgggt ggcagctag

19

SEQ ID NO: 671	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	6 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	7 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	8 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	10

-continued

	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

SEQUENCE: 671
ttagatgccca cccaccttgt t 21

SEQ ID NO: 672 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base	1	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

-continued

modified_base	12	mod_base = OTHER	
		note = 2'-fluoro modified nucleotide	
modified_base	13	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	14	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	15	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	16	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	17	mod_base = OTHER	
		note = 2'-fluoro modified nucleotide	
modified_base	18	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	19	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide connected to an	
		inverted thymidine via a 3'-3' bond	
SEQUENCE: 672			
gagccatctt tgcaacgca			19
SEQ ID NO: 673		moltype = RNA length = 21	
FEATURE		Location/Qualifiers	
source		1..21	
		mol_type = other RNA	
		organism = synthetic construct	
modified_base	1	mod_base = OTHER	
		note = Vinyl phosphonate 2'-O-methyl modified nucleotide	
		Phosphorothioate	
modified_base	2	mod_base = OTHER	
		note = 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	3	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	4	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	5	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	6	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	7	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	8	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	9	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	10	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	11	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	12	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	13	mod_base = OTHER	
		note = 2'-O-methyl modified nucleotide	
modified_base	14	mod_base = OTHER	
		note = 2'-fluoro modified nucleotide	

-continued

modified_base	15	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	19	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	21	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
SEQUENCE: 673		
tcggttgcaa agatggctct t		21
SEQ ID NO: 674		
FEATURE		
source		
	1..19	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1..2	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'-Fluoro modified nucleotide
modified_base	4..6	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	7..9	
		mod_base = OTHER
		note = 2'-Fluoro modified nucleotide
modified_base	10..11	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	12	
		mod_base = OTHER
		note = 2'-Fluoro modified nucleotide
modified_base	13..16	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'-Fluoro modified nucleotide
modified_base	18..19	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
misc_feature	19	
		note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 674		
gagccatctt tgcaacgcg		19
SEQ ID NO: 675		
FEATURE		
source		
	1..21	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1	
		mod_base = OTHER
		note = Vinyl phosphonate 2'-O-methyl modified nucleotide
		Phosphorothioate
modified_base	2	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

-continued

modified_base	4	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	5	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	6	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	7	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	8	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	9	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	10	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	11	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	12	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	14	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	15	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	19	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	21	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

SEQUENCE: 675
 tgcgttgcaa agatggctct t 21

SEQ ID NO: 676 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base	1..2	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	4..6	
		mod_base = OTHER
		note = 2'O-Methyl modified nucleotide
modified_base	7..9	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	10..11	
		mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 676
gagccatctt tgcaacgaa

19

SEQ ID NO: 677	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	6 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	7 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	8 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	10 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	11 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	13 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	15 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	16 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	17 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	18

-continued

	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 677		
tcggttgcaa agatggctct t		21
SEQ ID NO: 678	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-Fluoro modified nucleotide	
modified_base	18..19	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 678		
gagccatctt tgcaacaca		19
SEQ ID NO: 679	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	
	mod_base = OTHER	
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide	
	Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide Phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	7	

-continued

	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	8	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	9	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	10	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	11	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	13	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	14	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	15	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	16	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	18	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	19	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	20	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	21	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
SEQUENCE: 679		
tcggttgcaa agatggctct t		21
SEQ ID NO: 680	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1..2	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	4..6	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	7..9	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	10..11	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	12	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	
modified_base	13..16	
	mod_base = OTHER	
	note = 2'-O-Methyl modified nucleotide	
modified_base	17	
	mod_base = OTHER	
	note = 2' Fluoro modified nucleotide	

-continued

modified_base	18..19
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
misc_feature	19
	note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 680	
gagccatctt tgcaaagca	19
SEQ ID NO: 681	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide
	Phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	10
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	11
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	13
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide
modified_base	15
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	16
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	18
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	19
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	21
	mod_base = OTHER

-continued

note = 2'-O-methyl modified nucleotide

SEQUENCE: 681
tgcgttgcaa agatggctct t 21

SEQ ID NO: 682
FEATURE
source
moltype = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct

modified_base
1..2
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

modified_base
3
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base
4..6
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base
7..9
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base
10..11
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base
12
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base
13..16
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base
17
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base
18..19
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

misc_feature
19
note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 682
gagccatctt tgcagcgca 19

SEQ ID NO: 683
FEATURE
source
moltype = RNA length = 21
Location/Qualifiers
1..21
mol_type = other RNA
organism = synthetic construct

modified_base
1
mod_base = OTHER
note = Vinyl phosphonate 2'-O-methyl modified nucleotide
Phosphorothioate

modified_base
2
mod_base = OTHER
note = 2'-fluoro modified nucleotide Phosphorothioate

modified_base
3
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
4
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
5
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
6
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
7
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
8
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
9
mod_base = OTHER
note = 2'-O-methyl modified nucleotide

modified_base
10
mod_base = OTHER

-continued

modified_base	note = 2'-O-methyl modified nucleotide 11 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 13 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 14 mod_base = OTHER	
modified_base	note = 2'-fluoro modified nucleotide 15 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 16 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 18 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide 19 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide Phosphorothioate 20 mod_base = OTHER	
modified_base	note = 2'-O-methyl modified nucleotide Phosphorothioate 21 mod_base = OTHER	
SEQUENCE: 683		
tcggttgcaa agatggctct t		21
SEQ ID NO: 684	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1..2 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 684		
gagccatctt tgcagcgcg		19
SEQ ID NO: 685	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	

-continued

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source          1..21
                 mol_type = other RNA
                 organism = synthetic construct
modified_base   1
                 mod_base = OTHER
                 note = Vinyl phosphonate 2'-O-methyl modified nucleotide
                   Phosphorothioate
modified_base   2
                 mod_base = OTHER
                 note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base   3
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   4
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   5
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   6
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   7
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   8
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   9
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   10
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   11
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   12
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   13
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   14
                 mod_base = OTHER
                 note = 2'-fluoro modified nucleotide
modified_base   15
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   16
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   17
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   18
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide
modified_base   19
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base   20
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base   21
                 mod_base = OTHER
                 note = 2'-O-methyl modified nucleotide

SEQUENCE: 685
tgcgttgcaa agatggctct t

SEQ ID NO: 686      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1..2

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-continued

	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide
modified_base	4..6
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
modified_base	7..9
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide
modified_base	10..11
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide
modified_base	13..16
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
modified_base	17
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide
modified_base	18..19
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
misc_feature	19
	note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 686	
ttacggtatc tacattcta	19
SEQ ID NO: 687	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide
	Phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	4
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	5
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	6
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	7
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	8
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	9
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	10
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	11
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	12
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide
modified_base	13
	mod_base = OTHER
	note = 2'-O-methyl modified nucleotide

-continued

modified_base	14	
		mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	15	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	19	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	21	
		mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
SEQUENCE: 687		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 688		moltype = RNA length = 19
FEATURE		Location/Qualifiers
source	1..19	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1..2	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	3	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	4..6	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	7..9	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	10..11	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	12	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	13..16	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	17	
		mod_base = OTHER
		note = 2'Fluoro modified nucleotide
modified_base	18	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
modified_base	19	
		mod_base = OTHER
		note = 2'-O-Methyl modified nucleotide
misc_feature	19	
		note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 688		
ttacggtatc tacattctg		19
SEQ ID NO: 689		moltype = RNA length = 21
FEATURE		Location/Qualifiers
source	1..21	
		mol_type = other RNA
		organism = synthetic construct
modified_base	1	
		mod_base = OTHER
		note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate

-continued

modified_base	2	mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3..13	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14	mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	15..18	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	19..20	mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	21	mod_base = OTHER note = 2'O-Methyl modified nucleotide

SEQUENCE: 689
tagaatgtag ataccgtaat t 21

SEQ ID NO: 690 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base	1..2	mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3	mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	4..6	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	7..9	mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	10..11	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	12	mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	13..16	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	17	mod_base = OTHER note = 2'Fluoro modified nucleotide
modified_base	18..19	mod_base = OTHER note = 2'O-Methyl modified nucleotide

misc_feature 19
 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 690
tacggatatct acattcgat 19

SEQ ID NO: 691 moltype = RNA length = 21
FEATURE Location/Qualifiers
source 1..21
 mol_type = other RNA
 organism = synthetic construct

modified_base	1	mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate
modified_base	2	mod_base = OTHER note = 2'Fluoro modified nucleotide phosphorothioate
modified_base	3..13	mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14	mod_base = OTHER note = 2'Fluoro modified nucleotide

-continued

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modified_base      15..18
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      19..20
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      21
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
SEQUENCE: 691
tagaatgtag ataccgtaat t                               21

SEQ ID NO: 692      moltype = RNA length = 19
FEATURE            Location/Qualifiers
source             1..19
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1..2
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      3
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base      4..6
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      7..9
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base      10..11
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      12
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base      13..16
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      17
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base      18..19
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
misc_feature       19
                   note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 692
ttacggtatc tacattgta                               19

SEQ ID NO: 693      moltype = RNA length = 21
FEATURE            Location/Qualifiers
source             1..21
                   mol_type = other RNA
                   organism = synthetic construct
modified_base      1
                   mod_base = OTHER
                   note = Vinyl phosphonate 2'-O-methyl modified nucleotide
modified_base      2
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide phosphorothioate
modified_base      3..13
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      14
                   mod_base = OTHER
                   note = 2'Fluoro modified nucleotide
modified_base      15..18
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
modified_base      19..20
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base      21
                   mod_base = OTHER
                   note = 2'O-Methyl modified nucleotide
SEQUENCE: 693

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-continued

tagaatgtag ataccgtaat t	21
SEQ ID NO: 694	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = other RNA
	organism = synthetic construct
modified_base	1..2
	mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide phosphorothioate
	3
	mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide
	4..6
	mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide
	7..9
	mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide
	10..11
	mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide
	12
	mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide
	13..16
	mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide
	17
	mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide
	18..19
	mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide
	19
SEQUENCE: 694	note = Connected to an inverted thymidine via a 3'-3' bond
ttacggatc tacatgcta	19
SEQ ID NO: 695	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1
	mod_base = OTHER
	note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate
modified_base	2
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide phosphorothioate
modified_base	3..13
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
modified_base	14
	mod_base = OTHER
	note = 2'-Fluoro modified nucleotide
modified_base	15..18
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
modified_base	19..20
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	21
	mod_base = OTHER
	note = 2'-O-Methyl modified nucleotide
SEQUENCE: 695	
tagaatgtag ataccgtaat t	21
SEQ ID NO: 696	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = other RNA
	organism = synthetic construct
modified_base	1..2
	mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide phosphorothioate 3 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 4..6 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 7..9 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 696
ttacggtatc tacatgcta 19

SEQ ID NO: 697
FEATURE
source
mol_type = RNA length = 21
Location/Qualifiers
1..21
mol_type = other RNA
organism = synthetic construct

modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-Fluoro modified nucleotide phosphorothioate
modified_base	3..13 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	15..18 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	19..20 mod_base = OTHER note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	21 mod_base = OTHER note = 2'-O-Methyl modified nucleotide

SEQUENCE: 697
tagaatgtag ataccgtaat t 21

SEQ ID NO: 698
FEATURE
source
mol_type = RNA length = 19
Location/Qualifiers
1..19
mol_type = other RNA
organism = synthetic construct

modified_base	1..2 mod_base = OTHER note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	4..6 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	7..9 mod_base = OTHER

-continued

modified_base	note = 2' Fluoro modified nucleotide 10..11 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 12 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 13..16 mod_base = OTHER
modified_base	note = 2'O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2' Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond
SEQUENCE: 698	
ttacgggtatc tacagtcta	19
SEQ ID NO: 699	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate
modified_base	2 mod_base = OTHER note = 2' Fluoro modified nucleotide phosphorothioate
modified_base	3..13 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2' Fluoro modified nucleotide
modified_base	15..18 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	19..20 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	21 mod_base = OTHER note = 2'O-Methyl modified nucleotide
SEQUENCE: 699	
tagaatgtag ataccgtaat t	21
SEQ ID NO: 700	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1..2 mod_base = OTHER note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2' Fluoro modified nucleotide
modified_base	4..6 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	7..9 mod_base = OTHER note = 2' Fluoro modified nucleotide
modified_base	10..11 mod_base = OTHER note = 2'O-Methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2' Fluoro modified nucleotide
modified_base	13..16 mod_base = OTHER

-continued

modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER	
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 700		
ttacggtatc tacagtctg		19
SEQ ID NO: 701	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	3..13 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
modified_base	14 mod_base = OTHER note = 2'-Fluoro modified nucleotide	
modified_base	15..18 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
modified_base	19..20 mod_base = OTHER note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	21 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
SEQUENCE: 701		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 702	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = 2'-Fluoro modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2'-O-Methyl modified nucleotide phosphorothioate	
modified_base	3 mod_base = OTHER note = 2'-Fluoro modified nucleotide	
modified_base	4 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
modified_base	5 mod_base = OTHER note = 2'-Fluoro modified nucleotide	
modified_base	6 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
modified_base	7..9 mod_base = OTHER note = 2'-Fluoro modified nucleotide	
modified_base	10 mod_base = OTHER note = 2'-O-Methyl modified nucleotide	
modified_base	11 mod_base = OTHER note = 2'-Fluoro modified nucleotide	
modified_base	12..14 mod_base = OTHER	

-continued

modified_base	note = 2'-O-Methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-O-Methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-Fluoro modified nucleotide 18..19 mod_base = OTHER
misc_feature	note = 2'-O-Methyl modified nucleotide 19 note = Connected to an inverted thymidine via a 3'-3' bond

SEQUENCE: 702
caaggtgggt ggcacctag

19

SEQ ID NO: 703	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-Fluoro modified nucleotide phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	6..8 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	10 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	11..13 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	15 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	16 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	17 mod_base = OTHER note = 2'-O-Methyl modified nucleotide
modified_base	18 mod_base = OTHER note = 2'-Fluoro modified nucleotide
modified_base	19 mod_base = OTHER note = 2'-O-Methyl modified nucleotide phosphorothioate
modified_base	20 mod_base = OTHER note = 2'-Fluoro modified nucleotide phosphorothioate
modified_base	21 mod_base = OTHER note = 2'-O-Methyl modified nucleotide

SEQUENCE: 703

-continued

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ttagatgccca cccaccttgt t                                     21

SEQ ID NO: 704          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base          1..2
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base          3
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide
modified_base          4..6
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
modified_base          7..9
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide
modified_base          10..11
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
modified_base          12
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide
modified_base          13..16
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
modified_base          17
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide
modified_base          18..19
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
SEQUENCE: 704
ttacggtatc tacattcta                                     19

SEQ ID NO: 705          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base          1
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base          2
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide phosphorothioate
modified_base          3..13
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
modified_base          14
                        mod_base = OTHER
                        note = 2'Fluoro modified nucleotide
modified_base          15..18
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
modified_base          19..20
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base          21
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide
SEQUENCE: 705
tagaatgtag ataccgtaat t                                     21

SEQ ID NO: 706          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base          1..2
                        mod_base = OTHER
                        note = 2'O-Methyl modified nucleotide phosphorothioate
modified_base          3
                        mod_base = OTHER

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-continued

modified_base	note = 2' Fluoro modified nucleotide 4..6 mod_base = OTHER	
modified_base	note = 2' O-Methyl modified nucleotide 7..9 mod_base = OTHER	
modified_base	note = 2' Fluoro modified nucleotide 10..11 mod_base = OTHER	
modified_base	note = 2' O-Methyl modified nucleotide 12 mod_base = OTHER	
modified_base	note = 2' Fluoro modified nucleotide 13..16 mod_base = OTHER	
modified_base	note = 2' O-Methyl modified nucleotide 17 mod_base = OTHER	
modified_base	note = 2' Fluoro modified nucleotide 18..19 mod_base = OTHER	
modified_base	note = 2' O-Methyl modified nucleotide	
SEQUENCE: 706		
ttacggtatc tacattcta		19
SEQ ID NO: 707	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other RNA organism = synthetic construct	
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate	
modified_base	2 mod_base = OTHER note = 2' Fluoro modified nucleotide phosphorothioate	
modified_base	3..13 mod_base = OTHER note = 2' O-Methyl modified nucleotide	
modified_base	14 mod_base = OTHER note = 2' Fluoro modified nucleotide	
modified_base	15..18 mod_base = OTHER note = 2' O-Methyl modified nucleotide	
modified_base	19..20 mod_base = OTHER note = 2' O-Methyl modified nucleotide phosphorothioate	
modified_base	21 mod_base = OTHER note = 2' O-Methyl modified nucleotide	
SEQUENCE: 707		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 708	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90 mol_type = protein organism = synthetic construct	
SEQUENCE: 708		
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT		60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT		90
SEQ ID NO: 709	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90 mol_type = protein organism = synthetic construct	
SEQUENCE: 709		
MLPAPKQLVV SRVTEDSARL SWTAPDAAFD SFAIVYHEPR PDGEAIVLTV PGSCRSYDLT		60
GLKPGTEYEV VILGVKGGVH SYPLSAIFTT		90
SEQ ID NO: 710	moltype = AA length = 90	
FEATURE	Location/Qualifiers	
source	1..90	

-continued

mol_type = protein
organism = synthetic construct

SEQUENCE: 710
MLPAPKNLVV SRVTEDSARL SWTAPDAAFD SFFIGYLEPQ PPGEAIHLGV PGSCRSYDLT 60
GLKPGTEYNV TIQGVKGGFP SIPLFASFTT 90

SEQ ID NO: 711 moltype = RNA length = 19
FEATURE Location/Qualifiers
source 1..19
mol_type = other RNA
organism = synthetic construct

modified_base 1..2
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

modified_base 3
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 4..6
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 7..9
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 10..11
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 12
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 13..16
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 17
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 18..19
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

SEQUENCE: 711
tggacttcaa cctagacaa 19

SEQ ID NO: 712 moltype = RNA length = 20
FEATURE Location/Qualifiers
source 1..20
mol_type = other RNA
organism = synthetic construct

modified_base 1
mod_base = OTHER
note = Vinyl phosphonate 2'-O-methyl modified nucleotide phosphorothioate

modified_base 2
mod_base = OTHER
note = 2'-Fluoro modified nucleotide phosphorothioate

modified_base 3..13
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 14
mod_base = OTHER
note = 2'-Fluoro modified nucleotide

modified_base 15..16
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide

modified_base 19..20
mod_base = OTHER
note = 2'-O-Methyl modified nucleotide phosphorothioate

SEQUENCE: 712
tgtctagggtt gaagtcatt 20

SEQ ID NO: 713 moltype = length =
SEQUENCE: 713
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SEQ ID NO: 714 moltype = length =
SEQUENCE: 714
000

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SEQ ID NO: 715 SEQUENCE: 715 000	moltype = length =	
SEQ ID NO: 716 SEQUENCE: 716 000	moltype = length =	
SEQ ID NO: 717 SEQUENCE: 717 000	moltype = length =	
SEQ ID NO: 718 SEQUENCE: 718 000	moltype = length =	
SEQ ID NO: 719 SEQUENCE: 719 000	moltype = length =	
SEQ ID NO: 720 FEATURE source	moltype = AA length = 4 Location/Qualifiers 1..4 mol_type = protein organism = synthetic construct	
SEQUENCE: 720 GSGS		4
SEQ ID NO: 721 FEATURE source	moltype = AA length = 8 Location/Qualifiers 1..8 mol_type = protein organism = synthetic construct	
SEQUENCE: 721 GGGSGGGS		8
SEQ ID NO: 722 FEATURE source	moltype = AA length = 25 Location/Qualifiers 1..25 mol_type = protein organism = synthetic construct	
VARIANT	6..25 note = GGGGS repeats may be deleted	
SEQUENCE: 722 GGGSGGGGS GGGSGGGGS GGGGS		25
SEQ ID NO: 723 FEATURE source	moltype = AA length = 4 Location/Qualifiers 1..4 mol_type = protein organism = synthetic construct	
SEQUENCE: 723 APAP		4
SEQ ID NO: 724 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = synthetic construct	
SEQUENCE: 724 APAPAPAPAP		10
SEQ ID NO: 725 FEATURE source	moltype = AA length = 20 Location/Qualifiers 1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 725 APAPAPAPAP APAPAPAPAP		20
SEQ ID NO: 726 FEATURE source	moltype = AA length = 40 Location/Qualifiers 1..40 mol_type = protein organism = synthetic construct	
SEQUENCE: 726		

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APAPAPAPAP APAPAPAPAP APAPAPAPAP APAPAPAPAP	40
SEQ ID NO: 727	moltype = AA length = 29
FEATURE	Location/Qualifiers
source	1..29
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 727	
AEAAAKEAAA KEAAAKEAAA KEAAAKAAA	29
SEQ ID NO: 728	moltype = AA length = 25
FEATURE	Location/Qualifiers
source	1..25
	mol_type = protein
	organism = synthetic construct
VARIANT	6..25
	note = EAAAK repeats may be deleted
SEQUENCE: 728	
EAAAKEAAAK EAAAKEAAAK EAAAK	25
SEQ ID NO: 729	moltype = AA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 729	
EAAAKEAAAK EAAAKEAAAK	20
SEQ ID NO: 730	moltype = AA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 730	
GGGSGGGGS GGGSGGGGS	20
SEQ ID NO: 731	moltype = AA length = 10
FEATURE	Location/Qualifiers
source	1..10
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 731	
APAPAPAPAP	10
SEQ ID NO: 732	moltype = AA length = 5
FEATURE	Location/Qualifiers
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 732	
EAAAK	5
SEQ ID NO: 733	moltype = length =
SEQUENCE: 733	
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SEQ ID NO: 734	moltype = length =
SEQUENCE: 734	
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SEQ ID NO: 735	moltype = length =
SEQUENCE: 735	
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SEQ ID NO: 736	moltype = length =
SEQUENCE: 736	
000	
SEQ ID NO: 737	moltype = length =
SEQUENCE: 737	
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SEQ ID NO: 738	moltype = length =
SEQUENCE: 738	
000	

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SEQ ID NO: 739 SEQUENCE: 739 000	moltype = length =	
SEQ ID NO: 740 SEQUENCE: 740 000	moltype = length =	
SEQ ID NO: 741 SEQUENCE: 741 000	moltype = length =	
SEQ ID NO: 742 SEQUENCE: 742 000	moltype = length =	
SEQ ID NO: 743 SEQUENCE: 743 000	moltype = length =	
SEQ ID NO: 744 SEQUENCE: 744 000	moltype = length =	
SEQ ID NO: 745 SEQUENCE: 745 000	moltype = length =	
SEQ ID NO: 746 SEQUENCE: 746 000	moltype = length =	
SEQ ID NO: 747 SEQUENCE: 747 000	moltype = length =	
SEQ ID NO: 748 SEQUENCE: 748 000	moltype = length =	
SEQ ID NO: 749 SEQUENCE: 749 000	moltype = length =	
SEQ ID NO: 750 FEATURE source SEQUENCE: 750 RKRRRQRRR	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = synthetic construct	9
SEQ ID NO: 751 FEATURE source SEQUENCE: 751 RQIKIWFQNR RMKWKK	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = synthetic construct	16
SEQ ID NO: 752 FEATURE source SEQUENCE: 752 GWTLSAGYL LGKINKALAA LAKKIL	moltype = AA length = 26 Location/Qualifiers 1..26 mol_type = protein organism = synthetic construct	26
SEQ ID NO: 753 FEATURE source SEQUENCE: 753 KLALKLALKA LKAALKLA	moltype = AA length = 18 Location/Qualifiers 1..18 mol_type = protein organism = synthetic construct	18

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SEQ ID NO: 754	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 754		
KETWWETWWT EWSQPKKKRK V		21
SEQ ID NO: 755	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 755		
KDEL		4
SEQ ID NO: 756	moltype = AA length = 29	
FEATURE	Location/Qualifiers	
source	1..29	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 756		
WEAALAEALA ELAEHLAEAL AEALEALAA		29
SEQ ID NO: 757	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 757		
GDIMGEWGNE IFGAIAGFLG C		21
SEQ ID NO: 758	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 758		
GLFDIIKKIA ESP		13
SEQ ID NO: 759	moltype = AA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 759		
GALFLGWLGA AGSTMGAPKS KRKV		24
SEQ ID NO: 760	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 760		
AGYLLGKINL KALAALAKKI L		21
SEQ ID NO: 761	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
source	1..23	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 761		
LIRLWSHLIH IWFQNRRLKW KKK		23
SEQ ID NO: 762	moltype = AA length = 36	
FEATURE	Location/Qualifiers	
source	1..36	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 762		
GLFGAIAGFI ENGWEGMIDG RQIKIWFQNR RMKWKK		36
SEQ ID NO: 763	moltype = AA length = 26	
FEATURE	Location/Qualifiers	
source	1..26	
	mol_type = protein	

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SEQUENCE: 763	organism = synthetic construct	
FFKKLAHALH LLALLALHLA HALKKA		26
SEQ ID NO: 764	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 764		
LFEAIEGFIE NGWGMIDGWY G		21
SEQ ID NO: 765	moltype = AA length = 31	
FEATURE	Location/Qualifiers	
source	1..31	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 765		
LFEAIEGFIE NGWEGMIDGW YGRKKRRQRR R		31
SEQ ID NO: 766	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 766		
IGAVLKVLTT GLPALISWIK RKRQQ		25
SEQ ID NO: 767	moltype = AA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 767		
MKLAVTLTLV TLALSSSSAS A		21
SEQ ID NO: 768	moltype = length =	
SEQUENCE: 768		
000		
SEQ ID NO: 769	moltype = length =	
SEQUENCE: 769		
000		
SEQ ID NO: 770	moltype = length =	
SEQUENCE: 770		
000		
SEQ ID NO: 771	moltype = length =	
SEQUENCE: 771		
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SEQ ID NO: 772	moltype = length =	
SEQUENCE: 772		
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SEQ ID NO: 773	moltype = length =	
SEQUENCE: 773		
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SEQ ID NO: 774	moltype = length =	
SEQUENCE: 774		
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SEQ ID NO: 775	moltype = length =	
SEQUENCE: 775		
000		
SEQ ID NO: 776	moltype = length =	
SEQUENCE: 776		
000		
SEQ ID NO: 777	moltype = length =	
SEQUENCE: 777		
000		

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SEQ ID NO: 778 SEQUENCE: 778 000	moltype =	length =
SEQ ID NO: 779 SEQUENCE: 779 000	moltype =	length =
SEQ ID NO: 780 SEQUENCE: 780 000	moltype =	length =
SEQ ID NO: 781 SEQUENCE: 781 000	moltype =	length =
SEQ ID NO: 782 SEQUENCE: 782 000	moltype =	length =
SEQ ID NO: 783 SEQUENCE: 783 000	moltype =	length =
SEQ ID NO: 784 SEQUENCE: 784 000	moltype =	length =
SEQ ID NO: 785 SEQUENCE: 785 000	moltype =	length =
SEQ ID NO: 786 SEQUENCE: 786 000	moltype =	length =
SEQ ID NO: 787 SEQUENCE: 787 000	moltype =	length =
SEQ ID NO: 788 SEQUENCE: 788 000	moltype =	length =
SEQ ID NO: 789 SEQUENCE: 789 000	moltype =	length =
SEQ ID NO: 790 SEQUENCE: 790 000	moltype =	length =
SEQ ID NO: 791 SEQUENCE: 791 000	moltype =	length =
SEQ ID NO: 792 SEQUENCE: 792 000	moltype =	length =
SEQ ID NO: 793 SEQUENCE: 793 000	moltype =	length =
SEQ ID NO: 794 SEQUENCE: 794 000	moltype =	length =
SEQ ID NO: 795 SEQUENCE: 795 000	moltype =	length =
SEQ ID NO: 796 SEQUENCE: 796 000	moltype =	length =

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SEQ ID NO: 797 SEQUENCE: 797 000	moltype = length =	
SEQ ID NO: 798 SEQUENCE: 798 000	moltype = length =	
SEQ ID NO: 799 SEQUENCE: 799 000	moltype = length =	
SEQ ID NO: 800 SEQUENCE: 800 000	moltype = length =	
SEQ ID NO: 801 FEATURE source SEQUENCE: 801 caggggtgcg gtcttgcaa	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 802 FEATURE source SEQUENCE: 802 ggtgcggtct tgcaatagg	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 803 FEATURE source SEQUENCE: 803 gtgcggtctt gcaatagga	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 804 FEATURE source SEQUENCE: 804 agccatgcct ttaaaccgc	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 805 FEATURE source SEQUENCE: 805 atgccttta accgcactt	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 806 FEATURE source SEQUENCE: 806 tgggaggatg aattcgacc	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 807 FEATURE source SEQUENCE: 807 aacgcagtgc tcttcgaag	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA organism = synthetic construct	19
SEQ ID NO: 808 FEATURE source	moltype = RNA length = 19 Location/Qualifiers 1..19 mol_type = other RNA	

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SEQUENCE: 808	organism = synthetic construct	
ggacgaatgg ggcgacaac		19
SEQ ID NO: 809	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 809		
ggcgactgcc thtagcaac		19
SEQ ID NO: 810	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 810		
cacgctgctg gggcgctac		19
SEQ ID NO: 811	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 811		
tgcgctcacg tcttcacta		19
SEQ ID NO: 812	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 812		
ccttatactt ctttatcgc		19
SEQ ID NO: 813	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 813		
ctttatcgcc ggcgctat		19
SEQ ID NO: 814	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 814		
tcgcgggcg ctatgagtt		19
SEQ ID NO: 815	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 815		
gccgctatga gttctccaa		19
SEQ ID NO: 816	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 816		
ccgctatgag ttctccaac		19
SEQ ID NO: 817	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 817		
ggctcggctc aactatctg		19

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SEQ ID NO: 818	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 818		
gctcggctca actatctgc		19
SEQ ID NO: 819	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 819		
tcggctcaac tatctgctc		19
SEQ ID NO: 820	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 820		
ccagcgcgga ccaacaatt		19
SEQ ID NO: 821	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 821		
agcgcggaacc aacaatttc		19
SEQ ID NO: 822	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 822		
tcgggaggaa gctttatga		19
SEQ ID NO: 823	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 823		
acgcagcggc agtctttcc		19
SEQ ID NO: 824	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 824		
gaccaccatc cgccgaatc		19
SEQ ID NO: 825	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 825		
ccgaatcggc ctcttcaat		19
SEQ ID NO: 826	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 826		
aatcggcctc ttcaatagc		19
SEQ ID NO: 827	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 827	organism = synthetic construct	
caatagcagt gccgacagg		19
SEQ ID NO: 828	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 828		
aggagtttgt ccgtggctg		19
SEQ ID NO: 829	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 829		
cggctgagtg cacggttat		19
SEQ ID NO: 830	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 830		
tgggaatccc cagtatctc		19
SEQ ID NO: 831	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 831		
cagacccctc agcttacg		19
SEQ ID NO: 832	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 832		
ccctcagct tacggtatc		19
SEQ ID NO: 833	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 833		
ctcagcttac ggtatctac		19
SEQ ID NO: 834	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 834		
gcttacggtatctacattc		19
SEQ ID NO: 835	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 835		
ccggcggcag cgtatcatc		19
SEQ ID NO: 836	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 836		
agcgtatcat ccagcggaa		19

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SEQ ID NO: 837	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 837		
gcgtatcatc cagcggaac		19
SEQ ID NO: 838	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 838		
tacctaggcc ggtactata		19
SEQ ID NO: 839	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 839		
ctaggccggt actatatgt		19
SEQ ID NO: 840	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 840		
aggccggtac tatatgtct		19
SEQ ID NO: 841	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 841		
ggccggtact atatgtctg		19
SEQ ID NO: 842	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 842		
gccggtacta tatgtctgc		19
SEQ ID NO: 843	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 843		
cgtactata tgtctgcgc		19
SEQ ID NO: 844	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 844		
ggtactatat gtctgcgcg		19
SEQ ID NO: 845	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 845		
ctatatgtct gcgcgccac		19
SEQ ID NO: 846	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 846	organism = synthetic construct	
tatgtctgcg cgccacatg		19
SEQ ID NO: 847	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 847		
atgtctgcg cgccacatg		19
SEQ ID NO: 848	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 848		
tcgccctcgc tgtcacgac		19
SEQ ID NO: 849	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 849		
cggcgagcgc tacgatgag		19
SEQ ID NO: 850	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 850		
caaggaccgg cgcaacatc		19
SEQ ID NO: 851	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 851		
acatccgtgc accagagtg		19
SEQ ID NO: 852	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 852		
ctggcgagg agcgtaact		19
SEQ ID NO: 853	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 853		
gggcgaggag cgtaactaa		19
SEQ ID NO: 854	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 854		
gcgaggagcg taactaagt		19
SEQ ID NO: 855	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 855		
gaggagcgta actaagtcc		19

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SEQ ID NO: 856	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 856		
agtccgccaa acactccac		19
SEQ ID NO: 857	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 857		
ggcgatcaag tccagagcc		19
SEQ ID NO: 858	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 858		
ccctaacctg gcttatctc		19
SEQ ID NO: 859	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 859		
tgtgaaacca ctaggttct		19
SEQ ID NO: 860	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 860		
accactaggt tctaggtcc		19
SEQ ID NO: 861	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 861		
ttgcaagacc gcaccctgt t		21
SEQ ID NO: 862	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 862		
cctattgcaa gaccgcacct t		21
SEQ ID NO: 863	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 863		
tcctattgca agaccgcact t		21
SEQ ID NO: 864	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 864		
gcggttataa ggcattgctt t		21
SEQ ID NO: 865	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 865	organism = synthetic construct	
aagtgcggtt taaaggcatt t		21
SEQ ID NO: 866	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 866		
ggtcgaattc atcctcccat t		21
SEQ ID NO: 867	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 867		
cttcgaagag cactgcgttt t		21
SEQ ID NO: 868	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 868		
gttgctgccc cattcgctct t		21
SEQ ID NO: 869	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 869		
gttgctacag gcagtcgctt t		21
SEQ ID NO: 870	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 870		
gtagcgcccc agcagcgtgt t		21
SEQ ID NO: 871	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 871		
tagtgaagac gtgagcgcat t		21
SEQ ID NO: 872	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 872		
gcgataaaga agtataaggt t		21
SEQ ID NO: 873	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 873		
atagcggccg gcgataaagt t		21
SEQ ID NO: 874	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 874		
aactcatagc ggccggcgat t		21

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SEQ ID NO: 875	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 875		
ttggagaact catagcggct t		21
SEQ ID NO: 876	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 876		
gttgagaaac tcatagcggct t		21
SEQ ID NO: 877	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 877		
cagatagttg agccgagcct t		21
SEQ ID NO: 878	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 878		
gcagatagtt gagccgagct t		21
SEQ ID NO: 879	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 879		
gagcagatag ttgagccgat t		21
SEQ ID NO: 880	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 880		
aattgttggt ccgcgctggt t		21
SEQ ID NO: 881	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 881		
gaaattgttg gtccgcgctt t		21
SEQ ID NO: 882	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 882		
tcataaagct tcctccgat t		21
SEQ ID NO: 883	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 883		
ggaaagactg ccgctgcgtt t		21
SEQ ID NO: 884	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 884	organism = synthetic construct	
gattcggcgg atggtgtct t		21
SEQ ID NO: 885	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 885		
attgaagagg ccgattcgg t		21
SEQ ID NO: 886	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 886		
gctattgaag aggcgcgatt t		21
SEQ ID NO: 887	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 887		
cctgtcggca ctgctattgt t		21
SEQ ID NO: 888	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 888		
cagccacgga caaactcctt t		21
SEQ ID NO: 889	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 889		
ataaccgtgc actcagccgt t		21
SEQ ID NO: 890	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 890		
gagatactgg ggattcccat t		21
SEQ ID NO: 891	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 891		
ccgtaagctg aggggtctgt t		21
SEQ ID NO: 892	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 892		
gataccgtaa gctgaggggt t		21
SEQ ID NO: 893	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 893		
gtagataccg taagctgagt t		21

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SEQ ID NO: 894	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 894		
gaatgtagat accgtaagct t		21
SEQ ID NO: 895	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 895		
gatgatacgc tgccgccggt t		21
SEQ ID NO: 896	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 896		
ttccgctgga tgatacgett t		21
SEQ ID NO: 897	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 897		
gttccgctgg atgatacgt t		21
SEQ ID NO: 898	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 898		
tatagtaccg gcctagggtat t		21
SEQ ID NO: 899	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 899		
acatatagta ccggcctagt t		21
SEQ ID NO: 900	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 900		
agacatatag taccggcctt t		21
SEQ ID NO: 901	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 901		
cagacatata gtaccggcct t		21
SEQ ID NO: 902	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 902		
gcagacatat agtaccggct t		21
SEQ ID NO: 903	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 903	organism = synthetic construct	
gcgcagacat atagtaccgt t		21
SEQ ID NO: 904	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 904		
cgcgagaca tatagtacct t		21
SEQ ID NO: 905	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 905		
gtggcgcgca gacatatagt t		21
SEQ ID NO: 906	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 906		
catgtggcg gcagacatat t		21
SEQ ID NO: 907	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 907		
ccatgtggcg cgagacatt t		21
SEQ ID NO: 908	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 908		
gtcgtgacag cgagggcgat t		21
SEQ ID NO: 909	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 909		
ctcatcgtag cgctcgccgt t		21
SEQ ID NO: 910	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 910		
gatgttgcgc cggctcctgt t		21
SEQ ID NO: 911	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 911		
cactctggtg cacggatgtt t		21
SEQ ID NO: 912	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 912		
agttacgctc ctgcccagt t		21

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SEQ ID NO: 913	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 913		
ttagttacgc tctcgcct t		21
SEQ ID NO: 914	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 914		
acttagttac gctcctcgct t		21
SEQ ID NO: 915	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 915		
ggacttagtt acgctcctct t		21
SEQ ID NO: 916	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 916		
gtggagtgtt tggcggactt t		21
SEQ ID NO: 917	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 917		
ggctctggac ttgatcgct t		21
SEQ ID NO: 918	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 918		
ggaataagcc aggttagggt t		21
SEQ ID NO: 919	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 919		
agaacctagt ggttcacat t		21
SEQ ID NO: 920	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 920		
ggacctagaa ctagtggtt t		21
SEQ ID NO: 921	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 921		
caggggtgcg gtcttgcaa		19
SEQ ID NO: 922	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 922	organism = synthetic construct	
ggtgcggtct tgcaataga		19
SEQ ID NO: 923	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 923		
gtgcggtctt gcaatagga		19
SEQ ID NO: 924	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 924		
agccatgcct ttaaaccga		19
SEQ ID NO: 925	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 925		
atgcctttaa accgcacta		19
SEQ ID NO: 926	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 926		
tgggaggatg aattcgaca		19
SEQ ID NO: 927	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 927		
aacgcagtgc tcttcgaaa		19
SEQ ID NO: 928	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 928		
ggacgaatgg ggcgacaaa		19
SEQ ID NO: 929	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 929		
ggcgactgcc tgtagcaaa		19
SEQ ID NO: 930	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 930		
cacgctgctg gggcgctaa		19
SEQ ID NO: 931	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 931		
tcgctcacg tcttcacta		19

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SEQ ID NO: 932	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 932		
ccttatactt ctttatcga		19
SEQ ID NO: 933	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 933		
ctttatcgcc ggccgctaa		19
SEQ ID NO: 934	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 934		
tcgccggccg ctatgagta		19
SEQ ID NO: 935	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 935		
gccgctatga gttctccaa		19
SEQ ID NO: 936	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 936		
ccgctatgag ttctccaaa		19
SEQ ID NO: 937	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 937		
ggctcggctc aactatcta		19
SEQ ID NO: 938	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 938		
gctcggctca actatctga		19
SEQ ID NO: 939	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 939		
tcggctcaac tatctgcta		19
SEQ ID NO: 940	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 940		
ccagcgcgga ccaacaata		19
SEQ ID NO: 941	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 941	organism = synthetic construct	
agcgcgacc aacaattta		19
SEQ ID NO: 942	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 942		
tcgggaggaa gctttatga		19
SEQ ID NO: 943	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 943		
acgcagcggc agtctttca		19
SEQ ID NO: 944	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 944		
gaccaccatc cgccgaata		19
SEQ ID NO: 945	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 945		
ccgaatcggc ctcttcaaa		19
SEQ ID NO: 946	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 946		
aatcggcctc ttcaataga		19
SEQ ID NO: 947	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 947		
caatagcagt gccgacaga		19
SEQ ID NO: 948	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 948		
aggagtgtgt ccgtggcta		19
SEQ ID NO: 949	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 949		
cggtgagtg cacggttaa		19
SEQ ID NO: 950	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 950		
tgggaatccc cagtatcta		19

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SEQ ID NO: 951	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 951		
cagaccctc agcttacga		19
SEQ ID NO: 952	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 952		
cccctcagct tacggtata		19
SEQ ID NO: 953	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 953		
ctcagcttac ggtatctaa		19
SEQ ID NO: 954	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 954		
gcttacggta tctacatta		19
SEQ ID NO: 955	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 955		
ccggcggcag cgtatcata		19
SEQ ID NO: 956	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 956		
agcgtatcat ccagcggaa		19
SEQ ID NO: 957	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 957		
gcgtatcatc cagcggaaa		19
SEQ ID NO: 958	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 958		
tacctaggcc ggtactata		19
SEQ ID NO: 959	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 959		
ctaggccggt actatatga		19
SEQ ID NO: 960	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 960	organism = synthetic construct	
aggccggtac tatatgtca		19
SEQ ID NO: 961	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 961		
ggccggtact atatgtcta		19
SEQ ID NO: 962	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 962		
gccggtacta tatgtctga		19
SEQ ID NO: 963	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 963		
cgggtactata tgtctgcga		19
SEQ ID NO: 964	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 964		
ggtactatat gtctgcgca		19
SEQ ID NO: 965	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 965		
ctatatgtct gcgcgcaa		19
SEQ ID NO: 966	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 966		
tatgtctgcg cgccacata		19
SEQ ID NO: 967	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 967		
atgtctgcgc gccacatga		19
SEQ ID NO: 968	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 968		
tcgcctcgc tgtcacgaa		19
SEQ ID NO: 969	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 969		
cggcgagcgc tacgatgaa		19

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SEQ ID NO: 970	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 970		
caaggaccgg cgcaacata		19
SEQ ID NO: 971	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 971		
acatccgtgc accagagta		19
SEQ ID NO: 972	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 972		
ctgggcgagg agcgtaaca		19
SEQ ID NO: 973	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 973		
gggcgaggag cgtaactaa		19
SEQ ID NO: 974	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 974		
gcgaggagcg taactaaga		19
SEQ ID NO: 975	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 975		
gaggagcgta actaagtca		19
SEQ ID NO: 976	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 976		
agtccgccaa acactccaa		19
SEQ ID NO: 977	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 977		
ggcgatcaag tccagagca		19
SEQ ID NO: 978	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 978		
ccctaacctg gcttattca		19
SEQ ID NO: 979	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	

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SEQUENCE: 979	organism = synthetic construct	
tgtagaacca ctaggttca		19
SEQ ID NO: 980	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 980		
accactaggt tctaggtca		19
SEQ ID NO: 981	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 981		
ttgcaagacc gcaccctgt t		21
SEQ ID NO: 982	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 982		
tctattgcaa gaccgcacct t		21
SEQ ID NO: 983	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 983		
tcctattgca agaccgcacct t		21
SEQ ID NO: 984	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 984		
tcggtttaaa ggcattggctt t		21
SEQ ID NO: 985	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 985		
tagtgcggtt taaaggcatt t		21
SEQ ID NO: 986	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 986		
tgtagaatc atcctcccat t		21
SEQ ID NO: 987	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 987		
tttcgaagag cactgcgttt t		21
SEQ ID NO: 988	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 988		
tttgtagccc cattgcctt t		21

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SEQ ID NO: 989	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 989		
tttgctacag gcagtcgcct t		21
SEQ ID NO: 990	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 990		
ttagcgcccc agcagcgtgt t		21
SEQ ID NO: 991	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 991		
tagtgaagac gtgagcgcct t		21
SEQ ID NO: 992	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 992		
tcgataaaga agtataaggt t		21
SEQ ID NO: 993	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 993		
ttagcggccg gcgataaagt t		21
SEQ ID NO: 994	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 994		
tactcatagc ggccggcgat t		21
SEQ ID NO: 995	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 995		
ttggagaact catagcggct t		21
SEQ ID NO: 996	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 996		
tttgagaaac tcatagcggct t		21
SEQ ID NO: 997	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 997		
tagatagttg agccgagcct t		21
SEQ ID NO: 998	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 998	organism = synthetic construct	
tcagatagtt gagccgagct t		21
SEQ ID NO: 999	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 999		
tagcagatag ttgagccgat t		21
SEQ ID NO: 1000	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1000		
tattgttggt ccgcgctggt t		21
SEQ ID NO: 1001	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1001		
taaattgttg gtccgcgctt t		21
SEQ ID NO: 1002	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1002		
tcataaagct tcctccgat t		21
SEQ ID NO: 1003	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1003		
tgaaagactg ccgctgcgctt t		21
SEQ ID NO: 1004	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1004		
tattcggcgg atggtggtct t		21
SEQ ID NO: 1005	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1005		
tttgaagagg ccgattcgggt t		21
SEQ ID NO: 1006	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1006		
tctattgaag aggcggattt t		21
SEQ ID NO: 1007	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1007		
tctgtcgga ctgetattgt t		21

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SEQ ID NO: 1008	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1008		
tagccacgga caaactcctt t		21
SEQ ID NO: 1009	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1009		
ttaaccgtgc actcagccgt t		21
SEQ ID NO: 1010	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1010		
tagatactgg ggattcccat t		21
SEQ ID NO: 1011	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1011		
tcgtaagctg aggggtctgt t		21
SEQ ID NO: 1012	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1012		
tataccgtaa gctgaggggt t		21
SEQ ID NO: 1013	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1013		
ttagataccg taagctgagt t		21
SEQ ID NO: 1014	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1014		
taatgtagat accgtaagct t		21
SEQ ID NO: 1015	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1015		
tatgatacgc tgccgccggt t		21
SEQ ID NO: 1016	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1016		
ttccgctgga tgatacgctt t		21
SEQ ID NO: 1017	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 1017	organism = synthetic construct	
tttccgctgg atgatacgct t		21
SEQ ID NO: 1018	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1018		
tatagtagcg gcctaggtat t		21
SEQ ID NO: 1019	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1019		
tcatatagta ccggcctagt t		21
SEQ ID NO: 1020	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1020		
tgacatatag taccggcctt t		21
SEQ ID NO: 1021	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1021		
tagacatata gtaccggcct t		21
SEQ ID NO: 1022	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1022		
tcagacatat agtaccggct t		21
SEQ ID NO: 1023	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1023		
tcgcagacat atagtaccgt t		21
SEQ ID NO: 1024	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1024		
tgcgagaca tatagtacct t		21
SEQ ID NO: 1025	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1025		
ttggcgcgca gacatatagt t		21
SEQ ID NO: 1026	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1026		
tatgtggcgc gcagacatat t		21

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SEQ ID NO: 1027	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1027		
tcatgtggcg cgcagacatt t		21
SEQ ID NO: 1028	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1028		
ttcgtgacag cgagggcgat t		21
SEQ ID NO: 1029	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1029		
ttcatcgtag cgctcgccgt t		21
SEQ ID NO: 1030	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1030		
tatgttgccg cggtccttgt t		21
SEQ ID NO: 1031	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1031		
tactctgggtg cacggatgtt t		21
SEQ ID NO: 1032	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1032		
tgttacgctc ctgcgccagt t		21
SEQ ID NO: 1033	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1033		
ttagttacgc tctcgcctt t		21
SEQ ID NO: 1034	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1034		
tcttagttac gctcctcgct t		21
SEQ ID NO: 1035	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1035		
tgacttagtt acgctcctct t		21
SEQ ID NO: 1036	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	

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SEQUENCE: 1036	organism = synthetic construct	
ttggagtgtt tggcggactt t		21
SEQ ID NO: 1037	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1037		
tgctctggac ttgatcgctt t		21
SEQ ID NO: 1038	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1038		
tgaataagcc aggttagggt t		21
SEQ ID NO: 1039	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1039		
tgaacctagt gggttcacat t		21
SEQ ID NO: 1040	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1040		
tgacctagaa cctagtgggt t		21
SEQ ID NO: 1041	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1041		
tggaattcaa cctagacaa		19
SEQ ID NO: 1042	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1042		
ttgtctaggt tgaagtcacat t		21
SEQ ID NO: 1043	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1043		
ctgggaggat gaattcgaa		19
SEQ ID NO: 1044	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1044		
ttcgaattca tctccagct t		21
SEQ ID NO: 1045	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1045		
gggtgacaac tactatcta		19

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SEQ ID NO: 1046	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1046		
tagatagtag ttgtcacct t		21
SEQ ID NO: 1047	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1047		
ctgggaggat gaattcgaa		19
SEQ ID NO: 1048	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1048		
ttcgaattca tcctccagt t		21
SEQ ID NO: 1049	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1049		
caagtggtt ggcatctaa		19
SEQ ID NO: 1050	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1050		
ttagatgcca cccacctgt t		21
SEQ ID NO: 1051	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1051		
tccgcagcct ggatgatta		19
SEQ ID NO: 1052	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1052		
taatcatcca ggctgcggat t		21
SEQ ID NO: 1053	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
modified_base	1	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	2	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide Phosphorothioate	
modified_base	3	
	mod_base = OTHER	
	note = 2'-fluoro modified nucleotide	
modified_base	4	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	5	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	
modified_base	6	
	mod_base = OTHER	
	note = 2'-O-methyl modified nucleotide	

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modified_base      7
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      8
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     10
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     11
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     12
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     13
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     14
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     15
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     16
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     17
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base     18
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     19
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
source            1..19
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 1053
gggtgacaac tactatcta

SEQ ID NO: 1054    moltype = RNA length = 21
FEATURE
modified_base      1
                    Location/Qualifiers
                    mod_base = OTHER
                    note = Vinyl phosphonate 2'-O-methyl modified nucleotide
modified_base      2
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base      3
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base      4
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      5
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      6
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      7
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      8
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base      9
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base     10
                    mod_base = OTHER

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modified_base	note = 2'-O-methyl modified nucleotide 11 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	13 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	15 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	16 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	17 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	18 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	19 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	20 note = 2'-O-methyl modified nucleotide Phosphorothioate mod_base = OTHER
modified_base	21 note = 2'-O-methyl modified nucleotide Phosphorothioate mod_base = OTHER
source	1..21 mol_type = other RNA organism = synthetic construct
SEQUENCE: 1054	
tagatagtag ttgtcacccct t	21
SEQ ID NO: 1055	moltype = RNA length = 19
FEATURE	Location/Qualifiers
source	1..19 mol_type = other RNA organism = synthetic construct
modified_base	1 mod_base = OTHER note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-O-methyl modified nucleotide
misc_feature	2..3 note = Phosphorothioate linkage
modified_base	3 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	6 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	7 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	8 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	10 mod_base = OTHER

-continued

modified_base	note = 2'-O-methyl modified nucleotide 11 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 12 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 13 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 14 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 15 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 18 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 19 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
SEQUENCE: 1055	
ctgggaggat gaattcgaa	19
SEQ ID NO: 1056	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER
modified_base	note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide
modified_base	4 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
modified_base	5 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide
modified_base	6 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
modified_base	7 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide
modified_base	8 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
modified_base	9 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide
modified_base	10 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
modified_base	11 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide
modified_base	12 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide
modified_base	13 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide

-continued

modified_base	14	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	15	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	16	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	17	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	18	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	19	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	21	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide

SEQUENCE: 1056
 ttcgaattca tcctccagc t 21

SEQ ID NO: 1057 moltype = RNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other RNA
 organism = synthetic construct

modified_base	1	mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	2	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	3	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	4	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	5	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	6	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	7	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	8	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	9	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	10	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	11	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	12	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	14	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	15	mod_base = OTHER

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modified_base	note = 2'-fluoro modified nucleotide 16 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 17 mod_base = OTHER
modified_base	note = 2'-fluoro modified nucleotide 18 mod_base = OTHER
modified_base	note = 2'-O-methyl modified nucleotide 19 mod_base = OTHER
SEQUENCE: 1057	note = 2'-fluoro modified nucleotide
caaggtgggt ggcattctaa	19
SEQ ID NO: 1058	moltype = RNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other RNA
	organism = synthetic construct
modified_base	1 mod_base = OTHER note = Vinyl phosphonate 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	2 mod_base = OTHER note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	3 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	4 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	5 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	6 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	7 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	8 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	9 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	10 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	11 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	12 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	13 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	14 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	15 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	16 mod_base = OTHER note = 2'-fluoro modified nucleotide
modified_base	17 mod_base = OTHER note = 2'-O-methyl modified nucleotide
modified_base	18 mod_base = OTHER note = 2'-fluoro modified nucleotide

-continued

modified_base	19	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	20	mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	21	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
SEQUENCE: 1058		
ttagatgccca cccaccttgt t	21	
SEQ ID NO: 1059	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
modified_base	1	mod_base = OTHER
		note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base	2	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base	3	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	4	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	5	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	6	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	7	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	8	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	9	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	10	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	11	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	12	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	13	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	14	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	15	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	16	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	17	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
modified_base	18	mod_base = OTHER
		note = 2'-O-methyl modified nucleotide
modified_base	19	mod_base = OTHER
		note = 2'-fluoro modified nucleotide
SEQUENCE: 1059		
tccgcagcct ggatgatta	19	

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SEQ ID NO: 1060      moltype = RNA   length = 21
FEATURE              Location/Qualifiers
source               1..21
                    mol_type = other RNA
                    organism = synthetic construct
modified_base        1
                    mod_base = OTHER
                    note = Vinyl phosphonate 2'-O-methyl modified nucleotide
                    Phosphorothioate
modified_base        2
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base        3
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        4
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        5
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        6
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        7
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        8
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        9
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        10
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        11
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        12
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        13
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        14
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        15
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        16
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        17
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide
modified_base        18
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide
modified_base        19
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide Phosphorothioate
modified_base        20
                    mod_base = OTHER
                    note = 2'-fluoro modified nucleotide Phosphorothioate
modified_base        21
                    mod_base = OTHER
                    note = 2'-O-methyl modified nucleotide

SEQUENCE: 1060
taatcatcca ggctgcggat t

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21

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SEQ ID NO: 1061      moltype = AA   length = 22
FEATURE              Location/Qualifiers
source               1..22

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	mol_type = protein organism = synthetic construct	
SEQUENCE: 1061 RLIEDICLPR WGCLWEDDKD EL		22
SEQ ID NO: 1062 FEATURE source	moltype = AA length = 17 Location/Qualifiers 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 1062 MIRTLTLLSTL VAGALSK		17
SEQ ID NO: 1063 FEATURE source	moltype = AA length = 19 Location/Qualifiers 1..19 mol_type = protein organism = synthetic construct	
SEQUENCE: 1063 ILSSLTVTQL LRRHLQWIK		19
SEQ ID NO: 1064 FEATURE source	moltype = AA length = 20 Location/Qualifiers 1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 1064 MIRTLTLLSTL VAGALSKDEL		20
SEQ ID NO: 1065 FEATURE source	moltype = AA length = 25 Location/Qualifiers 1..25 mol_type = protein organism = synthetic construct	
VARIANT	6..25 note = GGGGS repeats may be deleted	
SEQUENCE: 1065 GGGSGGGGS GGGSGGGGS GGGGS		25
SEQ ID NO: 1066 FEATURE source	moltype = AA length = 40 Location/Qualifiers 1..40 mol_type = protein organism = synthetic construct	
VARIANT	3..40 note = AP repeats may be deleted	
SEQUENCE: 1066 APAPAPAPAP APAPAPAPAP APAPAPAPAP APAPAPAPAP		40
SEQ ID NO: 1067 FEATURE source	moltype = AA length = 40 Location/Qualifiers 1..40 mol_type = protein organism = synthetic construct	
VARIANT	3..40 note = AP repeats may be deleted	
SEQUENCE: 1067 APAPAPAPAP APAPAPAPAP APAPAPAPAP APAPAPAPAP		40
SEQ ID NO: 1068 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 1068 GGGS		5
SEQ ID NO: 1069 FEATURE source	moltype = AA length = 5 Location/Qualifiers 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 1069 GGGGA		5
SEQ ID NO: 1070	moltype = AA length = 6	

-continued

FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 1070		
HHHHHH		6
SEQ ID NO: 1071	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1071		
ttacggtatc tacattcta		19
SEQ ID NO: 1072	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1072		
tagaatgtag ataccgtaat t		21
SEQ ID NO: 1073	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 1073		
caaggtgggt ggcattcta		19
SEQ ID NO: 1074	moltype = RNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other RNA	
	organism = synthetic construct	
misc_feature	19	
	note = Connected to an inverted thymidine via a 3'-3' bond	
SEQUENCE: 1074		
gagccattct tgcaacaca		19
SEQ ID NO: 1075	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 1075		
tcggttgcaa agatggctct t		21

1-90. (canceled)

91. A method of treating a muscle disease, an immunological disease, a cancer, Pompe disease, or a glycogen storage disease in a subject in need thereof, the method comprising administering to the subject a pharmaceutical composition comprising a FN3 polypeptide domain linked by a linker to an siRNA molecule comprising a sense strand and an antisense strand, wherein:

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1071 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1072;

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1073 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1050; or

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1074 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEO ID NO: 1075.

92-97. (canceled)

98. The method of claim **91**, wherein administering the pharmaceutical composition delivers the siRNA molecule to a CD71 positive cell.

99. The method of claim **98**, wherein the cell is a tumor cell, a liver cell, a muscle cell, an immune cell, a dendritic cell, a heart cell, a cell of the CNS, or a cell inside the blood brain barrier.

100. The method of claim **98**, wherein the siRNA downregulates the expression of a target gene in the cell.

101. The method of claim **100**, wherein the downregulation of the expression of the target gene results in a reduction of about 99%, 90-99%, 50-90%, or 10-50%.

102. The method of claim **100**, wherein the target gene is GYS1.

103. The method of claim **102**, wherein the GYS1 has a mutation.

104. The method of claim **101**, wherein the reduced target gene and protein results in reduction in glycogen.

105. The method of claim **104**, wherein the glycogen reduction leads to improvement in Pompe disease or glycogen storage disease.

106. The method of claim **91**, wherein the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease (GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type II Diabetes/diabetic nephropathy, Aldolase A Deficiency GSD12, Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), adult polyglucosan body disease, Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11, Glucose transporter (GLUT2) deficiency, Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

107. The method of claim **91**, wherein the 5' end of the antisense strand further comprises a vinyl phosphonate modification and the 3' terminal nucleotide of the sense strand is attached to the linker.

108. The method of claim **107**, wherein:

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 706 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 707;

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 660 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 661; or

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 678 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 679.

109. The method of claim **107**, wherein:

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 704 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 705;

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 614 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 615; or

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 632 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 633.

110. The method of claim **91**, wherein the FN3 polypeptide domain binds to CD71.

111. The method of claim **110**, wherein the CD71 comprises the amino acid sequence of SEQ ID NO: 2 or SEQ ID NO: 5.

112. The method of claim **91**, wherein the sense strand of the siRNA molecule is linked to a cysteine residue in the FN3 polypeptide domain.

113. The method of claim **112**, wherein the cysteine residue in the FN3 polypeptide domain is located at a residue that corresponds to position 6, 8, 10, 11, 14, 15, 16, 20, 30, 34, 38, 40, 41, 45, 47, 48, 53, 54, 59, 60, 62, 64, 70, 88, 89, or 90 of a FN3 polypeptide domain comprising the amino acid sequence of SEQ ID NO: 445.

114. The method of claim **107**, wherein the linker comprises a molecule having the formula of $\text{Mal-C}_2\text{H}_4\text{C}(\text{O})(\text{NH})-(\text{CH}_2)_6$.

115. The method of claim **91**, wherein the FN3 polypeptide domain comprises an amino acid sequence at least 95% identical to the amino acid sequence of one of SEQ ID NOs: 273, 301-310, 312-572, 592-599, or 708-710.

116. The method of claim **91**, wherein the FN3 polypeptide domain comprises the amino acid sequence of any one of SEQ ID NOs: 273, 301-310, 312-572, 592-599, or 708-710.

117. The method of claim **91**, wherein the FN3 polypeptide domain comprises the amino acid sequence of SEQ ID NO: 509.

118. A method of treating a muscle disease, an immunological disease, a cancer, Pompe disease, or a glycogen storage disease in a subject in need thereof, the method comprising administering to the subject a pharmaceutical composition comprising a FN3 polypeptide domain linked by a linker to an siRNA molecule comprising a sense strand and an antisense strand, wherein the FN3 polypeptide domain comprises the amino acid sequence of SEQ ID NO: 509, and wherein:

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1071 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1072;

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1073 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1050; or

the sense strand is a modified oligonucleotide 19-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1074 and the antisense strand is a modified oligonucleotide 17-23 nucleotides in length that comprises the nucleobase sequence of SEQ ID NO: 1075.

119. The method of claim **118**, wherein administering the pharmaceutical composition delivers the siRNA molecule to a CD71 positive cell.

120. The method of claim **119**, wherein the CD71 positive cell is a tumor cell, a liver cell, a muscle cell, an immune cell, a dendritic cell, a heart cell, a cell of the CNS, or a cell inside the blood brain barrier.

121. The method of claim **119**, wherein the siRNA downregulates the expression of a target gene in the CD71 positive cell.

122. The method of claim **121**, wherein the downregulation of the expression of the target gene results in a reduction of about 99%, 90-99%, 50-90%, or 10-50%.

123. The method of claim **121**, wherein the target gene is GYS1.

124. The method of claim **123**, wherein the GYS1 has a mutation.

125. The method of claim **121**, wherein the reduced target gene and protein results in reduction in glycogen.

126. The method of claim **125**, wherein the glycogen reduction leads to improvement in Pompe disease or glycogen storage disease.

127. The method of claim **118**, wherein the glycogen storage disease is selected from the group consisting of Cori's disease or Forbes' disease (GSD3, Glycogen debranching enzyme (AGL) deficiency), McArdle disease

(GSD5, Muscle glycogen phosphorylase (PYGM) deficiency), type II Diabetes/diabetic nephropathy, Aldolase A Deficiency GSD12, Lafora Disease, hypoxia, Andersen disease (GSD4, Glycogen debranching enzyme (GBE1) deficiency), Tarui's Disease (GSD7, Muscle phosphofructokinase (PFKM) deficiency), adult polyglucosan body disease, Glycogen synthase (GYS2) deficiency (GSD0), Glucose-6-phosphatase (G6PC/SLC37A4) deficiency (GSD1, von Gierke's disease), Hers' disease (GSD6, Liver glycogen phosphorylase (PYGL) or Muscle phosphoglycerate mutase (PGAM2) deficiency), Phosphorylase kinase (PHKA2/PHKB/PHKG2/PHKA1) deficiency (GSD9), Phosphoglycerate mutase (PGAM2) deficiency (GSD10), Muscle lactate dehydrogenase (LDHA) deficiency (GSD11), Fanconi-Bickel syndrome (GSD11, Glucose transporter (GLUT2) deficiency, Aldolase A deficiency (GSD12), β -enolase (ENO3) deficiency (GSD13), and Glycogenin-1 (GYG1) deficiency (GSD15).

128. The method of claim **118**, wherein the 5' end of the antisense strand further comprises a vinyl phosphonate modification and the 3' terminal nucleotide of the sense strand is attached to the linker.

129. The method of claim **128**, wherein:

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 706 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 707;

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 660 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 661; or
the sense strand comprises the modified oligonucleotide of SEQ ID NO: 678 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 679.

130. The method of claim **128**, wherein:

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 704 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 705;

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 614 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 615; or

the sense strand comprises the modified oligonucleotide of SEQ ID NO: 632 and the antisense strand comprises the modified oligonucleotide of SEQ ID NO: 633.

* * * * *